



**US Army Corps
of Engineers®**
New York District

THIS IS AN UNRESTRICTED PROCUREMENT AISC CODE 237990
SAFETY PAYS

MAINTENANCE DREDGING

of

**NEW YORK & NEW JERSEY CHANNELS
- SEGUINE/WARD/OUTERBRIDGE
FEDERAL NAVIGATION PROJECT**

Solicitation &

Specifications

IFB No. W912DS-26-B-A014

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DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

SECTION 00 00 02

CHECK LIST FOR BIDDERS

04/21

PART 1 GENERAL

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SECTION 00 00 02

CHECK LIST FOR BIDDERS
04/21

PART 1 GENERAL

ATTACHED IS IFB FOR

**MAINTENANCE DREDGING
NEW YORK & NEW JERSEY CHANNEL,
SEGUINE/WARD/OUTERBRIDGE REACHES,
FEDERAL NAVIGATION PROJECT**

ALL INFORMATION REQUIRED BY THE TERMS OF THIS SOLICITATION MUST BE FURNISHED. MISTAKES OR OMISSIONS MAY RENDER YOUR BID INELIGIBLE FOR AWARD. IMPORTANT ITEMS FOR YOU TO CHECK ARE INCLUDED IN BUT NOT LIMITED TO THOSE LISTED BELOW. THIS INFORMATION IS FURNISHED ONLY TO ASSIST YOU IN SUBMITTING A PROPER BID.

___ HAVE YOU ACKNOWLEDGED ALL AMENDMENTS?

___ HAVE YOU COMPLETED THE "REPRESENTATIONS AND CERTIFICATIONS" (SECTION 00 45 00) PORTION OF THE SOLICITATION?

___ IS YOUR DUNS NUMBER LISTED ON THE STANDARD FORM 1442?

___ IS YOUR BID PROPERLY SIGNED?

___ A BID BOND IS REQUIRED. HAS YOUR SURETY PROVIDED YOU WITH A BID BOND ON STANDARD FORM 24 OR A SIMILAR FORM CONTAINING THE SAME LANGUAGE AS A STANDARD FORM 24?

___ IS YOUR BID GUARANTEE IN THE PROPER AMOUNT?

___ IS YOUR BID GUARANTEE PROPERLY SIGNED BY BOTH THE BIDDER AND SURETY AND ARE ALL REQUIRED SEALS AFFIXED?

___ IS THE NAME IN WHICH YOU SUBMITTED THE BID THE SAME ON YOUR BID AS ON THE BID BOND?

___ IS YOUR BID BOND INCLUDED WITH YOUR BID? (A LATE BID GUARANTEE IS TREATED THE SAME AS A LATE BID)

___ HAVE YOU ENSURED THAT YOU HAVE NOT RESTRICTED YOUR BID BY ALTERING THE PROVISIONS OF THE SOLICITATION?

___ WHEN REQUIRED, HAVE YOU ENTERED A UNIT PRICE FOR EACH BID ITEM? (THE SOLICITATION SPECIFICALLY STATES WHEN THIS IS NECESSARY.)

___ ARE DECIMALS IN YOUR PRICES IN THE PROPER PLACE? ARE YOUR FIGURES LEGIBLE?

___ IF YOU HAVE MADE ERASURES OR CORRECTIONS ON YOUR BID, ARE THEY INITIALED BY THE PERSON SIGNING THE BID?

___ DOES THE ENVELOPE CONTAINING YOUR BID PROPERLY IDENTIFY THAT IT IS A

SEALED BID AND DOES IT CONTAIN THE CORRECT SOLICITATION NUMBER AND BID
OPENING TIME?

___ WILL YOUR BID ARRIVE ON TIME? (SEE PARAGRAPH ENTITLED "LATE SUBMISSIONS,
MODIFICATIONS, AND WITHDRAWALS OF BIDS" IN THE INSTRUCTIONS, CONDITIONS,
AND NOTICES TO BIDDERS, SECTION 00 21 16 OF THE SOLICITATION.)

BIDDER IS RESPONSIBLE TO ENSURE BID IS SUBMITTED ON TIME.

-- End of Section --

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DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

DOCUMENT 00 01 15

LIST OF DRAWINGS

02/24

PART 1 GENERAL

1.1 SUMMARY

1.2 CONTRACT DRAWINGS, MAPS, & SPECIFICATIONS

-- End of Section Table of Contents --

DOCUMENT 00 01 15

LIST OF DRAWINGS
02/24

PART 1 GENERAL

1.1 SUMMARY

This section lists the drawings for the project pursuant to contract clause "DFARS 252.236-7001, Contract Drawings and Specifications."

1.2 CONTRACT DRAWINGS, MAPS, & SPECIFICATIONS

- a. The Government will provide to the Contractor, without charge, one set of contract drawings and specifications, except publications incorporated into the technical provisions by reference, in electronic or paper media as chosen by the Contracting Officer.
- b. The Contractor shall:
 - (1) Check all drawings furnished immediately upon receipt;
 - (2) Compare all drawings and verify the figures before laying out the work;
 - (3) Promptly notify the Contracting Officer or Contracting Officer's Representative of any discrepancies;
 - (4) Be responsible for any errors that might have been avoided by complying with this paragraph (b); and
 - (5) Reproduce and print contract drawings and specifications as needed.
- c. In general:
 - (1) Large-scale drawings shall in general govern over small-scale drawings. Information and data provided in contract specifications shall govern over similar information and data specified in contract drawings; and
 - (2) The Contractor shall follow figures marked on drawings in preference to scale measurements.
- d. Omissions from the drawings or specifications or the misdescription of details of work which are manifestly necessary to carry out the intent of the drawings and specifications, or that are customarily performed, shall not relieve the Contractor from performing such omitted or misdescribed details of work. The Contractor shall perform such details as if fully and correctly set forth and described in the drawings and specifications.
- e. The work shall conform to the following contract drawings and maps, all of which form a part of these specifications and are available in the office of the District Engineer, U.S. Army Corps of Engineers, New York District, 26 Federal Plaza, New York, N.Y.

10278-0090:

Contract drawings are as follows:

MAINTENANCE DREDGING NY & NJ CHANNELS- SEGUINE/WARD/OUTERBRIDGE REACHES	
DRAWING NO.	TITLE
G-001	COVER SHEET & INDEX
GI-101	PROJECT SITE PLAN & NOTES
C-101	DREDGING AREAS & SOUNDING MAP
C-102	DREDGING AREAS & SOUNDING MAP
C-103	DREDGING AREAS & SOUNDING MAP
C-104	DREDGING AREAS & SOUNDING MAP
C-105	DREDGING AREAS & SOUNDING MAP
C-106	DREDGING AREAS & SOUNDING MAP
C-107	DREDGING AREAS & SOUNDING MAP
C-108	DREDGING AREAS & SOUNDING MAP

f. The Contractor shall check all drawings furnished to him immediately upon their receipt and shall promptly notify the Contracting Officer or Contracting Officer's Representative of any discrepancies. Figures marked on drawings shall in general be followed in preference to scale measurements unless dimensions are specified. Large-scale drawings shall in general govern over small-scale drawings. Information and data provided in contract specifications shall govern over similar information and data specified in contract drawings. The Contractor shall compare all drawings and verify the figures before laying out the work and will be responsible for any errors which might have thereby been avoided (DOD FAR Suppl. 52.236-7002).

-- End of Section --

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DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

SECTION 00 10 00

SF 1442 AND BIDDING SCHEDULE

01/20

PART 1 GENERAL

- 1.1 SF 1442 SOLICITATION, OFFER, AND AWARD
- 1.2 BID SCHEDULE

-- End of Section Table of Contents --

SECTION 00 10 00

SF 1442 AND BIDDING SCHEDULE
01/20

PART 1 GENERAL

1.1 SF 1442 SOLICITATION, OFFER, AND AWARD

To be provided by Contracting Division.

1.2 BID SCHEDULE

To be provided by Contracting Division.

-- End of Section --

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DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

SECTION 00 21 13

INSTRUCTIONS, CONDITIONS, AND NOTICE TO BIDDERS

01/20

PART 1 GENERAL

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SECTION 00 21 13

INSTRUCTIONS, CONDITIONS, AND NOTICE TO BIDDERS
01/20

PART 1 GENERAL

To be provided by Contracting Division.

-- End of Section --

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DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

SECTION 00 45 00

REPRESENTATION AND CERTIFICATION

01/20

PART 1 GENERAL

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SECTION 00 45 00

REPRESENTATION AND CERTIFICATION
01/20

PART 1 GENERAL

To be provided by Contracting Division.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

SECTION 00 90 00

WAGE RATES

01/20

PART 1 GENERAL

1.1 NOTE

-- End of Section Table of Contents --

SECTION 00 90 00

WAGE RATES
01/20

PART 1 GENERAL

1.1 NOTE

Wage rates provided by Contracting Division.

A copy of the wage rate shall be displayed at a highly visible location on the dredge at all times for the duration of the project.

-- End of Section --

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DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

SECTION 00 90 01

LIST OF DOCUMENTS, EXHIBITS, AND OTHER ATTACHMENTS

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PART 2 PRODUCTS

PART 3 EXECUTION

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Project Location Reference Map

3.1.2 Attachment B Sample Daily Construction Quality Control Report

3.1.3 Attachment C Project Sign and Safety Sign

3.1.4 Attachment D Oil and Hazardous Material Operations

3.1.5 Attachment E Guideline for Preparing the Accident Prevention
Plan

3.1.6 Attachment F Daily Reports for Dredging

3.1.7 Attachment G USACE Transportation and Placement Log

3.1.8 Attachment H Daily Site Safety and Health Inspection Log

3.1.9 Attachment I Daily Whale and Turtle Report Logs, Incident
Report of Sturgeon

3.1.10 Attachment J Submittal Register

3.1.11 Attachment K Sampling and Testing Results (September 2025 EPA
Super-Memo)

3.1.12 Attachment L Federal Aviation Administration (FAA) Obstruction
Evaluation / Airport Airspace Analysis (OE/AAA) Desk Referenace
Guide

3.1.13 Attachment M XYZ Data of Plans and Specifications Survey

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SECTION 00 90 01

LIST OF DOCUMENTS, EXHIBITS, AND OTHER ATTACHMENTS
04/08

PART 1 GENERAL

1.1 SUBMITTALS

Not Used

PART 2 PRODUCTS

Not Used.

PART 3 EXECUTION

3.1 List of Attachments

The following attachments have been provided for reference to the bidder for use in performing the contract requirements.

3.1.1 Attachment A NY&NJ Channels- Seguine/Ward/Outerbridge Reaches Project Location Reference Map

3.1.2 Attachment B Sample Daily Construction Quality Control Report

3.1.3 Attachment C Project Sign and Safety Sign

3.1.4 Attachment D Oil and Hazardous Material Operations

3.1.5 Attachment E Guideline for Preparing the Accident Prevention Plan

3.1.6 Attachment F Daily Reports for Dredging

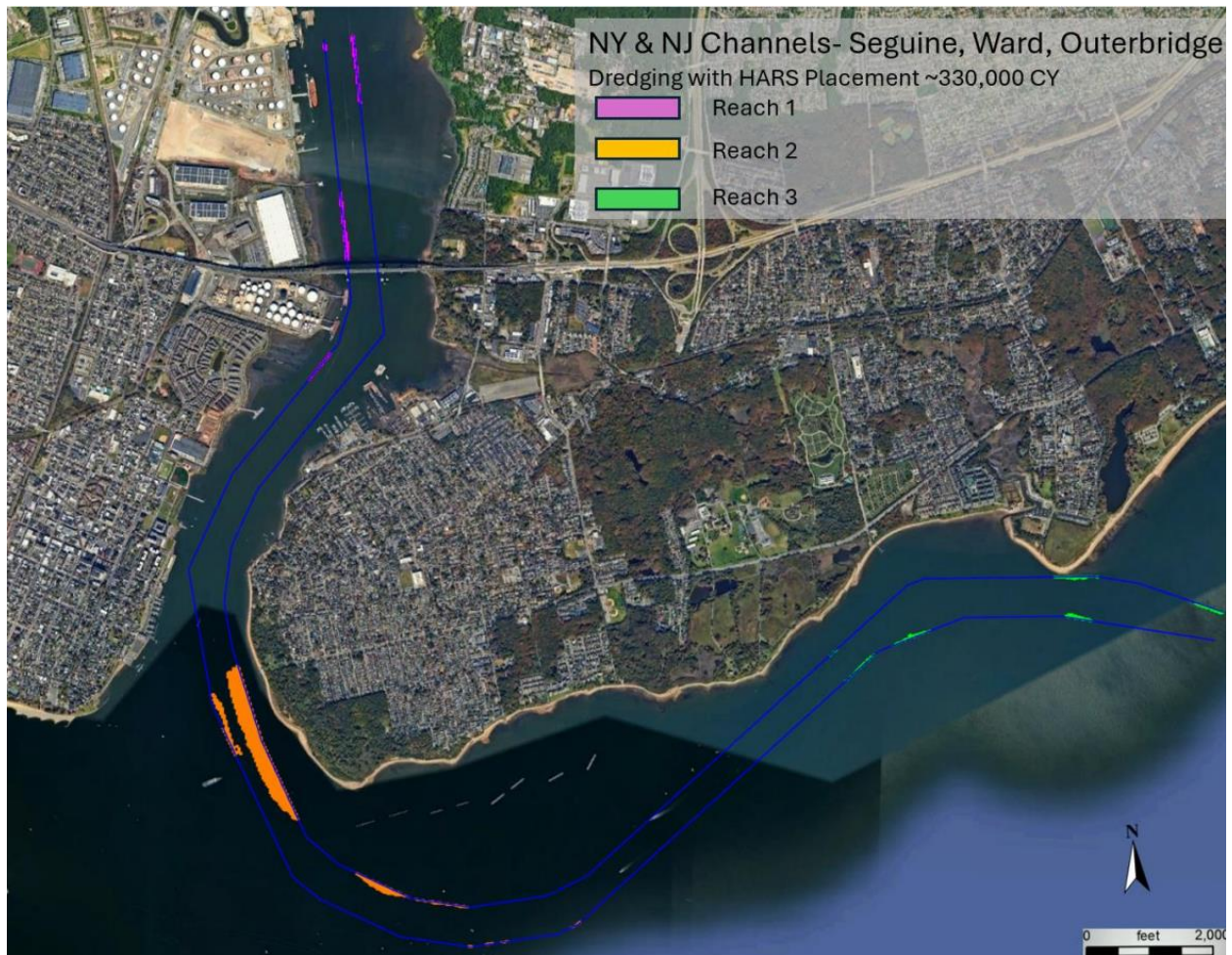
RMS Form 4267.

- 3.1.7 Attachment G USACE Transportation and Placement Log
- 3.1.8 Attachment H Daily Site Safety and Health Inspection Log
- 3.1.9 Attachment I Daily Whale and Turtle Report Logs, Incident Report of Sturgeon
- 3.1.10 Attachment J Submittal Register
- 3.1.11 Attachment K Sampling and Testing Results (September 2025 EPA Super-Memo)
- 3.1.12 Attachment L Federal Aviation Administration (FAA) Obstruction Evaluation / Airport Airspace Analysis (OE/AAA) Desk Referenace Guide
- 3.1.13 Attachment M XYZ Data of Plans and Specifications Survey

-- End of Section --

General Project Location Reference Map

NY&NJ Channels-Seguine/Ward/Outerbridge Reaches



CONTRACTORS QUALITY CONTROL REPORT (QCR) DAILY LOG OF CONSTRUCTION		REPORT NUMBER																	
		XXX																	
		DATE																	
		01 Sep 2026 - Monday																	
PROJECT Maintenance Dredging of NY&NJ Channels- Seguine/Ward/Outerbridge Reaches		CONTRACT NUMBER W912DS26CXXXX																	
CONTRACTOR Dredging Contractor Co.	WEATHER Weather Caused No Delay Temperature Min X F Max X F No Precipitation No Wind																		
QC Narratives XXX PREP/INITIAL DATES Preparatory inspections held today: XXX Initial inspections held today: XXX ACTIVITY START/FINISH XXX activities were started or finished today QC REQUIREMENTS XXX QC requirements were entered today ISSUED QA/QC DEFICIENCIES XXX QC Deficiency items were issued today CORRECTED QA/QC DEFICIENCY XXX QC Deficiency items were corrected today CONTRACTORS ON SITE XXX Contractors had their first or last day on the site																			
LABOR HOURS <table style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="text-align: left;">Crew</th> <th style="text-align: left;">Position</th> <th style="text-align: right;">Number of Employees</th> <th style="text-align: right;">Hours Worked</th> </tr> </thead> <tbody> <tr> <td>No labor hours to report</td> <td>N/A</td> <td style="text-align: right;">X.00</td> <td style="text-align: right;">X.00</td> </tr> <tr> <td colspan="2">Total hrs worked to date: XX.00</td> <td>Hrs worked MTD: XX.00</td> <td>Total: X.00</td> </tr> </tbody> </table>				Crew	Position	Number of Employees	Hours Worked	No labor hours to report	N/A	X.00	X.00	Total hrs worked to date: XX.00		Hrs worked MTD: XX.00	Total: X.00				
Crew	Position	Number of Employees	Hours Worked																
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Total hrs worked to date: XX.00		Hrs worked MTD: XX.00	Total: X.00																
EQUIPMENT HOURS <table style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="text-align: left;">Equipment Name</th> <th style="text-align: left;">Description</th> <th style="text-align: right;">Idle Hours</th> <th style="text-align: right;">Operating Hours</th> </tr> </thead> <tbody> <tr> <td colspan="2">No equipment hours to report</td> <td style="text-align: right;">X.00</td> <td style="text-align: right;">X.00</td> </tr> <tr> <td colspan="2">Total operating hrs to date: X,XXX.00</td> <td>Idle hrs MTD: XX.00</td> <td>Opr hrs MTD: XX.00</td> </tr> <tr> <td colspan="2"></td> <td>Total: X.00</td> <td>X.00</td> </tr> </tbody> </table>				Equipment Name	Description	Idle Hours	Operating Hours	No equipment hours to report		X.00	X.00	Total operating hrs to date: X,XXX.00		Idle hrs MTD: XX.00	Opr hrs MTD: XX.00			Total: X.00	X.00
Equipment Name	Description	Idle Hours	Operating Hours																
No equipment hours to report		X.00	X.00																
Total operating hrs to date: X,XXX.00		Idle hrs MTD: XX.00	Opr hrs MTD: XX.00																
		Total: X.00	X.00																
Page 1 of 2																			

CONTRACTORS QUALITY CONTROL REPORT (QCR) DAILY LOG OF CONSTRUCTION		REPORT NUMBER	
		XXX	
PROJECT		DATE	
		01 Sep 2026 - Monday	
Maintenance Dredging of NY&NJ Channels- Seguine/Ward/Outerbridge Reaches		CONTRACT NUMBER	
		W912DS26CXXXX	
MISHAP REPORTING No mishaps reported today			
CONTRACTOR CERTIFICATION On behalf of the contractor, I certify that this Report is complete and correct and all equipment and material used and work performed during this Reporting period are in compliance with the contract plans and specifications, to the best of my knowledge, except as noted above.			
QC REPRESENTATIVE SIGNATURE	DATE	SUPERINTENDENT'S INITIALS	DATE
Page 2 of 2			

Example Rendering Dimensions: (For Dimension Reference Only)



Example Rendering:



The use of signs to identify Corps managed or supervised design, construction, and rehabilitation projects - both for military and civil works - is an important part of efforts to keep the public informed of Corps work. For this purpose, a construction project sign package has been adopted. This package consists of two signs: one for project identification and the other to show on-the-job safety performance of the contractor.

These two signs are to be displayed side by side and mounted for reading by passing viewers. Exact placement location will be designated by the contracting officer's representative.

The panel sizes and graphic formats have been standardized for visual consistency throughout all Corps operations.

Panels are fabricated using HDO plywood or aluminum with dimensional lumber uprights and bracing. The sign faces are nonreflective vinyl.

All legends are to be die-cut or computer-cut in the sizes and typefaces specified and applied to the white panel background following the graphic formats shown on pages 16-2 and 16-3. The Communication Red panel on the left side of the construction project sign with Corps Signature (reverse version) is screen-printed onto the white background.

A display of these two signs is shown on the following two pages. Mounting and fabrication details are provided on page 16-4.

Special applications or situations not covered in these guidelines should be referred to the district Sign Program Manager.

Below are two samples of the Construction Project Identification sign showing how this panel is adaptable for use to identify either military (top) or civil works projects (bottom). The graphic format for this 4'x 6' sign panel follows the legend guidelines and layout as specified below. The large 4'x 4' section of the panel on the right is to be white with black legend. The 2'x 4' section of the sign on the left

with the full Corps Signature (reverse version) is to be screen-printed Communication Red on the white background. The designation of a sponsor in the area indicated is optional with Military or Civil Works construction signs. Signs may list one sponsoring entity. If agreement on a sponsor designation cannot be achieved, the area should be left blank.

This sign is to be placed with the Safety Performance sign shown on the following page. Mounting and fabrication details are provided on page 16-4.

Special applications or situations not covered in these guidelines should be referred to the district Sign Program Manager.

Legend Group 1: One- to two-line description of Corps relationship to project.

Color: White

Typeface: 1.25" Helvetica Regular

Maximum line length: 19"

Legend Group 2: Division or District Name (optional). Placed below 10.5" reverse

Signature (6" Castle).

Color: White

Typeface: 1.25" Helvetica Regular

Legend Group 2a: One- to three-line identification of Military or Civil Works sponsor (optional). Place below Corps Signature to cross-align with Group 5a-b.

Color: White

Typeface: 1.25" Helvetica Regular

Maximum line length: 19"

Legend Group 3: One- to three-line project title legend describes the work being done under this contract.

Color: Black

Typeface: 3" Helvetica Bold

Maximum line length: 42"

Legend Group 4: One- to two-line identification of project or facility (civil works) or name of sponsoring department (military).

Color: Black

Typeface: 1.5" Helvetica Regular

Maximum line length: 42"

Cross-align the first line of Legend Group 4 with the first line of the Corps Signature (US Army Corps) as shown.

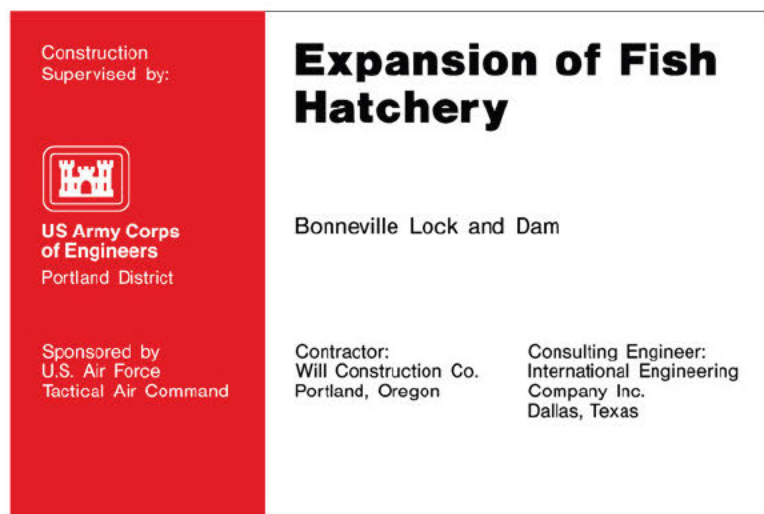
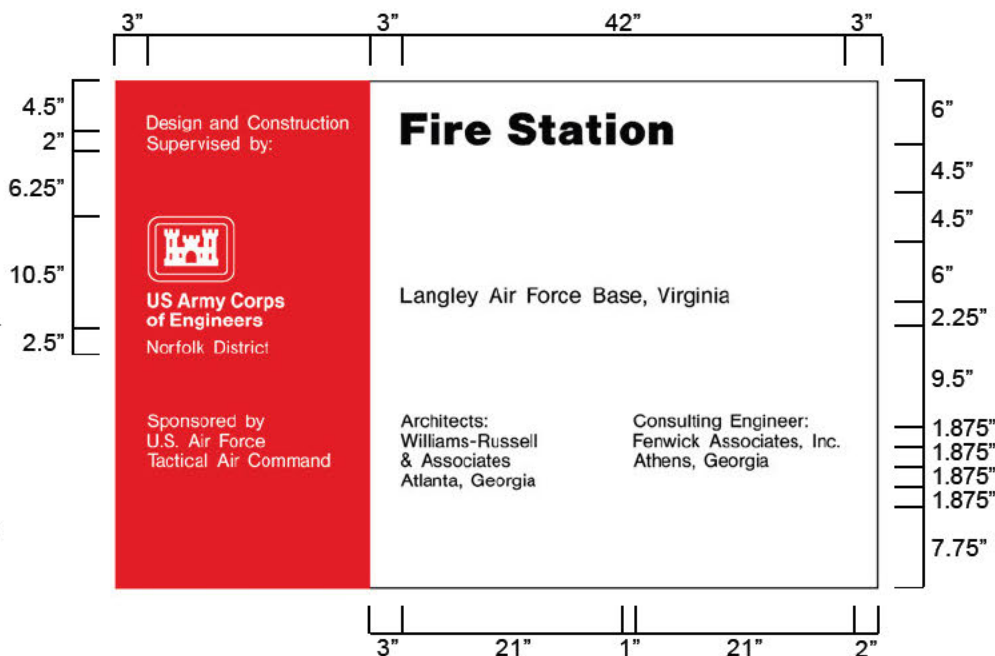
Legend Groups 5a-b: One- to five-line identification of prime contractors including: type (architect, general contractor, etc.), corporate or firm name, city, state. Use of Legend Group 5 is optional.

Color: Black

Typeface: 1.25" Helvetica Regular

Maximum line length: 21"

All typography is flush left and rag right, upper and lower case with initial capitals only as shown. Letter- and word-spacing to follow Corps standards as specified in Appendix D.



Sign Type	Legend Size (A)	Panel Size	Post Size	Specification Code	Mounting Height	Color Bkg/Lgd
CID-01	various	4'x6'	4"x4"	HDO-3	48"	WH-RD/BK

Each contractor's safety record is to be posted on Corps managed or supervised construction projects and mounted with the Construction Project Identification sign specified on page 16-2.

The graphic format, color, size and typefaces used on the sign are to be reproduced exactly as specified below. The

title with First Aid logo in the top section of the sign, and the performance record captions are standard for all signs of this type. Legend groups 2 and 3 below identify the project and the contractor and are to be placed on the sign as shown.

Safety record numbers are mounted on individual metal plates and are screw-

mounted to the background to allow for daily revisions to posted safety performance record.

Special applications or situations not covered in these guidelines should be referred to the district Sign Program Manager.

Legend Group 1: Standard two-line title "Safety is a Job Requirement" with 8" (outside diameter) Safety Green first aid logo.
Color: To match Pantone system 347
Typeface: 3" Helvetica Bold
Color: Black

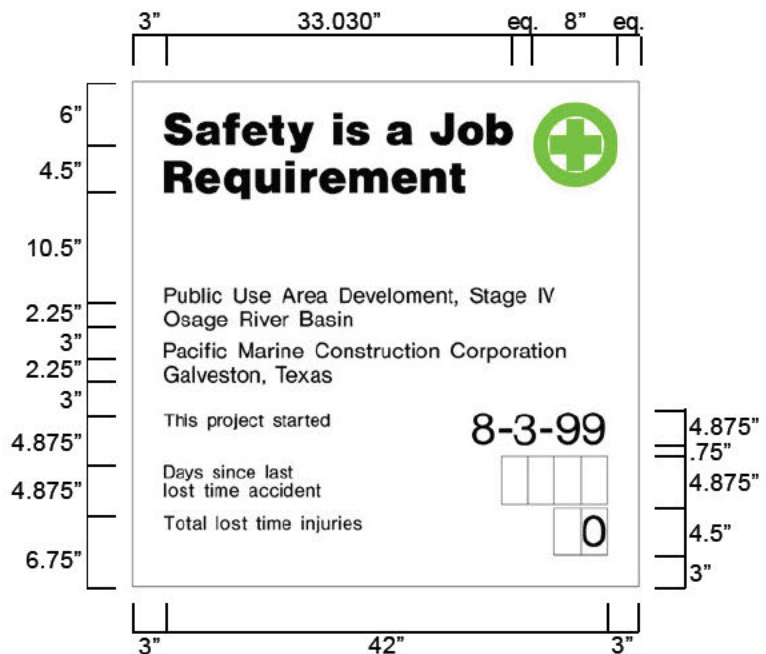
Legend Group 2: One- to two-line project title legend describes the work being done under this contract and name of host project.
Color: Black
Typeface: 1.5" Helvetica Regular
Maximum line length: 42"

Legend Group 3: One- to two-line identification: name of prime contractor and city, state address. Color: Black
Typeface: 1.5" Helvetica Regular
Maximum line length: 42"

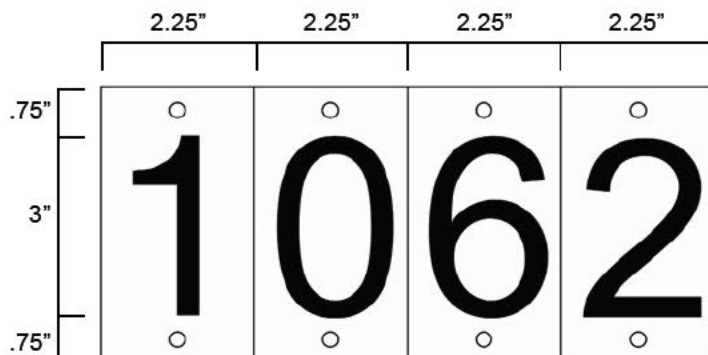
Legend Group 4: Standard safety record captions as shown.
Color: Black
Typeface: 1.25" Helvetica Regular

Replaceable numbers are to be mounted on white .060 aluminum plates and screw-mounted to background.
Color: Black
Typeface: 3" Helvetica Regular
Plate size: 2.5" x 4.5"

All typography is flush left and rag right, upper and lower case with initial capitals only as shown. Letter- and word-spacing to follow Corps standards as specified in Appendix D.



Sign Type	Legend Size (A)	Panel Size	Post Size	Specification Code	Mounting Height	Color Bkg/Lgd
CID-02	various	4'x4'	4"x4"	HDO-3	48"	WH/BK-SG



All Construction Project Identification signs and Safety Performance signs are to be fabricated and installed as described below. The signs are to be erected at a location designated by the contracting officer representative and shall conform to the size, format, and typographic standards shown on pages 16-2 and 16-3. Detailed specifica-

tions for HDO plywood panel preparation are provided in Appendix B.

Shown below the mounting diagram is a panel layout grid with spaces provided for project information. Photocopy this page and use as a worksheet when preparing sign legend orders.

For additional information on the proper method to prepare sign panel graphics, contact the district Sign Program Manager.

The sign panels are to be fabricated from .75" High Density Overlay Plywood. Panel preparation to follow HDO specifications provided in Appendix B.

Sign graphics to be prepared on a white nonreflective vinyl film with positionable adhesive backing.

All graphics except for the Communication Red background with Corps Signature on the project sign are to be die-cut or computer-cut nonreflective vinyl, prespaced legends prepared in the sizes and typefaces specified and applied to the background panel following the graphic formats shown on pages 16-2 and 16-3.

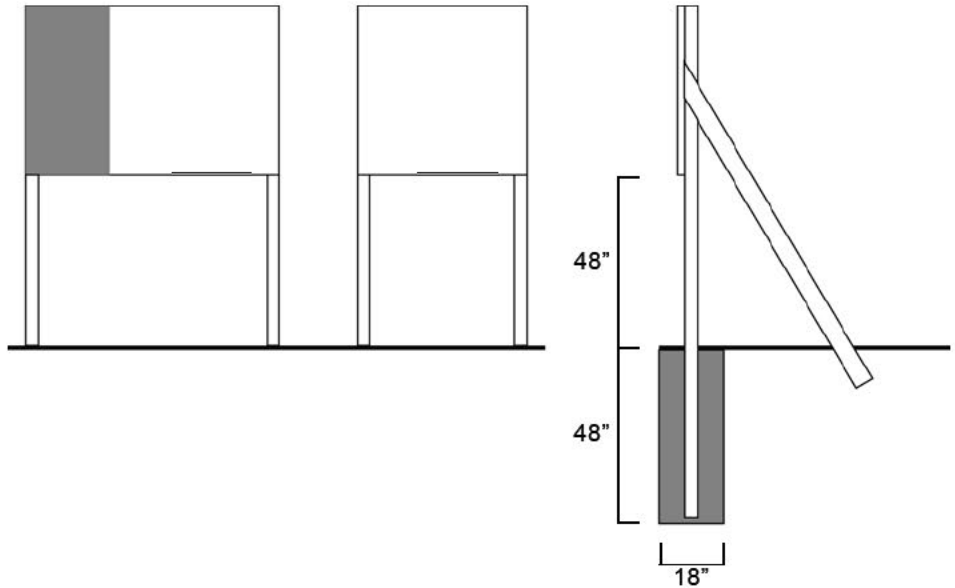
The 2'x 4' Communication Red panel (to match Pantone system 032) with full Corps Signature (reverse version) is to be screen-printed on the white background. Identification of the district or division may be applied under the signature with white cut vinyl letters prepared to Corps standards.

Drill and insert six (6) .375" T-nuts from the front face of the HDO sign panel. Position holes as shown. Flange of T-nut to be flush with sign face.

Apply graphic panel to prepared HDO plywood panel following manufacturers' instructions.

Sign uprights to be structural grade 4" x 4" treated Douglas Fir or Southern Yellow Pine, No.1 or better. Post to be 12' long. Drill six (6) .375" mounting holes in uprights to align with T-nuts in sign panel. Countersink (.5") back of hole to accept socket head cap screw (4" x .375").

Assemble sign panel and uprights. Imbed assembled sign panel and uprights in 4' hole. Local soil conditions and/or wind loading may require bolting additional 2" x 4" struts on inside face of uprights to reinforce installation as shown.



Construction Project Identification Sign
Legend Group 1: Corps Relationship

1. _____
2. _____

Legend Group 2: Division/District Name

1. _____
2. _____

Legend Group 2a: Military/Civil Works Sponsor

1. _____
2. _____

Legend Group 3: Project Title

1. _____
2. _____
3. _____

Legend Group 4: Facility Name

1. _____
2. _____

Legend Group 5: Contractor/A&E

1. _____
2. _____
3. _____
4. _____
5. _____

Legend Group 5b: Contractor/A&E

1. _____
2. _____
3. _____
4. _____
5. _____

Safety Performance Sign

Legend Group 2: Project Title

1. _____
2. _____

Legend Group 3: Contractor/A&E

1. _____
2. _____

The illustrations below show the various color configurations possible when using the Corps Signature on signs. Note that there are fewer possible ways to render the Signature on a sign panel than are specified for print applications (see *Graphic Standards Manual*, pages 1-5).

a) The most prominent use of the Signature will be on identification signs. For Standard Identification, post and panel signs, the reverse Signature is used; the Mark is Communication Red, the Signature type is white.



b) For large-scale Standard Identification signs of individual fabricated letters, the positive version is used; the Mark is Communication Red and the Signature typography is white (see page 5-7).



c) The Corps Participation Credit sign uses an all white reverse Signature on a Corps Brown background.



d) The header panels on Building Office Directories use a reverse Signature in white on Dark Grey.



e) Construction Project Identification signs use an all white reverse Signature on a Communication Red background.



f) Boundary signs use the positive Signature in black on a white background.



Within the Corps sign system there are five standard color palettes. Three have been developed by the Corps and include: 1) Recreation Area signs, 2) Lock, Dam and Waterway signs, and 3) Office Interior signs. Two color groups have been adopted from existing standards: 1) Traffic signs (MUTCD) and 2) Workplace Safety signs (ANSI). Each of these is illustrated on the following pages with descriptions for their use. The two-character color code is in parentheses immediately after the color. Additional color application instructions are included in each respective section.

Colors must conform to the standards presented on the following pages when preparing signs.

For many of the colors shown on the next five pages, a corresponding Federal Standard Color number is listed. These numbers refer to color samples contained in a fan deck titled *Federal Standard 595B Colors*. The fan deck is published by the General Services Administration, order number 7690-01-162-2210.

Shown below are the colors for use on Corps identification, directional, and recreation area signs.

Corps Brown (BR): Background for identification, directional, recreation, and symbol signs. The closest Federal Standard Color is 20095.



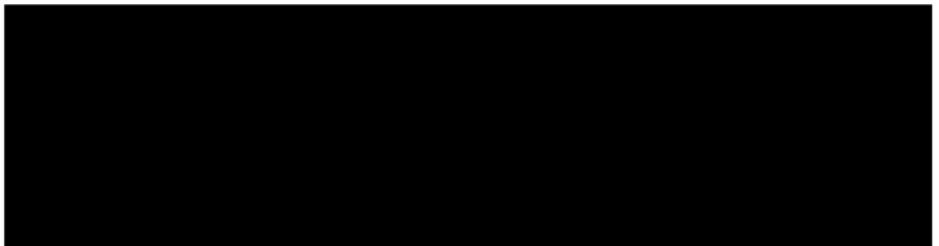
White (WH): Legend for identification, directional, and recreation signs. Background for boundary signs. The closest Federal Standard Color is 27925, but the match with the 595B fan deck is not exact.



Communication Red (CR): Corps Mark (Castle) on identification signs. The closest Federal Standard Color is 11350, but the match with the 595B fan deck is not exact. The Graphics Standards Manual (EP 310-1-6) specifies Communication Red shall match Pantone Red 032.



Black (BK): Legend and Signature for boundary signs. The closest Federal Standard Color is 17038.



The colors shown below are used on all safety signs as described in Section 11 of this manual.

Safety Red (SR): Federal Standard Color 11310, but the match with the 595B fan deck is not exact. Danger; warning of an immediate hazard.



Safety Yellow (SY): Federal Standard Color 13591, but the match with the 595B fan deck is not exact. Caution; warning of potential hazard.



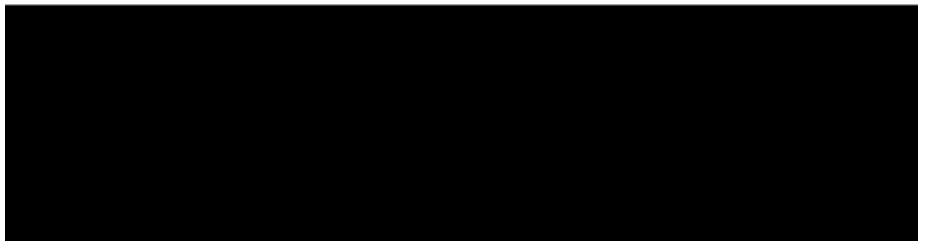
Safety Green (SG): Federal Standard Color 14109, but the match with the 595B fan deck is not exact. Notice; for safety.



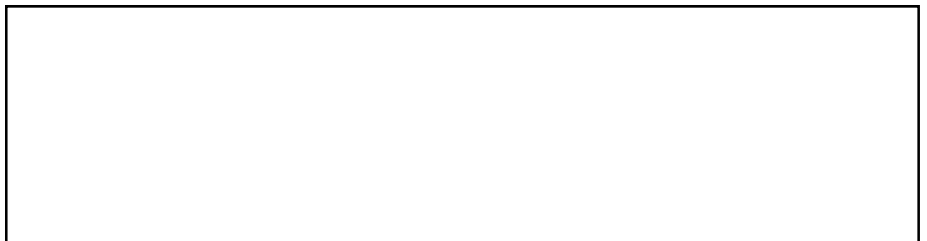
Safety Blue (SB): Federal Standard Color 15092. Information; general.



Black (SK): Federal Standard Color 17038. Directional and all descriptive legends.



White (SW): Federal Standard Color 27875. All sign backgrounds, except for Caution.



[Code of Federal Regulations]
[Title 33, Volume 2]
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[Page 314-316]

TITLE 33--NAVIGATION AND NAVIGABLE WATERS

CHAPTER I--COAST GUARD, DEPARTMENT OF HOMELAND SECURITY (CONTINUED)

PART 154_FACILITIES TRANSFERRING OIL OR HAZARDOUS MATERIAL IN BULK --Table of Contents

Subpart A_General

Sec. 154.105 Definitions.

As used in this part:

Barrel means a quantity of liquid equal to 42 U.S. gallons. Boundary Line means any of the lines described in 46 CFR part 7. Captain of the Port (COTP) means the U.S. Coast Guard officer commanding a Captain of the Port Zone described in Part 3 of this chapter, or that person's authorized representative.

Caretaker Status denotes a facility where all piping, hoses, loading arms, storage tanks, and related equipment in the marine transfer area are completely free of oil or hazardous materials, where these components have been certified as being gas free, where piping, hoses, and loading arms terminating near any body of water have been blanked, and where the facility operator has notified the COTP that the facility will be in caretaker status.

Commandant means the Commandant of the Coast Guard or an authorized representative.

Contiguous Zone means the entire zone established by the United States under Article 24 of the Convention on the Territorial Sea and the Contiguous Zone, but not extending beyond 12 miles from the baseline from which the breadth of the territorial sea is measured.

District Commander means the officer of the Coast Guard designated by the Commandant to command a Coast Guard District, as described in Part 3 of this chapter or an authorized representative.

Facility means either an onshore or offshore facility, except for an offshore facility operating under the jurisdiction of the Secretary of the Department of Interior, and includes, but is not limited to, structure, equipment, and appurtenances thereto, used or capable of being used to

transfer oil or hazardous materials to or from a vessel or public vessel. Also included are facilities that tank clean or strip and any floating structure that is used to support an integral part of the facility's operation. A facility includes federal, state, municipal, and private facilities.

Facility operator means the person who owns, operates, or is responsible for the operation of the facility. Hazardous material means a liquid material or substance, other than oil or liquefied gases, listed under 46 CFR 153.40 (a), (b), (c), or (e).

Marine transfer area means that part of a waterfront facility handling oil or hazardous materials in bulk between the vessel, or where the vessel moors, and the first manifold or shutoff valve on the pipeline encountered after the pipeline enters the secondary containment required under 40 CFR 112.7 or 49 CFR 195.264 inland of the terminal manifold or loading arm, or, in the absence of secondary containment, to the valve or manifold adjacent to the bulk storage tank, including the entire pier or wharf to which a vessel transferring oil or hazardous materials is moored.

MARPOL 73/78 means the International Convention for the Prevention of Pollution from Ships, 1973 (done at London, November 2, 1973) as modified by the Protocol of 1978 relating to the International Convention for the Prevention of Pollution from Ships, 1973 (done at London, February 17, 1978).

Mobile facility means any facility that can readily change location, such as a tank truck or tank car, other than a vessel or public vessel.

Monitoring device means any fixed or portable sensing device used to monitor for a discharge of oil or hazardous material onto the water, within or around a facility, and designed to notify operating personnel of a discharge of oil or hazardous material.

Officer in Charge, Marine Inspection (OCMI) means the U.S. Coast Guard officer commanding a Marine Inspection Zone described in Part 3 of this chapter, or an authorized representative.

Offshore facility means any facility of any kind located in, on, or under, any of the navigable waters of the United States, and any facility of any kind which is subject to the jurisdiction of the United States and is located in, on, or under any other waters, other than a vessel or a public vessel.

Oil means oil of any kind or in any form, including but not limited to, petroleum, fuel oil, sludge, oil refuse, and oil mixed with wastes other than dredged spoil.

Onshore facility means any facility (including, but not limited to, motor vehicles and rolling stock) of any kind located in, on, or under any land within the United States other than submerged land.

Person in charge means an individual designated as a person in charge of transfer operations under Sec. 154.710 (for facilities) or Sec. 155.700 (for vessels) of this chapter.

STCW means the International Convention on Standards of Training, Certification, and Watchkeeping for Seafarers, 1978.

Self-propelled tank vessel means a self-propelled tank vessel other than a tankship.

Tank barge means a non-self-propelled tank vessel.

Tankship means a self-propelled tank vessel constructed or adapted primarily to carry oil or hazardous material in bulk in the cargo spaces.

Tank vessel means a vessel that is constructed or adapted to carry, or that carries, oil or hazardous material in bulk as cargo or cargo residue, and that--

- (a) Is a vessel of the United States;
- (b) Operates on the navigable waters of the United States; or
- (c) Transfers oil or hazardous material in a port or place subject to the jurisdiction of the United States.

Transfer means any movement of oil or hazardous material to, from, or within a vessel by means of pumping, gravitation, or displacement. A transfer is considered to begin when the person in charge on the transferring vessel or facility and the person in charge on the receiving facility or vessel first meet to begin completing the declaration of inspection as required by Sec. 156.150 of this chapter. A transfer is considered to be complete when all the connections for the transfer have been uncoupled and secured with blanks or other closure devices and both of the persons in charge have completed the declaration of inspection to include the date and time the transfer was complete.

Vessel operator means a person who owns, operates, or is responsible for the operation of a vessel.

[CGD 75-124, 45 FR 7169, Jan. 31, 1980, as amended by CGD 86-034, 55 FR 36252, Sept. 4, 1990; CGD 79-116, 60 FR 17141, Apr. 4, 1995; CGD 93-056, 61 FR 41458, Aug. 8, 1996; 62 FR 3610, Jan. 24, 1997; CGD 79-116, 62 FR 25125, May 8, 1997]

TRANSFER OPERATIONS--Table of Contents

Subpart A_Oil and Hazardous Material Transfer Operations

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Subpart B_Special Requirements for Lightering of Oil and Hazardous Material Cargoes

156.200 Applicability.
156.205 Definitions.
156.210 General.
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156.220 Reporting of incidents.
156.225 Designation of lightering zones.
156.230 Factors considered in designating lightering zones.

Subpart C_Lightering Zones and Operational Requirements for the Gulf of Mexico

156.300 Designated lightering zones.
156.310 Prohibited areas.
156.320 Maximum operating conditions.
156.330 Operations.

Authority: 33 U.S.C. 1231, 1321(j); 46 U.S.C. 3703a, 3715; E.O.11735, 3 CFR 1971-1975 Comp., p. 793. Section 156.120(bb) and (ee) are also issued under 46 U.S.C. 3703.

This subpart applies to the transfer of oil or hazardous material on the navigable waters or contiguous zone of the United States to, from, or within each vessel with a capacity of 250 barrels or more; except that, this subpart does not apply to transfer operations within a public vessel.

[CGD 86-034, 55 FR 36255, Sept. 4, 1990]

Subpart A_Oil and Hazardous Material Transfer Operations

Sec. 156.105 Definitions.

Except as specifically stated in a section, the definitions in Sec. 154.105 of this chapter apply to this subpart.

[CGD 90-071a, 59 FR 53291, Oct. 21, 1994]

Subpart A_Oil and Hazardous Material Transfer Operations

Sec. 156.107 Alternatives.

(a) The COTP may consider and approve alternative procedures, methods, or equipment standards to be used by a vessel or facility operator in lieu of any requirements in this part if:

(1) Compliance with the requirement is economically or physically impractical;

(2) The vessel or facility operator submits a written request for the alternative at least 30 days before operations under the alternative are proposed, unless the COTP authorizes a shorter time; and

(3) The alternative provides an equivalent level of safety and protection from pollution by oil or hazardous material, which is documented in the request.

(b) The COTP takes final approval or disapproval action on any alternative requested, in writing, within 30 days of receipt of the request.

[CGD 75-124, 45 FR 7177, Jan. 31, 1980, as amended by CGD 86-034, 55 FR 36255, Sept. 4, 1990]

Subpart A_Oil and Hazardous Material Transfer Operations

Sec. 156.110 Exemptions.

(a) The Assistant Commandant for Marine Safety, Security and Environmental Protection, acting for the Commandant, may grant an exemption or partial exemption from compliance with any requirement in this part, and the District Commander may grant an exemption or partial exemption from compliance with any operating condition or requirement in subpart C of this part, if:

(1) The vessel or facility operator submits an application for exemption via the COTP at least 30 days before operations under the exemption are proposed, unless the COTP authorizes a shorter time; and

(2) It is determined, from the application, that:

(i) Compliance with a specific requirement is economically or physically impractical;
(ii) No alternative procedures, methods, or equipment standards exist that would provide an equivalent level of safety and protection from pollution by oil or hazardous material; and
(iii) The likelihood of oil or hazardous material being discharged as a result of the exemption is minimal.

(b) If requested, the applicant must submit any appropriate information, including an environmental and economic assessment of the effects of and reasons for the exemption and proposed procedures, methods or equipment standards.

(c) The exemption may specify the procedures, methods, or equipment standards that will apply.

(d) An exemption is granted or denied in writing. The decision of the Assistant Commandant for Marine Safety, Security and Environmental Protection is a final agency action.

[CGD 75-124, 45 FR 7177, Jan. 31, 1980, as amended by CGD 88-052, 53 FR 25122, July 1, 1988; CGD 86-034, 55 FR 36255, Sept. 4, 1990; CGD 93-081, 60 FR 45017, Aug. 29, 1995; CGD 96-026, 61 FR 33666, June 28, 1996; CGD 93-056, 61 FR 41461, Aug. 8, 1996; CGD 97-023, 62 FR 33364, June 19, 1997; USCG-2002-12471, 67 FR 41333, June 18, 2002]

Subpart A_Oil and Hazardous Material Transfer Operations

Sec. 156.111 Incorporation by reference.

(a) Certain material is incorporated by reference into this part with the approval of the Director of the Federal Register under 5 U.S.C. 552(a) and 1 CFR part 51. To enforce any edition other than that specified in paragraph (b) of this section, the Coast Guard must publish notice of the change in the Federal Register; and the material must be available to the public. All approved material is available for inspection at the U.S. Coast Guard, Office of Compliance (G-MOC), 2100 Second Street, SW, Washington, DC 20593-0001 and at the National Archives and Records Administration (NARA). For information on the availability of this material at NARA, call 202-741-6030, or go to: <http://www.archives.gov/federal--register/code--of--federal--regulations/ibr--locations.html>. All approved material is available from the sources indicated in paragraph (b) of this section.

(b) The material approved for incorporation by reference in this part and the sections affected are as follows:

Oil Companies International Marine Forum (OCIMF)
15th Floor, 96 Victoria Street, London SW1E 5JW, England.
Ship to Ship Transfer Guide (Petroleum), Second Edition, 1988--
156.330.

International Chamber of Shipping

12 Carthusian Street, London EC1M 6EB, England.
Guide to Helicopter/Ship Operations, Third Edition, 1989--156.330.

[CGD 93-081, 60 FR 45017, Aug. 29, 1995, as amended by CGD 96-026, 61 FR 33666, June 28, 1996; 69 FR 18803, Apr. 9, 2004]

Subpart A_Oil and Hazardous Material Transfer Operations

Sec. 156.112 Suspension order.

The COTP or OCMI may issue a suspension order to suspend transfer operations to the vessel or facility operator when the COTP or OCMI finds there is a condition requiring action to prevent the discharge or threat of discharge of oil or hazardous material, or when the COTP or OCMI is unable to verify compliance with the regulations through an inspection. A suspension order:

- (a) May be effective immediately;
- (b) Is issued in writing unless it is effective immediately and then it may be issued orally and followed up in writing;
- (c) Includes a statement of each condition requiring correction to--
 - (1) Prevent the discharge of oil or hazardous material; or
 - (2) Comply with Sec. 154.735 of this chapter; and
- (d) Is withdrawn when the COTP, OCMI, or District Commander, as applicable, determines that the condition requiring action to prevent the discharge or threat of discharge of oil or hazardous material has been corrected or no longer exists.

[CGD 75-124, 45 FR 7177, Jan. 31, 1980, as amended by CGD 86-034, 55 FR 36255, Sept. 4, 1990]

Subpart A_Oil and Hazardous Material Transfer Operations

Sec. 156.113 Compliance with suspension order.

(a) No vessel or facility operator to whom a suspension order has been issued may conduct transfer operations from the time the order is effective until that order is withdrawn by the applicable COTP, OCMI, or by the District Commander.

(b) The vessel or facility operator may request reconsideration of the suspension order either orally or in writing to the COTP or OCMI who issued it. The request may contain supporting documentation and evidence that the vessel or facility operator wishes to have considered.

(c) Any person not satisfied with a ruling made under the procedure contained in paragraph (b) of this section may appeal that ruling in writing, except as allowed under paragraph (e) of this section, to the Coast Guard District Commander of the district in which the suspension order was issued. The appeal may contain supporting documentation and evidence that the appellant wishes to have considered. The appeal does not stay the effect of the suspension order while the COTP or OCMI ruling is being reviewed. The District Commander issues a ruling after reviewing the appeal.

(d) The ruling by the District Commander is final agency action.

(e) If the delay in presenting a written appeal under paragraph (c)

of this section would have a significant adverse impact on the appellant, the appeal may initially be presented orally. If an initial presentation of the appeal is made orally, the appellant must submit the appeal in writing within five days of the oral presentation to the District Commander to whom the oral appeal was made, containing, at a minimum the basis for the appeal and a summary of the material presented orally.

[CGD 75-124, 45 FR 7177, Jan. 31, 1980, as amended by CGD 86-034, 55 FR 36255, Sept. 4, 1990]

Subpart A_Oil and Hazardous Material Transfer Operations

Sec. 156.115 Person in charge: Limitations.

(a) No person may serve as the person in charge of transfer operations on more than one vessel at a time during transfers between vessels or between two or more vessels and a facility unless authorized by the COTP.

(b) No person may serve as the person in charge of both a vessel and a facility during transfer operations unless authorized by the COTP.

[CGD 75-124, 45 FR 7177, Jan. 31, 1980, as amended by CGD 86-034, 55 FR 36255, Sept. 4, 1990]

Subpart A_Oil and Hazardous Material Transfer Operations

Sec. 156.118 Advance notice of transfer.

(a) The COTP may require a facility operator to notify the COTP of the time and place of each transfer operation at least 4 hours before it begins for facilities that:

- (1) Are mobile;
- (2) Are in a remote location;
- (3) Have a prior history of oil or hazardous material spills; or
- (4) Conduct infrequent transfer operations.

(b) In the case of a vessel to vessel transfer, the COTP may require a vessel operator of a lightering or fueling vessel to notify the COTP of the time and place of each transfer operation, as specified by the COTP, at least 4 hours before it begins.

(c) No person may conduct such transfer operations until advance notice has been given as specified by the COTP.

Note: The notification may be accomplished by submitting a written schedule, periodically updated to be current.

[CGD 75-124, 45 FR 7177, Jan. 31, 1980, as amended by CGD 86-034, 55 FR 36255, Sept. 4, 1990]

Subpart A_Oil and Hazardous Material Transfer Operations

Sec. 156.120 Requirements for transfer.

A transfer is considered to begin when the person in charge on the transferring vessel or facility and the person in charge on the receiving facility or vessel first meet to begin completing the declaration of inspection, as required by Sec. 156.150 of this part. No person shall conduct an oil or hazardous material transfer operation unless:

(a) The vessel's moorings are strong enough to hold during all expected conditions of surge, current, and weather and are long enough to allow adjustment for changes in draft, drift, and tide during the transfer operation;

(b) Transfer hoses and loading arms are long enough to allow the vessel to move to the limits of its moorings without placing strain on the hose, loading arm, or transfer piping system;

(c) Each hose is supported to prevent kinking or other damage to the hose and strain on its coupling.

(d) Each part of the transfer system is aligned to allow the flow of oil or hazardous material;

(e) Each part of the transfer system not necessary for the transfer operation is securely blanked or shut off;

(f) The end of each hose and loading arm that is not connected for the transfer of oil or hazardous material is blanked off using the closure devices required by Sec. Sec. 154.520 and 155.805 of this chapter;

(g) The transfer system is attached to a fixed connection on the vessel and the facility except that when a vessel is receiving fuel, an automatic back pressure shutoff nozzle may be used;

(h) Each overboard discharge or sea suction valve that is connected to the vessel's transfer or cargo tank system is sealed or lashed in the closed position; except when used to receive or discharge ballast in compliance with 33 CFR Part 157;

(i) Each transfer hose has no unrepaired loose covers, kinks, bulges, soft spots, or any other defect which would permit the discharge of oil or hazardous material through the hose material and no gouges, cuts, or slashes that penetrate the first layer of hose reinforcement ("reinforcement" means the strength members of the hose, consisting of fabric, cord and/or metal);

(j) Each hose or loading arm in use meets Sec. Sec. 154.500 and 154.510 of this chapter, respectively;

(k) Each connection meets Sec. 156.130;

(l) Any monitoring devices required by Sec. 154.525 of this chapter are installed and operating properly;

(m) The discharge containment equipment required by Sec. 154.545 of this chapter is readily accessible or deployed as applicable;

(n) The discharge containment required by Sec. Sec. 154.530, 155.310, and 155.320 of this chapter, as applicable, is in place and periodically drained to provide the required capacity;

(o) Each drain and scupper is closed by the mechanical means required by Sec. 155.310;

(p) All connections in the transfer system are leak free except that a component in the transfer system, such as the packing glands of a pump, may leak at a rate that does not exceed the capacity of the discharge containment provided during the transfer operation;

(q) The communications required by Sec. Sec. 154.560 and 155.785 of this chapter are operable for the transfer operation;

(r) The emergency means of shutdown required by Sec. Sec. 154.550 and 155.780 of this chapter, as applicable, is in position and operable;

(s) There is a person in charge on the transferring vessel or facility and the receiving vessel or facility except as otherwise authorized under Sec. 156.115;

(t) Each person in charge required by paragraph (s) of this section:

(1) Is at the site of the transfer operation and immediately available to the transfer personnel;

(2) Has in his or her possession a copy of the facility operations manual or vessel transfer procedures, as appropriate; and

(3) Conducts the transfer operation in accordance with the facility operations manual or vessel transfer procedures, as appropriate;

(u) The personnel required, under the facility operations manual and the vessel transfer procedures, to conduct the transfer operation:

(1) Are on duty; and

(2) Conduct the transfer operation in accordance with the facility operations manual or vessel transfer procedures, as appropriate;

(v) At least one person is at the site of the transfer operation who fluently speaks the language or languages spoken by both persons in charge;

(w) The person in charge of the transfer on the transferring vessel or facility and the person in charge of it on the receiving vessel or facility have held a conference, to ensure that each person in charge understands--

(1) The identity of the product to be transferred;

(2) The sequence of transfer operations;

(3) The transfer rate;

(4) The name or title and location of each person participating in the transfer operation;

(5) Details of the transferring and receiving systems including procedures to ensure that the transfer pressure does not exceed the maximum allowable working pressure (MAWP) for each hose assembly, loading arm and/or transfer pipe system;

(6) Critical stages of the transfer operation;

(7) Federal, state, and local rules that apply to the transfer of oil or hazardous material;

(8) Emergency procedures;

(9) Discharge containment procedures;

(10) Discharge reporting procedures;

(11) Watch or shift arrangement;

(12) Transfer shutdown procedures; and,

(13) If the persons use radios, a predetermined frequency for communications during the transfer, agreed upon by both.

(x) The person in charge of transfer operations on the transferring vessel or facility and the person in charge of transfer operations on the receiving vessel or facility agree to begin the transfer operation;

(y) Between sunset and sunrise the lighting required by Sec. Sec. 154.570 and 155.790 of this chapter is provided; and

(z) For transfer operations between tank barges from sunset to sunrise, lighting is provided as described in Sec. 155.790 of this chapter.

(aa) A transfer operation which includes collection of vapor emitted from a vessel's cargo tanks through a venting system not located on the vessel must have the following verified by the person in charge:

- (1) Each manual valve in the vapor collection system is correctly positioned to allow the collection of cargo vapor;
- (2) A vapor collection hose or arm is connected to the vessel's vapor connection;
- (3) The electrical insulating device required by Sec. 154.810(g) of this chapter or 46 CFR 39.40-3(c) is fitted between the facility vapor connection and the vessel vapor connection;
- (4) The initial loading rate and the maximum transfer rate are determined;
- (5) The maximum and minimum operating pressures at the facility vapor connection are determined;
- (6) The tank barge overfill control system, if installed, is connected to the facility, tested, and operating properly;
- (7) The following have been performed not more than 24 hours prior to the start of the transfer operation:
 - (i) Each alarm and automatic shutdown system required by subpart E of part 154 of this chapter and 46 CFR part 39 has been tested and found to be operating properly, and
 - (ii) Analyzers required by Sec. 154.820(a), Sec. 154.824 (d) and (e) of this chapter or 46 CFR 39.40-3(a) have been checked for calibration by use of a span gas;
- (8) Each vapor recovery hose has no unrepaired loose covers, kinks, bulges, soft spots, or any other defect which would permit the discharge of vapor through the hose material, and no external gouges, cuts, or slashes that penetrate the first layer of hose reinforcement; and
- (9) The oxygen content of the vessel's cargo tanks, if inerted, is at or below 8 percent by volume.
- (bb) If the transfer operation involves loading oil, as defined in Sec. 151.05 of this chapter, into a cargo tank, the overfill device required by Sec. 155.480 of this chapter is installed and operating properly.
- (cc) Smoking is not permitted in the facilities marine transfer area except in designated smoking areas.
- (dd) Welding, hot work operations and smoking are prohibited on vessels during the transfer of flammable or combustible materials, except that smoking may be permitted in accommodation areas designated by the master.
- (ee) Each tank level or pressure monitoring device required under 33 CFR 155.490 must be activated and monitored whenever the tank is not actively being subjected to cargo operations.

(Approved by the Office of Management and Budget under control number 2115-0506)

[CGD 75-124, 45 FR 7177, Jan. 31, 1980, as amended by CGD 88-102, 55 FR 25445, June 21, 1990; CGD 86-034, 55 FR 36255, Sept. 4, 1990; CGD 90-071a, 59 FR 53291, Oct. 21, 1994; CGD 93-056, 61 FR 41461, Aug. 8, 1996; CGD 79-116, 62 FR 25127, May 8, 1997; USCG-2001-9046, 67 FR 58524, Sept. 17, 2002]

Subpart A--Oil and Hazardous Material Transfer Operations

Sec. 156.125 Discharge cleanup.

- (a) Each person conducting the transfer operation shall stop the transfer operation whenever oil or hazardous material from any source is discharged:

- (1) In the transfer operation work area; or
- (2) Into the water or upon the adjoining shoreline in the transfer area.
- (b) Except as permitted under paragraph (c) of this section, no person may resume the transfer operation after it has been stopped under paragraph (a) of this section, unless:
 - (1) Oil or hazardous material discharged in the transfer operation work area is cleaned up; and
 - (2) Oil or hazardous material discharged into the water or upon the adjoining shoreline is cleaned up, or is contained and being cleaned up.
- (c) The COTP may authorize resuming the transfer operation if it is deemed appropriate.

[CGD 75-124, 45 FR 7177, Jan. 31, 1980, as amended by CGD 86-034, 55 FR 36255, Sept. 4, 1990]

Subpart A_Oil and Hazardous Material Transfer Operations

Sec. 156.130 Connection.

- (a) Each person who makes a connection for transfer operations shall:
 - (1) Use suitable material in joints and couplings to ensure a leak-free seal;
 - (2) Use a bolt in at least every other hole, and in no case less than four bolts, in each temporary bolted connection that uses a flange that meets American National Standards Institute (ANSI) standard flange requirements under Sec. 154.500(d)(2) of this chapter;
 - (3) Use a bolt in each hole in each temporary bolted connection that uses a flange other than one that meets ANSI standards;
 - (4) Use a bolt in each hole of each permanently connected flange;
 - (5) Use bolts of the correct size in each bolted connection; and
 - (6) Tighten each bolt and nut uniformly to distribute the load and sufficiently to ensure a leak free seal.
- (b) A person who makes a connection for transfer operations must not use any bolt that shows signs of strain or is elongated or deteriorated.
- (c) Except as provided in paragraph (d) of this section, no person may use a connection for transfer operations unless it is:
 - (1) A bolted or full threaded connection; or
 - (2) A quick-connect coupling acceptable to the Commandant.
- (d) No person may transfer oil or hazardous material to a vessel that has a fill pipe for which containment cannot practically be provided unless an automatic back pressure shutoff nozzle is used.

[CGD 75-124, 45 FR 7177, Jan. 31, 1980, as amended by CGD 86-034, 55 FR 36256, Sept. 4, 1990]

Subpart A_Oil and Hazardous Material Transfer Operations

Sec. 156.150 Declaration of inspection.

(a) No person may transfer oil or hazardous material to or from a vessel unless each person in charge, designated under Sec. Sec. 154.710 and 155.700 of this chapter, has filled out and signed the declaration of inspection form described in paragraph (c) of this section.

(b) No person in charge may sign the declaration of inspection unless he or she has determined by inspection, and indicated by initialling in the appropriate space on the declaration of inspection form, that the facility or vessel, as appropriate, meets Sec. 156.120.

(c) The declaration of inspection may be in any form but must contain at least:

(1) The name or other identification of the transferring vessel or facility and the receiving vessel or facility;

(2) The address of the facility or location of the transfer operation if not at a facility;

(3) The date and time the transfer operation is started;

(4) A list of the requirements in Sec. 156.120 with spaces on the form following each requirement for the person in charge of the vessel or facility to indicate by initialling that the requirement is met for the transfer operation; and

(5) A space for the date, time of signing, signature, and title of each person in charge during transfer operations on the transferring vessel or facility and a space for the date, time of signing, signature, and title of each person in charge during transfer operations on the receiving facility or vessel certifying that all tests and inspections have been completed and that they are both ready to begin transferring product; and

(6) The date and time the transfer operation is completed.

(d) The form for the declaration of inspection may incorporate the declaration-of-inspection requirements under 46 CFR 35.35-30.

(e) The vessel and facility persons in charge shall each have a signed copy of the declaration of inspection available for inspection by the COTP during the transfer operation.

(f) The operators of each vessel and facility engaged in the transfer operation shall retain a signed copy of the declaration of inspection on board the vessel or at the facility for at least 1 month from the date of signature.

[CGD 75-124, 45 FR 7177, Jan. 31, 1980, as amended by CGD 86-034, 55 FR 36256, Sept. 4, 1990; CGD 93-056, 61 FR 41461, Aug. 8, 1996]

Subpart A--Oil and Hazardous Material Transfer Operations

Sec. 156.160 Supervision by person in charge.

(a) No person may connect or disconnect a hose, top off a tank, or engage in any other critical procedures during the transfer operation unless the person in charge, required by Sec. 156.120(s), supervises that procedure.

(b) No person may start the flow of oil or hazardous material to or from a vessel unless instructed to do so by either person in charge.

(c) No person may transfer oil or hazardous material to or from a vessel unless each person in charge is in the immediate vicinity and immediately available to the transfer personnel.

[CGD 75-124, 45 FR 7177, Jan. 31, 1980, as amended by CGD 86-034, 55 FR 36256, Sept. 4, 1990]

Subpart A_Oil and Hazardous Material Transfer Operations

Sec. 156.170 Equipment tests and inspections.

(a) Except as provided in paragraph (d) of this section, no person may use any equipment listed in paragraph (c) of this section for transfer operations unless the vessel or facility operator, as appropriate, tests and inspects the equipment in accordance with paragraphs (b), (c) and (f) of this section and the equipment is in the condition specified in paragraph (c) of this section.

(b) During any test or inspection required by this section, the entire external surface of the hose must be accessible.

(c) For the purpose of paragraph (a) of this section:

(1) Each nonmetallic transfer hose must:

(i) Have no unrepaired loose covers, kinks, bulges, soft spots or any other defect which would permit the discharge of oil or hazardous material through the hose material, and no gouges, cuts or slashes that penetrate the first layer of hose reinforcement as defined in Sec. 156.120(i).

(ii) Have no external deterioration and, to the extent internal inspection is possible with both ends of the hose open, no internal deterioration;

(iii) Not burst, bulge, leak, or abnormally distort under static liquid pressure at least 1½ times the maximum allowable working pressure; and

(iv) Hoses not meeting the requirements of paragraph (c)(1)(i) of this section may be acceptable after a static liquid pressure test is successfully completed in the presence of the COTP. The test medium is not required to be water.

(2) Each transfer system relief valve must open at or below the pressure at which it is set to open;

(3) Each pressure gauge must show pressure within 10 percent of the actual pressure;

(4) Each loading arm and each transfer pipe system, including each metallic hose, must not leak under static liquid pressure at least 1½ times the maximum allowable working pressure; and

(5) Each item of remote operating or indicating equipment, such as a remotely operated valve, tank level alarm, or emergency shutdown device, must perform its intended function.

(d) No person may use any hose in underwater service for transfer operations unless the operator of the vessel or facility has tested and inspected it in accordance with paragraph (c)(1) or (c)(4) of this section, as applicable.

(e) The test fluid used for the testing required by this section is limited to liquids that are compatible with the hose tube as recommended by the hose manufacturer.

(f) The frequency of the tests and inspections required by this section must be:

(1) For facilities, annually or not less than 30 days prior to the first transfer conducted past one year from the date of the last tests and inspections;

(2) For a facility in caretaker status, not less than 30 days prior to the first transfer after the facility is removed from caretaker status; and

(3) For vessels, annually or as part of the biennial and mid-period inspections.

(g) If a facility or vessel collects vapor emitted from a vessel cargo tank with a vapor control system, the system must not be used unless the following tests and inspections are satisfactorily completed:

(1) Each vapor hose, vapor collection arm, pressure or vacuum relief valve, and pressure sensor is tested and inspected in accordance with paragraphs (b), (c), and (f) of this section;

(2) Each remote operating or indicating device is tested for proper operation in accordance with paragraph (f) of this section;

(3) Each detonation arrester required by Sec. 154.820, Sec. 154.826(a), and Sec. 154.828(a) of this chapter or 46 CFR 39.40-3(d), and each flame arrester required by Sec. 154.826(a), Sec. 154.828 (a) and (c) of this chapter has been inspected internally within the last year, or sooner if operational experience has shown that frequent clogging or rapid deterioration is likely; and

(4) Each hydrocarbon and oxygen analyzer required by Sec. 154.820(a) and Sec. 154.824 (d) and (e) of this chapter or 46 CFR 39.40-3(a) is calibrated:

(i) Within the previous two weeks, or

(ii) Within 24 hours prior to operation when the vapor control system is operated less frequently than once a week.

(h) Upon the request of the owner or operator, the COTP may approve alternative methods of compliance to the testing requirements of paragraph (c) of this section if the COTP determines that the alternative methods provide an equal level of protection.

(Approved by the Office of Management and Budget under control number 2115-0096)

[CGD 75-124, 45 FR 7177, Jan. 31, 1980, as amended by CGD 88-102, 55 FR 25445, June 21, 1990; CGD 86-034, 55 FR 36256, Sept. 4, 1990; CGD 93-056, 61 FR 41461, Aug. 8, 1996]

Subpart B--Special Requirements for Lightering of Oil and Hazardous Material Cargoes

Sec. 156.200 Applicability.

Source: CGD 78-180, 49 FR 11172, Mar. 26, 1984, unless otherwise noted.

This subpart applies to each vessel to be lightered and each service vessel engaged in a lightering operation in the marine environment beyond the baseline from which the territorial sea is measured when the oil or hazardous material lightered is destined for a port or place subject to the jurisdiction of the U.S. This subpart does not apply to lightering operations involving public vessels, or to the dedicated response vessels and vessels of opportunity in accordance with the National Contingency Plan (40 CFR parts 9 and 300) when conducting response activities. These rules are in addition to the rules of subpart A of this part, as well as the rules in the applicable sections of parts 151, 153, 155, 156, and 157 of this chapter.

[CGD 93-081, 60 FR 45017, Aug. 29, 1995]

Subpart B_Special Requirements for Lightering of Oil and Hazardous Material Cargoes

Sec. 156.205 Definitions.

(a) In addition to the terms defined in this section, the definitions in Sec. 154.105 of this chapter apply to this subpart and to subpart C.

(b) As used in this subpart and subpart C:

Lightering or Lightering operation means the transfer of a cargo of oil or a hazardous material in bulk from one vessel to another, including all phases of the operation from the beginning of the mooring operation to the departure of the service vessel from the vessel to be lightered, except when that cargo is intended only for use as fuel or lubricant aboard the receiving vessel.

Marine environment means--

(1) The navigable waters of the United States;

(2) The waters of an area over which the United States asserts exclusive fishery management authority; and

(3) The waters superadjacent to the Outer Continental Shelf of the United States.

Service vessel means the vessel which receives a cargo of oil or a hazardous material from another vessel in a lightering operation.

Vessel to be lightered means the vessel which transports a cargo of oil or a hazardous material to a place within the marine environment for transfer of that cargo to another vessel for further transport to a port or place subject to the jurisdiction of the United States. The term "vessel to be lightered" does not include drilling rigs, or offshore supply vessels transferring cargo intended for use as fuel or lubricant aboard the receiving vessel.

Work includes any administrative duties associated with the vessel whether performed on board the vessel or onshore.

[CGD 78-180, 49 FR 11172, Mar. 26, 1984, as amended by CGD 86-034, 55 FR 36256, Sept. 4, 1990; CGD 90-052, 58 FR 48436, Sept. 15, 1993; CGD 93-081, 60 FR 45017, Aug. 29, 1995]

Subpart B--Special Requirements for Lightering of Oil and Hazardous Material Cargoes

Sec. 156.210 General.

(a) No vessel may transfer oil or hazardous materials in a port or place subject to the jurisdiction of the United States, if the cargo has been lightered from another vessel, unless:

(1) The regulations in this subpart have been complied with;

(2) Both the vessel to be lightered and service vessel have, on board, at the time of transfer, a valid Certificate of Inspection, Certificate of Compliance, or a Tank Vessel Examination Letter, as would have been required under 46 U.S.C. 3710 or 3711, had the transfer taken place in a port or place subject to the jurisdiction of the United States;

(3) The delivering and receiving vessels have on board at the time of transfer, evidence that each vessel is operating in compliance with section 311(j) of the Federal Water Pollution Control Act (33 U.S.C. 1321(j)) and applicable regulations issued under the authority of section 311(j) in the form of a Declaration of Inspection as required by Sec. 156.150 and a vessel response plan if required under part 155 of this chapter; and

Note: Under 46 U.S.C. 3715, the delivering and receiving vessels must have on board at the time of transfer, a Certificate of Financial Responsibility that would be required if the transfer had taken place in a location subject to the jurisdiction of the U.S. Regulations concerning

Certificates of Financial Responsibility for vessels using the navigable waters of the U.S. are in part 130 of this chapter.

(4) The vessel to be lightered has on board, at the time of transfer, an International Oil Pollution Prevention (IOPP) Certificate or equivalent documentation of compliance with Annex I of the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto (MARPOL 73/78), as would be required by Part 151 of this chapter for vessels in navigable waters of the United States. The IOPP Certificate or documentation of compliance shall be that prescribed by Sec. 151.19 and 151.21 of this chapter, and shall be effective under the same timetable as specified in Sec. 151.19.

(b) Lightering operations involving hazardous materials, other than oil, may be conducted only with the specific approval of the Commandant. A request to lighter hazardous materials, other than oil, must be submitted to Commandant (G-M) prior to the planned beginning of lightering operations. The request must include the information described in Sec. 156.215(a) to the extent known, for the initial transfer, and the estimated frequency of subsequent lightering operations. After the entry into force of Annex II to MARPOL 73/78, vessels lightering hazardous materials shall carry an International Pollution Prevention Certificate for the Carriage of Noxious Liquid Substances in Bulk (1973), if required by Annex II to MARPOL 73/78, or equivalent documentation of compliance with the annex.

(c) In an emergency, the COTP, upon request, may authorize a deviation from any rule in this part if the COTP determines that its application will endanger persons, property, or the environment.

(d) On vessels conducting lightering operations in a designated lightering zone, a licensed individual or seaman may not work, except in an emergency or a drill, more than 15 hours in any 24-hour period, or more than 36 hours in any 72-hour period, including the 24-hour and 72-hour periods prior to commencing lightering operations.

[CGD 78-180, 49 FR 11172, Mar. 29, 1984, as amended by CGD 88-052, 53 FR 25122, July 1, 1988; CGD 90-052, 58 FR 48436, Sept. 15, 1993; CGD 93-081, 60 FR 45017, Aug. 29, 1995; USCG-2002-12471, 67 FR 41333, June 18, 2002]

Subpart B--Special Requirements for Lightering of Oil and Hazardous Material Cargoes

Sec. 156.215 Pre-arrival notices.

(a) The master, owner or agent of each vessel to be lightered must give at least 24 hours advance notice to the Captain of the Port nearest the lightering location or zone, prior to arrival in the lightering location or zone. This advance notice must include:

- (1) The vessel's name, call sign or official number, and registry;
- (2) The cargo type (if oil) or shipping name (if hazardous material) and approximate amount on board;
- (3) The number of transfers expected and the amount of cargo expected to be transferred during each transfer;
- (4) The lightering location or zone to be used;
- (5) The estimated time of arrival in the lightering location or zone;
- (6) The estimated duration of transfer operations; and

(7) The name and destination of service vessel(s).

(b) In the event the estimated time of arrival in the lightering location or zone changes by more than six hours, the Master, owner or agent of each vessel to be lightered must advise the Captain of the Port of this change as soon as possible.

(c) Where lightering is conducted as a result of collision, grounding, tank rupture or any similar emergency, immediate notice must be given to the Captain of the Port.

(d) In addition to the other requirements in this section, the master, owner, or agent of a vessel that requires a Tank Vessel Examination (TVE) or other special Coast Guard inspection in order to lighter in a designated lightering zone must request the TVE or other inspection from the cognizant Captain of the Port at least 72 hours prior to commencement of lightering operations.

[CGD 78-180, 49 FR 11172, Mar. 26, 1984, as amended by CGD 90-052, 58 FR 48437, Sept. 15, 1993; CGD 93-081, 60 FR 45017, Aug. 29, 1995]

Subpart B_Special Requirements for Lightering of Oil and Hazardous Material Cargoes

Sec. 156.220 Reporting of incidents.

(a) An immediate report must be made to the nearest Captain of the Port, by the service vessel, if fire, explosion, collision, grounding or any similar emergency, which poses a threat to the vessels involved, occurs during lightering.

(b) Any discharge of oil or hazardous material into the water shall be reported, by the service vessel, in accordance with the procedures specified in Sec. 151.15 of this chapter.

Subpart B--Special Requirements for Lightering of Oil and Hazardous Material Cargoes

Sec. 156.225 Designation of lightering zones.

The District Commander is delegated the authority to designate lightering zones and their operating requirements, where they are necessary for safety or environmental protection. When a lightering zone has been designated, lightering operations in a given geographic area may only be conducted within the designated lightering zone.

Subpart B_Special Requirements for Lightering of Oil and Hazardous Material Cargoes

Sec. 156.230 Factors considered in designating lightering zones.

The following factors are considered in designating a lightering zone:

(a) The findings of the environmental analysis or, if prepared, the Environmental Impact Statement;

- (b) The proximity of the zone to:
 - (1) Shipping lanes;
 - (2) Vessel traffic schemes or vessel separation systems;
 - (3) Anchorages;
 - (4) Fixed structures;
 - (5) Designated marine sanctuaries;
 - (6) Commercial and recreational fishing areas;
 - (7) Environmentally sensitive areas; and
 - (8) Designated units of the National Park System, National Wild and Scenic Rivers System, National Wilderness Preservation System, properties included on the National Register of Historic Places and National Registry of Natural Landmarks, and National Wildlife Refuge System.
- (c) The traditional use of areas for lightering operations;
- (d) The normal weather and sea conditions in the areas, and their effect on lightering operations, and the fate of possible cargo discharges;
- (e) The depth of water and underwater obstructions that may adversely impact anchorages and clearance of vessels;
- (f) Other relevant safety, environmental, or economic data.

U.S. Army Corps of Engineers (USACE)

ACCIDENT PREVENTION PLAN (APP)

For use of this form, see EM 385-1-1; the proponent agency is CESO.

Purpose: An APP is a written site-specific Safety & Occupational Health plan that documents project, contract, and job-specific potential hazards in the workplace, and the company policies, controls, and work practices that will be used to minimize those hazards. They are an integral part of the planning and risk management process. The APP must be developed by a Competent Person and reviewed and approved by the prime contractor and corporate safety official. Upon contractor approval, the plan must be submitted to the Contracting Officer (KO) or Contracting Officer Representative (COR) for review and acceptance prior to the performance of any work. No contract work can begin without a KO or COR accepted APP.

The completed APP must be posted on the SOH bulletin board, or a notice posted on the board stating the location of the plan.

PART 1 - SIGNATURES

Plan Preparer - For example: Qualified Person (QP), Competent Person (CP), or project Quality Control (QC).

1. Name:	2. Title:
3. Email Address:	4. Phone Number:
5. Signature	

Plan Approver - Corporate Safety Official

1. Name:	2. Title:
3. Email Address:	4. Phone Number:
5. Signature	

Plan Concurrence - For example: Industrial hygienist, Project Manager, Superintendent, Site Safety & Health Officer, Quality Control, etc.

1. Name:	2. Title:
3. Email Address:	4. Phone Number:
5. Signature:	

PART 2: PROJECT INFORMATION

1. Project Name:	
2. Project Address (attach map in Appendix 1):	
3. Estimated Project Start Date:	4. Estimated Project Completion Date:
5. Project Description / Description of work to be performed:	

PART 3: PRIME CONTRACTOR INFORMATION

1. Prime Contractor Name:

2. Contract Number:

3. Project Manager Name:

4. Quality Control (QC) Manager Name:

5. Contractor Corporate Safety Official Name:

6. Primary Site Safety & Health Officer (SSHO) Name:

7. Alternate Site Safety & Health Officer (SSHO) Name(s):

PART 4: SOH COMMITMENT AND POLICY

1. Provide a statement of safety and health policy detailing commitment to providing a safe and healthful workplace for all employees on this project.

2. Provide a statement outlining the Contractor's safety program goals and objectives.

3. Identify policies and procedure regarding non-compliance with safety requirements.

4. Provide written procedures for holding managers and supervisors accountable for safety.

5. Prime contractor is responsible for the implementation and enforcement of the SOH program for contractor employees, subcontractors, suppliers, and visitors at the site of work?

☐ Yes☐ No

6. Prime contractor will require subcontractors, suppliers, and visitors to comply with the APP and EM 385-1-1?

☐ Yes☐ No

7. No work will be performed unless a designated SSHO is present on the site of work.

☐ Yes☐ No

8. A list of all known subcontractor(s) and supplier(s) are listed in Appendix 2.

☐ Yes☐ No

9. If all subcontractor(s) and supplier(s) are not known at the time of initial APP submittal, Appendix 2 will be updated and resubmitted to the KO/COR for acceptance prior to the start of any work activities.

☐ Yes☐ No

10. All subcontractors will have a copy of the APP, applicable AHAs and will be required to comply with them.

☐ Yes☐ No

11. Subcontractors and suppliers at the site of work will abide by the EM 385-1-1, this APP, and all AHAs.

☐ Yes☐ No

12. This APP and AHAs will be available to all workers at the site of work.

☐ Yes☐ No

PART 5: TRAINING

1. All employees will be trained according to applicable EM 385-1-1, federal, state, and local requirements prior to the commencement of work. ☐ Yes ☐ No
2. The following personnel are identified and designated as personnel responsible for SOH on this project. Add the following documentation to Appendix 3 of this worksheet: 1) SSHO designation letter (ENG Form 6282), 2) instructor signed OSHA 30-hour card (or course completion certificate if within 90 days of having completed the OSHA 30-hour training), and 3) proof of 24 hours of competency of SOH training within the last 3 years for the personnel identified below. (EM 385-1-1, Chapter 2, 2-3 Personnel Qualification/Training):
(Check and list names of personnel being designated)
- ☐ Primary SSHO Name:
- ☐ Alternate SSHO Name(s):
3. QP(s) and CP(s) required by EM 385-1-1 (for example, excavation, confined space, scaffolding, fall protection, etc.) for this project will be included in Appendix 3. Include 1) List of names or designation letters and 2) proof of qualification/competency.

PART 6: SOH INSPECTIONS

1. All applicable inspections required by EM 385-1-1, contract, federal, state, and local requirements will be completed. ☐ Yes ☐ No
2. The following personnel are assigned to conduct inspections, the type of inspection and frequency for the worksite. At minimum the SSHO must conduct daily site inspections.
- | Who will conduct inspection? | Frequency? | What type of inspection? |
|------------------------------|------------|--------------------------|
| | | |
3. Employees that conduct inspections will have technical proficiency needed to perform inspections. Proof of inspector's training/qualification is included in Appendix 3. ☐ Yes ☐ No
4. Deficiencies will be documented and tracked as required by this EM 385-1-1? ☐ Yes ☐ No
6. Will any external inspections be required (for example, USCG, OSHA, etc.)? ☐ Yes ☐ No
If yes, describe:

PART 7: ACCIDENT REPORTING AND INVESTIGATION

1. Exposure data (man-hours worked) will be reported to KO/COR as required by the contract? ☐ Yes ☐ No
2. All accidents will be reported as soon as possible according to EM 385-1-1? ☐ Yes ☐ No
3. All accidents occurring incidentally to an operation, project, or facility for which the EM 385-1-1 is applicable will be thoroughly investigated and analyzed. Corrective actions to prevent reoccurrence will be implemented as soon as reasonably possible. The KO/COR will be notified when corrective actions are completed. ☐ Yes ☐ No
4. Accident notification will be provided to the KO/COR as stated below: ☐ Yes ☐ No

ACCIDENT TYPE	KO/COR NOTIFICATION TIME
Fatality, permanent and partial disability, or property damage over \$600,000	Immediately, no later than (NLT) 8 hours
All other accidents and near misses	Immediately, NLT 24 hours

PART 8: SOH OVERSIGHT

1. A risk management process that meets or exceeds the requirement of the EM 385-1-1 will be implemented on this project site?	☐ Yes	☐ No
2. Hazardous conditions and unsafe acts will be immediately corrected?	☐ Yes	☐ No
3. APP and AHAs will be prepared by a QP or CP according to the EM 385-1-1 prior to the start of any work activities on the site of work (as much as the information can be known at that point in time).	☐ Yes	☐ No
4. The plans and AHAs will be updated throughout the life of the project to include changes in personnel, equipment, conditions, etc. Additional revisions must be incorporated as necessary to reflect changing site conditions, construction methods, personnel roles and responsibilities and construction schedules.	☐ Yes	☐ No
5. No activity/work will be started at the site of work until the APP is reviewed and submitted to the KO/COR for acceptance, with the site-specific plans and procedures required by the EM 385-1-1 to complete the project?	☐ Yes	☐ No
6. APP and AHAs will be provided to and reviewed by all workers involved in the activity prior to work?	☐ Yes	☐ No
7. Workers will sign AHA(s) to concur with identified job steps and controls needed to mitigate hazards?	☐ Yes	☐ No

PART 9: SEVERE WEATHER PLAN

1. The SSHO or designated onsite personnel is responsible for checking the weather conditions at a minimum of twice a day?	☐ Yes	☐ No
2. What severe weather forecast will require the SSHO to monitor weather conditions continuously? Describe.		
3. What are the training requirements for severe weather precautions and actions?		
4. Identify area of retreat or other actions to be taken such as evacuation, shelter in place, etc.		
5. Identify procedures to stop outdoor activities if extreme weather conditions to include thunderstorms are in the area.		

PART 10: ACTIVITY HAZARD ANALYSIS (RISK MANAGEMENT)

1. Major phases of work anticipated:

2. Definable Features of Work (DFOW)

Note. An activity-specific AHA will be submitted to the KO/COR for each DFOW prior to any work being performed.

3. List equipment to be used:

4. Anticipated High-Risk Activities: (See examples below, check all that apply):

☐ Electrical and/or Hazardous Energy	☐ Confined Space
☐ Working at Heights	☐ Excavations & Trenching
☐ Scaffolding	☐ Other: _____
☐ Diving	☐ Other: _____
☐ Load Handling Equipment	☐ Other: _____
☐ Occupational Exposure	☐ Other: _____

5. Detailed project-specific hazards and controls will be provided utilizing AHAs for each activity/DFOW.	☐ Yes	☐ No
6. No work will begin on an activity/DFOW until the initial AHA addressing the project-specific hazards has been developed, reviewed, and accepted by the KO/COR. (See EM 385-1-1 Chapter 2 section 2-6).	☐ Yes	☐ No
7. AHAs are living documents and are intended to be created, utilized in the field, and updated (by the workers) as necessary.	☐ Yes	☐ No
8. Workers/crews will have accepted AHA(s) that reflects current site conditions, personnel, equipment, control measures, etc. while the work is being performed?	☐ Yes	☐ No
9. AHA will be used by the contractor personnel to assure work is being performed consistent with the AHA?	☐ Yes	☐ No
10. Once the activity has been completed, the AHA will be available and kept onsite for the length of the contract.	☐ Yes	☐ No
11. AHAs for unknown work activities will be submitted to the KO/COR prior to work and attached to Appendix 4 of this document.	☐ Yes	☐ No
12. Who has the authority to accept residual risk on AHA(s)? (List below)		

Category of Risk	Approval Authority (Name and Title)
Extremely High Risk	
High Risk	
Medium Risk	
Low Risk	

PART 11: REQUIRED PROJECT SITE-SPECIFIC PLANS AND PROCEDURES

- | | | |
|---|------------------------------|-----------------------------|
| 1. Based on a risk assessment of contracted activities and on mandatory OSHA compliance programs, the contractor will ensure that all applicable SOH risks and associated compliance plans are in place. | ☐ Yes | ☐ No |
| 2. Include all project-specific compliance plans, as applicable to the work being performed, and as identified below. The plans will incorporate project-wide procedures to control hazards to which the employees of all project employers may be exposed. | ☐ Yes | ☐ No |
| 3. These plans and procedures will be coordinated with all project employers and be included in project-specific emergency response and evacuation procedures. | ☐ Yes | ☐ No |

4. Address each of the plans in the outline below in the order that they are listed.
- Option 1: If a plan is applicable, develop and attach it to Appendix 5. Submit to the KO/COR with this worksheet for acceptance prior to work.
- Option 2: If a plan is applicable, but all of the minimum required information for the plan is not yet known, provide justification next to each plan title, and provide an expected date the plan will be created, reviewed, and attached to Appendix 5. Submit to the KO/COR with this worksheet for acceptance prior to work.
- Option 3: If a plan is not applicable because of the nature of the work to be performed, provide justification next to each plan title.

Minimum plans required by EM 385-1-1	Reference EM 385-1-1 paragraph:	Select Option	Justification for Option 2 and Option 3
Exposure Control Plan	3-7.a		
AED Program	3-7.b		
Construction Site Plan	4-7.a		
Access/Haul Road Plan	4-7.b		
Hearing Conservation Program	5-7.a		
Respiratory Protection Program	5-7.b		
Hazard Communication (HazCom) Program	6-7.a		
Process Safety Management (PSM) Program	6-7.b		
Lead Compliance Plan	6-7.c		
Asbestos Management Plan	6-7.d		
Ionizing Radiation Safety Plan	6-7.e		
Radiofrequency Plan	6-7.e		
Abrasive Blasting Plan	6-7.f		
Thermal Stress Management Plan	6-7.g		
Mold Remediation Plan	6-7.h		
Silica Compliance Plan	6-7.i		
Fatigue Management Plan	6-7.j		
Night Operations Lighting Plan	7-7		
Traffic Control Plan	8-7		
Fire Prevention Plan	9-7.a		
Wildland Fire Management Plan	9-7.b		
Housekeeping Plan (Welding and Cutting)	10-7		
Electrical Safety Plan	11-7.a		
Assured Equipment Grounding Conductor Program	11-7.b		
Site-Specific Energy Control Plan	12-7.a		
Standard Lift Plan	16-7.a & b		

Standard Lift Plan (Floating Plant)	16-7.a & c		
Critical Lift Plan	16-7.a & d		
Pile Driving Plan	16-7.a & e		
Structural Demolition Plan	17-7		
Soft Demolition Plan	17-7		
Emergency Plan for Marine Activities	19-7.a		
Severe Weather Plan for Marine Activities	19-7.b		
Float Plans	19-7.c		
Tow Plans	19-7.d		
Fall Protection and Prevention Plan	21-7.a		
Rescue Plan	21-7.b		
Scaffold Work Plan	22-7.b		
Rope Access Work Plan	23-7		
Excavation and Trenching Plan	25-7		
Compressed Air Work Plan	26-7.a & b		
Contingency Response Plan	26-7.a & c		
Emergency Rescue Plans and Equipment	26-7.a & d		
Ventilation and Dust Control Plan	26-7.a & e		
Fire Prevention and Protection Plan	26-7.a & f		
Formwork and Shoring Plan	27-7.a		
Precast Concrete Plan	27-7.b		
Masonry Erection Plan	27-7.c		
Steel Erection Plan	28-7		
Explosive Site Safety Plan	29-7.a		
Master Blasting Plan	29-7.b		
Vibration Monitoring Plan	29-7.c		
Safe Practices Plan	30-7.a & b		
Dive Operations Plan	30-7.a & c		
Emergency Management Manual	30-7.a & d		
Snorkeling Plan	30-7.a & e		
Tree Maintenance and Removal Plan	31-7		
Construction Safety and Phasing Plan	32-7.a		
Safety Plan Compliance Document	32-7.b		
Aviation Pre-Accident Plan	33-7.f		
Confined Space Plan	34-7.b & c		
Housekeeping Plan (Sanitation)	35-7.a		
Extermination Plan	35-7.b		
Site Safety and Health Plan	36-7.a		
Comprehensive Work Plan	36-7.b		
Emergency Response Plan	36-7.c		

PART 12: PROJECT SITE COORDINATION

1. Provide procedures for coordinating SOH activities with other employers on the site of work:

PART 13: REQUIRED APPENDICES

Indicate if each of the following items are, or are not, attached to this plan.

Appendix 1. Project Map	☐ Yes	☐ No
Appendix 2. Subcontractor/Suppliers	☐ Yes	☐ No
Appendix 3. Personal Qualifications and Designations	☐ Yes	☐ No
Appendix 4. Activity Hazard Analysis	☐ Yes	☐ No
Appendix 5. Site-specific plans and procedures	☐ Yes	☐ No
Appendix 6. Any additional plans, procedures, etc. can be added to this appendix and will be coordinated with the KO/COR.	☐ Yes	☐ No

QUALITY CONTROL REPORT-BUCKET/PIPELINE										PAGE NO. Page 1 of X	
CONTRACT NO. W912DS26CXXXX NA		CONTRACT TITLE Maintenance Dredging of			CONTRACTOR			REPORT NO. X			
CHARACTER OF REPORT		☒ MAINTENANCE ☐ NEW WORK ENVIRONMENT					DATE X/XX/2026				
DREDGE	NAME AND TYPE XXX				SHIFTS PER DAY			DAYS PER WEEK			
	SIZE		HORSEPOWER			NO. OF CREW MEMBERS					
	PIPELINE	DIPPER OR BUCKET	DREDGE PUMP	SUCTION PIPE JET	CUTTER OR BUCKET	DREDGE	SHORE	OTHER	TOTAL		
LOCATION/CHANNEL OF WORK											
LOCATION OF WORK		REACH DREDGED: STATION TO STATION			DISPOSAL AREA USE		QTY DEPOSITED GROSS (CY)		CUMULATIVE AND QTY DEPOSIT FOR DA (CY)		
CHARACTER OF MATERIAL (%)		GRAVEL %	SAND %	CLAY %	MUD %	SILT %	HARDPAN %	STONE %	SHELL %	OTHER %	
CHANNEL CONDITION		AVERAGE DEPTH			BEFORE DREDGING			AFTER DREDGING			
RIVER/TIDE STAGE	MIN	0	TIME	00:00	MAX	0	TIME	00:00	GAGE LOCATION		
	MIN	0	TIME	00:00	MAX	0	TIME	00:00	GAGE DATUM		
WEATHER CONDITION		WEATHER Weather Caused No Delay			TEMP (min/max) F / F		VISIBILITY MI		WIND MPH		
WORK PERFORMED					DISTRIBUTION OF TIME AND MILES RUN						
ITEM		UNIT		QUANTITY						HR.	MIN.
AVERAGE WIDTH OF CUT		FT				EFFECTIVE WORKING TIME					
TOTAL ADVANCED THIS PERIOD		FT				PUMPING OR DREDGING					
TOTAL ADVANCE PREVIOUSLY		FT				PCT OF EFFECTIVE TIME					
						BOOSTER (IN LINE)					
TOTAL ADVANCE TO DATE		FT				NON-EFFECTIVE WORKING TIME (CHARGEABLE COST OF WORK)					
SCOWS LOADED		NUMBER				HANDLING PIPE LINES					
AVERAGE LOAD PER SCOW		CY				HANDLING ANCHOR LINES					
FLOATING PIPE 0		SHORE PIPE 0		SUBMERGED PIPE		CLEARING PUMP AND PUMP LINES					
						CLEARING CUTTER AND SUCTION HEAD					
						WAITING FOR SCOWS					
TOTAL LENGTH OF DISCHARGE PIPE		FT				TO AND FROM WHARF OR ANCHORAGE					
CUBIC YARDS REMOVED		GROSS		CREDITED		CHANGING LOCATION OF PLANT ON JOB					
AMOUNT DREDGE THIS DATE						LOSS DUE TO OPPOSING NATURAL ELEMENTS					
AMOUNT PREVIOUSLY REPORTED						SHORE LINE AND SHORE WORK					
TOTAL AMOUNT DREDGED TO DATE						WAITING FOR BOOSTER					
AMOUNT DREDGED PER PUMPING/CUTTING HR						MINOR OPERATING REPAIRS (EXPLAIN IN REMARKS)					
						WAITING FOR ATTENDANT PLANT					
						PREPARATION AND MAKING UP TOW					
						TRANSFERRING PLANT BETWEEN WORKS					
COMMODITY		CONSUMED		CREDITED		LAY TIME OFF SHIFT AND SATURDAYS					
ITEM	UNIT	QUANTITY		QUANTITY		SUNDAYS AND HOLIDAYS					
FUEL	BBL					FIRE DRILL					
ELECTRICITY	KW					MOVING OUT OF WAY OF TRAFFIC					
LUBRICANTS	GAL					MISCELLANEOUS (EXPLAIN IN REMARKS)					
MOB DATE		DREDGE START		DEMOB DATE		TOTAL NON-EFFECTIVE TIME					
						PCT. OF NON-EFFECTIVE					
						TOTAL EFFECTIVE AND NON-EFFECTIVE TIME				24	0

QUALITY CONTROL REPORT-BUCKET/PIPELINE					PAGE NO. Page 2 of 4		
CONTRACT NO. W912DS26CXXXX NA		CONTRACT TITLE Maintenance Dredging of Arthur Kill 50'		CONTRACTOR		REPORT NO. XX	
DREDGE		XXX				DATE X/XX/2026	
ATTENDANT PLANT				LOST TIME (NOT CHARGEABLE TO COST OF WORK)			
ITEM	NAME OR NUMBER	H.P.	HOURS		HR	MIN	
				MAJOR REPAIRS			
				CESSATION			
				COLLISIONS			
				MISCELLANEOUS			
				TOTAL LOST TIME			
				PERCENTAGE OF TOTAL TIME			
				TOTAL TIME IN PERIOD	24	0	

QC NARRATIVE(S)

CONTRACTOR CERTIFICATION		On behalf of the contractor, I certify that this Report is complete and correct and all equipment and material used and work performed during this Reporting period are in compliance with the contract plans and specifications, to the best of my knowledge, except as noted above.		
QC REPRESENTATIVE'S SIGNATURE		DATE	SUPERINTENDENT'S INITIALS	DATE

TRANSPORTATION AND PLACEMENT LOG
CONTRACT NUMBER
PROJECT NAME

DATE: _____

LOAD NUMBER: _____

INSPECTOR: _____

WEATHER CONDITIONS: _____ **WIND SPEED:** _____ **VISIBILITY:** _____

TUG NAME: _____ **SCOW NAME:** _____

DESCRIPTION OF MATERIAL: _____

	<u>Date</u>	<u>Time</u>
LOADING STARTED:	_____	_____
LOADING COMPLETED:	_____	_____
SCOW DEPARTED DREDGING SITE:	_____	_____
SCOW ARRIVED DEWATERING SITE:	_____	_____
SCOW DEPARTED DEWATERING SITE:	_____	_____
SCOW ARRIVED PROCESSING SITE:	_____	_____
SCOW DEPARTED PROCESSING SITE:	_____	_____

COMMENTS: _____

Inspector Signature/date

DAILY SITE SAFETY & HEALTH INSPECTION LOG

Project:

Date: _____

Dredges, Scows, and other Work Areas Checked:

Names of All Employees in the Work area

_____	_____
_____	_____
_____	_____
_____	_____

Personal Protective Equipment and Work Equipment in Use

Special Safety and Health Issues & Notes

Name of Safety Personnel

Signature of Safety Personnel

TRANSPORTATION AND PLACEMENT LOG
CONTRACT NUMBER
PROJECT NAME

DATE: _____

LOAD NUMBER: _____

INSPECTOR: _____

WEATHER CONDITIONS: _____ **WIND SPEED:** _____ **VISIBILITY:** _____

TUG NAME: _____ **SCOW NAME:** _____

DESCRIPTION OF MATERIAL: _____

	<u>Date</u>	<u>Time</u>
LOADING STARTED:	_____	_____
LOADING COMPLETED:	_____	_____
SCOW DEPARTED DREDGING SITE:	_____	_____
SCOW ARRIVED DEWATERING SITE:	_____	_____
SCOW DEPARTED DEWATERING SITE:	_____	_____
SCOW ARRIVED PROCESSING SITE:	_____	_____
SCOW DEPARTED PROCESSING SITE:	_____	_____

COMMENTS: _____

Inspector Signature/date

DAILY SITE SAFETY & HEALTH INSPECTION LOG

Project:

Date: _____

Dredges, Scows, and other Work Areas Checked:

Names of All Employees in the Work area

_____	_____
_____	_____
_____	_____
_____	_____

Personal Protective Equipment and Work Equipment in Use

Special Safety and Health Issues & Notes

Name of Safety Personnel

Signature of Safety Personnel

DAILY WHALE REPORTING LOG

PROJECT: MAINTENANCE DREDGING, _____

1. DATE: _____

2. WHALE SIGHTED: YES _____ NO _____

3. TYPE OF WHALE: _____

4. TIME: _____

5. NUMBER OF WHALES SIGHTED:

ADULT _____ JUVENILE _____

6. NUMBER OF WHALES INJURED:

ADULT _____ JUVENILE _____ WORK-RELATED: YES _____ NO _____

7. NUMBER OF WHALES KILLED:

ADULT _____ JUVENILE _____ WORK-RELATED: YES _____ NO _____

8. LOCATION: _____

9. REMARKS: _____

10. SIGNATURE: _____

11. TITLE: _____

TURTLE OBSERVATION REPORTING LOG

PROJECT: MAINTENANCE DREDGING, _____

LOAD
NUMBER: _____ DATE _____ TIME _____

LOCATION IN CHANNEL: LATITUDE _____ LONGITUDE _____

WEATHER
CONDITIONS _____

PORT BASKET CONTENTS

TURTLE OR TURTLE PARTS PRESENT: YES _____ NO _____

COMMENTS AND OTHER OBSERVATIONS

BRIDGE WATCH: TIME _____ LOCATION _____

NUMBER OF TURTLES SIGHTED _____

OBSERVER'S NAME

Sea Turtle Handling and Resuscitation

It is unlikely that sea turtles will survive entrainment in a hopper dredge, as the turtles found in the dragheads are usually dead, dying, or dismantled. However, the procedures for handling live sea turtles follow in case the unlikely event should occur.

Please photograph all turtles (alive or dead) and turtle parts found during dredging activities and complete the Incident Report of Sea Turtle Take.

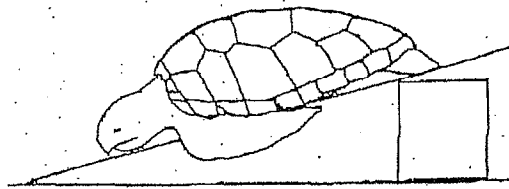
Live sea turtles

When a sea turtle is found in the dredge gear, observe it for activity and potential injuries.

- ▶ If the turtle is actively moving, it should be retained onboard until evaluated for injuries by a permitted rehabilitation facility. Due to the potential for internal injuries associated with hopper entrainment, it is necessary to transport the live turtle to the nearest rehabilitation facility as soon as possible, following these steps:
 - 1) Contact the nearest rehabilitation facility to inform them of the incident.
 - 2) Keep the turtle shaded and moist (e.g., with a water-soaked towel over the eyes, carapace, and flippers).
 - 3) Contact the crew boat to pick up the turtle as soon as possible from the dredge (within 12 to 24 hours maximum). The crew boat should be aware of the potential for such an incident to occur and should develop an appropriate protocol for transporting live sea turtles.
 - 4) Transport the live turtle to the closest permitted rehabilitation facility able to handle such a case.

Do not assume that an inactive turtle is dead. The onset of rigor mortis and/or rotting flesh are often the only definite indications that a turtle is dead. Releasing a comatose turtle into any amount of water will drown it, and a turtle may recover once its lungs have had a chance to drain.

- ▶ If a turtle appears to be comatose (unconscious), contact the designated stranding/rehabilitation personnel immediately. Once the rehabilitation personnel has been informed of the incident, attempts should be made to revive the turtle at once. Sea turtles have been known to revive up to 24 hours after resuscitation procedures have been followed. The resuscitation regulations can be found at 50 CFR 223.206(d)(1).
 - Place the animal on its bottom shell (plastron) so that the turtle is right side up and elevate the hindquarters at least 6 inches for a period of 4 up to 24 hours. The degree of elevation depends on the size of the turtle; greater elevations are required for larger turtles.
 - Periodically, rock the turtle gently left to right and right to left by holding the outer edge of the shell (carapace) and lifting one side about 3 inches then alternate to the other side.
 - Periodically, gently conduct one of the above reflex tests to see if there is a response.
 - Keep the turtle in a safe, contained place, shaded, and moist (e.g., with a water-soaked towel over the eyes, carapace, and flippers) and observe it for up to 24 hours.
 - If the turtle begins actively moving, retain the turtle until the appropriate rehabilitation personnel can evaluate the animal. The rehabilitation facility should eventually release the animal in a manner that minimizes the chances of re-impingement and potential harm to the animal (i.e., from cold stunning).
 - Turtles that fail to move within several hours (up to 24) must be handled in the manner described in Appendix B-II-E, or transported to a suitable facility for necropsy (if the condition of the sea turtle allows and the rehabilitation facility wants to necropsy the animal).



Stranding/rehabilitation contacts

- In New York, the stranding network contact is Kim Durham, Riverhead Foundation for Marine Research, 431 East Main Street, Riverhead, NY (631-369-9829), and New Jersey contacts include Bob Schoelkopf at the Marine Mammal Stranding Center, P.O. Box 773, Brigantine, NJ (609-266-0538).
- Dana Hartely, (NOAA Fisheries Stranding Network Coordinator: (508) 495-2090) should also be contacted immediately for any marine mammal injuries or mortalities.

ENDANGERED SPECIES OBSERVER FORM

Daily Report

Date: _____

Geographic Site: _____

Location: Lat/Long _____ Vessel Name _____

Weather conditions: _____

Water temperature: Surface _____ Below midwater (if known) _____

Condition of screening apparatus: _____

Incidents involving endangered or threatened species? (Circle) Yes No
(If yes, fill out Incident Report of Sea Turtle/Shortnose Sturgeon Mortality)

Comments (type of material, biological specimens, unusual circumstances,
etc.): _____

Observer's Name: _____

Observer's Signature: _____

BRIDGE WATCH SUMMARY

Species # of Sightings # of Animals Comments

_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

Incident Report of Sea Turtle Take

Species _____ Date _____ Time (specimen found) _____

Geographic Site _____

Location: Lat/Long _____

Vessel Name _____ Load # _____

Begin load time _____ End load time _____

Begin dump time _____ End dump time _____

Sampling method _____

Condition of screening _____ Location where

specimen recovered _____

Draghead deflector used? YES / NO Rigid deflector draghead? YES / NO

Condition of deflector _____

Weather conditions _____

Water temp: Surface _____ Below midwater (if known) _____

Species Information: (please designate cm/m or inches.)

Head width _____ Plastron length _____

Straight carapace length (or total length) _____

Straight carapace width _____

Curved carapace length _____

Curved carapace width _____

Condition of specimen/description of animal (please complete attached
diagram) _____

Turtle Decomposed: NO SLIGHTLY MODERATELY SEVERELY

Turtle tagged: YES / NO Please record all tag numbers. Tag # _____

Photograph attached: YES / NO

(please label species, date, geographic site and vessel name on back of photograph)

Comments/other (include justification on how species was identified)

Observer's Name _____

Observer's Signature _____

Incident Report of Sea Turtle Take

Draw wounds, abnormalities, tag locations on diagram and briefly describe below.

Description of animal:

Incident Report of Atlantic Sturgeon Take

Photographs should be taken and the following information should be collected from all sturgeon (alive and dead) found in association with the project.

Date _____ Time (specimen found) _____

Geographic Site _____

Location: Lat/Long _____

Vessel Name _____ Load # _____

Begin load time _____ End load time _____

Begin dump time _____ End dump time _____

Sampling method _____

Condition of screening _____

Location where specimen recovered _____

Draghead deflector used? YES NO Rigid deflector draghead? YES NO

Condition of deflector _____

Weather conditions _____

Water temp: Surface _____ Below midwater (if known) _____

Species Information: *(please designate cm/m or inches.)*

Fork length (or total length) _____ Weight _____

Condition of specimen/description of animal _____

Fish Decomposed: NO SLIGHTLY MODERATELY SEVERELY

Fish tagged: YES / NO Please record all tag numbers. Tag # _____

Genetic sample taken: YES NO

Photograph attached: YES / NO

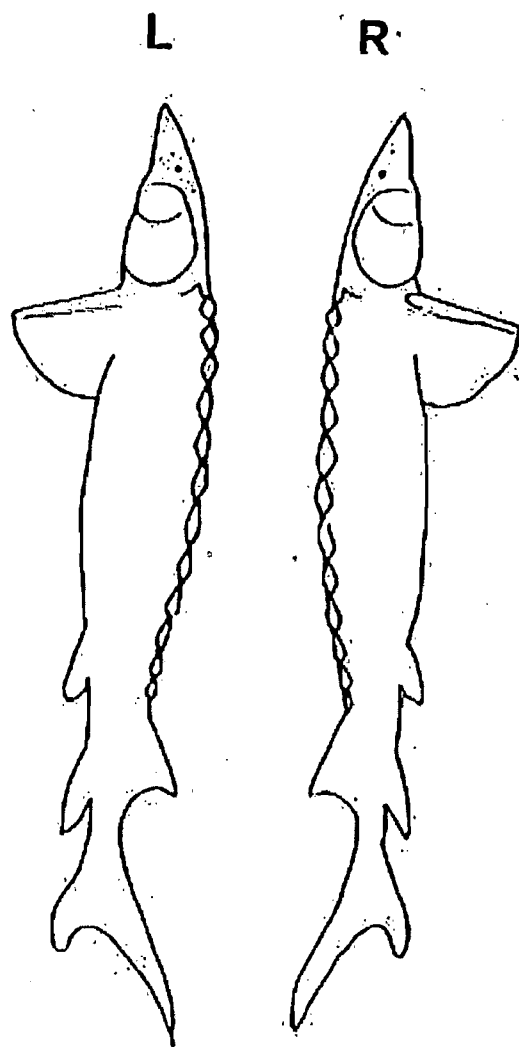
(please label species, date, geographic site and vessel name on back of photograph)

Comments/other (include justification on how species was identified)

Observer's Name _____

Observer's Signature _____

Draw wounds, abnormalities, tag locations on diagram and briefly describe below



Description of fish condition:

SUBMITTAL REGISTER

TITLE AND LOCATION												CONTRACT NO.					
Maint. Dredging of NY&NJ Channels -Seguine Ward Outerbridge Federal Navigation Project						CONTRACTOR											
ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		APPROVING AUTHORITY						REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		00 73 00	SD-01 Preconstruction Submittals														
			Order of work	1.35													
			Accident prevention plan	1.35	G												
			Quality control plan	1.35	G												
			Letter appointing superintendent	1.35													
			List of subcontractors	1.35													
			Environmental protection plan	1.35	G												
			Progress schedule	1.35	G												
			Submittal register	1.35													
			List of proposed products	1.35	G												
			Work plan	1.35													
		01 30 00	SD-01 Preconstruction Submittals														
			Project Schedule		G												
			SD-03 Product Data														
			Monthly Progress Report	1.4													
			Project Schedule Status Report	1.4													
		01 33 00	SD-01 Preconstruction Submittals														
			Submittal Register														
		01 35 26	SD-01 Preconstruction Submittals														
			Accident Prevention Plan (APP)	1.6	G												
			RO														
			Activity Hazard Analysis (AHA)	1.7	G												
			Crane Critical Lift Plan	1.6.1	G												
			Crane Operators	1.5.1.3	G												
			SD-06 Test Reports														
			Notifications and Reports	1.11													

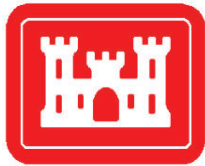
CONTRACT NO.

TITLE AND LOCATION

Maint. Dredging of NY&NJ Channels -Seguine Ward Outerbridge Federal Navigation Project

CONTRACTOR

[illegible]



**US ARMY CORPS
OF ENGINEERS**
NEW YORK DISTRICT



September 16, 2025

MEMO FOR THE RECORD

SUBJECT: Review of Compliance with the Requirements of 40 CFR 227.6 and 227.27, and Site Designation Provisions of 40 CFR 228.15 for NY&NJ Channels, Seguine Point Bend, Ward Point Bend to Outerbridge Crossing Federal Navigation Project Reach 1

**PATRICIA
PECHKO**

Digitally signed by
PATRICIA PECHKO
Date: 2025.09.16 13:09:16
-04'00'

FROM: Mark Reiss, Chief
Dredging, Sediments, and Oceans Section
Water Division
EPA Region 2

**VEGA.KELLY.LOU
ISE.1237379132**

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VEGA.KELLY.LOUISE.1237379132
Date: 2025.09.16 14:45:08 -04'00'

Kelly Vega, Chief
Dredged Material Management Section
Operations Division, New York District
U.S. Army Corps of Engineers

I. SUMMARY

The USACE is proposing to dredge the NY&NJ Channels, Seguine Point Bend, Ward Point Bend to Outerbridge Crossing Federal Navigation Project (herein referred to as SWO FNP) located in New York with placement of the dredged material as Remediation Material at the Historic Area Remediation Site (HARS) in the Atlantic Ocean.



This memorandum provides comprehensive review and analysis of SWO FNP test results for Reach 1. This memorandum addresses compliance with the regulatory testing criteria of 40 CFR Sections 227.6 and 227.27, and the requirements of the rule establishing the Historic Area Remediation Site (HARS) set out in Section 228.15(d)(6). These requirements hereinafter are referred to as the “Regulations.” Project dredged material meets the non-testing regulatory criteria of Subparts B, C, D, and E of 40 CFR § 227 pertaining to: environmental impacts, need, impact on esthetic, recreational and economic values, and other uses of the ocean (EPA/USACE 2019).

As discussed in the HARS rulemaking preamble (see 62 Fed. Reg. 46142 (August 29, 1997) and 62 Fed. Reg. 26267 (May 13, 1997)) and its accompanying documentation, the need to remediate the Historic Area Remediation Site is amply supported by the presence in the HARS of toxic effects, dioxin bioaccumulation exceeding Category I levels in worm tissue, as well as TCDD/PCB contamination in area lobster stocks. Individual elements of the aforementioned data do not prove that sediments within the Study Area are imminent hazards to the New York Bight Apex ecosystem, living resources, or human health. However, the collective evidence presents cause for concern, justifies a need for remediation, indicates that the site is Impact Category I (see 40 CFR 228.10), and necessitates that the site should be managed to reduce impacts to acceptable levels (see 40 CFR 228.11(c)). Further information on the conditions in the Study Area and the surveys performed may be found in the Supplemental Environmental Impact Statement (EPA, 1997c).

This evaluation confirms that: 1) all tests required under the Regulations were conducted, 2) this project meets the criteria at 40 CFR Section 227.6 for trace contaminants and Section 227.27 for Limiting Permissible Concentration (LPC), and 3) the dredged material is Category I under U.S. Environmental Protection Agency (EPA) Region 2/U.S. Army Corps of Engineers, NY District (CENAN) guidance and is suitable for placement at the HARS as Remediation Material.

II. PROJECT DESCRIPTION

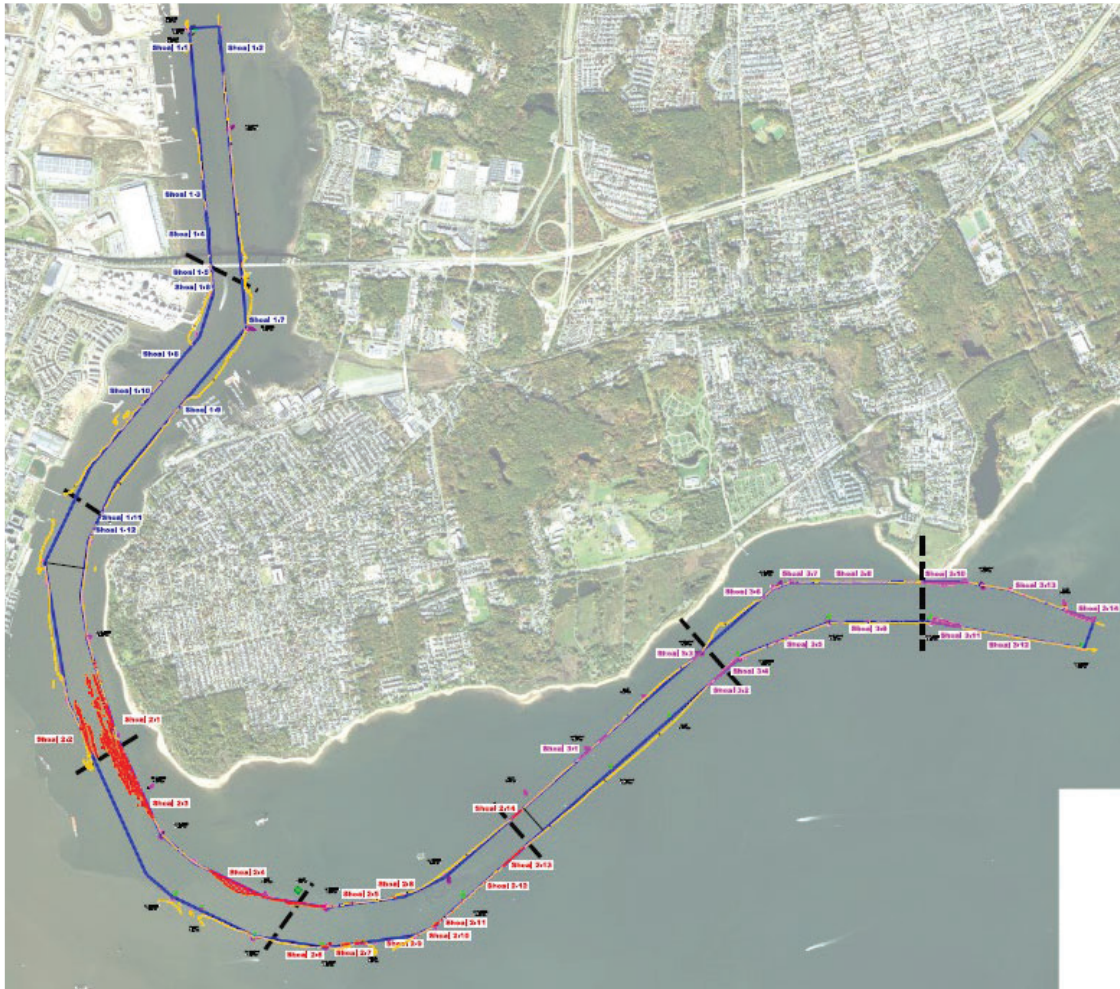
The proposal is to dredge shoals that extend along the western, southern, and eastern sides of Staten Island, NY, from north of Outerbridge Crossing to east of Seguine Point in three reaches that have been composited for testing within the SWO FNP to a depth of -35’ MLLW with an additional 2’ of allowable overdredge with placement of approximately 106,485 cy of (based on an August 8, 2024 bathymetric survey) at the HARS. The project has been sampled and tested as three separate reaches for the purposes of bioassay testing (see sampling plan; CENAN 2024) as described below:

Reach 1, shown below in blue hatching has an estimated volume of 5,585 cy. The project sediments from this reach are represented by 7 core samples composited for the purpose of bioassay testing.

Reach 2, shown below in red hatching has an estimated volume of 87,034 cy. The project sediments from this reach are represented by 9 core samples composited for the purpose of bioassay testing.

Reach 3, shown below in purple hatching has an estimated volume of 13,866 cy. The project sediments from this reach are represented by 5 core samples composited for the purpose of bioassay testing.

This memorandum provides comprehensive review and analysis of Reach 1.



III. REGULATORY REQUIREMENTS

For dredged material to be suitable for use as Remediation Material, it must conform to the Regulations. The Marine Protection, Research, and Sanctuaries Act (MPRSA) or “The Act” prohibits dumping of materials into the ocean except as authorized by EPA or, in the case of dredged materials, by the U.S. Army Corps of Engineers (USACE). Section 102 of the Act directs the EPA to establish and apply criteria for reviewing and evaluating permit applications (33 U.S.C. Section 1412). EPA has adopted such criteria in the Regulations. 40 CFR Section 227.6(a) lists constituents that are prohibited from being placed in the ocean unless only present as trace contaminants in material otherwise suitable for dumping (hereinafter referred to as “listed constituents”). Section 227.27 addresses compliance with the LPC. See also, Section 227.13(c).

Section 227.6(b) states that these listed constituents are considered to be present as trace contaminants only when they are present in such forms and amounts that the “dumping of the

materials will not cause significant undesirable effects, including the possibility of danger associated with their bioaccumulation in marine organisms.” The Regulations set forth criteria for determining the potential for significant undesirable effects in Section 227.6(c). To be found environmentally acceptable for ocean placement, the liquid phase must not contain any of the listed constituents in concentrations that would exceed applicable marine water quality criteria after allowance for initial mixing (Section 227.6(c)(1)). For the suspended particulate phase (Section 227.6(c)(2)) and the solid phase (Section 227.6(c)(3)), bioassay results must not indicate occurrence of significant mortality or significant adverse sublethal effects due to the ocean placement of dredged material containing the listed constituents.

Section 227.27 of the Regulations addresses the LPC. For the liquid phase, Section 227.27(a) provides that the LPC is that concentration which does not exceed applicable marine water quality criteria after initial mixing, or when there are no applicable marine water criteria, that concentration of material after initial mixing, would not exceed 0.01 of a concentration shown to be acutely toxic to appropriate sensitive marine organisms in a bioassay carried out in accordance with procedures approved by EPA and USACE. For the suspended particulate phase and the solid phase, Section 227.27(b) provides that the LPC is that concentration of material which will not cause unreasonable acute or chronic toxicity or other sublethal adverse effects based on results of bioassays using appropriate sensitive organisms and conducted according to procedures that have been approved by EPA and USACE, and which will not cause accumulation of toxic materials in the human food chain.

The HARS encompasses an area which includes the Mud Dump Site (MDS), and which has exhibited the potential for adverse ecological impacts and has been identified for remediation. The site will be remediated with uncontaminated dredged material (i.e., dredged material that meets current Category I standards and will not cause significant undesirable effects including through bioaccumulation) (hereinafter referred to as “Remediation Material” or “Material for Remediation”). Under 40 CFR 228.15 (d)(6)(v)(A) the site will be managed to reduce impacts within the Priority Remediation Area (PRA) to acceptable levels in accordance with 40 CFR Section 228.11(c). Use of the site is restricted to dredged material suitable for use as Material for Remediation. This material shall be selected to ensure it will not cause significant undesirable effects including through bioaccumulation or unacceptable toxicity in accordance with 40 CFR 227.6.

Section 228.15(d)(6) of the Regulations describes the locations where material from NY/NJ Harbor and surrounding areas may be placed in the HARS, if it is suitable as Remediation Material. The HARS consists of nine PRAs, a Buffer Zone, and a No Discharge Zone. Under 228.15(d)(6) placement of Remediation Material is limited to the PRAs.

IV. GUIDANCE FOR TESTING AND EVALUATION OF DREDGED MATERIAL

The discussion below describes how the material proposed to be dredged from the SWO FNP Reach 1 was evaluated for placement at the HARS and evaluated for compliance with the requirements of 40 CFR 227.6, 227.27, and 228.15(d)(6). Testing of the material was conducted following procedures approved by EPA and USACE, and contained in the joint EPA/USACE national guidance “Evaluation of Dredged Material Proposed for Ocean Dumping - Testing

Manual” (the “Green Book,” EPA/USACE, 1991), and the regional implementation manual developed by the EPA Region 2 and CENAN (EPA/CENAN, 2016).

These test results were analyzed in accordance with the Regulations to ensure that the proposed placement meets the criteria of Part 227 and the requirements of 228.15(d)(6). As explained in the preamble to the HARS Rule, Remediation Material is uncontaminated dredged material (i.e., dredged material that meets current Category I standards and will not cause significant undesirable effects including through bioaccumulation). The determination of whether materials meeting the Part 227 criteria are assigned to Category I is based upon technical and scientific judgment as set out below.

Applying the EPA Region 2/CENAN guidance to this project, the material would be Category I if it meets the Part 227 criteria (including the requirements regarding acute toxicity) and:

- bioaccumulation test results do not exceed the Regional Matrix Values for cadmium, mercury, total PCBs (clam), and total DDT (USACE 1981), and are below the regional Category I guidance values for dioxin (EPA 1997a), and
- bioaccumulation test results do not exceed the HARS-Specific Polychlorinated Biphenyl (PCB) Worm Tissue Criterion defined in 40 CFR 228.15(d)(6)(v)(E), and
- bioaccumulation test results for the other bioaccumulative chemicals of concern identified in EPA/CENAN (2016) do not indicate a potential for undesirable effects using conservative assessment techniques.

Sediments that meet this Category I definition are suitable for placement at the HARS as Remediation Material as they will improve sediment conditions at the HARS to reduce impacts to acceptable levels in accordance with 40 CFR Section 228.11(c). Sediments that do not meet this definition are not suitable for placement at the HARS.

V. RESULTS OF EVALUATION OF THE MATERIAL

A. Evaluation of the liquid phase

The liquid phase of the material was evaluated for compliance with Sections 227.6(c)(1) and 227.27(a). There are applicable marine water quality criteria for constituents in the material, including listed constituents, and the applicable marine water quality criteria would not be exceeded after initial mixing. In addition, liquid phase bioassays run as part of the suspended particulate phase on three appropriate sensitive marine organisms, show that after initial mixing (as determined under 40 CFR 227.29(a)(2)), the liquid phase of the material would not exceed a toxicity threshold of 0.01 of a concentration shown to be acutely toxic to appropriate sensitive marine organisms. Accordingly, it is concluded that the liquid phase of the material complies with 40 CFR 227.6(c)(1) and 227.27(a). The specific test results and technical analysis of the data underlying this conclusion are described and evaluated in CENAN memo dated 2025.

B. Evaluation of the suspended particulate phase

The suspended particulate phase of the material was evaluated for compliance with Sections 227.6(c)(2) and 227.27(b). Bioassay testing of the suspended particulate phase of the material has been conducted using three appropriately sensitive marine organisms: inland silversides (*Menidia beryllina*), mysid shrimp (*Americamysis bahia*, formerly *Mysidopsis bahia*), and the planktonic larvae of the blue mussel (*Mytilus galloprovincialis*). That information shows that when placed in the HARS and after initial mixing (as determined under 40 CFR 227.29(a)(2)), the suspended particulate phase of this material would not exceed a toxicity threshold of 0.01 of a concentration shown to be acutely toxic in the laboratory bioassays (LC/EC₅₀), and thus would not result in significant mortality. The material may be discharged on either incoming or outgoing tides. However, based on a dump volume of 3500 cubic yards, discharge may occur only at distances of at least 450 feet from the northern, southern, eastern, and western boundaries of the HARS, independent of the direction of tidal transport at the time of disposal. Based on a dump volume of 6000 cubic yards, discharge may occur only at distances of at least 600 feet from the northern, southern, eastern, and western boundaries of the HARS, independent of the direction of tidal transport at the time of disposal. The specific test results and technical analysis of the data underlying this conclusion are described in CENAN 2025.

The factor of 0.01 was applied to ensure that there would be no significant adverse sublethal effects. Moreover, the fact that after placement, the suspended particulate phase would only exist in the environment for a short time, means the suspended particulate phase would not cause significant undesirable effects, including the possibility of danger associated with bioaccumulation, since these impacts require longer exposure durations (see EPA, 1994). Accordingly, it is concluded that the suspended phase of the material complies with 40 CFR 227.6(c)(2) and 227.27(b).

C. Evaluation of the solid phase

The solid phase of the material was evaluated for compliance with Sections 227.6(c)(3) and 227.27(b). This evaluation was made using the results of two specific types of evaluations on the solid phase of the material, one focusing on the acute (ten-day) toxicity of the material, and the other focusing on the potential for contaminant bioaccumulation in test organisms exposed to the dredged material as compared to those same endpoints in test organisms exposed to a reference sediment. Both types of tests were conducted following procedures approved by EPA and the USACE (EPA/USACE 1991).

Reference sediment serves as a point of comparison to identify potential effects of contaminants in the dredged material (EPA/USACE, 1991). Exposing test organisms to this sediment allows for the prediction of toxicity and bioaccumulated contaminant levels that would be expected to result in the test organisms were they “in the wild” at the area from which the reference sediment was taken. The reference sediment used for this project was collected at the Reference Site, in an area of clean, sandy sediments located in the New York Bight near the HARS, where the sediments are unaffected by prior dredged material disposal.

To make a statistically valid determination that exposure to the project sediment does or does not cause greater toxicity or bioaccumulation than the reference sediment, laboratory exposures to several sub-samples (i.e., replicates) of the dredged material and reference sediment are run concurrently. Means and standard deviations are calculated for each set of exposures and compared using standard statistical approaches to determine, with 95% confidence, whether

exposure to project sediment results in significantly higher toxicity or bioaccumulation than exposure to reference sediment. Throughout this memorandum, statements regarding project sediment having “greater” or “less” toxicity or bioaccumulation are referring to calculated differences which are statistically significant at the 95% confidence level.

A statistically significant difference between test and reference bioaccumulation is not itself a quantitative prediction that an impact would occur in the field. A key to understanding bioaccumulation and potential adverse impacts is the fact that bioaccumulation does not necessarily result in an effect. At low exposure (concentration and duration), bioaccumulation may cause no harm. On the other hand, as exposure and/or bioaccumulation increases, the potential for adverse effects increases.

The following sections address the results of solid phase tests and further analyze compliance with the regulatory criteria of Sections 227.6(c)(3), 227.27(b), and 228.15 and with EPA Region 2/CENAN guidance.

Solid phase toxicity evaluation

Ten-day acute toxicity tests were conducted on project materials using mysids (*Americamysis bahia*) and amphipods (*Ampelisca abdita*), which are appropriate sensitive benthic marine organisms. These organisms are good predictors of adverse effects to benthic marine communities (see EPA 1996a). The results of these tests show that the solid phase of the material would not cause significant mortality (statistically different project sediment-exposed mortality exceeding mortality in the reference sediment exposures by 10% for mysid shrimp or 20% for amphipods) and meets the solid phase toxicity criteria of Sections 227.6 and 227.27.

Solid phase bioaccumulation evaluation

28-day tests were conducted on the solid phase of the project material to assess the potential for bioaccumulation of contaminants of concern identified in the project sampling plan (CENAN 2018) using two appropriate sensitive benthic marine organisms, sand worm (*Alitta virens*, formerly *Nereis virens*) and bent-nosed clam (*Macoma nasuta*). These species are good representatives of the phylogenetically diverse base of the marine food chain. Contaminants of concern were identified from Squibb, *et al.*, (1991), a report prepared for the Toxics Work Group of the NY/NJ Harbor Estuary Program to identify and characterize contaminants in NY/NJ Harbor. Compounds from that list, including organic compounds with the potential to bioaccumulate (K_{ow} of approximately 4 or greater), are included on the bioaccumulation testing list when expected to be present in project sediments based upon contaminant inputs and previous sediment sampling results.

There are FDA Action Levels for seven compounds (aldrin, dieldrin, a-chlordane, heptachlor, heptachlor epoxide, PCBs, and mercury; see Table 4, Column 18). Bioaccumulation of any of these contaminants to levels exceeding an FDA Action Level (see EPA/USACE 1991) results in a conclusion that ocean disposal of the dredged material would cause significant adverse effects and exceed the LPC as per 40 CFR 227.27. Therefore, dredged material causing bioaccumulation of any of these contaminants to levels exceeding FDA Action Levels is not suitable for use as Remediation Material at the HARS. None of the contaminants for which there are FDA Action Levels exceed such thresholds in the tissues of organisms exposed to sediments from this project.

When bioaccumulation of a specific contaminant or class of contaminants in organisms exposed to project sediments is not significantly greater than bioaccumulation in organisms exposed to reference sediments, that contaminant is concluded to be present in forms and amounts that do not result in bioaccumulation that would cause unreasonable degradation of the environment or human health, or significant adverse effects. These contaminants are not considered further in the evaluation unless they contribute to a mixture that is evaluated against a single combined guideline value or criterion (i.e., total PCBs, PAH mixtures, total DDT, total chlordane and total endosulfans). In cases where bioaccumulation levels are statistically greater (at the 95% confidence limit) in project sediment-exposed organism than in reference sediment-exposed organisms, further evaluation for potential effects is warranted to evaluate any risk that might be associated with accumulated contaminants.

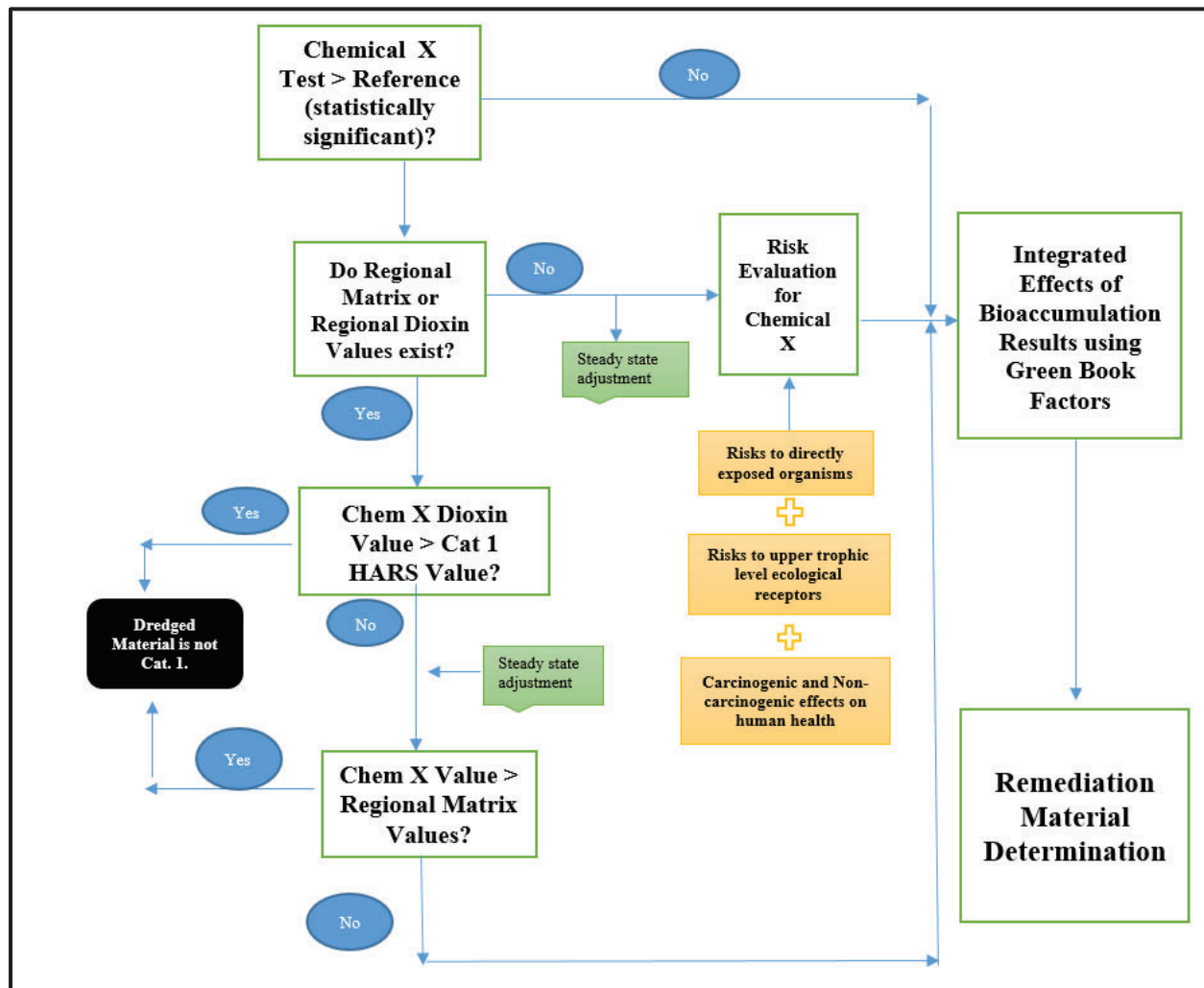
EPA Region 2 and CENAN use a framework to evaluate project sediment bioaccumulation results that follows national guidance described in the Green Book (EPA/USACE 1991). This guidance outlines a series of evaluations to consider risks associated with project sediment bioaccumulation results that exceed reference. The framework used for HARS-suitability determinations is shown in Figure 1 and includes the following evaluations:

- *Comparison of the 28-day bioaccumulation results to Regional Dioxin Values.* Dioxins and furans that bioaccumulate to greater than reference are evaluated by comparing accumulated concentrations to Regional Dioxin Values (EPA 1997a) (listed in Table 4, Column 20 of Dioxin sheet). 28-day bioaccumulation test results that equal or exceed the Regional Dioxin Values for 2,3,7,8-TCDD or total toxic equivalency (TEQ) of combined congeners indicate that the sediment is not Category I dredged material and is not suitable for use as Remediation Material at the HARS.
- *Comparison of the 28-day bioaccumulation results to Regional Matrix Values.* PCBs, DDTs, cadmium and mercury that bioaccumulate to concentrations greater than reference are evaluated by comparing 'steady state'-adjusted concentrations to Regional Matrix Values (USACE 1981) and the HARS-Specific PCB Worm Tissue Criterion (40 CFR 228.15(d)(6)(v)(E)) (listed in Table 4, Column 20; steady state adjustment is discussed in Section 1 of Appendix A). Bioaccumulation test results that exceed the Regional Matrix Values or HARS-Specific PCB Worm Tissue Criterion indicate that the sediment is not Category I dredged material and is not suitable for use as Remediation Material at the HARS.
- Risk-based assessments are conducted to evaluate concentrations of compounds that bioaccumulate above reference for which Regional Dioxin Values or Regional Matrix Values do not exist. These evaluations are made using 'steady state'-adjusted concentrations and include comparisons to risk values protective of benthic organisms and upper level trophic receptors, including humans.
- The final evaluation uses all available information and results of the individual chemical evaluations to evaluate the solid phase of the dredged material as a whole.

The remainder of this memorandum documents the assessment conducted using the described framework to evaluate risks associated with contaminants that bioaccumulated to greater than

reference levels in tissues of organisms exposed to SWO FNP Reach 1 sediment. Bioaccumulation test results for this project are summarized in Table 4.

Figure 1. EPA Region 2/CENAN Framework for Evaluating Bioaccumulation Test Results for SWO FNP



Solid Phase Bioaccumulation Evaluation-Comparison to Reference Sediment Bioaccumulation

Contaminants for which bioaccumulation from the dredged material in tissue was statistically greater than the reference in the clam and/or worm are indicated by an asterisk in the applicable column of Table 1.

Table 1. Statistically Significant Bioaccumulation of Contaminants in Organisms Exposed to Project Sediments vs. Reference Sediments

CONTAMINANT	CLAMS	WORMS
PAHs		
Acenaphthene	*	
Acenaphthylene	*	*
Anthracene	*	*
Fluorene	*	*
Naphthalene	*	
Phenanthrene		*
Benzo(a)anthracene**	*	*
Benzo(a)pyrene**	*	*
Benzo(g,h,i)perylene	*	*
Benzo(b)fluoranthene**	*	*
Benzo(k)fluoranthene**	*	*
Chrysene**	*	*
Dibenzo(a,h)anthracene**	*	
Fluoranthene	*	*
Indeno(1,2,3-cd)pyrene**	*	*
Pyrene	*	*
PESTICIDES		
Dieldrin	*	*
a-Chlordane	*	*
Trans nonachlor	*	*
Endosulfan I	*	*
Endosulfan II	*	*
Endosulfan sulfate	*	*
4,4-DDT		*
4,4-DDD	*	*
2,4-DDD	*	*
4,4-DDE	*	*
2,4-DDE	*	*
TOTAL PCBs	*	*
1,4-Dichlorobenzene	*	
METALS		
Arsenic	*	
Chromium (total)	*	*
Copper	*	*
Lead	*	*
Nickel	*	*
Zinc	*	
DIOXINS/FURANS		
2,3,7,8,-TCDD		*
1,2,3,7,8-PeCDD		*
1,2,3,4,6,7,8-HpCDD	*	*
OCDD	*	*
2,3,7,8-TCDF		*
2,3,4,7,8-PeCDF		*
1,2,3,4,7,8-HxCDF	*	
1,2,3,6,7,8-HxCDF	*	
2,3,4,6,7,8-HxCDF	*	
1,2,3,4,6,7,8-HpCDF		*
OCDF	*	

Solid Phase Bioaccumulation Evaluation - Comparison to Regional Dioxin Values

Bioaccumulation test results were below all Regional Dioxin Values (EPA 1997a) for both test organisms. Therefore, no further evaluation of these contaminants is required in accordance with EPA Region 2/CENAN guidance.

Solid Phase Bioaccumulation Evaluation- Comparison to Regional Matrix Values and the HARS-Specific PCB Worm Tissue Criterion

Bioaccumulation test results (adjusted for steady state, as described in Section 1 of Appendix A) were below all Regional Matrix Values (USACE 1981) for both test organisms and the mean 'steady state'-adjusted worm PCB tissue concentration was below the HARS-Specific PCB Worm Tissue Criterion (40 CFR 228.15(d)(6)(v)(E)). Therefore, no further evaluation of these contaminants is required in accordance with EPA Region 2/CENAN guidance.

Solid Phase Bioaccumulation Evaluation- Risk-based Evaluations

Risk associated with compounds that bioaccumulated to greater than reference in test organisms exposed to project sediment for which Regional Matrix Values (USACE 1981) or Regional Dioxin Values (EPA 1997a) do not exist are evaluated using risk-based assessments, as discussed below. Contaminants that fall into this group for this project include: PAHs, dieldrin, a-chlordane, trans nonachlor, endosulfan I, endosulfan II, endosulfan sulfate, 1,4-dichlorobenzene, arsenic, total chromium, copper, lead, nickel, silver and zinc.

Contaminants bioaccumulated from dredged material can present risks to organisms that are directly exposed to the sediment and to species at upper trophic levels of the marine food web (e.g., predatory fish). Human exposure can potentially result via consumption of fish that has been dietarily exposed to HARS contaminants. Risk to directly exposed ecological receptors, risk to dietarily exposed ecological receptors, and human risk are examined separately in the EPA Region 2/CENAN framework.

a. Risks to Directly Exposed Organisms

Metals, Pesticides, and Industrial Chemicals

Water Quality Criterion Tissue Levels (WQCTL) are established for aldrin, dieldrin, total chlordane, total endosulfan, 1,4-dichlorobenzene, arsenic chromium, copper, lead, nickel, silver and zinc (see Appendix, Section 2, A). For this project, none of the WQCTLs were exceeded.

PAHs

To assess direct exposure risks of PAHs, a direct comparison is made of total measured PAH tissue residue to the Critical Body Residue (CBR) associated with narcosis, which is the primary non-cancer effect of PAHs. The CBR approach, described by McCarty (1991), was used to evaluate the potential impacts of total PAHs accumulated in the dredged material bioaccumulation test organisms and is discussed in the Appendix. As shown in Table 4, total PAH levels in tissues from the dredged material bioaccumulation test were below levels at which chronic adverse effects might be expected from a narcotic mode of action in sensitive aquatic organisms (i.e., fish) as estimated by the CBR.

Therefore, these bioaccumulation test results do not indicate a potential for undesirable effects to directly exposed organisms resulting from use of this material as Remediation Material at the HARS.

b. Risks to Upper Trophic Level Ecological Receptors

To properly assess risks to upper trophic level species (including humans) the efficiency with which a contaminant is transferred through the diet and the propensity of those compounds to accumulate to higher concentrations in the tissues of the consumer must be considered. The potential movement of contaminants through the food chain was considered and appropriate trophic transfer factors applied to adjust the data accordingly to reflect the propensity of specific contaminants or contaminant groups to accumulate to higher (or lower) concentrations in upper trophic level organisms (including humans). Data and/or methods used to develop these trophic transfer adjustment factors are discussed in the Appendix.

Effects of Mutagenic, Carcinogenic and Teratogenic PAHs

Applying an uncertainty factor (UF) of 10 to account for interspecies differences in sensitivities and a trophic transfer factor of 0.1 to the no-effects level for BaP calculated from Hannah, *et al.* (1982), as discussed in the Appendix results in a no-effect level for BaP of approximately 8,000 for BaP only ppb in benthic tissue, which is considerably greater than the highest steady state-adjusted tissue concentration of BaP measured in organisms exposed to project sediment (23.8 ppb for BaP only and 33.3 ppb for total BaP equivalents). Project tissue concentrations would still be below this no-effect level if the higher trophic transfer factor (0.23) reported by McElroy, *et al.* (1991) was used.

Hall and Oris (1991) reported on experiments that exposed fathead minnows to anthracene during long-term exposures and observed adverse effects on reproduction. The paper reported that a concentration of anthracene in the tissue of the egg in the range of 3,750 to 8,000 ppb resulted in no significant effects on egg hatching or survivorship. Using the same approach for accounting for species-to-species uncertainty and food chain transfer yields a conservative benthic tissue level of 3,750 ppb. Anthracene tissue concentrations from the project bioaccumulation tests are well below this level.

As stated above, PAHs accumulated to greater than reference in test organisms exposed to project sediments. However, the most relevant aquatic effects information reviewed indicates that the highest tissue PAH levels accumulated in the dredged material bioaccumulation tests are below mutagenic, carcinogenic, and teratogenic no-effect levels.

c. Consideration of Potential Carcinogenic and Non-carcinogenic Effects on Human Health

Benthic tissue screening levels for human carcinogenic risk and non-carcinogenic hazard were developed for those contaminants that do not have Regional Matrix Values (USACE 1981) or Regional Dioxin Values (EPA 1997a). The approach used to derive these screening levels (described in the Appendix) uses standard EPA risk-assessment approaches and assumptions (fish and shellfish consumption rates; whole body to fillet contaminant ratios) to back calculate a fish tissue concentration that corresponds to specific levels of cancer risk or non-cancer hazard. The fish tissue concentration is then converted to a corresponding benthic prey concentration using the trophic transfer factors discussed in the Appendix.

Because the analysis implicitly incorporates various conservative assumptions regarding exposure (e.g., 100% of consumed seafood is fish taken from the HARS; fish has fed exclusively at HARS long enough prior to capture to be at equilibrium with benthic prey tissue contaminant concentrations), the resulting benthic tissue concentrations represent conservative estimates of

risk, or what are in effect plausible upper-bound estimates. Thus, the true risk is highly unlikely to be greater than estimated and could be much lower.

Table 4, Column 14 lists conservative human cancer protection levels in benthic organisms for the chemicals which are known or suspected carcinogens that correspond to a human cancer risk level of 1×10^{-4} . None of the human health cancer protection levels were exceeded in the bioaccumulation test results.

The potential for non-cancer impacts can be expressed as a hazard quotient (HQ), which is the ratio of the average daily intake divided by the toxicological reference dose for the chemical. Table 4, Column 15 includes the noncancer protection levels in benthic organisms for the chemicals requiring further analysis that are known to cause, or suspected of causing, non-carcinogenic effects, that would result in a human HQ equal to unity (i.e., 1). If the HQ is less than unity, an adverse noncarcinogenic effect is highly unlikely to occur. The higher the HQ, the more likely that an adverse noncarcinogenic effect will occur as a result of exposure to the contaminant in the dredged material after placement.

The concentrations of all contaminants in test organisms' tissues were below their respective non-cancer protection level (HQs for all contaminants were less than unity).

D. Evaluation of Bioaccumulation Results for Dredged Material as a Whole

As a final step in the evaluative process, it is appropriate to go beyond assessing the individual test results to evaluate the bioaccumulation test results as a whole to provide an opportunity for an integrated assessment of the individual test results (see Figure 1, Box d). EPA/USACE (1991) provides additional factors to be considered in making an integrated assessment of dredged material test results. The recommended factors are:

Magnitude of toxicity and number and phylogenetic diversity of species exhibiting greater mortality in the dredged material than in the reference material

As discussed above, project dredged material would not cause significant mortality (statistically different project sediment-exposed mortality exceeding mortality in the reference sediment exposures by 10% for mysid shrimp or 20% for amphipods) and meets the solid phase toxicity criteria of Sections 227.6 and 227.27.

Number and magnitude of contaminants for which bioaccumulation from the dredged material is statistically greater than bioaccumulation from the reference material and number and phylogenetic diversity of species in which bioaccumulation from the dredged material is statistically greater than bioaccumulation from the reference material

The magnitudes by which compounds bioaccumulated to levels greater than reference are shown in Table 2.

Table 2. Magnitude of Bioaccumulation of Contaminants in Organisms Exposed to Project Sediments vs. Reference Sediments

CONTAMINANT	CLAMS	WORMS
PAHs		
Acenaphthene	2.2	
Acenaphthylene	5.1	2.8
Anthracene	4.1	7.8
Fluorene	1.2	1.4
Naphthalene	1.7	
Phenanthrene		1.6
Benzo(a)anthracene**	11.1	14.4
Benzo(a)pyrene**	15.4	11.1
Benzo(g,h,i)perylene	18.7	10.5
Benzo(b)fluoranthene**	17.2	14.6
Benzo(k)fluoranthene**	12.2	15.6
Chrysene**	7.7	17.0
Dibenzo(a,h)anthracene**	20.5	
Fluoranthene	3.8	7.5
Indeno(1,2,3-cd)pyrene**	22.4	4.4
Pyrene	12.7	26.3
PESTICIDES		
Dieldrin	9.2	5.6
a-Chlordane	10.6	6.7
Trans nonachlor	24.5	1.7
Endosulfan I	38.3	7.3
Endosulfan II	12.1	3.3
Endosulfan sulfate	15.7	6.0
4,4-DDT		2.8
4,4-DDD	29.7	27.9
2,4-DDD	16.0	8.9
4,4-DDE	38.7	24.7
2,4-DDE	66.1	8.9
TOTAL PCBs	15.0	4.2
1,4-Dichlorobenzene	3.0	
METALS		
Arsenic	1.2	
Chromium (total)	2.6	1.4
Copper	1.6	1.2
Lead	3.6	1.8
Nickel	1.4	1.3
Zinc	1.4	
DIOXINS/FURANS		
2,3,7,8-TCDD		1.6
1,2,3,7,8-PeCDD		2.1
1,2,3,4,6,7,8-HpCDD	3.4	2.5
OCDD	11.9	3.2
2,3,7,8-TCDF		2.2
2,3,4,7,8-PeCDF		3.1
1,2,3,4,7,8-HxCDF	2.0	
1,2,3,6,7,8-HxCDF	1.5	
2,3,4,6,7,8-HxCDF	1.6	
1,2,3,4,6,7,8-HpCDF		3.4
OCDF	2.5	

All contaminants except for two PAHs and five pesticides accumulated to less than twenty times reference in *M. nasuta*, and all contaminants except for one PAH and two pesticides accumulated to less than twenty times reference in *A. virens*.

Exceeding reference concentrations is common when reference concentrations are very low or “non-detect.” Because the HARS reference sediment is greater than 90% sand and has only trace levels of contamination, reference-exposed organisms tend to exhibit very low or undetectable levels of bioaccumulation.

Toxicological importance of the contaminants whose bioaccumulation from the dredged material exceeds that from the reference material, including their propensity to biomagnify within aquatic food webs

Although some of the contaminants that bioaccumulated to levels greater than reference in organisms exposed to the project sediment can be toxicologically important, they did not accumulate to toxicologically important concentrations. Levels of these contaminants were all below the acceptable human health risk range and acceptable aquatic effects range using conservative approaches and analyses as described above to evaluate those test results, including through their potential transfer through the food web.

Magnitude by which contaminants whose bioaccumulation from the dredged material exceeds that from the reference material also exceed the concentrations found in comparable species living in the vicinity of the proposed site.

Organisms collected from a broad area of the sea floor in the vicinity of (but not inside) the HARS have been collected and analyzed for tissue concentration for bioaccumulative contaminants of concern (Charles and Muramoto 1990; USACE 1994; EPA 1996f; EPA 1997b). These data provide a measure of the tissue residues for organisms living outside the HARS. Table 4, Columns 16 and 17 summarize the most recent background data.

The magnitude by which compounds that bioaccumulated to levels greater than background levels (where background values exist) are reported in Table 3. All other compounds accumulated in test organisms to steady state levels below background.

Table 3. Magnitude of Bioaccumulation above Background in Organisms Exposed to Project Sediments

CONTAMINANT	CLAMS	WORMS
PAHs		
Chrysene**		1.2
PESTICIDES		
Dieldrin		7.8
Endosulfan II		1.7
Endosulfan sulfate		2.4
4,4-DDT	2.6	
4,4-DDD		1.4

4,4-DDE	2.8	
2,4-DDE		1.6
METALS		
Zinc	1.8	1.0
DIOXINS		

All contaminants except for two pesticides accumulated to less than two times background in *M. nasuta*, and all contaminants except for two pesticides accumulated to less than two times background in *A. virens*.

When bioaccumulation in organisms exposed to project sediments is not greater than tissue concentrations in organisms from the vicinity of the remediation site (the background levels), this indicates that placement of the material would not result in bioaccumulation above existing ambient levels in the general area and thus does not have a potential to cause undesirable effects. However, bioaccumulation in organisms exposed to project sediments greater than these levels is not necessarily predictive of adverse effects. It may, for example, reflect extremely low background levels.

Although some of the contaminants that bioaccumulated to levels greater than background in organisms exposed to the project sediment can be toxicologically important, they did not accumulate to toxicologically important concentrations. Levels of these contaminants were all below the acceptable human health risk range and acceptable aquatic effects range using conservative approaches and analyses as described above to evaluate those test results, including through their potential transfer through the food web.

VI. OVERALL CONCLUSION ON THE PROPOSED PROJECT

The evaluation of the testing results performed above indicates that the material does not have a potential to cause undesirable effects to aquatic marine biota due to chronic adverse effects including such effects due to mutagenic, carcinogenic, or teratogenic contaminants, or to human health due to cancer or non-cancer effects from the individual contaminants. Evaluation of the solid phase bioaccumulation test results for the dredged material as a whole would not indicate a different outcome than that shown by the individual test results themselves; i.e., that the material does not have the potential to cause undesirable effects due to bioaccumulation.

Based upon this review of the results of testing of the sediments proposed for dredging and ocean placement from SWO FNP Reach 1, the material meets the criteria for acceptability for ocean placement as described in Sections 227.6, and 227.27 of the Regulations, is Category I under EPA Region 2/CENAN guidance and is suitable for placement at the HARS under Section 228.15(d)(6) as Remediation Material.

Table 4. Bioaccumulation and Comparison Data for SWO FNP Reach 1

Col. 1 Sample I.D.	PROJECT DATA										COMPARISON DATA						
	Col. 2 [10] Reference	Col. 3 [10] Reference	Col. 4 [10] Test Sed.	Col. 5 Col. 6 [10] [1] Test Sed.	Col. 7 Col. 8 [1] Conv.Fac. clam/worm	Col. 9 Test Sed. SS	Col. 10 Test Sed. SS	Col. 11 Carcinogenic TEF [3]	Col. 12 Test Sed. Bap Tox. Equiv. Conc.(clam)[3] (ug/Kg)	Col. 13 Test Sed. Bap Tox. Equiv. Conc.(worm)[3] (ug/Kg)	Col. 14 Human Health Cancer (10E-4) Level[4] (ug/Kg)	Col. 15 Human Health Non-Cancer Level (HQ=1) (ug/Kg)[9]	Col. 16 Background (clam) (ug/Kg)	Col. 17 Background (worm) (ug/Kg)	Col. 18 FDA Limits [6] (ug/Kg)	Col. 19 Ecological Non-Specific Effects Level (ug/Kg)	Col. 20 Regional Matrix (ug/Kg)
Compound	(clam) (ug/Kg)	(worm) (ug/Kg)	(clam) (ug/Kg)	(worm) (ug/Kg)	[2]	(clam) (ug/Kg)	(worm) (ug/Kg)										
PAHs																	
Acenaphthene	0.144	0.115	0.32	*	1/1	0.32	0.134						8,775,000	8.1	0.5		[13]
Acenaphthylene	0.084	0.078	0.432	*	1/1	0.432	0.222						[5]	4.6	1.3		[13]
Anthracene	0.266	0.029	1.10	*	1/1	1.10	0.225						43,605,000	10.0	1.6		3,750
Fluorene	0.266	0.066	0.341	*	1/1	0.341	0.090						5,805,000	7.4	0.3		[13]
Naphthalene	0.549	0.61	0.91	*	1/1	0.91	0.673						[5]	26.4	4.5		[13]
Phenanthrene	1.83	0.226	1.92		1/1	1.92	0.358						43,605,000	32.7	4.7		[13]
Benzo(a)anthracene**	0.767	0.030	8.50	*	2/2	17.00	0.884	0.1	1.70	0.086			[5]	25.1	4.8		[13]
Benzo(b)pyrene**	0.773	0.094	11.90	*	2/2	23.80	2.080	1	23.8	2.080			[5]	24.2	7.6		8,000 [7]
Benzo(g,h,i)perylene	0.344	0.057	6.44	*	3/3	19.30	1.790						[5]	22.2	7.6		[13]
Benzo(k)fluoranthene**	0.760	0.063	13.10	*	2/2	26.20	1.840	0.01	2.62	0.184			[5]	40.6	16.5		[13]
Benzo(e)fluoranthene**	0.987	0.072	12.00	*	2/2	24.00	2.240	0.01	0.240	0.022			[5]	81.6	5.6		[13]
Chrysene**	1.85	0.237	14.30	*	2/2	28.6	0.001	0.0286	0.0080				[5]	29.4	6.7		[13]
Dibenz(a,h)anthracene**	0.059	0.0470	1.210	*	2/2	2.420	0.170	1	2.42	0.17			[5]	8.5	1.1		[13]
Fluoranthene	3.81	0.439	14.5	*	1/1	14.5	3.30						5,805,000	43.8	10.6		[13]
Indeno(1,2,3-cd)pyrene**	0.223	0.0530	4.99	*	3/3	15.00	0.702	0.1	1.500	0.070			[5]	16.6	4.4		[13]
Pyrene	2.77	0.375	35.1	*	1/1	35.1	9.87						4,387,000	51.3	26.6		[13]
TOTAL PAHs						211	32.6		33.3	2.6 (BaP eqv.)					104.60		40,000 [8]
PESTICIDES																	
Aldrin	0.014U	0.014U	0.014U			0.014	0.014				33		167	0.9	0.1	300	299 [9]
Dieldrin	0.026	0.069	0.238	*	2/2	0.476	0.778				65		518		0.1	300	4,37 [9]
a-Chlordane	0.017	0.034	0.180	*	2/2	0.360	0.454								0.7	300	
Trans nonachlor	0.028	0.131	0.686	*	2/2	1.370	0.440								0.5		
Heptachlor	0.011U	0.011U	0.011U		2/2	0.011	0.778								0.05	300	
Heptachlor epoxide	0.017U	0.025	0.017U		1/1	0.009	0.009								0.2	300	
Total Residual						1.750	1.68				114		135			1.7	64# [9]
Chlordane/Heptachlor																	
Endosulfan I	0.015U	0.015U	0.574	*	1/1	0.574	0.110				[5]				0.2		
Endosulfan II	0.022	0.051	0.266	*	1/1	0.266	0.166				[5]				0.1		
Endosulfan sulfate	0.012	0.040	0.188	*	1/1	0.188	0.240				[5]				0.1		
Total Endosulfans						1.030	0.516					87,000			0.4	2.86# [9]	
4,4-DDT	0.062	0.015	0.140		11/11	1.540	0.462								1.0		
2,4-DDT	0.017U	0.017U	0.017U		2/2	0.017	0.017								0.4		
4,4-DDD	0.065	0.067	1.930	*	3/3	5.79	5.61								3.9		
2,4-DDD	0.040	0.072	0.640	*	2/2	1.280	1.280								1.4		
4,4-DDE	0.126	0.047	4.880	*	2/2	9.76	2.320								4.3		
2,4-DDE	0.009U	0.009U	0.595	*	2/2	1.190	0.160								0.1		
Total DDT						19.60	9.85								11.1		40
TOTAL PCBs	2.78	9.8	41.7	*	1/2	41.7	82.2										
1,4-Dichlorobenzene	0.101	0.109	0.306	*	1/1	0.306	0.109				60,000		[5]	106.6	88.1	2,000	100(clam) 113(worm) 11,820 [9]

FOOTNOTES FOR TABLE 4:

**: Carcinogenic PAHs.

#: Levels represent the conservative level of protection for the sum of the related compounds and their metabolites.

na: Not Available

1. An “*” in this column indicates that the analyte concentration in the test sediment-exposed organisms are statistically greater than in those exposed to reference sediment. Means and statistical comparisons were determined using 1/2 the detection limit to estimate concentrations of analytes that were below the detection limit.
2. Conversion factors from 28-day bioaccumulation results to steady state (see discussion in Appendix)
3. Toxic equivalencies (TEFs) for the carcinogenic PAHs are from EPA (1993); Dioxin TEFs are from: EPA (1989).
4. This value represents the 10^{-4} cancer risk level for the carcinogenic PAHs. The total concentration of carcinogenic PAHs is expressed in BaP equivalents (see discussion in the text of the memo).
5. Cancer risk factor or reference dose are not assigned by EPA in IRIS (EPA 1995).
6. FDA limits are from EPA/USACE (1991).
7. This value represents the benthic level expected to result in a no-effect level for possible mutagenic and teratogenic effects in fish from exposure to BaP, which is the most toxic PAH.
8. This value represents the non-specific narcosis effects level (see discussion in Appendix). This value is compared to the sum of all PAHs measured.
9. Calculations are included in Appendix.
10. Means of five tissue replicates calculated using 1/2 detection limits to estimate concentrations of analytes that were not detected; “U” indicates that all five replicates were not detected.
11. Levels are based on the Regional Dioxin Values (EPA 1997a).
12. Level is the sum of TEQs of all dioxin congeners other than 2,3,7,8-TCDD.
13. For this PAH, the no-effect level for possible mutagenic and teratogenic effects in fish is estimated from exposure to BaP, which is the most toxic PAH.
14. Cancer and non-cancer protection levels, based on inorganic arsenic as contained in EPA’s IRIS database, are not appropriate for evaluating the potential human health impacts of arsenic bioaccumulation from dredged material, and therefore, are not included in Table 4 (see discussion in Appendix).

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Appendix

I. CORRECTION OF 28-DAY BIOACCUMULATION RESULTS TO ACCOUNT FOR STEADY STATE (Column 8-10)

Bioaccumulation tests were conducted using 28-day exposure of appropriate sensitive benthic marine organisms to sediment. Following placement of the material at the HARS, exposures to any contaminants would continue beyond 28 days, therefore, assessment of risks associated with accumulations must consider the extent to which the 28-day measured tissue concentration approximate the 'steady state' concentration of each contaminant that could occur in organisms at the HARS after extended exposure. Therefore, comparisons to guidance values or criteria for given contaminants that reflect 'steady state' conditions (as opposed to comparisons to other 28-day tissue concentrations such as comparison to reference) may require that a correction factor be applied to the 28-day concentration to account for incomplete attainment of steady state accumulation.

Steady state may be defined operationally as the lack of any significant difference (ANOVA, $\alpha = 0.05$) among tissue residues taken at three consecutive sampling intervals (Lee, *et al.* 1989). The 28-day test exposure period was selected as appropriate because most chemicals of concern will reach at least 80% of steady state in benthic marine organisms within that time frame (Boese and Lee 1992). For the few chemicals that may not meet steady state tissue concentrations in 28 days, a factor is used to adjust the data to steady state.

Steady state considerations for specific contaminants (or contaminant classes) and the derivation of correction factors (listed in Column 8) to account for incomplete uptake are discussed in the remainder of this section. Steady state-adjusted concentrations for clams and worms are presented in Columns 9 and 10, respectively.

A. Metals

In general, metals bioaccumulate more rapidly than organics and 28-day tests are sufficient to evaluate potential effects (see EPA/USACE 1991), for example, arsenic (Naqvi, *et al.* 1990; Riedel, *et al.* 1987; Oladimeji, *et al.* 1984).

Cadmium and mercury are not regulated in marine organisms as are essential metals, and, thus, no adjustment for steady state is applicable. Bioaccumulation of these metals is affected by many complex factors, and is essentially linear (Dethlefsen 1978; Giesy, *et al.* 1980; V-Balogh and Salanka 1984). Therefore, there are no adjustments that can be made to reproduce steady state. The Regional Matrix Values for cadmium and mercury, therefore, do not represent steady state conditions and so 28-day test results are used to compare to the Regional Matrix Values.

B. Organic Contaminants

a. Pesticides and 1,4-Dichlorobenzene

Uptake of non-polar organic contaminants from food is highly dependent on their hydrophobicity, a property measured by the octanol/water partition coefficient, K_{ow} . The higher the value of K_{ow} , the longer it takes to reach steady state in benthic marine organisms. Organochlorine compounds that have $\log K_{ow} > 6$ (i.e., aldrin, dieldrin, heptachlor, trans nonachlor, and alpha-chlordane), may not attain steady-state within 28 days of exposure to sediment.

Information contained in Boese and Lee (1992) indicates that aldrin, dieldrin, and α -chlordane achieve at least 50% of steady-state in 28-day bioaccumulation tests. Heptachlor and trans nonachlor, were not addressed in Boese and Lee (1992). However, the fraction of steady state achieved for these compounds after 28-days was estimated using the equations contained in McFarland (1995). Results of such calculations indicate heptachlor and trans nonachlor reach approximately 50% of steady state after 28 days. Calculating the fraction of steady state achieved for the remaining pesticides with $\log K_{ow} < 6$ (heptachlor epoxide) and for 1,4-dibenzochlorene ($\log K_{ow} = 3.44$) indicates that these compounds reach steady state within 28 days, so no adjustment is necessary.

The degree to which DDT and its metabolites reach steady state in test organisms after 28 days was evaluated using data reported by Lee *et al.* (1994). To assess the rate of bioaccumulation, they exposed the clam *Macoma nasuta*, to sediments collected from the vicinity of the United Heckathorn Superfund site in Richmond California to study tissue residues and uptake kinetics associated with exposure to pesticide-contaminated sediments. Results of the study indicate that one parent compound, 4,4-DDT, bioaccumulates much more slowly than 2,4-DDT and the DDT metabolites. The results range from approximately 9 percent of steady state after 28 days for 4,4-DDT, to 55 percent of steady state after 28 days for 2,4-DDT (Lee, et al. 1994). Based on these results, 28-day test tissue concentrations of DDT compounds and their metabolites are adjusted by multiplying the 28-day result by a factor of 11 for 4,4-DDT, a factor of three for 4,4-DDD, and a factor of two for 2,4-DDT and the remaining DDT metabolites before they are compared to Regional Matrix Values (USACE 1981).

b. PCBs

The degree to which PCBs reach steady state in test organisms after 28 days was evaluated using data reported by Lee, *et al.* (1994). To assess the rate of bioaccumulation of PCBs and other compounds, Rubinstein, *et al.* (1990) and Pruell, *et al.* (1993) exposed three species of organisms, the grass shrimp *Palaemonetes pugio*; the sandworm *Nereis virens*, and the clam *Macoma nasuta*, to sediments collected from the Passaic River, N.J. Sub-samples of the exposed organisms were removed on various days into the study including days 0, 10, 28, 42, 84, and 180.

For the clam tissue, the variance in the concentrations on day 28 and day 84 (by which point the maximum concentration had been reached) overlap, thus indicating that the two are not statistically different and the bioaccumulation on day 28 is at or very close to steady state. Thus, the clam bioaccumulation for the project sediments using 28-day exposures is acceptable for use as steady state tissue levels.

For the worm tissue, variances for days 28 and 180 do not overlap, thus indicating that steady state was probably not reached in 28 days. Variance in the data makes it difficult to quantify a real difference; means for days 28 and 180 from Rubinstein *et al.* (1990) (approximately 1,750 ng/g (nanograms per gram or parts per billion, ppb) for 28 days, and 3,000 ng/g for 180 days) suggest approximately 58% of steady state would have been reached in 28 days. On this basis, worm project data are conservatively adjusted by a factor of two to calculate a steady state tissue concentration for comparison to the HARS-Specific PCB Worm Tissue Criterion (40 CFR 228.15(d)(6)(v)(E)).

c. PAHs

The degree to which PAHs reach steady state in test organisms after 28 days was evaluated using data reported by McFarland (1995) and Meador, *et al.* (1995). The time required for a given PAH to attain a steady-state concentration following exposure to bedded sediments (t_{ss}) is determined primarily by the log K_{ow} of the compound in question (McFarland 1995; Meador, *et al.* 1995). Meador, *et al.* (1995) reviewed nine studies that investigated the attainment of steady-state tissue concentrations of PAHs by various marine invertebrates. In each case, tissue concentrations approached steady state within several days to two weeks after initiating exposure to both low molecular weight PAHs and high molecular weight PAHs. McFarland (1995) estimated t_{ss} values ranging from 3.5 to 326 days for 15 PAHs based on their hydrophobicities. The estimated steady-state concentration of the sum total of the 15 PAHs analyzed by McFarland for sediments collected from typical harbor areas revealed that the mean concentration attained after 28-day bioaccumulation tests was approximately 86% of steady-state. McFarland (1995) concluded that 28-day tests are likely to reflect steady-state. Despite this conclusion, steady state total PAH concentrations were estimated by summing individual PAH concentrations that were conservatively adjusted by a factor of 1, 2, or 3 based on their hydrophobicities and McFarland (1995).

II. CONSIDERATION OF FOOD-CHAIN TRANSFER

To properly assess risks to upper trophic level species (including humans) the efficiency with which a contaminant is transferred through the diet and the propensity of those compounds to accumulate to higher concentrations in the tissues of the consumer must be considered. The potential movement of contaminants through the food chain was considered and appropriate trophic transfer factors applied to adjust the data accordingly to reflect the propensity of specific contaminants or contaminant groups to accumulate to higher (or lower) concentrations in upper trophic level organisms (including humans). Data and/or methods used to develop these trophic transfer adjustment factors are discussed in the remainder of this section.

Metals

Trophic transfer of most metals is not sufficient to qualify as biomagnification (Brown and Neff 1993). The lack of observed biomagnification for such metals as arsenic, cadmium, lead, mercury, and zinc is the result of incomplete absorption of metals across the gut, rapid excretion, and dilution in muscle, which represents a large part of the total body weight of most marine animals (Fowler 1982; Suedel *et al.* 1994). For purposes of conducting the human health and ecological evaluations below, a conservative trophic transfer coefficient equal to one will be used for these non-biomagnifying metals (Suedel *et al.* 1994 and references cited therein).

Pesticides and Industrial Chemicals

The potential for these chemicals to biomagnify was also evaluated. Although organic contaminants with values of $\log K_{ow} > 4$ tend to biomagnify in the marine food chain, studies (USACE 1995) have shown that this is not true for higher molecular weight compounds such as the most highly chlorinated PCBs or for easily metabolized compounds such as PAHs. Those organic compounds which are not efficiently excreted, such as certain pesticides, can biomagnify in the food chain. For the organic constituents with a potential to biomagnify in the marine food chain, trophic transfer factors were calculated, using the approach described by Gobas (1993). The values are summarized in Attachment B.

PAHs

With regard to the potential for biomagnification of PAHs, feeding studies show that assimilation rates from ingested food are extremely low, e.g., more than 98% of the target contaminant remained in an undigested form in fish gut 48 hours after feeding squid containing radio-labeled benzo[a]pyrene to young cod (Corner, *et al.* 1976) and juvenile Atlantic herring (Whittle, *et al.* 1977). PAH metabolites are also transferred through the marine food chain; however, they are absorbed even less efficiently than their parent compounds (McElroy and Sisson 1989; McElroy, *et al.* 1991). Up to 99% of the PAH compounds taken up by fish are metabolized and excreted into bile, the usual elimination mode, within 24 hours of uptake (Varanasi, *et al.* 1989). Similar results are described in Brown and Neff (1993) who evaluated various studies describing trophic transfer. The studies cited in Brown and Neff (1993) indicate a trophic transfer rate for BaP from invertebrates to fish of between 0.02 and 0.23 times the concentration in the ingested invertebrates (Corner, *et al.* 1976, O'Connor, *et al.*, 1988, McElroy, *et al.* 1991). This was considered when assessing the ecological and human health effects of the project material as discussed below.

III. CONSIDERATION OF ECOLOGICAL EFFECTS

A. Potential for ecological effects based on Water Quality Criteria (Column 19)

The potential for ecological impacts due to bioaccumulation of several compounds of concern was evaluated by calculating a Water Quality Criterion Tissue Level (WQCTL). The WQCTL is calculated by multiplying the Clean Water Act Section 304(a)(1) Federal water quality criterion chronic value (CV) for the chemical by the empirically determined bioconcentration factor (BCF) for the chemical for a representative marine organism (Lee, *et al.* 1989). A BCF is the ratio of the concentration of a contaminant in an organism to the concentration of the contaminant in water. Thus, the WQCTL represents the tissue concentration that would be expected in an organism exposed to water containing the chemical at the CV concentration. The CV level is set to protect 95% of all tested organisms included in the water quality criterion database, thus representing a conservative level of protection (EPA 1985b). Table 4 lists the calculated WQCTLs. Sources of CVs and BCFs are from EPA ambient water quality criteria documents (EPA 1980b, 1980c, 1980d, 1980e, 1980f, 1984a, 1984b, 1985a, 1985c, 1986, 1987b and 1992a) and Calabrese (1984; for silver). Calculations are shown in Attachment A.

Several pesticides were evaluated based on the sum of their primary constituents and associated metabolites (e.g., total chlordane, total endosulfan, and total DDT). Alpha(trans)-chlordane, trans

nonachlor, heptachlor and heptachlor epoxide represent the primary components of technical chlordane and its metabolites found in the tissue of aquatic organisms (Jarman, *et. al.* 1996; Verschueren 1983; Sweeney, *et. al.* 1993). These constituents are summed as total chlordane as is consistent with current practice for chlordane (Jarman, *et. al.* 1996) and total DDT. The WQCTL for total chlordane was calculated using the WQC for chlordane as a conservative level of protection. While water quality criteria exist, and WQCTLs can be calculated, for heptachlor (133 ppb) and chlordane (64 ppb), the total chlordane is compared to the WQCTL for chlordane in order to be more environmentally conservative. The chlordane WQCTL provides a conservative level of protection as indicated by published residue effects levels (Sweeney, *et. al.* 1993; Bauman, *et. al.* 1987; Feroz, *et. al.* 1979; Parrish, *et. al.* 1976). Consistent with the above approach, the tissue concentration for endosulfan I, endosulfan II and endosulfan sulfate were also summed as total endosulfan and compared to the WQCTL for total endosulfan.

The WQCTLs were also calculated for all metals of concern which do not have Matrix values. For total chromium, the WQCTL was calculated based on chromium (VI), which is substantially more toxic than chromium (III) and elemental chromium in order to provide a conservative level of environmental protection.

B. Potential for ecological effects based on PAH toxicity (Column 19).

The Critical Body Residue (CBR) approach described by McCarty (1991) was used to derive values for use in evaluating the potential impacts of PAHs accumulated in the dredged material bioaccumulation test organisms. CBRs are concentrations of chemical residues in organisms which elicit a deleterious biological response associated with narcosis, which is the primary non-cancer effect of PAHs. Narcotic responses measured can be acute (e.g., immobilization or death) or chronic endpoints (e.g., reduced reproduction, fecundity or growth). CBRs are represented as the ratio of the mass of toxicant to the mass of the organism, such as millimoles or micrograms of toxicant per kilogram (mmole or ug/kg) of organism. For the narcosis endpoint, each molecule of individual PAH congeners is generally equipotent, thus the total PAH concentration is compared to the CBR. A 400-ppb dose of naphthalene would elicit a similar toxicity response as a 400-ppb dose of fluorene; if both chemicals are present together at these concentrations, then the combined dose would be approximately equal to 800 ppb of either contaminant alone.

McCarty (1991) states that an average critical body residue of 400,000 - 1,200,000 ppb can be used as an estimate for acute effects for a narcosis-producing chemical (e.g., PAHs) on fish populations. (Note: McCarty reports the CBR in units of millimoles per kilogram; this value has been converted to ppb for PAHs using the average molecular weight of the PAHs analyzed in the bioaccumulation test). Chronic effect critical body residues can be estimated by applying an acute to chronic ratio of 10 to the acute CBR (McCarty 1986; Call, *et al.* 1985). Therefore, the chronic critical body residue for PAHs can be estimated at 40,000 - 120,000 ppb of PAHs in organism tissue, and Table 4 thus uses the 40,000-ppb level.

These CBRs were based on fish data. The use of CBRs based on fish toxicity represents a conservative estimate of potential toxicity due to exposure to dredged material because: (1) it is extremely unlikely that a fish would get its whole diet from the HARS, and (2) fish are generally more sensitive than the benthic organisms in direct contact with the dredged material placed at

the HARS (e.g., Landrum, *et al.* (1988) estimated an acute CBR for crustaceans of 800,000 ppb - 42,000,000 ppb).

C. Potential ecological impacts of mutagenic, carcinogenic and teratogenic PAHs (Column 19)

EPA and the USACE reviewed eleven scientific journal articles that reported the results of laboratory experiments that sought to relate the concentration of a PAH contaminant(s) in water, as injected doses, or tissue concentrations, to mutagenic, carcinogenic, teratogenic and/or reproductive effects to fish. These studies all used fish species which are among the most sensitive organisms in the marine environment to exhibit the above effects (EPA 1996c). In addition, most of these studies focused on the PAH most believed to cause such effects for which there is data, benzo(a)pyrene (BaP). One study (Breteler 1984) discussed the possible sources and distribution of PAHs in the Hudson/Raritan estuary, and ranked the threat of PAHs to aquatic biota and humans. The main threat was believed to be carcinogenicity, with a greater threat ranking assigned to humans than biota. However, Breteler (1984) did not provide specific effects-based levels that could be used in the following analysis. Two articles evaluated the effects of crude oil, and thus were not useful for evaluating the effects of specific PAHs measured in the bioaccumulation test (Rice, *et al.* 1987; Lee, *et al.* 1981). Three studies considered the effects of specific PAHs, but did not synoptically measure tissue concentrations in the organisms (Ward, *et al.* 1981; Holcombe, *et al.* 1983; Finger, *et al.* 1985) and were not used, because the lack of tissue data for these studies makes their utility in evaluating the tissue concentration resulting from the dredged material bioaccumulation tests highly uncertain.

The remaining five papers reported measured tissue concentrations and observed reproductive effects in organisms exposed to PAH-spiked water. One article reported the tissue concentrations of adult fish and the observed effect on survival and health of the fish's offspring. Hose, *et al.* (1981) reported that adult English sole injected with benzo(a)pyrene (BaP) accumulated the chemical in the gonad and mature gametes. The amount of BaP taken up by the ovary ranged from 16,800 to 49,700 ppb. Two samples of ripe eggs contained 51,200 and 263,000 ppb of BaP and its metabolites. No adverse effects were reported for these concentrations. Hose, *et al.* (1981) also reported the results of injecting female flathead sole with BaP. Adverse effects to egg hatching success were reported for each female. The paper does not report tissue concentrations in either the parent fish or the egg of the flathead sole. Effects on reproductive success were reported but could only be correlated with the external dose injected into the parent. Therefore, since concentrations and effects were not synoptic in this report, it was not useful in the evaluation of the dredged material bioaccumulation results.

Three papers reported the results of experiments which measured fish egg or alevin concentrations of BaP and associated reproductive or carcinogenic effects (Hose, *et al.* 1982; Hannah, *et al.* 1982; Black, *et al.* 1988). Hose, *et al.* (1982) exposed three species of sole, sand sole, English sole, and flathead sole, to BaP-spiked water. Tissue concentrations of 2,100 ppb were measured in sand sole on day 6 (24 hours after hatching) and were associated with reduced hatching success. However, we did not consider the results to be appropriate for use in setting effects levels because they may have been compromised by the methods of replication used in the experimental design.

Another study considered the carcinogenicity of BaP in rainbow trout resulting from embryo microinjection (Black, *et al.* 1988). A statistically significant number of liver neoplasms was found at a concentration of approximately 200,000 ppb, with non-significant effects at up to one half that concentration. Therefore, using the above across-species UF of 10 and trophic transfer factor of 0.1 results in an aquatic no-effect level of 100,000 ppb.

Hannah, *et al.* (1982) estimated a concentration of BaP in tissue that caused abnormalities in development of rainbow trout eggs, using aqueous exposures and actual measured tissue concentrations in alevin tissues. An exposure to a 2.4 ppb mean aqueous BaP concentration accumulated an average of 12,340 ppb in alevins. This concentration was associated with an increase in percentage of abnormalities from approximately 6% at lower water concentrations (0.08, 0.21, 0.37 and 1.48 ppb) to approximately 13% at higher concentrations. From 0.08 to 1.48 ppb in the water, there were no increasing effects exhibited; therefore, the effects were apparently “real” (i.e., significantly greater than the threshold effect level of 6%) only at the aqueous 2.4 ppb concentration. The Hannah, *et al.* (1982) study is considered the most reliable study for this evaluation since it used exposure series and measured tissue concentrations associated with observed effects, and therefore allows for the calculation of a no-effects level directly from the measured results.

In applying these studies to evaluations of dredged material, consideration must be given to uncertainties in converting these kinds of results to concentrations protective of other biota. Three uncertainties needing to be considered are: (1) those associated with converting effect to no-effect concentrations, (2) across-species uncertainties, and (3) uncertainties in estimating the dose of contaminants to which the organism is exposed. These uncertainties are discussed below.

With respect to uncertainty when converting effect to no-effect concentrations, an uncertainty factor of one order of magnitude is often used when only an effect measure is reported. However, in Hannah, *et al.* (1982), the no-effect level can be estimated to be the next lowest concentration below the lowest-observed effect level, since the range of concentrations below this level did not exhibit significantly different responses. In this case, the no-effect level occurred at the water exposure concentration of 1.48 ppb. Although a tissue concentration was not measured at the 1.48 ppb water concentration, it can be calculated from the concentration measured at the effect level (i.e., the no-effect water concentration (1.48 ppb) is close to 65 percent of the observed effect concentration (2.4 ppb) so the no-effect tissue concentration should be about 65 percent of the lowest-observable effect tissue concentration ($0.65 \times 12,340 \text{ ppb} = 8,021 \text{ ppb}$)). Thus, a factor to adjust these data from lowest observed effect tissue concentration to the calculated no-observed effect tissue concentration is obtained directly from the data.

There can also be uncertainty as to the proximity to the site of toxic action in the organism that a dose or concentration is measured, and with respect to species-to-species variability. Hannah (1982) reported dose concentrations in the tissue and, therefore, there is no need to account for variability associated with the large uncertainties encountered in typical water-only exposure studies where the actual concentration at the site of toxic action is unknown. When measured in the tissues, as was done for this project, concentrations of narcotic chemicals causing effects (i.e., critical body residues, CBRs) in aquatic organisms are reported to range only from 1.4 to 21 umoles/g wet weight (a factor of about one order of magnitude) for organisms as diverse as insects, crustaceans, and fish (McCarty, *et al.* 1992). Therefore, from a tissue concentration

perspective, the species-to-species uncertainty factor appropriate for both total PAHs operating as narcotics and individual PAHs having teratogenic effects would be one order of magnitude, or a value of 10 (EPA 1996d).

In summary, a factor of 10 (representing species to species uncertainty) is an appropriate UF to use in these evaluations. Also, as described Appendix Sections I and II, Brown and Neff (1993) show that trophic transfer of PAHs up the food chain to fish decrease tissue levels by over an order of magnitude. Given this data and the fact that these studies included fish that spent 100% of their time feeding in the test sediment, whereas this would be highly unlikely to occur at an ocean site, a trophic transfer factor of 0.1 is used in this analysis. Applying this UF of 10 and a trophic transfer factor of 0.1 to the no-effects level for BaP calculated from Hannah, *et al.* (1982), as discussed above (8,021 ppb) results in a no-effect level for BaP of approximately 8,000 ppb in benthic tissue.

IV. CONSIDERATION OF POTENTIAL EFFECTS ON HUMAN HEALTH (Columns 14 and 15)

Human effects screening levels were developed with risk-based methods using conservative estimates of exposure to assess whether these contaminants would accumulate to levels in fish and shellfish that could lead to significant adverse effects to humans. The approach assessed consumption of fish and shellfish to derive conservative estimates of contaminant concentrations in benthic tissue protective of human health using the following EPA standard risk-assessment assumptions: 70-kilogram adult eats 6.5 grams of fish and shellfish per day over a 70-year lifetime. This assessment considered potential for both cancer and non-cancer effects in humans. EPA IRIS (EPA 1995) and effects information from EPA's National Toxics Rule (EPA 1992b) were used in the human health assessment to calculate acceptable levels in fish and shellfish to protect human health. Trophic transfer factors, as discussed earlier, were then used to convert these fish and shellfish levels into benthic tissue concentrations.

For regulatory purposes, EPA utilizes 10^{-4} to 10^{-6} (one in ten thousand to one in one million) as an acceptable incremental risk range for activities with potential for causing cancer in human beings (EPA 1980a; EPA 1988; EPA 1987a; Thomas 1987; EPA 1991). EPA considers a cancer risk within this range to be safe and protective of public health. This is supported by the World Health Organization Guidelines for Drinking Water Quality (WHO 1993), where it selected a 10^{-5} guideline value, and then explained that the application could vary by a factor of ten (e.g., 10^{-4} to 10^{-6}). Since this analysis uses conservative methods, the results represent conservative estimates of risk, or what are in effect plausible upper-bound estimates. Thus, the true risk is highly unlikely to be greater than estimated and could be much lower.

Table 4, Column 14 lists human cancer protection levels in benthic organisms for chemicals which are known or suspected carcinogens that would lead to a human cancer risk level of 10^{-4} . For PAHs, this analysis used BaP-equivalents derived from the toxic equivalence factor for each carcinogenic PAH (from EPA (1993); note: these factors are listed in Column 11 of Table 4 for each of the compounds).

The potential for non-cancer impacts can be expressed as a hazard quotient (HQ), which is the ratio of the average daily intake divided by the toxicological reference dose for the chemical. If the HQ is less than unity (e.g., 1), an adverse noncarcinogenic effect is highly unlikely to occur. If the HQ exceeds unity, an adverse health impact may occur. The higher the HQ, the more likely that an adverse noncarcinogenic effect will occur as a result of exposure to the contaminant in the dredged material after placement. Table 4, Column 15 lists noncancer protection levels in benthic organisms for the chemicals that are known to cause, or suspected of causing, non-carcinogenic effects, that would result in a human HQ equal to unity. Those numbers were derived using the conservative assumptions and source materials described in the introductory paragraph to this section.

For the following compounds, the following special considerations were used in evaluating the results in Table 4.

a. Metals

No reference dose has been established for lead. EPA has adopted a blood lead level of 10ug/dl as the level of concern and EPA policies are that regulatory programs should seek to minimize the number of children with blood lead levels above a target of 10 ug/dl (Final Rule for Lead and Copper NPDWR, 56FR26468, June 7, 1991), and this value was used to calculate the effects level in Table 4 (see EPA 1996e).

When interpreting the importance of arsenic tissue concentrations for human health, consideration was given to the arsenic form present (i.e., inorganic vs. organic). Arsenic is found in marine organisms as an organic complex which includes such compounds as arsenobetaine and arsenocholine (Abel and Axiak 1991). Organic arsenic in the tissues of aquatic organisms is not metabolized by predators or humans and is readily eliminated from the body through excretion (Hrudey *et al.* 1995). As a result, the toxicity of organic arsenic ingested from seafood is low and appears to pose no significant hazard (Abernathy and Ohanian 1992). For this reason, cancer and non-cancer protection levels, based on inorganic arsenic as contained in EPA's IRIS database, are not appropriate for evaluating the potential human health impacts of arsenic bioaccumulation from dredged material, and therefore, are not included in Table 4.

b. Pesticides

Alpha(trans)-chlordane, trans nonachlor, heptachlor and heptachlor epoxide represent the primary components of technical chlordane and its metabolites found in the tissue of aquatic organisms (Jarman, *et. al.* 1996; Verschueren 1983; Sweeney, *et. al.* 1993). These constituents are summed as total chlordane as is consistent with current practice for chlordane (Jarman, *et. al.* 1996) and other pesticides (e.g., total DDT). Total chlordane is evaluated using the 10^{-4} cancer risk level and non-cancer level for heptachlor epoxide, which has the greatest potency of the chlordane constituents or metabolites. Similarly, endosulfan I, endosulfan II, and endosulfan sulfate are summed and the total is compared to the conservative non-cancer protection level for endosulfan.

c. PAHs

For PAHs, this analysis used BaP-equivalents derived from the toxic equivalence factor for each carcinogenic PAH (from EPA (1993); note: these factors are listed in column 11 of Table 4 for each of the compounds).

ATTACHMENT A

Tissue concentration is calculated using BCF * Water Quality Criteria (WQC) ambient aqueous concentration, and assuming the conversion factor of 1 kg=1L.

Compound	Ambient Conc. (ug/L) ¹	BCF	Tissue Conc. (ug/Kg)	Remarks
Aldrin	0.13	2,300	299	WQC was reduced by a factor of 10 to account for chronic effects; BCF estimate is based on Dieldrin since Aldrin rapidly transformed to Dieldrin in the
Dieldrin	0.0019	2,300	4.37	BCF is based on 1.1% lipid level for marine fish, Spot (<i>Leiostomus xanthurus</i>).
Total Chlordane	0.004	16,000	64	Total Chlordane includes alpha-chlordane, trans nonachlor, heptachlor, heptachlor epoxide; WQC for Chlordane is used for Total Chlordane; BCF is based
Total Endosulfan	0.0087	328	2.85	Total Endosulfan includes endosulfan I, endosulfan II and endosulfan sulfate; WQC for Endosulfan is used for Total Endosulfan; BCF is based on 3.6% lipid
1,4-Dichlorobenzene	197	60	11820	Ambient conc. is based on lowest observed effect level (LOEL) for saltwater species from WQC, and reduced by a factor of 10 to account for chronic
Arsenic	36	350	12600	Ambient conc. is based on the saltwater criteria continuous conc. for arsenic (III); BCF is based on the Eastern Oyster (<i>Crassostrea virginica</i>).
Chromium	50	236	11800	Ambient conc. is based on Chromium (VI) since it is substantially more toxic than Chromium (III); BCF is based on polychaete worm.
Copper	2.9	3,300	9570	WQC is based on a hardness value of 100; BCF is based on soft shell clam.
Lead	8.5	1,400	11900	Ambient conc. is based on saltwater criteria continuous conc.; BCF is based on the Eastern Oyster (<i>Crassostrea virginica</i>).
Nickel	8.3	458	3802	Ambient conc. is based on saltwater criteria continuous conc.; BCF is based on the Eastern Oyster (<i>Crassostrea virginica</i>).
Silver	0.23	6,500	1495	Water Quality Criterion (WQC) was reduced by a factor of 10 to account for chronic effects; BCF is based on the Blue Mussel (<i>Mytilus edulis</i>).
Zinc	86	17,640	1517040	Ambient conc. is based on saltwater criteria continuous conc.; BCF is based on Eastern Oyster (<i>Crassostrea virginica</i>).

¹.The following documents were used to obtain the water quality criteria values.

Calabrese, A. 1984. "Effects of Long-Term Exposure to Silver and Copper on Growth, Bioaccumulation and Histopathology in the Blue Mussel (*Mytilus edulis*).” *Mar. Envir. Res.* 1, 253-274.

EPA. 1980b. Ambient Water Quality Criteria for Aldrin/Dieldrin; EPA 440/5-80-019; December 1980.

EPA. 1980c. Ambient Water Quality Criteria for Chlordane; EPA 440/5-80-027; October 1980.

EPA. 1980d. Ambient Water Quality Criteria for Heptachlor; EPA 440/5-80-052; October 1980.

EPA. 1980e. Ambient Water Quality Criteria for Endosulfan; EPA 440/5-80-046; October 1980.

EPA. 1980f. Ambient Water Quality Criteria for Dichlorobenzenes; EPA 440/5-80-039; October 1980.

EPA. 1984a. Ambient Water Quality Criteria for Lead - 1984; EPA 440/5-84-027; January 1985.

EPA. 1984b. Ambient Water Quality Criteria for Copper - 1984; EPA 440/5-84-031; January 1985.

EPA. 1985a. Ambient Water Quality Criteria for Chromium - 1984; EPA 440/5-84-029; January 1985.

EPA. 1985c. Ambient Water Quality Criteria for Arsenic - 1984; EPA 440/5-84-033; January 1985.

EPA. 1986. Ambient Water Quality Criteria for Nickel - 1986; EPA 440/5-86-004; September 1986.

EPA. 1987b. Ambient Water Quality Criteria for Zinc - 1987; EPA 440/5-87-003.

EPA. 1992a. Draft Ambient Water Quality Criteria for Silver.

ATTACHMENT B

Benthic Cancer Protection Level Calculations for the Protection of Human Health from the Consumption of Fish Exposed to Dredged Material at the Historic Area Remediation Site

Basis^{fn1}: $10^{-4} \text{ Benthic Tissue Level (ug/kg)} = \frac{[10^{-4} \text{ Conc. in Fish}] \times [\text{Whole Body/fillet Factor (1.35)}]^{fn5}}{\text{Trophic Transfer Factor}^{fn2}}$ $10^{-4} \text{ Conc. in Fish (ug/kg)} = \frac{\text{Toxicological Dose (ug/day)}}{[\text{Seafood Consumption (6.5 g/day)}]^{fn3} \times [10^{-3} \text{ kg/g}]}$ $\text{Toxicological Dose (ug/day)} = \frac{[\text{Risk Level (10}^{-4}\text{)}] \times [\text{Body Weight (70 kg)}]^{fn3} \times [10^3 \text{ ug/mg}]}{\text{Potency Factor, } q_1^* \text{ (kg-day/mg)}^{fn4}}$				
	Cancer Potency Factor q_1^* (kg-day/mg)	Acceptable Concentration in Fish (ug/kg)	Trophic Transfer Factor	Benthic Protection Level (ug/kg)
Pesticides				
Aldrin	17	63	2.6	33
Chlordane	1.3	828	2.3	486
Dieldrin	16	67	1.4	65
Heptachlor	4.5	239	2.7	120
Heptachlor epoxide	9.1	118	1.4	114
Industrial Organics				
1,4- Dichlorobenzene	0.024	44,872	1	60,577
PAHs				
Benzo(a)pyrene	7.3	147	0.1	2,000
METALS				
Arsenic	1.5	718	3	323

ATTACHMENT C

**Benthic Non-Cancer Protection Level Calculations for the Protection
of Human Health from the Consumption of Fish Exposed to Dredged Material
at the Historic Area Remediation Site**

Basis ^{fn1} :				
$\text{Benthic Tissue Level (ug/kg)} = \frac{[\text{Conc. in Fish}] \times [\text{Whole Body/fillet Factor (1.35)}]^{fn5}}{\text{Trophic Transfer Factor}^{fn2}}$ $\text{Conc. in Fish (ug/kg)} = \frac{\text{Toxicological Dose (ug/day)}}{[\text{Seafood Consumption (6.5 g/day)}^{fn3}] \times [10^{-3}\text{kg/g}]}$ $\text{Toxicological Dose (ug/day)} = [\text{Reference dose}^{fn4}] \times [\text{Body Weight (70 kg)}^{fn3}]$				
	Reference Dose RfD (ug/kg-day)	Acceptable Concentration in Seafood (ug/kg)	Trophic Transfer Factor	Benthic Protection Level (ug/kg)
Metals				
Arsenic	0.3	3,231	3	1,454
Chromium	5	54,000	1	73,000
Copper	37.1	400,000	1	540,000
Nickel	20	215,000	1	290,000
Silver	5	54,000	1	73,000
Zinc	300	3,230,769	1	4,361,538
Pesticides				
Aldrin	0.03	323	2.6	167
Chlordane	0.06	592	2.3	350
Dieldrin	0.05	538	1.4	518
Endosulfan	6	64,615	1	87,231
Heptachlor	0.5	5,385	2.7	2,692
Heptachlor epoxide	0.013	140	1.4	135
PAHs				
Acenaphthene	60	650,000	0.1	8,775,000
Fluorene	40	430,000	0.1	5,805,000
Phenanthrene	300	3,230,000	0.1	43,605,000
Anthracene	300	3,230,000	0.1	43,605,000
Fluoranthene	40	430,000	0.1	5,805,000
Pyrene	30	325,000	0.1	4,387,000

NOTES:

^{fn1} Human health cancer and non-cancer assessments adapted from *Guidance for Assessing Chemical Contaminant Data for use in Fish Advisories: Volume II: Risk Assessment and Fish Consumption Limits*. U.S. Environmental Protection Agency, EPA823-B-94-004, Office of Science and Technology, Washington, DC, June 1994.

^{fn2} Trophic transfer factors were calculated by Mr. Lawrence Burkhard, EPA Duluth, using the food chain transfer model developed by Gobas (1993).

^{fn3} Default values were taken from EPA's national toxics rule for setting water quality criteria, EPA (1992b).

^{fn4} Cancer potency factors and non-cancer reference doses are taken from EPA (1995).

^{fn5} The acceptable concentration in seafood is defined on the basis of the fillet or edible portion for humans. Trophic transfer, however, was defined based on whole-body characteristics, including lipid concentrations. Experience in New York State indicates a whole body to fillet ration ranging from 1.2 to 1.5 is applicable to lipophilic substances. The mid-range value of 1.35 is used in this analysis.

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**US ARMY CORPS
OF ENGINEERS**
NEW YORK DISTRICT



September 16, 2025

MEMO FOR THE RECORD

SUBJECT: Review of Compliance with the Requirements of 40 CFR 227.6 and 227.27, and Site Designation Provisions of 40 CFR 228.15 for NY&NJ Channels, Seguine Point Bend, Ward Point Bend to Outerbridge Crossing Federal Navigation Project Reach 2

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FROM: Mark Reiss, Chief
Dredging, Sediments, and Oceans Section
Water Division
EPA Region 2

Kelly Vega, Chief
Dredged Material Management Section
Operations Division, New York District
U.S. Army Corps of Engineers

I. SUMMARY

The USACE is proposing to dredge the NY&NJ Channels, Seguine Point Bend, Ward Point Bend to Outerbridge Crossing Federal Navigation Project (herein referred to as SWO FNP) located in New York with placement of the dredged material as Remediation Material at the Historic Area Remediation Site (HARS) in the Atlantic Ocean.



This memorandum provides comprehensive review and analysis of SWO FNP test results for Reach 2. This memorandum addresses compliance with the regulatory testing criteria of 40 CFR Sections 227.6 and 227.27, and the requirements of the rule establishing the Historic Area Remediation Site (HARS) set out in Section 228.15(d)(6). These requirements hereinafter are referred to as the “Regulations.” Project dredged material meets the non-testing regulatory criteria of Subparts B, C, D, and E of 40 CFR § 227 pertaining to: environmental impacts, need, impact on esthetic, recreational and economic values, and other uses of the ocean (EPA/USACE 2019).

As discussed in the HARS rulemaking preamble (see 62 Fed. Reg. 46142 (August 29, 1997) and 62 Fed. Reg. 26267 (May 13, 1997)) and its accompanying documentation, the need to remediate the Historic Area Remediation Site is amply supported by the presence in the HARS of toxic effects, dioxin bioaccumulation exceeding Category I levels in worm tissue, as well as TCDD/PCB contamination in area lobster stocks. Individual elements of the aforementioned data do not prove that sediments within the Study Area are imminent hazards to the New York Bight Apex ecosystem, living resources, or human health. However, the collective evidence presents cause for concern, justifies a need for remediation, indicates that the site is Impact Category I (see 40 CFR 228.10), and necessitates that the site should be managed to reduce impacts to acceptable levels (see 40 CFR 228.11(c)). Further information on the conditions in the Study Area and the surveys performed may be found in the Supplemental Environmental Impact Statement (EPA, 1997c).

This evaluation confirms that: 1) all tests required under the Regulations were conducted, 2) this project meets the criteria at 40 CFR Section 227.6 for trace contaminants and Section 227.27 for Limiting Permissible Concentration (LPC), and 3) the dredged material is Category I under U.S. Environmental Protection Agency (EPA) Region 2/U.S. Army Corps of Engineers, NY District (CENAN) guidance and is suitable for placement at the HARS as Remediation Material.

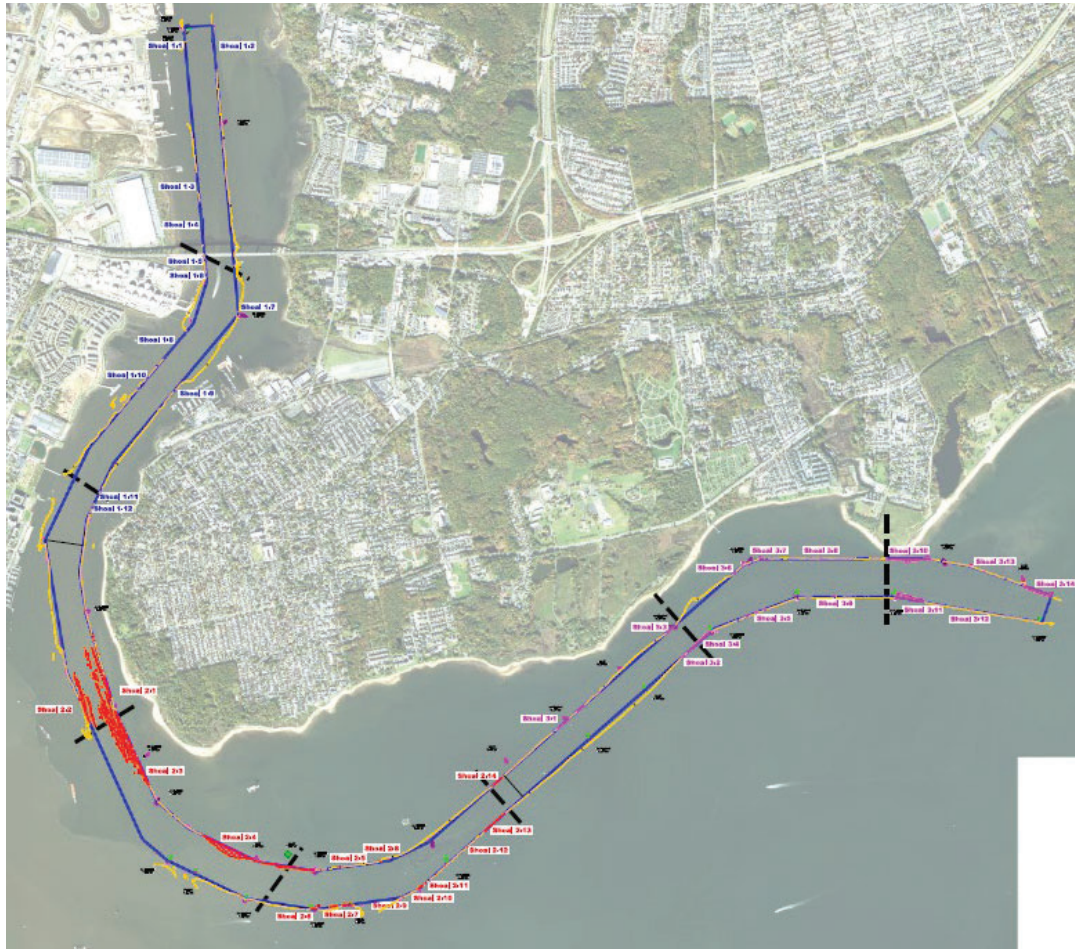
II. PROJECT DESCRIPTION

The proposal is to dredge shoals that extend along the western, southern, and eastern sides of Staten Island, NY, from north of Outerbridge Crossing to east of Seguine Point in three reaches that have been composited for testing within the SWO FNP to a depth of -35’ MLLW with an additional 2’ of allowable overdredge with placement of approximately 106,485 cy of (based on an August 8, 2024 bathymetric survey) at the HARS. The project has been sampled and tested as three separate reaches for the purposes of bioassay testing (see sampling plan; CENAN 2024) as described below:

Reach 1, shown below in blue hatching has an estimated volume of 5,585 cy. The project sediments from this reach are represented by 7 core samples composited for the purpose of bioassay testing.

Reach 2, shown below in red hatching has an estimated volume of 87,034 cy. The project sediments from this reach are represented by 9 core samples composited for the purpose of bioassay testing.

Reach 3, shown below in purple hatching has an estimated volume of 13,866 cy. The project sediments from this reach are represented by 5 core samples composited for the purpose of bioassay testing.



III. REGULATORY REQUIREMENTS

For dredged material to be suitable for use as Remediation Material, it must conform to the Regulations. The Marine Protection, Research, and Sanctuaries Act (MPRSA) or “The Act” prohibits dumping of materials into the ocean except as authorized by EPA or, in the case of dredged materials, by the U.S. Army Corps of Engineers (USACE). Section 102 of the Act directs the EPA to establish and apply criteria for reviewing and evaluating permit applications (33 U.S.C. Section 1412). EPA has adopted such criteria in the Regulations. 40 CFR Section 227.6(a) lists constituents that are prohibited from being placed in the ocean unless only present as trace contaminants in material otherwise suitable for dumping (hereinafter referred to as “listed constituents”). Section 227.27 addresses compliance with the LPC. See also, Section 227.13(c).

Section 227.6(b) states that these listed constituents are considered to be present as trace contaminants only when they are present in such forms and amounts that the “dumping of the materials will not cause significant undesirable effects, including the possibility of danger

associated with their bioaccumulation in marine organisms.” The Regulations set forth criteria for determining the potential for significant undesirable effects in Section 227.6(c). To be found environmentally acceptable for ocean placement, the liquid phase must not contain any of the listed constituents in concentrations that would exceed applicable marine water quality criteria after allowance for initial mixing (Section 227.6(c)(1)). For the suspended particulate phase (Section 227.6(c)(2)) and the solid phase (Section 227.6(c)(3)), bioassay results must not indicate occurrence of significant mortality or significant adverse sublethal effects due to the ocean placement of dredged material containing the listed constituents.

Section 227.27 of the Regulations addresses the LPC. For the liquid phase, Section 227.27(a) provides that the LPC is that concentration which does not exceed applicable marine water quality criteria after initial mixing, or when there are no applicable marine water criteria, that concentration of material after initial mixing, would not exceed 0.01 of a concentration shown to be acutely toxic to appropriate sensitive marine organisms in a bioassay carried out in accordance with procedures approved by EPA and USACE. For the suspended particulate phase and the solid phase, Section 227.27(b) provides that the LPC is that concentration of material which will not cause unreasonable acute or chronic toxicity or other sublethal adverse effects based on results of bioassays using appropriate sensitive organisms and conducted according to procedures that have been approved by EPA and USACE, and which will not cause accumulation of toxic materials in the human food chain.

The HARS encompasses an area which includes the Mud Dump Site (MDS), and which has exhibited the potential for adverse ecological impacts and has been identified for remediation. The site will be remediated with uncontaminated dredged material (i.e., dredged material that meets current Category I standards and will not cause significant undesirable effects including through bioaccumulation) (hereinafter referred to as “Remediation Material” or “Material for Remediation”). Under 40 CFR 228.15 (d)(6)(v)(A) the site will be managed to reduce impacts within the Priority Remediation Area (PRA) to acceptable levels in accordance with 40 CFR Section 228.11(c). Use of the site is restricted to dredged material suitable for use as Material for Remediation. This material shall be selected to ensure it will not cause significant undesirable effects including through bioaccumulation or unacceptable toxicity in accordance with 40 CFR 227.6.

Section 228.15(d)(6) of the Regulations describes the locations where material from NY/NJ Harbor and surrounding areas may be placed in the HARS, if it is suitable as Remediation Material. The HARS consists of nine PRAs, a Buffer Zone, and a No Discharge Zone. Under 228.15(d)(6) placement of Remediation Material is limited to the PRAs.

IV. GUIDANCE FOR TESTING AND EVALUATION OF DREDGED MATERIAL

The discussion below describes how the material proposed to be dredged from the SWO FNP Reach 2 was evaluated for placement at the HARS and evaluated for compliance with the requirements of 40 CFR 227.6, 227.27, and 228.15(d)(6). Testing of the material was conducted following procedures approved by EPA and USACE, and contained in the joint EPA/USACE national guidance “Evaluation of Dredged Material Proposed for Ocean Dumping - Testing Manual” (the “Green Book;” EPA/USACE, 1991), and the regional implementation manual developed by the EPA Region 2 and CENAN (EPA/CENAN, 2016).

These test results were analyzed in accordance with the Regulations to ensure that the proposed placement meets the criteria of Part 227 and the requirements of 228.15(d)(6). As explained in the preamble to the HARS Rule, Remediation Material is uncontaminated dredged material (i.e., dredged material that meets current Category I standards and will not cause significant undesirable effects including through bioaccumulation). The determination of whether materials meeting the Part 227 criteria are assigned to Category I is based upon technical and scientific judgment as set out below.

Applying the EPA Region 2/CENAN guidance to this project, the material would be Category I if it meets the Part 227 criteria (including the requirements regarding acute toxicity) and:

- bioaccumulation test results do not exceed the Regional Matrix Values for cadmium, mercury, total PCBs (clam), and total DDT (USACE 1981), and are below the regional Category I guidance values for dioxin (EPA 1997a), and
- bioaccumulation test results do not exceed the HARS-Specific Polychlorinated Biphenyl (PCB) Worm Tissue Criterion defined in 40 CFR 228.15(d)(6)(v)(E), and
- bioaccumulation test results for the other bioaccumulative chemicals of concern identified in EPA/CENAN (2016) do not indicate a potential for undesirable effects using conservative assessment techniques.

Sediments that meet this Category I definition are suitable for placement at the HARS as Remediation Material as they will improve sediment conditions at the HARS to reduce impacts to acceptable levels in accordance with 40 CFR Section 228.11(c). Sediments that do not meet this definition are not suitable for placement at the HARS.

V. RESULTS OF EVALUATION OF THE MATERIAL

A. Evaluation of the liquid phase

The liquid phase of the material was evaluated for compliance with Sections 227.6(c)(1) and 227.27(a). There are applicable marine water quality criteria for constituents in the material, including listed constituents, and the applicable marine water quality criteria would not be exceeded after initial mixing. In addition, liquid phase bioassays run as part of the suspended particulate phase on three appropriate sensitive marine organisms, show that after initial mixing (as determined under 40 CFR 227.29(a)(2)), the liquid phase of the material would not exceed a toxicity threshold of 0.01 of a concentration shown to be acutely toxic to appropriate sensitive marine organisms. Accordingly, it is concluded that the liquid phase of the material complies with 40 CFR 227.6(c)(1) and 227.27(a). The specific test results and technical analysis of the data underlying this conclusion are described and evaluated in CENAN memo dated 2025.

B. Evaluation of the suspended particulate phase

The suspended particulate phase of the material was evaluated for compliance with Sections 227.6(c)(2) and 227.27(b). Bioassay testing of the suspended particulate phase of the material

has been conducted using three appropriately sensitive marine organisms: inland silversides (*Menidia beryllina*), mysid shrimp (*Americamysis bahia*, formerly *Mysidopsis bahia*), and the planktonic larvae of the Mediterranean mussel (*Mytilus galloprovincialis*). That information shows that when placed in the HARS and after initial mixing (as determined under 40 CFR 227.29(a)(2)), the suspended particulate phase of this material would not exceed a toxicity threshold of 0.01 of a concentration shown to be acutely toxic in the laboratory bioassays (LC/EC₅₀), and thus would not result in significant mortality. The material may be discharged on either incoming or outgoing tides. However, based on a dump volume of 3500 cubic yards, discharge may occur only at distances of at least 900 feet from the northern, southern, eastern, and western boundaries of the HARS, independent of the direction of tidal transport at the time of disposal. Based on a dump volume of 6000 cubic yards, discharge may occur only at distances of at least 1350 feet from the northern, southern, eastern, and western boundaries of the HARS, independent of the direction of tidal transport at the time of disposal. The specific test results and technical analysis of the data underlying this conclusion are described in CENAN 2025.

The factor of 0.01 was applied to ensure that there would be no significant adverse sublethal effects. Moreover, the fact that after placement, the suspended particulate phase would only exist in the environment for a short time, means the suspended particulate phase would not cause significant undesirable effects, including the possibility of danger associated with bioaccumulation, since these impacts require longer exposure durations (see EPA, 1994). Accordingly, it is concluded that the suspended phase of the material complies with 40 CFR 227.6(c)(2) and 227.27(b).

C. Evaluation of the solid phase

The solid phase of the material was evaluated for compliance with Sections 227.6(c)(3) and 227.27(b). This evaluation was made using the results of two specific types of evaluations on the solid phase of the material, one focusing on the acute (ten-day) toxicity of the material, and the other focusing on the potential for contaminant bioaccumulation in test organisms exposed to the dredged material as compared to those same endpoints in test organisms exposed to a reference sediment. Both types of tests were conducted following procedures approved by EPA and the USACE (EPA/USACE 1991).

Reference sediment serves as a point of comparison to identify potential effects of contaminants in the dredged material (EPA/USACE, 1991). Exposing test organisms to this sediment allows for the prediction of toxicity and bioaccumulated contaminant levels that would be expected to result in the test organisms were they “in the wild” at the area from which the reference sediment was taken. The reference sediment used for this project was collected at the Reference Site, in an area of clean, sandy sediments located in the New York Bight near the HARS, where the sediments are unaffected by prior dredged material disposal.

To make a statistically valid determination that exposure to the project sediment does or does not cause greater toxicity or bioaccumulation than the reference sediment, laboratory exposures to several sub-samples (i.e., replicates) of the dredged material and reference sediment are run concurrently. Means and standard deviations are calculated for each set of exposures and compared using standard statistical approaches to determine, with 95% confidence, whether exposure to project sediment results in significantly higher toxicity or bioaccumulation than exposure to reference sediment. Throughout this memorandum, statements regarding project

sediment having “greater” or “less” toxicity or bioaccumulation are referring to calculated differences which are statistically significant at the 95% confidence level.

A statistically significant difference between test and reference bioaccumulation is not itself a quantitative prediction that an impact would occur in the field. A key to understanding bioaccumulation and potential adverse impacts is the fact that bioaccumulation does not necessarily result in an effect. At low exposure (concentration and duration), bioaccumulation may cause no harm. On the other hand, as exposure and/or bioaccumulation increases, the potential for adverse effects increases.

The following sections address the results of solid phase tests and further analyze compliance with the regulatory criteria of Sections 227.6(c)(3), 227.27(b), and 228.15 and with EPA Region 2/CENAN guidance.

Solid phase toxicity evaluation

Ten-day acute toxicity tests were conducted on project materials using mysids (*Americamysis bahia*) and amphipods (*Ampelisca abdita*), which are appropriate sensitive benthic marine organisms. These organisms are good predictors of adverse effects to benthic marine communities (see EPA 1996a). The results of these tests show that the solid phase of the material would not cause significant mortality (statistically different project sediment-exposed mortality exceeding mortality in the reference sediment exposures by 10% for mysid shrimp or 20% for amphipods) and meets the solid phase toxicity criteria of Sections 227.6 and 227.27.

Solid phase bioaccumulation evaluation

28-day tests were conducted on the solid phase of the project material to assess the potential for bioaccumulation of contaminants of concern identified in the project sampling plan (CENAN 2018) using two appropriate sensitive benthic marine organisms, sand worm (*Alitta virens*, formerly *Nereis virens*) and bent-nosed clam (*Macoma nasuta*). These species are good representatives of the phylogenetically diverse base of the marine food chain. Contaminants of concern were identified from Squibb, *et al.*, (1991), a report prepared for the Toxics Work Group of the NY/NJ Harbor Estuary Program to identify and characterize contaminants in NY/NJ Harbor. Compounds from that list, including organic compounds with the potential to bioaccumulate (K_{ow} of approximately 4 or greater), are included on the bioaccumulation testing list when expected to be present in project sediments based upon contaminant inputs and previous sediment sampling results.

There are FDA Action Levels for seven compounds (aldrin, dieldrin, a-chlordane, heptachlor, heptachlor epoxide, PCBs, and mercury; see Table 4, Column 18). Bioaccumulation of any of these contaminants to levels exceeding an FDA Action Level (see EPA/USACE 1991) results in a conclusion that ocean disposal of the dredged material would cause significant adverse effects and exceed the LPC as per 40 CFR 227.27. Therefore, dredged material causing bioaccumulation of any of these contaminants to levels exceeding FDA Action Levels is not suitable for use as Remediation Material at the HARS. None of the contaminants for which there are FDA Action Levels exceed such thresholds in the tissues of organisms exposed to sediments from this project.

When bioaccumulation of a specific contaminant or class of contaminants in organisms exposed to project sediments is not significantly greater than bioaccumulation in organisms exposed to reference sediments, that contaminant is concluded to be present in forms and amounts that do

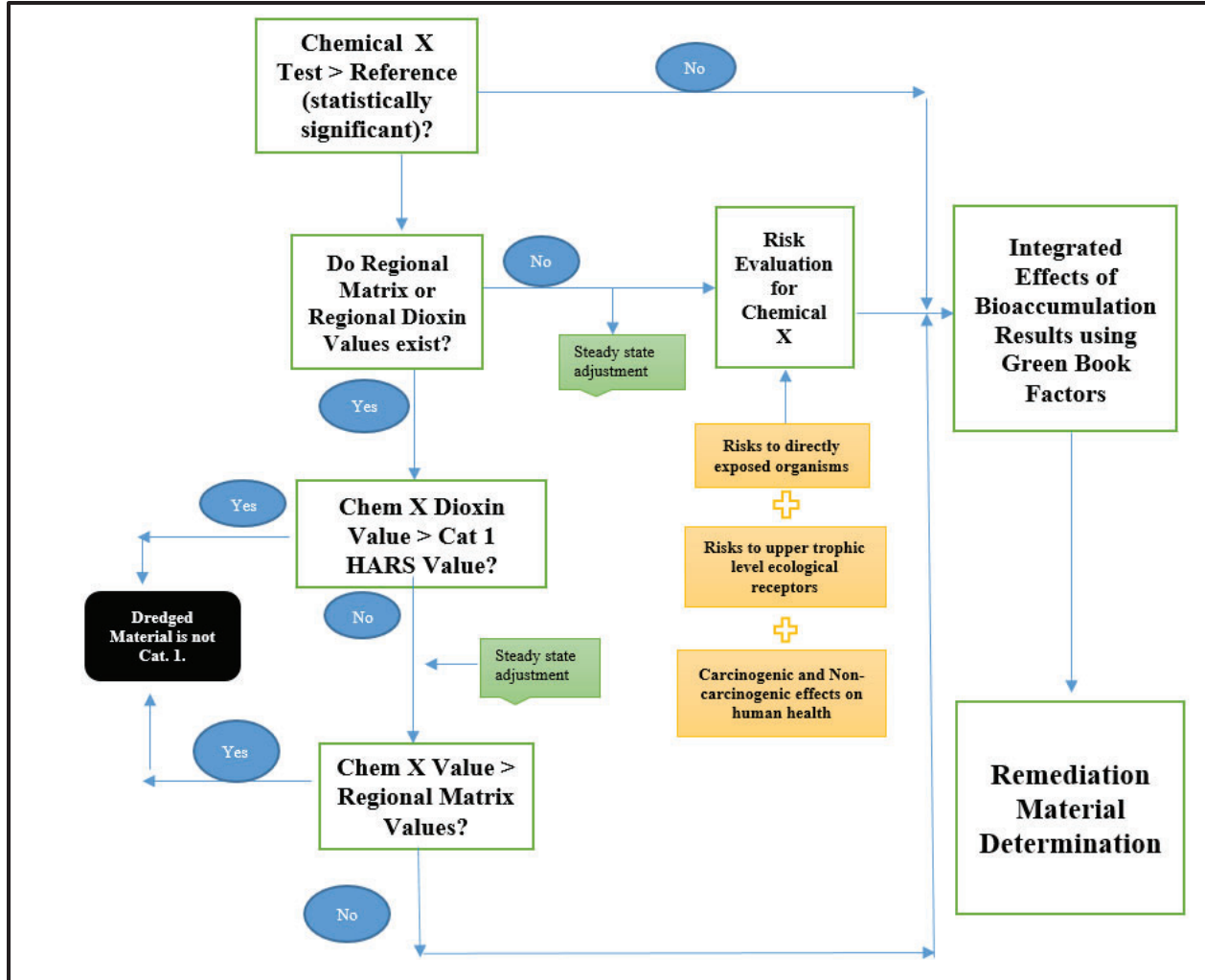
not result in bioaccumulation that would cause unreasonable degradation of the environment or human health, or significant adverse effects. These contaminants are not considered further in the evaluation unless they contribute to a mixture that is evaluated against a single combined guideline value or criterion (i.e., total PCBs, PAH mixtures, total DDT, total chlordane and total endosulfan). In cases where bioaccumulation levels are statistically greater (at the 95% confidence limit) in project sediment-exposed organism than in reference sediment-exposed organisms, further evaluation for potential effects is warranted to evaluate any risk that might be associated with accumulated contaminants.

EPA Region 2 and CENAN use a framework to evaluate project sediment bioaccumulation results that follows national guidance described in the Green Book (EPA/USACE 1991). This guidance outlines a series of evaluations to consider risks associated with project sediment bioaccumulation results that exceed reference. The framework used for HARS-suitability determinations is shown in Figure 1 and includes the following evaluations:

- *Comparison of the 28-day bioaccumulation results to Regional Dioxin Values.* Dioxins and furans that bioaccumulate to greater than reference are evaluated by comparing accumulated concentrations to Regional Dioxin Values (EPA 1997a) (listed in Table 4, Column 20 of Dioxin sheet). 28-day bioaccumulation test results that equal or exceed the Regional Dioxin Values for 2,3,7,8-TCDD or total toxic equivalency (TEQ) of combined congeners indicate that the sediment is not Category I dredged material and is not suitable for use as Remediation Material at the HARS.
- *Comparison of the 28-day bioaccumulation results to Regional Matrix Values.* PCBs, DDTs, cadmium and mercury that bioaccumulate to concentrations greater than reference are evaluated by comparing 'steady state'-adjusted concentrations to Regional Matrix Values (USACE 1981) and the HARS-Specific PCB Worm Tissue Criterion (40 CFR 228.15(d)(6)(v)(E)) (listed in Table 4, Column 20; steady state adjustment is discussed in Section 1 of Appendix A). Bioaccumulation test results that exceed the Regional Matrix Values or HARS-Specific PCB Worm Tissue Criterion indicate that the sediment is not Category I dredged material and is not suitable for use as Remediation Material at the HARS.
- Risk-based assessments are conducted to evaluate concentrations of compounds that bioaccumulate above reference for which Regional Dioxin Values or Regional Matrix Values do not exist. These evaluations are made using 'steady state'-adjusted concentrations and include comparisons to risk values protective of benthic organisms and upper level trophic receptors, including humans.
- The final evaluation uses all available information and results of the individual chemical evaluations to evaluate the solid phase of the dredged material as a whole.

The remainder of this memorandum documents the assessment conducted using the described framework to evaluate risks associated with contaminants that bioaccumulated to greater than reference levels in tissues of organisms exposed to SWO FNP Reach 2 sediment. Bioaccumulation test results for this project are summarized in Table 4.

Figure 1. EPA Region 2/CENAN Framework for Evaluating Bioaccumulation Test Results for SWO FNP



Solid Phase Bioaccumulation Evaluation-Comparison to Reference Sediment Bioaccumulation

Contaminants for which bioaccumulation from the dredged material in tissue was statistically greater than the reference in the clam and/or worm are indicated by an asterisk in the applicable column of Table 1.

Table 1. Statistically Significant Bioaccumulation of Contaminants in Organisms Exposed to Project Sediments vs. Reference Sediments

CONTAMINANT	CLAMS	WORMS
PAHs		
Acenaphthene	*	*
Acenaphthylene	*	*
Anthracene	*	*
Fluorene	*	*
Naphthalene	*	
Phenanthrene	*	
Benzo(a)anthracene**	*	*
Benzo(a)pyrene**	*	*
Benzo(g,h,i)perylene	*	*
Benzo(b)fluoranthene**	*	*
Benzo(k)fluoranthene**	*	*
Chrysene**	*	*
Dibenzo(a,h)anthracene**	*	
Fluoranthene	*	*
Indeno(1,2,3-cd)pyrene**	*	*
Pyrene	*	*
PESTICIDES		
Dieldrin	*	*
a-Chlordane	*	*
Trans nonachlor	*	*
Endosulfan I	*	*
Endosulfan II	*	*
Endosulfan sulfate	*	
4,4-DDD	*	*
2,4-DDD	*	*
4,4-DDE	*	*
2,4-DDE	*	*
TOTAL PCBs	*	*
1,4-Dichlorobenzene	*	
METALS		
Chromium (total)	*	
Copper	*	*
Lead	*	
Nickel	*	
DIOXINS/FURANS		
1,2,3,4,6,7,8-HpCDD	*	*
OCDD	*	*
2,3,7,8-TCDF		*
2,3,4,7,8-PeCDF		*
2,3,4,6,7,8-HxCDF		*
1,2,3,4,6,7,8-HpCDF		*

Solid Phase Bioaccumulation Evaluation - Comparison to Regional Dioxin Values

Bioaccumulation test results were below all Regional Dioxin Values (EPA 1997a) for both test organisms. Therefore, no further evaluation of these contaminants is required in accordance with EPA Region 2/CENAN guidance.

Solid Phase Bioaccumulation Evaluation- Comparison to Regional Matrix Values and the HARS-Specific PCB Worm Tissue Criterion

Bioaccumulation test results (adjusted for steady state, as described in Section 1 of Appendix A) were below all Regional Matrix Values (USACE 1981) for both test organisms and the mean 'steady state'-adjusted worm PCB tissue concentration was below the HARS-Specific PCB Worm Tissue Criterion (40 CFR 228.15(d)(6)(v)(E)). Therefore, no further evaluation of these contaminants is required in accordance with EPA Region 2/CENAN guidance.

Solid Phase Bioaccumulation Evaluation- Risk-based Evaluations

Risk associated with compounds that bioaccumulated to greater than reference in test organisms exposed to project sediment for which Regional Matrix Values (USACE 1981) or Regional Dioxin Values (EPA 1997a) do not exist are evaluated using risk-based assessments, as discussed below. Contaminants that fall into this group for this project include PAHs, dieldrin, a-chlordane, trans nonachlor, endosulfan I, endosulfan II, endosulfan sulfate, 1,4-dichlorobenzene, total chromium, copper, lead, and nickel.

Contaminants bioaccumulated from dredged material can present risks to organisms that are directly exposed to the sediment and to species at upper trophic levels of the marine food web (e.g., predatory fish). Human exposure can potentially result via consumption of fish that has been dietarily exposed to HARS contaminants. Risk to directly exposed ecological receptors, risk to dietarily exposed ecological receptors, and human risk are examined separately in the EPA Region 2/CENAN framework.

a. Risks to Directly Exposed Organisms

Metals, Pesticides, and Industrial Chemicals

Water Quality Criterion Tissue Levels (WQCTL) are established for aldrin, dieldrin, total chlordane, total endosulfan, 1,4-dichlorobenzene, arsenic chromium, copper, lead, nickel, silver and zinc (see Appendix, Section 2, A). For this project, none of the WQCTLs were exceeded.

PAHs

To assess direct exposure risks of PAHs, a direct comparison is made of total measured PAH tissue residue to the Critical Body Residue (CBR) associated with narcosis, which is the primary non-cancer effect of PAHs. The CBR approach, described by McCarty (1991), was used to evaluate the potential impacts of total PAHs accumulated in the dredged material bioaccumulation test organisms and is discussed in the Appendix. As shown in Table 4, total PAH levels in tissues from the dredged material bioaccumulation test were below levels at which chronic adverse effects might be expected from a narcotic mode of action in sensitive aquatic organisms (i.e., fish) as estimated by the CBR.

Therefore, these bioaccumulation test results do not indicate a potential for undesirable effects to directly exposed organisms resulting from use of this material as Remediation Material at the HARS.

b. Risks to Upper Trophic Level Ecological Receptors

To properly assess risks to upper trophic level species (including humans) the efficiency with which a contaminant is transferred through the diet and the propensity of those compounds to accumulate to higher concentrations in the tissues of the consumer must be considered. The potential movement of contaminants through the food chain was considered and appropriate trophic transfer factors applied to adjust the data accordingly to reflect the propensity of specific contaminants or contaminant groups to accumulate to higher (or lower) concentrations in upper trophic level organisms (including humans). Data and/or methods used to develop these trophic transfer adjustment factors are discussed in the Appendix.

Effects of Mutagenic, Carcinogenic and Teratogenic PAHs

Applying an uncertainty factor (UF) of 10 to account for interspecies differences in sensitivities and a trophic transfer factor of 0.1 to the no-effects level for BaP calculated from Hannah, *et al.* (1982), as discussed in the Appendix results in a no-effect level for BaP of approximately 8,000 for BaP only ppb in benthic tissue, which is considerably greater than the highest steady state-adjusted tissue concentration of BaP measured in organisms exposed to project sediment (12.1 ppb for BaP only and 17.3 ppb for total BaP equivalents). Project tissue concentrations would still be below this no-effect level if the higher trophic transfer factor (0.23) reported by McElroy, *et al.* (1991) was used.

Hall and Oris (1991) reported on experiments that exposed fathead minnows to anthracene during long-term exposures and observed adverse effects on reproduction. The paper reported that a concentration of anthracene in the tissue of the egg in the range of 3,750 to 8,000 ppb resulted in no significant effects on egg hatching or survivorship. Using the same approach for accounting for species-to-species uncertainty and food chain transfer yields a conservative benthic tissue level of 3,750 ppb. Anthracene tissue concentrations from the project bioaccumulation tests are well below this level.

As stated above, PAHs accumulated to greater than reference in test organisms exposed to project sediments. However, the most relevant aquatic effects information reviewed indicates that the highest tissue PAH levels accumulated in the dredged material bioaccumulation tests are below mutagenic, carcinogenic, and teratogenic no-effect levels.

c. Consideration of Potential Carcinogenic and Non-carcinogenic Effects on Human Health

Benthic tissue screening levels for human carcinogenic risk and non-carcinogenic hazard were developed for those contaminants that do not have Regional Matrix Values (USACE 1981) or Regional Dioxin Values (EPA 1997a). The approach used to derive these screening levels (described in the Appendix) uses standard EPA risk-assessment approaches and assumptions (fish and shellfish consumption rates; whole body to fillet contaminant ratios) to back calculate a fish tissue concentration that corresponds to specific levels of cancer risk or non-cancer hazard. The fish tissue concentration is then converted to a corresponding benthic prey concentration using the trophic transfer factors discussed in the Appendix.

Because the analysis implicitly incorporates various conservative assumptions regarding exposure (e.g., 100% of consumed seafood is fish taken from the HARS; fish has fed exclusively at HARS long enough prior to capture to be at equilibrium with benthic prey tissue contaminant concentrations), the resulting benthic tissue concentrations represent conservative estimates of

risk, or what are in effect plausible upper-bound estimates. Thus, the true risk is highly unlikely to be greater than estimated and could be much lower.

Table 4, Column 14 lists conservative human cancer protection levels in benthic organisms for the chemicals which are known or suspected carcinogens that correspond to a human cancer risk level of 1×10^{-4} . None of the human health cancer protection levels were exceeded in the bioaccumulation test results.

The potential for non-cancer impacts can be expressed as a hazard quotient (HQ), which is the ratio of the average daily intake divided by the toxicological reference dose for the chemical. Table 4, Column 15 includes the noncancer protection levels in benthic organisms for the chemicals requiring further analysis that are known to cause, or suspected of causing, non-carcinogenic effects, that would result in a human HQ equal to unity (i.e., 1). If the HQ is less than unity, an adverse noncarcinogenic effect is highly unlikely to occur. The higher the HQ, the more likely that an adverse noncarcinogenic effect will occur as a result of exposure to the contaminant in the dredged material after placement.

The concentrations of all contaminants in test organisms' tissues were below their respective non-cancer protection level (HQs for all contaminants were less than unity).

D. Evaluation of Bioaccumulation Results for Dredged Material as a Whole

As a final step in the evaluative process, it is appropriate to go beyond assessing the individual test results to evaluate the bioaccumulation test results as a whole to provide an opportunity for an integrated assessment of the individual test results (see Figure 1, Box d). EPA/USACE (1991) provides additional factors to be considered in making an integrated assessment of dredged material test results. The recommended factors are:

Magnitude of toxicity and number and phylogenetic diversity of species exhibiting greater mortality in the dredged material than in the reference material

As discussed above, project dredged material would not cause significant mortality (statistically different project sediment-exposed mortality exceeding mortality in the reference sediment exposures by 10% for mysid shrimp or 20% for amphipods) and meets the solid phase toxicity criteria of Sections 227.6 and 227.27.

Number and magnitude of contaminants for which bioaccumulation from the dredged material is statistically greater than bioaccumulation from the reference material and number and phylogenetic diversity of species in which bioaccumulation from the dredged material is statistically greater than bioaccumulation from the reference material

The magnitudes by which compounds bioaccumulated to levels greater than reference are shown in Table 2.

Table 2. Magnitude of Bioaccumulation of Contaminants in Organisms Exposed to Project Sediments vs. Reference Sediments

CONTAMINANT	CLAMS	WORMS
PAHs		
Acenaphthene	2.5	1.6
Acenaphthylene	4.2	2.2
Anthracene	3.2	4.3
Fluorene	1.7	1.4
Naphthalene	1.7	
Phenanthrene	1.6	
Benzo(a)anthracene**	6.6	5.9
Benzo(a)pyrene**	7.8	3.1
Benzo(g,h,i)perylene	9.7	4.1
Benzo(b)fluoranthene**	7.8	4.7
Benzo(k)fluoranthene**	6.5	5.7
Chrysene**	4.9	8.4
Dibenzo(a,h)anthracene**	9.6	
Fluoranthene	3.0	5.0
Indeno(1,2,3-cd)pyrene**	11.5	1.4
Pyrene	5.7	8.7
PESTICIDES		
Dieldrin	5.6	3.7
α-Chlordane	5.7	4.6
Trans nonachlor	5.5	1.5
Endosulfan I	11.4	2.9
Endosulfan II	7.2	2.2
Endosulfan sulfate	7.1	
4,4-DDD	10.9	10.9
2,4-DDD	5.7	4.3
4,4-DDE	14.9	8.3
2,4-DDE	17.9	3.1
TOTAL PCBs	6.8	2.1
1,4-Dichlorobenzene	2.2	
METALS		
Chromium (total)	2.2	
Copper	1.4	1.2
Lead	2.8	
Nickel	1.2	
DIOXINS		
1,2,3,4,6,7,8-HpCDD	1.8	2.7
OCDD	9.7	4.4
2,3,7,8-TCDF		2.9
2,3,4,7,8-PeCDF		3.3
2,3,4,6,7,8-HxCDF		2.2
1,2,3,4,6,7,8-HpCDF		4.1

All contaminants except for one PAH and four pesticides accumulated to less than ten times reference in *M. nasuta*, and all contaminants except for one pesticide accumulated to less than ten times reference in *A. virens*.

Exceeding reference concentrations is common when reference concentrations are very low or “non-detect.” Because the HARS reference sediment is greater than 90% sand and has only trace levels of contamination, reference-exposed organisms tend to exhibit very low or undetectable levels of bioaccumulation.

Toxicological importance of the contaminants whose bioaccumulation from the dredged material exceeds that from the reference material, including their propensity to biomagnify within aquatic food webs

Although some of the contaminants that bioaccumulated to levels greater than reference in organisms exposed to the project sediment can be toxicologically important, they did not accumulate to toxicologically important concentrations. Levels of these contaminants were all below the acceptable human health risk range and acceptable aquatic effects range using conservative approaches and analyses as described above to evaluate those test results, including through their potential transfer through the food web.

Magnitude by which contaminants whose bioaccumulation from the dredged material exceeds that from the reference material also exceed the concentrations found in comparable species living in the vicinity of the proposed site.

Organisms collected from a broad area of the sea floor in the vicinity of (but not inside) the HARS have been collected and analyzed for tissue concentration for bioaccumulative contaminants of concern (Charles and Muramoto 1990; USACE 1994; EPA 1996f; EPA 1997b). These data provide a measure of the tissue residues for organisms living outside the HARS. Table 4, Columns 16 and 17 summarize the most recent background data.

The magnitude by which compounds that bioaccumulated to levels greater than background levels (where background values exist) are reported in Table 3. All other compounds accumulated in test organisms to steady state levels below background.

Table 3. Magnitude of Bioaccumulation above Background in Organisms Exposed to Project Sediments

CONTAMINANT	CLAMS	WORMS
PAHs		
PESTICIDES		
Dieldrin		5.1
Endosulfan II		1.1
4,4-DDE	1.1	
METALS		
DIOXINS		

All contaminants accumulated to less than two times background in *M. nasuta*, and all contaminants except for one pesticide accumulated to less than two times background in *A. virens*.

When bioaccumulation in organisms exposed to project sediments is not greater than tissue concentrations in organisms from the vicinity of the remediation site (the background levels),

this indicates that placement of the material would not result in bioaccumulation above existing ambient levels in the general area and thus does not have a potential to cause undesirable effects. However, bioaccumulation in organisms exposed to project sediments greater than these levels is not necessarily predictive of adverse effects. It may, for example, reflect extremely low background levels.

Although some of the contaminants that bioaccumulated to levels greater than background in organisms exposed to the project sediment can be toxicologically important, they did not accumulate to toxicologically important concentrations. Levels of these contaminants were all below the acceptable human health risk range and acceptable aquatic effects range using conservative approaches and analyses as described above to evaluate those test results, including through their potential transfer through the food web.

VI. OVERALL CONCLUSION ON THE PROPOSED PROJECT

The evaluation of the testing results performed above indicates that the material does not have a potential to cause undesirable effects to aquatic marine biota due to chronic adverse effects including such effects due to mutagenic, carcinogenic, or teratogenic contaminants, or to human health due to cancer or non-cancer effects from the individual contaminants. Evaluation of the solid phase bioaccumulation test results for the dredged material as a whole would not indicate a different outcome than that shown by the individual test results themselves; i.e., that the material does not have the potential to cause undesirable effects due to bioaccumulation.

Based upon this review of the results of testing of the sediments proposed for dredging and ocean placement from SWO FNP Reach 2, the material meets the criteria for acceptability for ocean placement as described in Sections 227.6, and 227.27 of the Regulations, is Category I under EPA Region 2/CENAN guidance and is suitable for placement at the HARS under Section 228.15(d)(6) as Remediation Material.

Table 4. Bioaccumulation and Comparison Data for SWO FNP Reach 2

COMPARISON DATA																		
PROJECT DATA																		
Col. 1	Col. 2 [10]	Col. 3 [10]	Col. 4 [10]	Col. 5 [Col. 6 [10]	Col. 7/Col. 8	Col. 9	Col. 10	Col. 11	Col. 12	Col. 13	Col. 14	Col. 15	Col. 16	Col. 17	Col. 18	Col. 19	Col. 20	
Sample I.D.	Reference	Test Sed.	[1]	Test Sed.	[1]	Test Sed.	Test Sed.	Test Sed.	Test Sed.	Test Sed.	Test Sed.	Human Health	Human Health	Background	Background	FDA	Ecological	
		(clam)	(clam)	(clam)	(clam)	(clam)	SS	Carcinogenic	BaP Tox. Equiv.	BaP Tox. Equiv.	BaP Tox. Equiv.	Cancer [10E-4]	Non-Cancer	(clam)	(worm)	Limits [6]	Non-Specific	Regional
		(ug/Kg)	(ug/Kg)	(ug/Kg)	[2]	(ug/Kg)	(worm)	TEF [3]	(ug/Kg)	Conc.(clam)[3]	Conc.(worm)[3]	Level[4]	Level (HQ=1)	(ug/Kg)	(ug/Kg)	(ug/Kg)	Effects Level	Matrix
Compound																	(ug/Kg)	(ug/Kg)
PAHs																		
Acenaphthene	0.144	0.115	0.36	*	0.187	*	0.36	0.187					8,775,000	8.1	0.5		[13]	
Acenaphthylene	0.084	0.078	0.356	*	0.169	*	0.356	0.169				[5]	4.6	1.3			[13]	
Anthracene	0.266	0.029	0.84	*	0.125	*	0.84	0.125				43,605,000	10.0	1.6		3,750		
Fluorene	0.286	0.066	0.485	*	0.091	*	0.485	0.091				5,805,000	7.4	0.3			[13]	
Naphthalene	0.549	0.61	0.92	*	0.691	*	0.92	0.691				[5]	26.4	4.5			[13]	
Phenanthrene	1.83	0.226	2.90	*	0.273	*	2.90	0.273				43,605,000	32.7	4.7			[13]	
Benz[<i>a</i>]anthracene**	0.767	0.030	5.04	*	0.177	*	10.10	0.354	0.1	1.01	0.035	[5]	25.1	4.8			[13]	
Benz[<i>a</i>]pyrene**	0.773	0.094	6.03	*	0.289	*	12.10	0.578	1	12.1	0.578	[5]	24.2	7.6		8,000 [7]		
Benz[<i>ghi</i>]perylene	0.344	0.067	3.33	*	0.232	*	9.99	0.696				[5]	22.2	7.6			[13]	
Benz[<i>b</i>]fluoranthene**	0.760	0.063	5.93	*	0.298	*	11.90	0.596	0.1	1.19	0.060	[5]	40.6	16.5			[13]	
Benz[<i>k</i>]fluoranthene**	0.987	0.072	6.40	*	0.412	*	12.80	0.824	0.01	0.128	0.008	[5]	81.6	5.6			[13]	
Chrysene**	1.85	0.237	9.05	*	2.00	*	18.1	4.00	0.001	0.0181	0.0040	[5]	29.4	6.7			[13]	
Dibenz[<i>a,h</i>]anthracene**	0.059	0.0470	0.565	*	0.0470	*	1.130	0.047	1	1.13	0.05	[5]	8.5	1.1			[13]	
Fluoranthene	3.81	0.439	11.4	*	2.18	*	11.4	2.18		1		5,805,000	43.8	10.6			[13]	
Indeno[1,2,3- <i>cd</i>]pyrene**	0.223	0.0530	2.56	*	0.073	*	7.68	0.219	0.1	0.768	0.022	[5]	16.6	4.4			[13]	
Pyrene	2.77	0.375	15.8	*	3.27	*	15.8	3.27				4,387,000	51.3	26.6			[13]	
TOTAL PAHs							117	14.3		17.3	0.8	2,000 (BaP eqv.)		104.60		40,000 [8]		
PESTICIDES																		
Aldrin	0.0140	0.0140	0.0140		0.0140		0.014	0.014				33		0.1	300	299 [9]		
Dieldrin	0.026	0.069	0.146	*	0.257	*	0.292	0.514				65		0.1	300	4.37 [9]		
a-Chlordane	0.017	0.034	0.097	*	0.156	*	0.194	0.312						0.7	300			
Trans nonachlor	0.028	0.131	0.154	*	0.192	*	0.308	0.384						0.5				
Heptachlor	0.0110	0.0110	0.068		0.257		0.136	0.514						0.05	300			
Heptachlor epoxide	0.0170	0.025	0.0170		0.0170		0.009	0.009						0.2	300			
Total Residual																		
Chlordane/Heptachlor							0.647	1.22				114	135	1.7		64# [9]		
Endosulfan I	0.0150	0.0150	0.171	*	0.043	*	0.171	0.043				[5]		0.2				
Endosulfan II	0.022	0.051	0.158	*	0.114	*	0.158	0.114				[5]		0.1				
Endosulfan sulfate	0.012	0.040	0.085	*	0.0120		0.085	0.006				[5]		0.1				
Total Endosulfans							0.414	0.163					87,000	0.4		2.86# [9]		

Project Data																		COMPARISON DATA									
Col. 2 [10]	Col. 3 [10]	Col. 4 [10]	Col. 5 Col. 6 [10]	Col. 7	Col. 8	Col. 9	Col. 10	Col. 11	Col. 12	Col. 13	Col. 14	Col. 15	Col. 16	Col. 17	Col. 18	Col. 19	Col. 20										
Reference	Reference	Test Sed.	Test Sed. [1]	Test Sed.	[1]	Test Sed.	Test Sed.	Test Sed.	Test Sed.	Test Sed.	Human Health Cancer (10E-4)	Human Health Non-Cancer Level (HC=1)	Background (clam)	Background (worm)	FDA Limits [6]	Ecological Non specific	Regional Dioxin										
(clam)	(worm)	(clam)	(worm)	(clam)	[2]	(clam)	(worm)	TEF [3]	(clam)	(worm)	Level	Level (HC=1)	TEQ (pptr)	TEQ (pptr)	(pptr)	Effects Level	Value										
(pptr)	(pptr)	(pptr)	(pptr)	(pptr)		(pptr)	(pptr)		(pptr)	(pptr)							[1*]										
DIOXINS																											
2,3,7,8-TCDD	0.086U	0.094U	0.091U					1	0.046	0.068			1.73	2.50			1										
1,2,3,7,8-PeCDD	0.05U	0.064U	0.062U					0.5	0.016	0.032				0.41													
1,2,3,4,7,8-HxCDD	0.082U	0.105U	0.108U					0.1	0.005	0.01055				0.10													
1,2,3,6,7,8-HxCDD	0.079U	0.1U	0.106U					0.1	0.005	0.01045				0.17													
1,2,3,7,8,9-HxCDD	0.083U	0.105U	0.11U					0.1	0.006	0.0108				0.08													
1,2,3,4,6,7,8-HpCDD	0.126U	0.175U	0.233	*				0.01	0.00233	0.00466				0.13													
OCDD	0.47	1.440	4.540	*				0.001	0.00454	0.00639				0.11													
2,3,7,8-TCDF	0.126U	0.214	0.179U					0.1	0.00895	0.0629				0.27													
1,2,3,7,8-PeCDF	0.043U	0.054U	0.06U					0.05	0.0015	0.002975				0.05													
2,3,4,7,8-PeCDF	0.039U	0.047U	0.056U		*			0.5	0.014	0.0785				0.39													
1,2,3,4,7,8-HxCDF	0.034U	0.056U	0.043U					0.1	0.00215	0.00505				0.08													
1,2,3,6,7,8-HxCDF	0.033U	0.055U	0.043U					0.1	0.00215	0.0051				0.09													
1,2,3,7,8,9-HxCDF	0.046U	0.079U	0.058U					0.1	0.0029	0.0071				0.11													
2,3,4,6,7,8-HxCDF	0.033U	0.055U	0.043U		*			0.1	0.00215	0.0123				0.08													
1,2,3,4,6,7,8-HpCDF	0.061U	0.087	0.068U		*			0.01	0.00034	0.00361				0.04													
1,2,3,4,7,8,9-HpCDF	0.08U	0.079U	0.062U					0.01	0.00031	0.000645				0.02													
OCDF	0.113U	0.201U	0.147					0.001	0.000147	0.0001005				0.01													
TEQs - non 2,3,7,8-TCDD																											
TEQs (all)									0.073467	0.2531305				2.14			4.5 [12]										
									0.119467	0.3211305			1.73	4.64			10										

FOOTNOTES FOR TABLE 4:

**: Carcinogenic PAHs.

#: Levels represent the conservative level of protection for the sum of the related compounds and their metabolites.

na: Not Available

1. An “*” in this column indicates that the analyte concentration in the test sediment-exposed organisms are statistically greater than in those exposed to reference sediment. Means and statistical comparisons were determined using 1/2 the detection limit to estimate concentrations of analytes that were below the detection limit.
2. Conversion factors from 28-day bioaccumulation results to steady state (see discussion in Appendix)
3. Toxic equivalencies (TEFs) for the carcinogenic PAHs are from EPA (1993); Dioxin TEFs are from: EPA (1989).
4. This value represents the 10⁻⁴ cancer risk level for the carcinogenic PAHs. The total concentration of carcinogenic PAHs is expressed in BaP equivalents (see discussion in the text of the memo).
5. Cancer risk factor or reference dose are not assigned by EPA in IRIS (EPA 1995).
6. FDA limits are from EPA/USACE (1991).
7. This value represents the benthic level expected to result in a no-effect level for possible mutagenic and teratogenic effects in fish from exposure to BaP, which is the most toxic PAH.
8. This value represents the non-specific narcosis effects level (see discussion in Appendix). This value is compared to the sum of all PAHs measured.
9. Calculations are included in Appendix.
10. Means of five tissue replicates calculated using 1/2 detection limits to estimate concentrations of analytes that were not detected; “U” indicates that all five replicates were not detected.
11. Levels are based on the Regional Dioxin Values (EPA 1997a).
12. Level is the sum of TEQs of all dioxin congeners other than 2,3,7,8-TCDD.
13. For this PAH, the no-effect level for possible mutagenic and teratogenic effects in fish is estimated from exposure to BaP, which is the most toxic PAH.
14. Cancer and non-cancer protection levels, based on inorganic arsenic as contained in EPA’s IRIS database, are not appropriate for evaluating the potential human health impacts of arsenic bioaccumulation from dredged material, and therefore, are not included in Table 4 (see discussion in Appendix).

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Appendix

I. CORRECTION OF 28-DAY BIOACCUMULATION RESULTS TO ACCOUNT FOR STEADY STATE (Column 8-10)

Bioaccumulation tests were conducted using 28-day exposure of appropriate sensitive benthic marine organisms to sediment. Following placement of the material at the HARS, exposures to any contaminants would continue beyond 28 days, therefore, assessment of risks associated with accumulations must consider the extent to which the 28-day measured tissue concentration approximate the 'steady state' concentration of each contaminant that could occur in organisms at the HARS after extended exposure. Therefore, comparisons to guidance values or criteria for given contaminants that reflect 'steady state' conditions (as opposed to comparisons to other 28-day tissue concentrations such as comparison to reference) may require that a correction factor be applied to the 28-day concentration to account for incomplete attainment of steady state accumulation.

Steady state may be defined operationally as the lack of any significant difference (ANOVA, $\alpha = 0.05$) among tissue residues taken at three consecutive sampling intervals (Lee, *et al.* 1989). The 28-day test exposure period was selected as appropriate because most chemicals of concern will reach at least 80% of steady state in benthic marine organisms within that time frame (Boese and Lee 1992). For the few chemicals that may not meet steady state tissue concentrations in 28 days, a factor is used to adjust the data to steady state.

Steady state considerations for specific contaminants (or contaminant classes) and the derivation of correction factors (listed in Column 8) to account for incomplete uptake are discussed in the remainder of this section. Steady state-adjusted concentrations for clams and worms are presented in Columns 9 and 10, respectively.

A. Metals

In general, metals bioaccumulate more rapidly than organics and 28-day tests are sufficient to evaluate potential effects (see EPA/USACE 1991), for example, arsenic (Naqvi, *et al.* 1990; Riedel, *et al.* 1987; Oladimeji, *et al.* 1984).

Cadmium and mercury are not regulated in marine organisms as are essential metals, and, thus, no adjustment for steady state is applicable. Bioaccumulation of these metals is affected by many complex factors, and is essentially linear (Dethlefsen 1978; Giesy, *et al.* 1980; V-Balogh and Salanka 1984). Therefore, there are no adjustments that can be made to reproduce steady state. The Regional Matrix Values for cadmium and mercury, therefore, do not represent steady state conditions and so 28-day test results are used to compare to the Regional Matrix Values.

B. Organic Contaminants

a. Pesticides and 1,4-Dichlorobenzene

Uptake of non-polar organic contaminants from food is highly dependent on their hydrophobicity, a property measured by the octanol/water partition coefficient, K_{ow} . The higher the value of K_{ow} , the longer it takes to reach steady state in benthic marine organisms. Organochlorine compounds that have $\log K_{ow} > 6$ (i.e., aldrin, dieldrin, heptachlor, trans nonachlor, and alpha-chlordane), may not attain steady-state within 28 days of exposure to sediment.

Information contained in Boese and Lee (1992) indicates that aldrin, dieldrin, and α -chlordane achieve at least 50% of steady-state in 28-day bioaccumulation tests. Heptachlor and trans nonachlor, were not addressed in Boese and Lee (1992). However, the fraction of steady state achieved for these compounds after 28-days was estimated using the equations contained in McFarland (1995). Results of such calculations indicate heptachlor and trans nonachlor reach approximately 50% of steady state after 28 days. Calculating the fraction of steady state achieved for the remaining pesticides with $\log K_{ow} < 6$ (heptachlor epoxide) and for 1,4-dibenzochlorene ($\log K_{ow} = 3.44$) indicates that these compounds reach steady state within 28 days, so no adjustment is necessary.

The degree to which DDT and its metabolites reach steady state in test organisms after 28 days was evaluated using data reported by Lee *et al.* (1994). To assess the rate of bioaccumulation, they exposed the clam *Macoma nasuta*, to sediments collected from the vicinity of the United Heckathorn Superfund site in Richmond California to study tissue residues and uptake kinetics associated with exposure to pesticide-contaminated sediments. Results of the study indicate that one parent compound, 4,4-DDT, bioaccumulates much more slowly than 2,4-DDT and the DDT metabolites. The results range from approximately 9 percent of steady state after 28 days for 4,4-DDT, to 55 percent of steady state after 28 days for 2,4-DDT (Lee, et al. 1994). Based on these results, 28-day test tissue concentrations of DDT compounds and their metabolites are adjusted by multiplying the 28-day result by a factor of 11 for 4,4-DDT, a factor of three for 4,4-DDD, and a factor of two for 2,4-DDT and the remaining DDT metabolites before they are compared to Regional Matrix Values (USACE 1981).

b. PCBs

The degree to which PCBs reach steady state in test organisms after 28 days was evaluated using data reported by Lee, *et al.* (1994). To assess the rate of bioaccumulation of PCBs and other compounds, Rubinstein, *et al.* (1990) and Pruell, *et al.* (1993) exposed three species of organisms, the grass shrimp *Palaemonetes pugio*; the sandworm *Nereis virens*, and the clam *Macoma nasuta*, to sediments collected from the Passaic River, N.J. Sub-samples of the exposed organisms were removed on various days into the study including days 0, 10, 28, 42, 84, and 180.

For the clam tissue, the variance in the concentrations on day 28 and day 84 (by which point the maximum concentration had been reached) overlap, thus indicating that the two are not statistically different and the bioaccumulation on day 28 is at or very close to steady state. Thus, the clam bioaccumulation for the project sediments using 28-day exposures is acceptable for use as steady state tissue levels.

For the worm tissue, variances for days 28 and 180 do not overlap, thus indicating that steady state was probably not reached in 28 days. Variance in the data makes it difficult to quantify a real difference; means for days 28 and 180 from Rubinstein *et al.* (1990) (approximately 1,750 ng/g (nanograms per gram or parts per billion, ppb) for 28 days, and 3,000 ng/g for 180 days) suggest approximately 58% of steady state would have been reached in 28 days. On this basis, worm project data are conservatively adjusted by a factor of two to calculate a steady state tissue concentration for comparison to the HARS-Specific PCB Worm Tissue Criterion (40 CFR 228.15(d)(6)(v)(E)).

c. PAHs

The degree to which PAHs reach steady state in test organisms after 28 days was evaluated using data reported by McFarland (1995) and Meador, *et al.* (1995). The time required for a given PAH to attain a steady-state concentration following exposure to bedded sediments (t_{ss}) is determined primarily by the log K_{ow} of the compound in question (McFarland 1995; Meador, *et al.* 1995). Meador, *et al.* (1995) reviewed nine studies that investigated the attainment of steady-state tissue concentrations of PAHs by various marine invertebrates. In each case, tissue concentrations approached steady state within several days to two weeks after initiating exposure to both low molecular weight PAHs and high molecular weight PAHs. McFarland (1995) estimated t_{ss} values ranging from 3.5 to 326 days for 15 PAHs based on their hydrophobicities. The estimated steady-state concentration of the sum total of the 15 PAHs analyzed by McFarland for sediments collected from typical harbor areas revealed that the mean concentration attained after 28-day bioaccumulation tests was approximately 86% of steady-state. McFarland (1995) concluded that 28-day tests are likely to reflect steady-state. Despite this conclusion, steady state total PAH concentrations were estimated by summing individual PAH concentrations that were conservatively adjusted by a factor of 1, 2, or 3 based on their hydrophobicities and McFarland (1995).

II. CONSIDERATION OF FOOD-CHAIN TRANSFER

To properly assess risks to upper trophic level species (including humans) the efficiency with which a contaminant is transferred through the diet and the propensity of those compounds to accumulate to higher concentrations in the tissues of the consumer must be considered. The potential movement of contaminants through the food chain was considered and appropriate trophic transfer factors applied to adjust the data accordingly to reflect the propensity of specific contaminants or contaminant groups to accumulate to higher (or lower) concentrations in upper trophic level organisms (including humans). Data and/or methods used to develop these trophic transfer adjustment factors are discussed in the remainder of this section.

Metals

Trophic transfer of most metals is not sufficient to qualify as biomagnification (Brown and Neff 1993). The lack of observed biomagnification for such metals as arsenic, cadmium, lead, mercury, and zinc is the result of incomplete absorption of metals across the gut, rapid excretion, and dilution in muscle, which represents a large part of the total body weight of most marine animals (Fowler 1982; Suedel *et al.* 1994). For purposes of conducting the human health and ecological evaluations below, a conservative trophic transfer coefficient equal to one will be used for these non-biomagnifying metals (Suedel *et al.* 1994 and references cited therein).

Pesticides and Industrial Chemicals

The potential for these chemicals to biomagnify was also evaluated. Although organic contaminants with values of $\log K_{ow} > 4$ tend to biomagnify in the marine food chain, studies (USACE 1995) have shown that this is not true for higher molecular weight compounds such as the most highly chlorinated PCBs or for easily metabolized compounds such as PAHs. Those organic compounds which are not efficiently excreted, such as certain pesticides, can biomagnify in the food chain. For the organic constituents with a potential to biomagnify in the marine food chain, trophic transfer factors were calculated, using the approach described by Gobas (1993). The values are summarized in Attachment B.

PAHs

With regard to the potential for biomagnification of PAHs, feeding studies show that assimilation rates from ingested food are extremely low, e.g., more than 98% of the target contaminant remained in an undigested form in fish gut 48 hours after feeding squid containing radio-labeled benzo[a]pyrene to young cod (Corner, *et al.* 1976) and juvenile Atlantic herring (Whittle, *et al.* 1977). PAH metabolites are also transferred through the marine food chain; however, they are absorbed even less efficiently than their parent compounds (McElroy and Sisson 1989; McElroy, *et al.* 1991). Up to 99% of the PAH compounds taken up by fish are metabolized and excreted into bile, the usual elimination mode, within 24 hours of uptake (Varanasi, *et al.* 1989). Similar results are described in Brown and Neff (1993) who evaluated various studies describing trophic transfer. The studies cited in Brown and Neff (1993) indicate a trophic transfer rate for BaP from invertebrates to fish of between 0.02 and 0.23 times the concentration in the ingested invertebrates (Corner, *et al.* 1976, O'Connor, *et al.*, 1988, McElroy, *et al.* 1991). This was considered when assessing the ecological and human health effects of the project material as discussed below.

III. CONSIDERATION OF ECOLOGICAL EFFECTS

A. Potential for ecological effects based on Water Quality Criteria (Column 19)

The potential for ecological impacts due to bioaccumulation of several compounds of concern was evaluated by calculating a Water Quality Criterion Tissue Level (WQCTL). The WQCTL is calculated by multiplying the Clean Water Act Section 304(a)(1) Federal water quality criterion chronic value (CV) for the chemical by the empirically determined bioconcentration factor (BCF) for the chemical for a representative marine organism (Lee, *et al.* 1989). A BCF is the ratio of the concentration of a contaminant in an organism to the concentration of the contaminant in water. Thus, the WQCTL represents the tissue concentration that would be expected in an organism exposed to water containing the chemical at the CV concentration. The CV level is set to protect 95% of all tested organisms included in the water quality criterion database, thus representing a conservative level of protection (EPA 1985b). Table 4 lists the calculated WQCTLs. Sources of CVs and BCFs are from EPA ambient water quality criteria documents (EPA 1980b, 1980c, 1980d, 1980e, 1980f, 1984a, 1984b, 1985a, 1985c, 1986, 1987b and 1992a) and Calabrese (1984; for silver). Calculations are shown in Attachment A.

Several pesticides were evaluated based on the sum of their primary constituents and associated metabolites (e.g., total chlordane, total endosulfan, and total DDT). Alpha(trans)-chlordane, trans

nonachlor, heptachlor and heptachlor epoxide represent the primary components of technical chlordane and its metabolites found in the tissue of aquatic organisms (Jarman, *et. al.* 1996; Verschueren 1983; Sweeney, *et. al.* 1993). These constituents are summed as total chlordane as is consistent with current practice for chlordane (Jarman, *et. al.* 1996) and total DDT. The WQCTL for total chlordane was calculated using the WQC for chlordane as a conservative level of protection. While water quality criteria exist, and WQCTLs can be calculated, for heptachlor (133 ppb) and chlordane (64 ppb), the total chlordane is compared to the WQCTL for chlordane in order to be more environmentally conservative. The chlordane WQCTL provides a conservative level of protection as indicated by published residue effects levels (Sweeney, *et. al.* 1993; Bauman, *et. al.* 1987; Feroz, *et. al.* 1979; Parrish, *et. al.* 1976). Consistent with the above approach, the tissue concentration for endosulfan I, endosulfan II and endosulfan sulfate were also summed as total endosulfan and compared to the WQCTL for total endosulfan.

The WQCTLs were also calculated for all metals of concern which do not have Matrix values. For total chromium, the WQCTL was calculated based on chromium (VI), which is substantially more toxic than chromium (III) and elemental chromium in order to provide a conservative level of environmental protection.

B. Potential for ecological effects based on PAH toxicity (Column 19).

The Critical Body Residue (CBR) approach described by McCarty (1991) was used to derive values for use in evaluating the potential impacts of PAHs accumulated in the dredged material bioaccumulation test organisms. CBRs are concentrations of chemical residues in organisms which elicit a deleterious biological response associated with narcosis, which is the primary non-cancer effect of PAHs. Narcotic responses measured can be acute (e.g., immobilization or death) or chronic endpoints (e.g., reduced reproduction, fecundity or growth). CBRs are represented as the ratio of the mass of toxicant to the mass of the organism, such as millimoles or micrograms of toxicant per kilogram (mmole or ug/kg) of organism. For the narcosis endpoint, each molecule of individual PAH congeners is generally equipotent, thus the total PAH concentration is compared to the CBR. A 400-ppb dose of naphthalene would elicit a similar toxicity response as a 400-ppb dose of fluorene; if both chemicals are present together at these concentrations, then the combined dose would be approximately equal to 800 ppb of either contaminant alone.

McCarty (1991) states that an average critical body residue of 400,000 - 1,200,000 ppb can be used as an estimate for acute effects for a narcosis-producing chemical (e.g., PAHs) on fish populations. (Note: McCarty reports the CBR in units of millimoles per kilogram; this value has been converted to ppb for PAHs using the average molecular weight of the PAHs analyzed in the bioaccumulation test). Chronic effect critical body residues can be estimated by applying an acute to chronic ratio of 10 to the acute CBR (McCarty 1986; Call, *et al.* 1985). Therefore, the chronic critical body residue for PAHs can be estimated at 40,000 - 120,000 ppb of PAHs in organism tissue, and Table 4 thus uses the 40,000-ppb level.

These CBRs were based on fish data. The use of CBRs based on fish toxicity represents a conservative estimate of potential toxicity due to exposure to dredged material because: (1) it is extremely unlikely that a fish would get its whole diet from the HARS, and (2) fish are generally more sensitive than the benthic organisms in direct contact with the dredged material placed at

the HARS (e.g., Landrum, *et al.* (1988) estimated an acute CBR for crustaceans of 800,000 ppb - 42,000,000 ppb).

C. Potential ecological impacts of mutagenic, carcinogenic and teratogenic PAHs (Column 19)

EPA and the USACE reviewed eleven scientific journal articles that reported the results of laboratory experiments that sought to relate the concentration of a PAH contaminant(s) in water, as injected doses, or tissue concentrations, to mutagenic, carcinogenic, teratogenic and/or reproductive effects to fish. These studies all used fish species which are among the most sensitive organisms in the marine environment to exhibit the above effects (EPA 1996c). In addition, most of these studies focused on the PAH most believed to cause such effects for which there is data, benzo(a)pyrene (BaP). One study (Breteler 1984), discussed the possible sources and distribution of PAHs in the Hudson/Raritan estuary, and ranked the threat of PAHs to aquatic biota and humans. The main threat was believed to be carcinogenicity, with a greater threat ranking assigned to humans than biota. However, Breteler (1984) did not provide specific effects-based levels that could be used in the following analysis. Two articles evaluated the effects of crude oil, and thus were not useful for evaluating the effects of specific PAHs measured in the bioaccumulation test (Rice, *et al.* 1987; Lee, *et al.* 1981). Three studies considered the effects of specific PAHs, but did not synoptically measure tissue concentrations in the organisms (Ward, *et al.* 1981; Holcombe, *et al.* 1983; Finger, *et al.* 1985) and were not used, because the lack of tissue data for these studies makes their utility in evaluating the tissue concentration resulting from the dredged material bioaccumulation tests highly uncertain.

The remaining five papers reported measured tissue concentrations and observed reproductive effects in organisms exposed to PAH-spiked water. One article reported the tissue concentrations of adult fish and the observed effect on survival and health of the fish's offspring. Hose, *et al.* (1981) reported that adult English sole injected with benzo(a)pyrene (BaP) accumulated the chemical in the gonad and mature gametes. The amount of BaP taken up by the ovary ranged from 16,800 to 49,700 ppb. Two samples of ripe eggs contained 51,200 and 263,000 ppb of BaP and its metabolites. No adverse effects were reported for these concentrations. Hose, *et al.* (1981) also reported the results of injecting female flathead sole with BaP. Adverse effects to egg hatching success were reported for each female. The paper does not report tissue concentrations in either the parent fish or the egg of the flathead sole. Effects on reproductive success were reported but could only be correlated with the external dose injected into the parent. Therefore, since concentrations and effects were not synoptic in this report, it was not useful in the evaluation of the dredged material bioaccumulation results.

Three papers reported the results of experiments which measured fish egg or alevin concentrations of BaP and associated reproductive or carcinogenic effects (Hose, *et al.* 1982; Hannah, *et al.* 1982; Black, *et al.* 1988). Hose, *et al.* (1982) exposed three species of sole, sand sole, English sole, and flathead sole, to BaP-spiked water. Tissue concentrations of 2,100 ppb were measured in sand sole on day 6 (24 hours after hatching) and were associated with reduced hatching success. However, we did not consider the results to be appropriate for use in setting effects levels because they may have been compromised by the methods of replication used in the experimental design.

Another study considered the carcinogenicity of BaP in rainbow trout resulting from embryo microinjection (Black, *et al.* 1988). A statistically significant number of liver neoplasms was found at a concentration of approximately 200,000 ppb, with non-significant effects at up to one half that concentration. Therefore, using the above across-species UF of 10 and trophic transfer factor of 0.1 results in an aquatic no-effect level of 100,000 ppb.

Hannah, *et al.* (1982) estimated a concentration of BaP in tissue that caused abnormalities in development of rainbow trout eggs, using aqueous exposures and actual measured tissue concentrations in alevin tissues. An exposure to a 2.4 ppb mean aqueous BaP concentration accumulated an average of 12,340 ppb in alevins. This concentration was associated with an increase in percentage of abnormalities from approximately 6% at lower water concentrations (0.08, 0.21, 0.37 and 1.48 ppb) to approximately 13% at higher concentrations. From 0.08 to 1.48 ppb in the water, there were no increasing effects exhibited, therefore, the effects were apparently “real” (i.e., significantly greater than the threshold effect level of 6%) only at the aqueous 2.4 ppb concentration. The Hannah, *et al.* (1982) study is considered the most reliable study for this evaluation since it used exposure series and measured tissue concentrations associated with observed effects, and therefore allows for the calculation of a no-effects level directly from the measured results.

In applying these studies to evaluations of dredged material, consideration must be given to uncertainties in converting these kinds of results to concentrations protective of other biota. Three uncertainties needing to be considered are: (1) those associated with converting effect to no-effect concentrations, (2) across-species uncertainties, and (3) uncertainties in estimating the dose of contaminants to which the organism is exposed. These uncertainties are discussed below.

With respect to uncertainty when converting effect to no-effect concentrations, an uncertainty factor of one order of magnitude is often used when only an effect measure is reported. However, in Hannah, *et al.* (1982), the no-effect level can be estimated to be the next lowest concentration below the lowest-observed effect level, since the range of concentrations below this level did not exhibit significantly different responses. In this case, the no-effect level occurred at the water exposure concentration of 1.48 ppb. Although a tissue concentration was not measured at the 1.48 ppb water concentration, it can be calculated from the concentration measured at the effect level (i.e., the no-effect water concentration (1.48 ppb) is close to 65 percent of the observed effect concentration (2.4 ppb) so the no-effect tissue concentration should be about 65 percent of the lowest-observable effect tissue concentration ($0.65 \times 12,340 \text{ ppb} = 8,021 \text{ ppb}$)). Thus, a factor to adjust these data from lowest observed effect tissue concentration to the calculated no-observed effect tissue concentration is obtained directly from the data.

There can also be uncertainty as to the proximity to the site of toxic action in the organism that a dose or concentration is measured, and with respect to species-to-species variability. Hannah (1982) reported dose concentrations in the tissue and, therefore, there is no need to account for variability associated with the large uncertainties encountered in typical water-only exposure studies where the actual concentration at the site of toxic action is unknown. When measured in the tissues, as was done for this project, concentrations of narcotic chemicals causing effects (i.e., critical body residues, CBRs) in aquatic organisms are reported to range only from 1.4 to 21 umoles/g wet weight (a factor of about one order of magnitude) for organisms as diverse as insects, crustaceans, and fish (McCarty, *et al.* 1992). Therefore, from a tissue concentration

perspective, the species-to-species uncertainty factor appropriate for both total PAHs operating as narcotics and individual PAHs having teratogenic effects would be one order of magnitude, or a value of 10 (EPA 1996d).

In summary, a factor of 10 (representing species to species uncertainty) is an appropriate UF to use in these evaluations. Also, as described Appendix Sections I and II, Brown and Neff (1993) show that trophic transfer of PAHs up the food chain to fish decrease tissue levels by over an order of magnitude. Given this data and the fact that these studies included fish that spent 100% of their time feeding in the test sediment, whereas this would be highly unlikely to occur at an ocean site, a trophic transfer factor of 0.1 is used in this analysis. Applying this UF of 10 and a trophic transfer factor of 0.1 to the no-effects level for BaP calculated from Hannah, *et al.* (1982), as discussed above (8,021 ppb) results in a no-effect level for BaP of approximately 8,000 ppb in benthic tissue.

IV. CONSIDERATION OF POTENTIAL EFFECTS ON HUMAN HEALTH (Columns 14 and 15)

Human effects screening levels were developed with risk-based methods using conservative estimates of exposure to assess whether these contaminants would accumulate to levels in fish and shellfish that could lead to significant adverse effects to humans. The approach assessed consumption of fish and shellfish to derive conservative estimates of contaminant concentrations in benthic tissue protective of human health using the following EPA standard risk-assessment assumptions: 70-kilogram adult eats 6.5 grams of fish and shellfish per day over a 70-year lifetime. This assessment considered potential for both cancer and non-cancer effects in humans. EPA IRIS (EPA 1995) and effects information from EPA's National Toxics Rule (EPA 1992b) were used in the human health assessment to calculate acceptable levels in fish and shellfish to protect human health. Trophic transfer factors, as discussed earlier, were then used to convert these fish and shellfish levels into benthic tissue concentrations.

For regulatory purposes, EPA utilizes 10^{-4} to 10^{-6} (one in ten thousand to one in one million) as an acceptable incremental risk range for activities with potential for causing cancer in human beings (EPA 1980a; EPA 1988; EPA 1987a; Thomas 1987; EPA 1991). EPA considers a cancer risk within this range to be safe and protective of public health. This is supported by the World Health Organization Guidelines for Drinking Water Quality (WHO 1993), where it selected a 10^{-5} guideline value, and then explained that the application could vary by a factor of ten (e.g., 10^{-4} to 10^{-6}). Since this analysis uses conservative methods, the results represent conservative estimates of risk, or what are in effect plausible upper-bound estimates. Thus, the true risk is highly unlikely to be greater than estimated and could be much lower.

Table 4, Column 14 lists human cancer protection levels in benthic organisms for chemicals which are known or suspected carcinogens that would lead to a human cancer risk level of 10^{-4} . For PAHs, this analysis used BaP-equivalents derived from the toxic equivalence factor for each carcinogenic PAH (from EPA (1993); note: these factors are listed in Column 11 of Table 4 for each of the compounds).

The potential for non-cancer impacts can be expressed as a hazard quotient (HQ), which is the ratio of the average daily intake divided by the toxicological reference dose for the chemical. If the HQ is less than unity (e.g., 1), an adverse noncarcinogenic effect is highly unlikely to occur. If the HQ exceeds unity, an adverse health impact may occur. The higher the HQ, the more likely that an adverse noncarcinogenic effect will occur as a result of exposure to the contaminant in the dredged material after placement. Table 4, Column 15 lists noncancer protection levels in benthic organisms for the chemicals that are known to cause, or suspected of causing, non-carcinogenic effects, that would result in a human HQ equal to unity. Those numbers were derived using the conservative assumptions and source materials described in the introductory paragraph to this section.

For the following compounds, the following special considerations were used in evaluating the results in Table 4.

a. Metals

No reference dose has been established for lead. EPA has adopted a blood lead level of 10ug/dl as the level of concern and EPA policies are that regulatory programs should seek to minimize the number of children with blood lead levels above a target of 10 ug/dl (Final Rule for Lead and Copper NPDWR, 56FR26468, June 7, 1991), and this value was used to calculate the effects level in Table 4 (see EPA 1996e).

When interpreting the importance of arsenic tissue concentrations for human health, consideration was given to the arsenic form present (i.e., inorganic vs. organic). Arsenic is found in marine organisms as an organic complex which includes such compounds as arsenobetaine and arsenocholine (Abel and Axiak 1991). Organic arsenic in the tissues of aquatic organisms is not metabolized by predators or humans and is readily eliminated from the body through excretion (Hrudey *et al.* 1995). As a result, the toxicity of organic arsenic ingested from seafood is low and appears to pose no significant hazard (Abernathy and Ohanian 1992). For this reason, cancer and non-cancer protection levels, based on inorganic arsenic as contained in EPA's IRIS database, are not appropriate for evaluating the potential human health impacts of arsenic bioaccumulation from dredged material, and therefore, are not included in Table 4.

b. Pesticides

Alpha(trans)-chlordane, trans nonachlor, heptachlor and heptachlor epoxide represent the primary components of technical chlordane and its metabolites found in the tissue of aquatic organisms (Jarman, *et. al.* 1996; Verschueren 1983; Sweeney, *et. al.* 1993). These constituents are summed as total chlordane as is consistent with current practice for chlordane (Jarman, *et. al.* 1996) and other pesticides (e.g., total DDT). Total chlordane is evaluated using the 10^{-4} cancer risk level and non-cancer level for heptachlor epoxide, which has the greatest potency of the chlordane constituents or metabolites. Similarly, endosulfan I, endosulfan II, and endosulfan sulfate are summed and the total is compared to the conservative non-cancer protection level for endosulfan.

c. PAHs

For PAHs, this analysis used BaP-equivalents derived from the toxic equivalence factor for each carcinogenic PAH (from EPA (1993); note: these factors are listed in column 11 of Table 4 for each of the compounds).

ATTACHMENT A

Tissue concentration is calculated using BCF * Water Quality Criteria (WQC) ambient aqueous concentration, and assuming the conversion factor of 1 kg=1L.

Compound	Ambient Conc. (ug/L) ¹	BCF	Tissue Conc. (ug/Kg)	Remarks
Aldrin	0.13	2,300	299	WQC was reduced by a factor of 10 to account for chronic effects; BCF estimate is based on Dieldrin since Aldrin rapidly transformed to Dieldrin in the
Dieldrin	0.0019	2,300	4.37	BCF is based on 1.1% lipid level for marine fish, Spot (<i>Leiostomus xanthurus</i>).
Total Chlordane	0.004	16,000	64	Total Chlordane includes alpha-chlordane, trans nonachlor, heptachlor, heptachlor epoxide; WQC for Chlordane is used for Total Chlordane; BCF is based
Total Endosulfan	0.0087	328	2.85	Total Endosulfan includes endosulfan I, endosulfan II and endosulfan sulfate; WQC for Endosulfan is used for Total Endosulfan; BCF is based on 3.6% lipid
1,4-Dichlorobenzene	197	60	11820	Ambient conc. is based on lowest observed effect level (LOEL) for saltwater species from WQC, and reduced by a factor of 10 to account for chronic
Arsenic	36	350	12600	Ambient conc. is based on the saltwater criteria continuous conc. for arsenic (III); BCF is based on the Eastern Oyster (<i>Crassostrea virginica</i>).
Chromium	50	236	11800	Ambient conc. is based on Chromium (VI) since it is substantially more toxic than Chromium (III); BCF is based on polychaete worm.
Copper	2.9	3,300	9570	WQC is based on a hardness value of 100; BCF is based on soft shell clam.
Lead	8.5	1,400	11900	Ambient conc. is based on saltwater criteria continuous conc.; BCF is based on the Eastern Oyster (<i>Crassostrea virginica</i>).
Nickel	8.3	458	3802	Ambient conc. is based on saltwater criteria continuous conc.; BCF is based on the Eastern Oyster (<i>Crassostrea virginica</i>).
Silver	0.23	6,500	1495	Water Quality Criterion (WQC) was reduced by a factor of 10 to account for chronic effects; BCF is based on the Blue Mussel (<i>Mytilus edulis</i>).
Zinc	86	17,640	1517040	Ambient conc. is based on saltwater criteria continuous conc.; BCF is based on Eastern Oyster (<i>Crassostrea virginica</i>).

¹.The following documents were used to obtain the water quality criteria values.

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EPA. 1980f. Ambient Water Quality Criteria for Dichlorobenzenes; EPA 440/5-80-039; October 1980.

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EPA. 1984b. Ambient Water Quality Criteria for Copper - 1984; EPA 440/5-84-031; January 1985.

EPA. 1985a. Ambient Water Quality Criteria for Chromium - 1984; EPA 440/5-84-029; January 1985.

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ATTACHMENT B

Benthic Cancer Protection Level Calculations for the Protection of Human Health from the Consumption of Fish Exposed to Dredged Material at the Historic Area Remediation Site

Basis^{fn1}: $10^{-4} \text{ Benthic Tissue Level (ug/kg)} = \frac{[10^{-4} \text{ Conc. in Fish}] \times [\text{Whole Body/fillet Factor (1.35)}]^{fn5}}{\text{Trophic Transfer Factor}^{fn2}}$ $10^{-4} \text{ Conc. in Fish (ug/kg)} = \frac{\text{Toxicological Dose (ug/day)}}{[\text{Seafood Consumption (6.5 g/day)}]^{fn3} \times [10^{-3} \text{ kg/g}]}$ $\text{Toxicological Dose (ug/day)} = \frac{[\text{Risk Level (10}^{-4}\text{)}] \times [\text{Body Weight (70 kg)}]^{fn3} \times [10^3 \text{ ug/mg}]}{\text{Potency Factor, } q_1^* \text{ (kg-day/mg)}^{fn4}}$				
	Cancer Potency Factor q_1^* (kg-day/mg)	Acceptable Concentration in Fish (ug/kg)	Trophic Transfer Factor	Benthic Protection Level (ug/kg)
Pesticides				
Aldrin	17	63	2.6	33
Chlordane	1.3	828	2.3	486
Dieldrin	16	67	1.4	65
Heptachlor	4.5	239	2.7	120
Heptachlor epoxide	9.1	118	1.4	114
Industrial Organics				
1,4- Dichlorobenzene	0.024	44,872	1	60,577
PAHs				
Benzo(a)pyrene	7.3	147	0.1	2,000
METALS				
Arsenic	1.5	718	3	323

ATTACHMENT C

**Benthic Non-Cancer Protection Level Calculations for the Protection
of Human Health from the Consumption of Fish Exposed to Dredged Material
at the Historic Area Remediation Site**

Basis ^{fn1} :				
$\text{Benthic Tissue Level (ug/kg)} = \frac{[\text{Conc. in Fish}] \times [\text{Whole Body/fillet Factor (1.35)}]^{fn5}}{\text{Trophic Transfer Factor}^{fn2}}$ $\text{Conc. in Fish (ug/kg)} = \frac{\text{Toxicological Dose (ug/day)}}{[\text{Seafood Consumption (6.5 g/day)}^{fn3}] \times [10^{-3}\text{kg/g}]}$ $\text{Toxicological Dose (ug/day)} = [\text{Reference dose}^{fn4}] \times [\text{Body Weight (70 kg)}^{fn3}]$				
	Reference Dose RfD (ug/kg-day)	Acceptable Concentration in Seafood (ug/kg)	Trophic Transfer Factor	Benthic Protection Level (ug/kg)
Metals				
Arsenic	0.3	3,231	3	1,454
Chromium	5	54,000	1	73,000
Copper	37.1	400,000	1	540,000
Nickel	20	215,000	1	290,000
Silver	5	54,000	1	73,000
Zinc	300	3,230,769	1	4,361,538
Pesticides				
Aldrin	0.03	323	2.6	167
Chlordane	0.06	592	2.3	350
Dieldrin	0.05	538	1.4	518
Endosulfan	6	64,615	1	87,231
Heptachlor	0.5	5,385	2.7	2,692
Heptachlor epoxide	0.013	140	1.4	135
PAHs				
Acenaphthene	60	650,000	0.1	8,775,000
Fluorene	40	430,000	0.1	5,805,000
Phenanthrene	300	3,230,000	0.1	43,605,000
Anthracene	300	3,230,000	0.1	43,605,000
Fluoranthene	40	430,000	0.1	5,805,000
Pyrene	30	325,000	0.1	4,387,000

NOTES:

^{fn1} Human health cancer and non-cancer assessments adapted from *Guidance for Assessing Chemical Contaminant Data for use in Fish Advisories: Volume II: Risk Assessment and Fish Consumption Limits*. U.S. Environmental Protection Agency, EPA823-B-94-004, Office of Science and Technology, Washington, DC, June 1994.

^{fn2} Trophic transfer factors were calculated by Mr. Lawrence Burkhard, EPA Duluth, using the food chain transfer model developed by Gobas (1993).

^{fn3} Default values were taken from EPA's national toxics rule for setting water quality criteria, EPA (1992b).

^{fn4} Cancer potency factors and non-cancer reference doses are taken from EPA (1995).

^{fn5} The acceptable concentration in seafood is defined on the basis of the fillet or edible portion for humans. Trophic transfer, however, was defined based on whole-body characteristics, including lipid concentrations. Experience in New York State indicates a whole body to fillet ration ranging from 1.2 to 1.5 is applicable to lipophilic substances. The mid-range value of 1.35 is used in this analysis.

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**US ARMY CORPS
OF ENGINEERS**
NEW YORK DISTRICT



REGION 2

September 16, 2025

MEMO FOR THE RECORD

SUBJECT: Review of Compliance with the Requirements of 40 CFR 227.6 and 227.27, and Site Designation Provisions of 40 CFR 228.15 for NY&NJ Channels, Seguine Point Bend, Ward Point Bend to Outerbridge Crossing Federal Navigation Project Reach 3

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Dredging, Sediments, and Oceans Section
Water Division
EPA Region 2

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Kelly Vega, Chief
Dredged Material Management Section
Operations Division, New York District
U.S. Army Corps of Engineers

I. SUMMARY

The USACE is proposing to dredge the NY&NJ Channels, Seguine Point Bend, Ward Point Bend to Outerbridge Crossing Federal Navigation Project (herein referred to as SWO FNP) located in New York with placement of the dredged material as Remediation Material at the Historic Area Remediation Site (HARS) in the Atlantic Ocean.



This memorandum provides comprehensive review and analysis of SWO FNP test results for Reach 3. This memorandum addresses compliance with the regulatory testing criteria of 40 CFR Sections 227.6 and 227.27, and the requirements of the rule establishing the Historic Area Remediation Site (HARS) set out in Section 228.15(d)(6). These requirements hereinafter are referred to as the “Regulations.” Project dredged material meets the non-testing regulatory criteria of Subparts B, C, D, and E of 40 CFR § 227 pertaining to: environmental impacts, need, impact on esthetic, recreational and economic values, and other uses of the ocean (EPA/USACE 2019).

As discussed in the HARS rulemaking preamble (see 62 Fed. Reg. 46142 (August 29, 1997) and 62 Fed. Reg. 26267 (May 13, 1997)) and its accompanying documentation, the need to remediate the Historic Area Remediation Site is amply supported by the presence in the HARS of toxic effects, dioxin bioaccumulation exceeding Category I levels in worm tissue, as well as TCDD/PCB contamination in area lobster stocks. Individual elements of the aforementioned data do not prove that sediments within the Study Area are imminent hazards to the New York Bight Apex ecosystem, living resources, or human health. However, the collective evidence presents cause for concern, justifies a need for remediation, indicates that the site is Impact Category I (see 40 CFR 228.10), and necessitates that the site should be managed to reduce impacts to acceptable levels (see 40 CFR 228.11(c)). Further information on the conditions in the Study Area and the surveys performed may be found in the Supplemental Environmental Impact Statement (EPA, 1997c).

This evaluation confirms that: 1) all tests required under the Regulations were conducted, 2) this project meets the criteria at 40 CFR Section 227.6 for trace contaminants and Section 227.27 for Limiting Permissible Concentration (LPC), and 3) the dredged material is Category I under U.S. Environmental Protection Agency (EPA) Region 2/U.S. Army Corps of Engineers, NY District (CENAN) guidance and is suitable for placement at the HARS as Remediation Material.

II. PROJECT DESCRIPTION

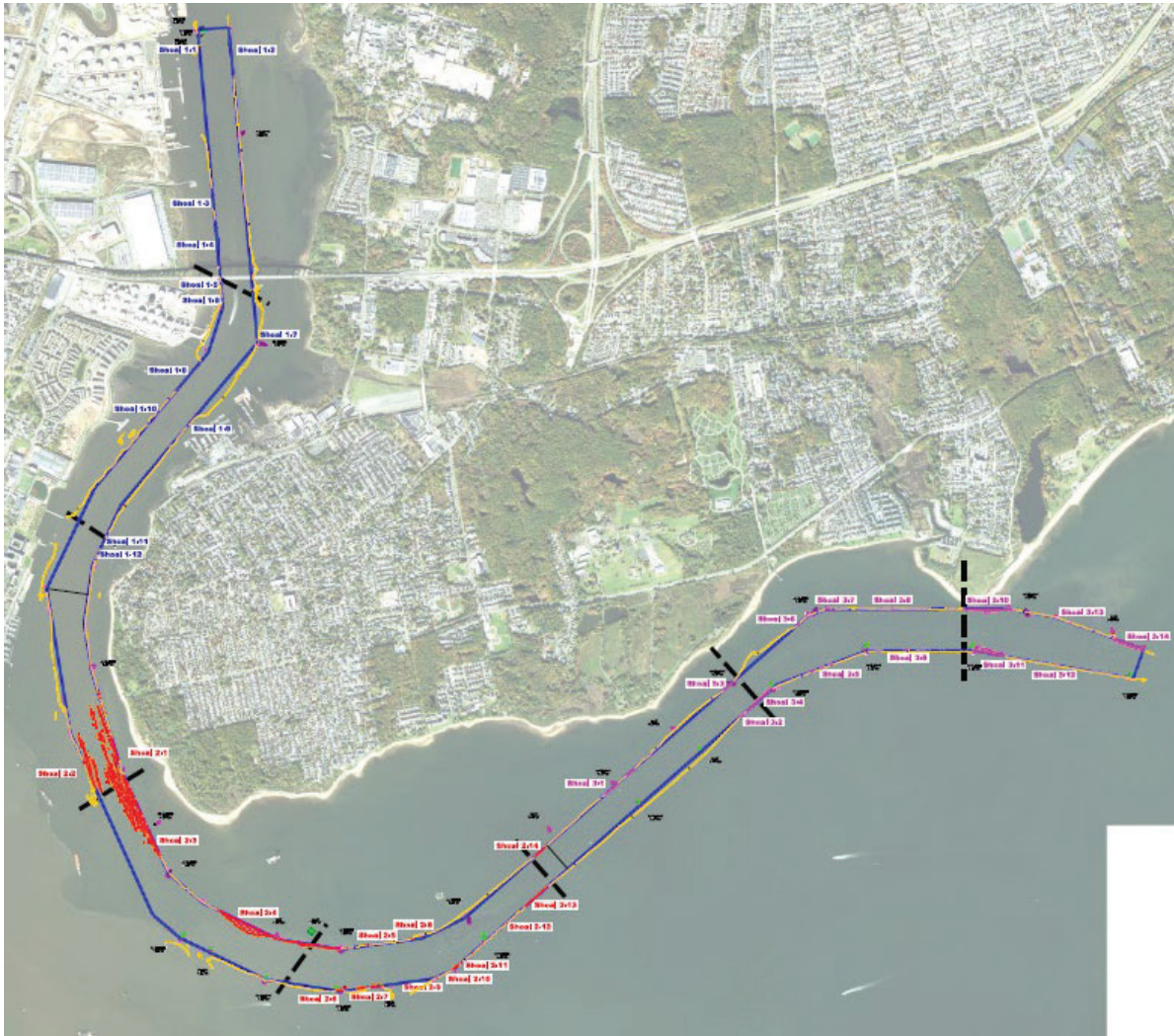
The proposal is to dredge shoals that extend along the western, southern, and eastern sides of Staten Island, NY, from north of Outerbridge Crossing to east of Seguine Point in three reaches that have been composited for testing within the SWO FNP to a depth of -35’ MLLW with an additional 2’ of allowable overdredge with placement of approximately 106,485 cy of (based on an August 8, 2024 bathymetric survey) at the HARS. The project has been sampled and tested as three separate reaches for the purposes of bioassay testing (see sampling plan; CENAN 2024) as described below:

Reach 1, shown below in blue hatching has an estimated volume of 5,585 cy. The project sediments from this reach are represented by 7 core samples composited for the purpose of bioassay testing.

Reach 2, shown below in red hatching has an estimated volume of 87,034 cy. The project sediments from this reach are represented by 9 core samples composited for the purpose of bioassay testing.

Reach 3, shown below in purple hatching has an estimated volume of 13,866 cy. The project sediments from this reach are represented by 5 core samples composited for the purpose of bioassay testing.

This memorandum provides comprehensive review and analysis of Reach 3.



III. REGULATORY REQUIREMENTS

For dredged material to be suitable for use as Remediation Material, it must conform to the Regulations. The Marine Protection, Research, and Sanctuaries Act (MPRSA) or “The Act” prohibits dumping of materials into the ocean except as authorized by EPA or, in the case of dredged materials, by the U.S. Army Corps of Engineers (USACE). Section 102 of the Act directs the EPA to establish and apply criteria for reviewing and evaluating permit applications (33 U.S.C. Section 1412). EPA has adopted such criteria in the Regulations. 40 CFR Section 227.6(a) lists constituents that are prohibited from being placed in the ocean unless only present as trace contaminants in material otherwise suitable for dumping (hereinafter referred to as “listed constituents”). Section 227.27 addresses compliance with the LPC. See also, Section 227.13(c).

Section 227.6(b) states that these listed constituents are considered to be present as trace contaminants only when they are present in such forms and amounts that the “dumping of the materials will not cause significant undesirable effects, including the possibility of danger associated with their bioaccumulation in marine organisms.” The Regulations set forth criteria for determining the potential for significant undesirable effects in Section 227.6(c). To be found environmentally acceptable for ocean placement, the liquid phase must not contain any of the listed constituents in concentrations that would exceed applicable marine water quality criteria after allowance for initial mixing (Section 227.6(c)(1)). For the suspended particulate phase (Section 227.6(c)(2)) and the solid phase (Section 227.6(c)(3)), bioassay results must not indicate occurrence of significant mortality or significant adverse sublethal effects due to the ocean placement of dredged material containing the listed constituents.

Section 227.27 of the Regulations addresses the LPC. For the liquid phase, Section 227.27(a) provides that the LPC is that concentration which does not exceed applicable marine water quality criteria after initial mixing, or when there are no applicable marine water criteria, that concentration of material after initial mixing, would not exceed 0.01 of a concentration shown to be acutely toxic to appropriate sensitive marine organisms in a bioassay carried out in accordance with procedures approved by EPA and USACE. For the suspended particulate phase and the solid phase, Section 227.27(b) provides that the LPC is that concentration of material which will not cause unreasonable acute or chronic toxicity or other sublethal adverse effects based on results of bioassays using appropriate sensitive organisms and conducted according to procedures that have been approved by EPA and USACE, and which will not cause accumulation of toxic materials in the human food chain.

The HARS encompasses an area which includes the Mud Dump Site (MDS), and which has exhibited the potential for adverse ecological impacts and has been identified for remediation. The site will be remediated with uncontaminated dredged material (i.e., dredged material that meets current Category I standards and will not cause significant undesirable effects including through bioaccumulation) (hereinafter referred to as “Remediation Material” or “Material for Remediation”). Under 40 CFR 228.15 (d)(6)(v)(A) the site will be managed to reduce impacts within the Priority Remediation Area (PRA) to acceptable levels in accordance with 40 CFR Section 228.11(c). Use of the site is restricted to dredged material suitable for use as Material for Remediation. This material shall be selected to ensure it will not cause significant undesirable effects including through bioaccumulation or unacceptable toxicity in accordance with 40 CFR 227.6.

Section 228.15(d)(6) of the Regulations describes the locations where material from NY/NJ Harbor and surrounding areas may be placed in the HARS, if it is suitable as Remediation Material. The HARS consists of nine PRAs, a Buffer Zone, and a No Discharge Zone. Under 228.15(d)(6) placement of Remediation Material is limited to the PRAs.

IV. GUIDANCE FOR TESTING AND EVALUATION OF DREDGED MATERIAL

The discussion below describes how the material proposed to be dredged from the SWO FNP Reach 3 was evaluated for placement at the HARS and evaluated for compliance with the requirements of 40 CFR 227.6, 227.27, and 228.15(d)(6). Testing of the material was conducted following procedures approved by EPA and USACE, and contained in the joint EPA/USACE

national guidance “Evaluation of Dredged Material Proposed for Ocean Dumping - Testing Manual” (the “Green Book;” EPA/USACE, 1991), and the regional implementation manual developed by the EPA Region 2 and CENAN (EPA/CENAN, 2016).

These test results were analyzed in accordance with the Regulations to ensure that the proposed placement meets the criteria of Part 227 and the requirements of 228.15(d)(6). As explained in the preamble to the HARS Rule, Remediation Material is uncontaminated dredged material (i.e., dredged material that meets current Category I standards and will not cause significant undesirable effects including through bioaccumulation). The determination of whether materials meeting the Part 227 criteria are assigned to Category I is based upon technical and scientific judgment as set out below.

Applying the EPA Region 2/CENAN guidance to this project, the material would be Category I if it meets the Part 227 criteria (including the requirements regarding acute toxicity) and:

- bioaccumulation test results do not exceed the Regional Matrix Values for cadmium, mercury, total PCBs (clam), and total DDT (USACE 1981), and are below the regional Category I guidance values for dioxin (EPA 1997a), and
- bioaccumulation test results do not exceed the HARS-Specific Polychlorinated Biphenyl (PCB) Worm Tissue Criterion defined in 40 CFR 228.15(d)(6)(v)(E), and
- bioaccumulation test results for the other bioaccumulative chemicals of concern identified in EPA/CENAN (2016) do not indicate a potential for undesirable effects using conservative assessment techniques.

Sediments that meet this Category I definition are suitable for placement at the HARS as Remediation Material as they will improve sediment conditions at the HARS to reduce impacts to acceptable levels in accordance with 40 CFR Section 228.11(c). Sediments that do not meet this definition are not suitable for placement at the HARS.

V. RESULTS OF EVALUATION OF THE MATERIAL

A. Evaluation of the liquid phase

The liquid phase of the material was evaluated for compliance with Sections 227.6(c)(1) and 227.27(a). There are applicable marine water quality criteria for constituents in the material, including listed constituents, and the applicable marine water quality criteria would not be exceeded after initial mixing. In addition, liquid phase bioassays run as part of the suspended particulate phase on three appropriate sensitive marine organisms, show that after initial mixing (as determined under 40 CFR 227.29(a)(2)), the liquid phase of the material would not exceed a toxicity threshold of 0.01 of a concentration shown to be acutely toxic to appropriate sensitive marine organisms. Accordingly, it is concluded that the liquid phase of the material complies with 40 CFR 227.6(c)(1) and 227.27(a). The specific test results and technical analysis of the data underlying this conclusion are described and evaluated in CENAN memo dated 2025.

B. Evaluation of the suspended particulate phase

The suspended particulate phase of the material was evaluated for compliance with Sections 227.6(c)(2) and 227.27(b). Bioassay testing of the suspended particulate phase of the material has been conducted using three appropriately sensitive marine organisms: inland silversides (*Menidia beryllina*), mysid shrimp (*Americamysis bahia*, formerly *Mysidopsis bahia*), and the planktonic larvae of the Mediterranean mussel (*Mytilus galloprovincialis*). That information shows that when placed in the HARS and after initial mixing (as determined under 40 CFR 227.29(a)(2)), the suspended particulate phase of this material would not exceed a toxicity threshold of 0.01 of a concentration shown to be acutely toxic in the laboratory bioassays (LC/EC₅₀) and thus would not result in significant mortality. The material may be discharged on either incoming or outgoing tides. However, based on a dump volume of 3500 cubic yards, discharge may occur only at distances of at least 1200 feet from the northern, southern, eastern, and western boundaries of the HARS, independent of the direction of tidal transport at the time of disposal. Based on a dump volume of 6000 cubic yards, discharge may occur only at distances of at least 1350 feet from the northern, southern, eastern, and western boundaries of the HARS, independent of the direction of tidal transport at the time of disposal. The specific test results and technical analysis of the data underlying this conclusion are described in CENAN 2025.

The factor of 0.01 was applied to ensure that there would be no significant adverse sublethal effects. Moreover, the fact that after placement, the suspended particulate phase would only exist in the environment for a short time, means the suspended particulate phase would not cause significant undesirable effects, including the possibility of danger associated with bioaccumulation, since these impacts require longer exposure durations (see EPA, 1994). Accordingly, it is concluded that the suspended phase of the material complies with 40 CFR 227.6(c)(2) and 227.27(b).

C. Evaluation of the solid phase

The solid phase of the material was evaluated for compliance with Sections 227.6(c)(3) and 227.27(b). This evaluation was made using the results of two specific types of evaluations on the solid phase of the material, one focusing on the acute (ten-day) toxicity of the material, and the other focusing on the potential for contaminant bioaccumulation in test organisms exposed to the dredged material as compared to those same endpoints in test organisms exposed to a reference sediment. Both types of tests were conducted following procedures approved by EPA and the USACE (EPA/USACE 1991).

Reference sediment serves as a point of comparison to identify potential effects of contaminants in the dredged material (EPA/USACE, 1991). Exposing test organisms to this sediment allows for the prediction of toxicity and bioaccumulated contaminant levels that would be expected to result in the test organisms were they “in the wild” at the area from which the reference sediment was taken. The reference sediment used for this project was collected at the Reference Site, in an area of clean, sandy sediments located in the New York Bight near the HARS, where the sediments are unaffected by prior dredged material disposal.

To make a statistically valid determination that exposure to the project sediment does or does not cause greater toxicity or bioaccumulation than the reference sediment, laboratory exposures to several sub-samples (i.e., replicates) of the dredged material and reference sediment are run concurrently. Means and standard deviations are calculated for each set of exposures and

compared using standard statistical approaches to determine, with 95% confidence, whether exposure to project sediment results in significantly higher toxicity or bioaccumulation than exposure to reference sediment. Throughout this memorandum, statements regarding project sediment having “greater” or “less” toxicity or bioaccumulation are referring to calculated differences which are statistically significant at the 95% confidence level.

A statistically significant difference between test and reference bioaccumulation is not itself a quantitative prediction that an impact would occur in the field. A key to understanding bioaccumulation and potential adverse impacts is the fact that bioaccumulation does not necessarily result in an effect. At low exposure (concentration and duration), bioaccumulation may cause no harm. On the other hand, as exposure and/or bioaccumulation increases, the potential for adverse effects increases.

The following sections address the results of solid phase tests and further analyze compliance with the regulatory criteria of Sections 227.6(c)(3), 227.27(b), and 228.15 and with EPA Region 2/CENAN guidance.

Solid phase toxicity evaluation

Ten-day acute toxicity tests were conducted on project materials using mysids (*Americamysis bahia*) and amphipods (*Ampelisca abdita*), which are appropriate sensitive benthic marine organisms. These organisms are good predictors of adverse effects to benthic marine communities (see EPA 1996a). The results of these tests show that the solid phase of the material would not cause significant mortality (statistically different project sediment-exposed mortality exceeding mortality in the reference sediment exposures by 10% for mysid shrimp or 20% for amphipods) and meets the solid phase toxicity criteria of Sections 227.6 and 227.27.

Solid phase bioaccumulation evaluation

28-day tests were conducted on the solid phase of the project material to assess the potential for bioaccumulation of contaminants of concern identified in the project sampling plan (CENAN 2018) using two appropriate sensitive benthic marine organisms, sand worm (*Alitta virens*, formerly *Nereis virens*) and bent-nosed clam (*Macoma nasuta*). These species are good representatives of the phylogenetically diverse base of the marine food chain. Contaminants of concern were identified from Squibb, *et al.*, (1991), a report prepared for the Toxics Work Group of the NY/NJ Harbor Estuary Program to identify and characterize contaminants in NY/NJ Harbor. Compounds from that list, including organic compounds with the potential to bioaccumulate (K_{ow} of approximately 4 or greater), are included on the bioaccumulation testing list when expected to be present in project sediments based upon contaminant inputs and previous sediment sampling results.

There are FDA Action Levels for seven compounds (aldrin, dieldrin, a-chlordane, heptachlor, heptachlor epoxide, PCBs, and mercury; see Table 4, Column 18). Bioaccumulation of any of these contaminants to levels exceeding an FDA Action Level (see EPA/USACE 1991) results in a conclusion that ocean disposal of the dredged material would cause significant adverse effects and exceed the LPC as per 40 CFR 227.27. Therefore, dredged material causing bioaccumulation of any of these contaminants to levels exceeding FDA Action Levels is not suitable for use as Remediation Material at the HARS. None of the contaminants for which there are FDA Action Levels exceed such thresholds in the tissues of organisms exposed to sediments from this project.

When bioaccumulation of a specific contaminant or class of contaminants in organisms exposed to project sediments is not significantly greater than bioaccumulation in organisms exposed to reference sediments, that contaminant is concluded to be present in forms and amounts that do not result in bioaccumulation that would cause unreasonable degradation of the environment or human health, or significant adverse effects. These contaminants are not considered further in the evaluation unless they contribute to a mixture that is evaluated against a single combined guideline value or criterion (i.e., total PCBs, PAH mixtures, total DDT, total chlordane and total endosulfans). In cases where bioaccumulation levels are statistically greater (at the 95% confidence limit) in project sediment-exposed organism than in reference sediment-exposed organisms, further evaluation for potential effects is warranted to evaluate any risk that might be associated with accumulated contaminants.

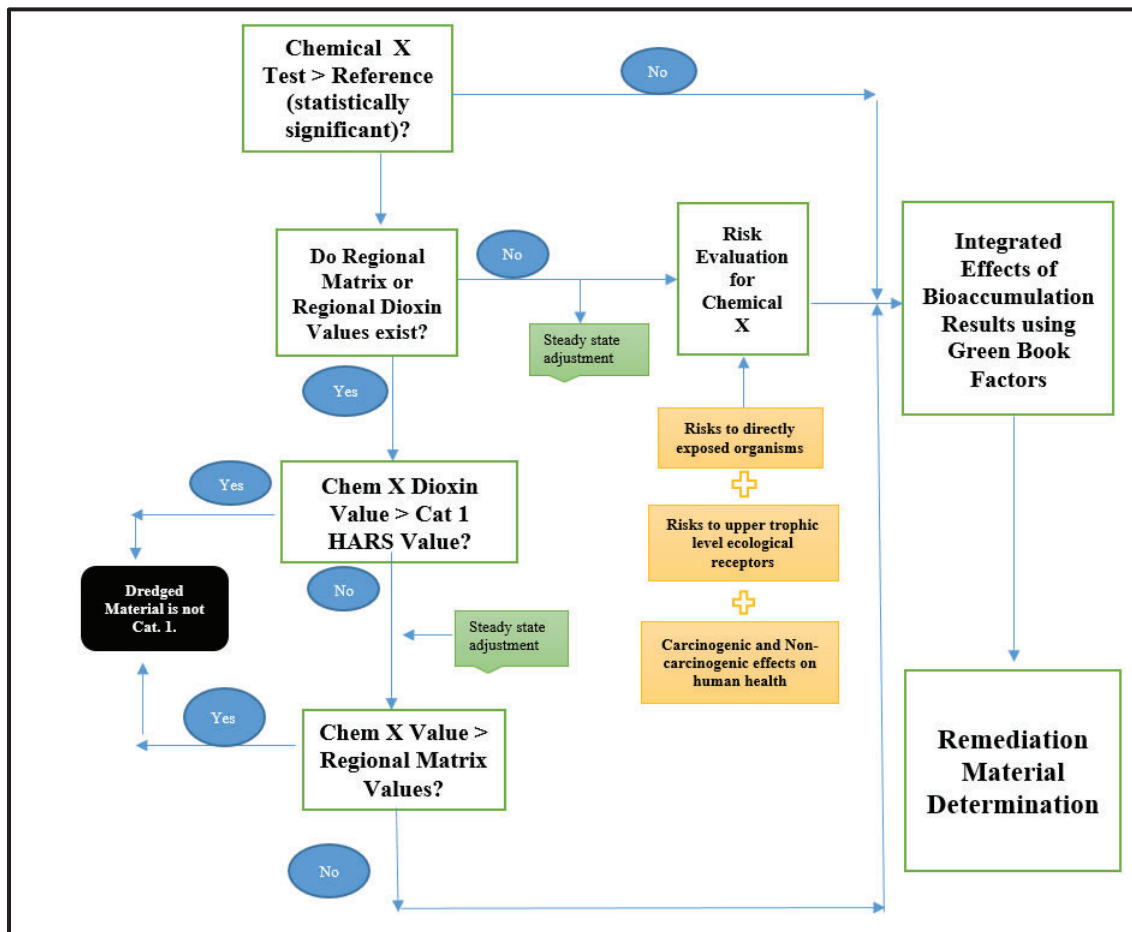
EPA Region 2 and CENAN use a framework to evaluate project sediment bioaccumulation results that follows national guidance described in the Green Book (EPA/USACE 1991). This guidance outlines a series of evaluations to consider risks associated with project sediment bioaccumulation results that exceed reference. The framework used for HARS-suitability determinations is shown in Figure 1 and includes the following evaluations:

- *Comparison of the 28-day bioaccumulation results to Regional Dioxin Values.* Dioxins and furans that bioaccumulate to greater than reference are evaluated by comparing accumulated concentrations to Regional Dioxin Values (EPA 1997a) (listed in Table 4, Column 20 of Dioxin sheet). 28-day bioaccumulation test results that equal or exceed the Regional Dioxin Values for 2,3,7,8-TCDD or total toxic equivalency (TEQ) of combined congeners indicate that the sediment is not Category I dredged material and is not suitable for use as Remediation Material at the HARS.
- *Comparison of the 28-day bioaccumulation results to Regional Matrix Values.* PCBs, DDTs, cadmium and mercury that bioaccumulate to concentrations greater than reference are evaluated by comparing 'steady state'-adjusted concentrations to Regional Matrix Values (USACE 1981) and the HARS-Specific PCB Worm Tissue Criterion (40 CFR 228.15(d)(6)(v)(E)) (listed in Table 4, Column 20; steady state adjustment is discussed in Section 1 of Appendix A). Bioaccumulation test results that exceed the Regional Matrix Values or HARS-Specific PCB Worm Tissue Criterion indicate that the sediment is not Category I dredged material and is not suitable for use as Remediation Material at the HARS.
- Risk-based assessments are conducted to evaluate concentrations of compounds that bioaccumulate above reference for which Regional Dioxin Values or Regional Matrix Values do not exist. These evaluations are made using 'steady state'-adjusted concentrations and include comparisons to risk values protective of benthic organisms and upper level trophic receptors, including humans.
- The final evaluation uses all available information and results of the individual chemical evaluations to evaluate the solid phase of the dredged material as a whole.

The remainder of this memorandum documents the assessment conducted using the described framework to evaluate risks associated with contaminants that bioaccumulated to greater than

reference levels in tissues of organisms exposed to SWO FNP Reach 3 sediment. Bioaccumulation test results for this project are summarized in Table 4.

Figure 1. EPA Region 2/CENAN Framework for Evaluating Bioaccumulation Test Results for SWO FNP



Solid Phase Bioaccumulation Evaluation-Comparison to Reference Sediment Bioaccumulation

Contaminants for which bioaccumulation from the dredged material in tissue was statistically greater than the reference in the clam and/or worm are indicated by an asterisk in the applicable column of Table 1.

Table 1. Statistically Significant Bioaccumulation of Contaminants in Organisms Exposed to Project Sediments vs. Reference Sediments

CONTAMINANT	CLAMS	WORMS
PAHs		
Acenaphthene	*	*
Acenaphthylene	*	*
Anthracene	*	*
Fluorene	*	*
Naphthalene	*	
Phenanthrene	*	*
Benzo(a)anthracene**	*	*
Benzo(a)pyrene**	*	*
Benzo(g,h,i)perylene	*	*
Benzo(b)fluoranthene**	*	*
Benzo(k)fluoranthene**	*	*
Chrysene**	*	*
Dibenzo(a,h)anthracene**	*	
Fluoranthene	*	*
Indeno(1,2,3-cd)pyrene**	*	*
Pyrene	*	*
PESTICIDES		
Dieldrin	*	*
a-Chlordane	*	*
Trans nonachlor		*
Endosulfan I	*	
Endosulfan II	*	*
Endosulfan sulfate	*	*
4,4-DDD	*	*
2,4-DDD	*	*
4,4-DDE	*	*
2,4-DDE	*	
TOTAL PCBs	*	*
1,4-Dichlorobenzene	*	
METALS		
Arsenic	*	
Chromium (total)	*	
Copper		*
Lead	*	*
Nickel	*	
Silver	*	
Zinc	*	
DIOXINS/FURANS		
OCDD	*	*
2,3,7,8-TCDF		*
2,3,4,7,8-PeCDF		*

Solid Phase Bioaccumulation Evaluation - Comparison to Regional Dioxin Values

Bioaccumulation test results were below all Regional Dioxin Values (EPA 1997a) for both test organisms. Therefore, no further evaluation of these contaminants is required in accordance with EPA Region 2/CENAN guidance.

Solid Phase Bioaccumulation Evaluation- Comparison to Regional Matrix Values and the HARS-Specific PCB Worm Tissue Criterion

Bioaccumulation test results (adjusted for steady state, as described in Section 1 of Appendix A) were below all Regional Matrix Values (USACE 1981) for both test organisms and the mean 'steady state'-adjusted worm PCB tissue concentration was below the HARS-Specific PCB Worm Tissue Criterion (40 CFR 228.15(d)(6)(v)(E)). Therefore, no further evaluation of these contaminants is required in accordance with EPA Region 2/CENAN guidance.

Solid Phase Bioaccumulation Evaluation- Risk-based Evaluations

Risk associated with compounds that bioaccumulated to greater than reference in test organisms exposed to project sediment for which Regional Matrix Values (USACE 1981) or Regional Dioxin Values (EPA 1997a) do not exist are evaluated using risk-based assessments, as discussed below. Contaminants that fall into this group for this project include:

PAHs, dieldrin, a-chlordane, trans nonachlor, endosulfan I, endosulfan II, endosulfan sulfate, 1,4-dichlorobenzene, arsenic, total chromium, lead, nickel, silver and zinc.

Contaminants bioaccumulated from dredged material can present risks to organisms that are directly exposed to the sediment and to species at upper trophic levels of the marine food web (e.g., predatory fish). Human exposure can potentially result via consumption of fish that has been dietarily exposed to HARS contaminants. Risk to directly exposed ecological receptors, risk to dietarily exposed ecological receptors, and human risk are examined separately in the EPA Region 2/CENAN framework.

a. Risks to Directly Exposed Organisms

Metals, Pesticides, and Industrial Chemicals

Water Quality Criterion Tissue Levels (WQCTL) are established for aldrin, dieldrin, total chlordane, total endosulfan, 1,4-dichlorobenzene, arsenic chromium, copper, lead, nickel, silver and zinc (see Appendix, Section 2, A). For this project, none of the WQCTLs were exceeded.

PAHs

To assess direct exposure risks of PAHs, a direct comparison is made of total measured PAH tissue residue to the Critical Body Residue (CBR) associated with narcosis, which is the primary non-cancer effect of PAHs. The CBR approach, described by McCarty (1991), was used to evaluate the potential impacts of total PAHs accumulated in the dredged material bioaccumulation test organisms and is discussed in the Appendix. As shown in Table 4, total PAH levels in tissues from the dredged material bioaccumulation test were below levels at which chronic adverse effects might be expected from a narcotic mode of action in sensitive aquatic organisms (i.e., fish) as estimated by the CBR.

Therefore, these bioaccumulation test results do not indicate a potential for undesirable effects to directly exposed organisms resulting from use of this material as Remediation Material at the HARS.

b. Risks to Upper Trophic Level Ecological Receptors

To properly assess risks to upper trophic level species (including humans) the efficiency with which a contaminant is transferred through the diet and the propensity of those compounds to accumulate to higher concentrations in the tissues of the consumer must be considered. The potential movement of contaminants through the food chain was considered and appropriate trophic transfer factors applied to adjust the data accordingly to reflect the propensity of specific contaminants or contaminant groups to accumulate to higher (or lower) concentrations in upper trophic level organisms (including humans). Data and/or methods used to develop these trophic transfer adjustment factors are discussed in the Appendix.

Effects of Mutagenic, Carcinogenic and Teratogenic PAHs

Applying an uncertainty factor (UF) of 10 to account for interspecies differences in sensitivities and a trophic transfer factor of 0.1 to the no-effects level for BaP calculated from Hannah, *et al.* (1982), as discussed in the Appendix results in a no-effect level for BaP of approximately 8,000 for BaP only ppb in benthic tissue, which is considerably greater than the highest steady state-adjusted tissue concentration of BaP measured in organisms exposed to project sediment (10 ppb for BaP only and 14.4 ppb for total BaP equivalents). Project tissue concentrations would still be below this no-effect level if the higher trophic transfer factor (0.23) reported by McElroy, *et al.* (1991) was used.

Hall and Oris (1991) reported on experiments that exposed fathead minnows to anthracene during long-term exposures and observed adverse effects on reproduction. The paper reported that a concentration of anthracene in the tissue of the egg in the range of 3,750 to 8,000 ppb resulted in no significant effects on egg hatching or survivorship. Using the same approach for accounting for species-to-species uncertainty and food chain transfer yields a conservative benthic tissue level of 3,750 ppb. Anthracene tissue concentrations from the project bioaccumulation tests are well below this level.

As stated above, PAHs accumulated to greater than reference in test organisms exposed to project sediments. However, the most relevant aquatic effects information reviewed indicates that the highest tissue PAH levels accumulated in the dredged material bioaccumulation tests are below mutagenic, carcinogenic, and teratogenic no-effect levels.

c. Consideration of Potential Carcinogenic and Non-carcinogenic Effects on Human Health

Benthic tissue screening levels for human carcinogenic risk and non-carcinogenic hazard were developed for those contaminants that do not have Regional Matrix Values (USACE 1981) or Regional Dioxin Values (EPA 1997a). The approach used to derive these screening levels (described in the Appendix) uses standard EPA risk-assessment approaches and assumptions (fish and shellfish consumption rates; whole body to fillet contaminant ratios) to back calculate a fish tissue concentration that corresponds to specific levels of cancer risk or non-cancer hazard. The fish tissue concentration is then converted to a corresponding benthic prey concentration using the trophic transfer factors discussed in the Appendix.

Because the analysis implicitly incorporates various conservative assumptions regarding exposure (e.g., 100% of consumed seafood is fish taken from the HARS; fish has fed exclusively at HARS long enough prior to capture to be at equilibrium with benthic prey tissue contaminant concentrations), the resulting benthic tissue concentrations represent conservative estimates of

risk, or what are in effect plausible upper-bound estimates. Thus, the true risk is highly unlikely to be greater than estimated and could be much lower.

Table 4, Column 14 lists conservative human cancer protection levels in benthic organisms for the chemicals which are known or suspected carcinogens that correspond to a human cancer risk level of 1×10^{-4} . None of the human health cancer protection levels were exceeded in the bioaccumulation test results.

The potential for non-cancer impacts can be expressed as a hazard quotient (HQ), which is the ratio of the average daily intake divided by the toxicological reference dose for the chemical. Table 4, Column 15 includes the noncancer protection levels in benthic organisms for the chemicals requiring further analysis that are known to cause, or suspected of causing, non-carcinogenic effects, that would result in a human HQ equal to unity (i.e., 1). If the HQ is less than unity, an adverse noncarcinogenic effect is highly unlikely to occur. The higher the HQ, the more likely that an adverse noncarcinogenic effect will occur as a result of exposure to the contaminant in the dredged material after placement.

The concentrations of all contaminants in test organisms' tissues were below their respective non-cancer protection level (HQs for all contaminants were less than unity).

D. Evaluation of Bioaccumulation Results for Dredged Material as a Whole

As a final step in the evaluative process, it is appropriate to go beyond assessing the individual test results to evaluate the bioaccumulation test results as a whole to provide an opportunity for an integrated assessment of the individual test results (see Figure 1, Box d). EPA/USACE (1991) provides additional factors to be considered in making an integrated assessment of dredged material test results. The recommended factors are:

Magnitude of toxicity and number and phylogenetic diversity of species exhibiting greater mortality in the dredged material than in the reference material

As discussed above, project dredged material would not cause significant mortality (statistically different project sediment-exposed mortality exceeding mortality in the reference sediment exposures by 10% for mysid shrimp or 20% for amphipods) and meets the solid phase toxicity criteria of Sections 227.6 and 227.27.

Number and magnitude of contaminants for which bioaccumulation from the dredged material is statistically greater than bioaccumulation from the reference material and number and phylogenetic diversity of species in which bioaccumulation from the dredged material is statistically greater than bioaccumulation from the reference material

The magnitudes by which compounds bioaccumulated to levels greater than reference are shown in Table 2.

Table 2. Magnitude of Bioaccumulation of Contaminants in Organisms Exposed to Project Sediments vs. Reference Sediments

CONTAMINANT	CLAMS	WORMS
PAHs		
Acenaphthene	2.8	2.1
Acenaphthylene	3.7	2.1
Anthracene	3.0	4.6
Fluorene	2.0	1.7
Naphthalene	1.7	
Phenanthrene	2.0	1.7
Benzo(a)anthracene**	5.3	11.6
Benzo(a)pyrene**	6.5	4.9
Benzo(g,h,i)perylene	8.1	5.8
Benzo(b)fluoranthene**	6.2	5.9
Benzo(k)fluoranthene**	5.4	8.1
Chrysene**	3.6	7.2
Dibenzo(a,h)anthracene**	7.7	
Fluoranthene	2.8	5.0
Indeno(1,2,3-cd)pyrene**	9.8	2.9
Pyrene	5.1	8.1
PESTICIDES		
Dieldrin	3.4	2.5
α-Chlordane	2.3	2.4
Trans nonachlor		1.1
Endosulfan I	4.4	
Endosulfan II	3.9	1.5
Endosulfan sulfate	3.2	1.9
4,4-DDD	3.1	3.5
2,4-DDD	3.2	2.3
4,4-DDE	6.6	6.5
2,4-DDE	5.3	
TOTAL PCBs	5.7	2.2
1,4-Dichlorobenzene	2.3	
METALS		
Arsenic	1.2	
Chromium (total)	2.1	
Copper		1.1
Lead	3.0	6.0
Nickel	1.3	
Silver	1.4	
Zinc	1.1	
DIOXINS/FURANS		
OCDD	6.9	1.6
2,3,7,8-TCDF		2.1
2,3,4,7,8-PeCDF		1.7

All contaminants accumulated to less than ten times reference in *M. nasuta*, and all contaminants except for one PAH accumulated to less than four times reference in *A. virens*.

Exceeding reference concentrations is common when reference concentrations are very low or “non-detect.” Because the HARS reference sediment is greater than 90% sand and has only trace levels of contamination, reference-exposed organisms tend to exhibit very low or undetectable levels of bioaccumulation.

Toxicological importance of the contaminants whose bioaccumulation from the dredged material exceeds that from the reference material, including their propensity to biomagnify within aquatic food webs

Although some of the contaminants that bioaccumulated to levels greater than reference in organisms exposed to the project sediment can be toxicologically important, they did not accumulate to toxicologically important concentrations. Levels of these contaminants were all below the acceptable human health risk range and acceptable aquatic effects range using conservative approaches and analyses as described above to evaluate those test results, including through their potential transfer through the food web.

Magnitude by which contaminants whose bioaccumulation from the dredged material exceeds that from the reference material also exceed the concentrations found in comparable species living in the vicinity of the proposed site.

Organisms collected from a broad area of the sea floor in the vicinity of (but not inside) the HARS have been collected and analyzed for tissue concentration for bioaccumulative contaminants of concern (Charles and Muramoto 1990; USACE 1994; EPA 1996f; EPA 1997b). These data provide a measure of the tissue residues for organisms living outside the HARS. Table 4, Columns 16 and 17 summarize the most recent background data.

The magnitude by which compounds that bioaccumulated to levels greater than background levels (where background values exist) are reported in Table 3. All other compounds accumulated in test organisms to steady state levels below background.

Table 3. Magnitude of Bioaccumulation above Background in Organisms Exposed to Project Sediments

CONTAMINANT	CLAMS	WORMS
PAHs		
PESTICIDES		
Dieldrin		3.5
METALS		
Zinc	1.4	
DIOXINS		

All contaminants except for one metal accumulated to less than two times background in *M. nasuta*, and all contaminants except for one pesticide accumulated to less than four times background in *A. virens*.

When bioaccumulation in organisms exposed to project sediments is not greater than tissue concentrations in organisms from the vicinity of the remediation site (the background levels), this indicates that placement of the material would not result in bioaccumulation above existing ambient levels in the general area and thus does not have a potential to cause undesirable effects.

However, bioaccumulation in organisms exposed to project sediments greater than these levels is not necessarily predictive of adverse effects. It may, for example, reflect extremely low background levels.

Although some of the contaminants that bioaccumulated to levels greater than background in organisms exposed to the project sediment can be toxicologically important, they did not accumulate to toxicologically important concentrations. Levels of these contaminants were all below the acceptable human health risk range and acceptable aquatic effects range using conservative approaches and analyses as described above to evaluate those test results, including through their potential transfer through the food web.

VI. OVERALL CONCLUSION ON THE PROPOSED PROJECT

The evaluation of the testing results performed above indicates that the material does not have a potential to cause undesirable effects to aquatic marine biota due to chronic adverse effects including such effects due to mutagenic, carcinogenic, or teratogenic contaminants, or to human health due to cancer or non-cancer effects from the individual contaminants. Evaluation of the solid phase bioaccumulation test results for the dredged material as a whole would not indicate a different outcome than that shown by the individual test results themselves; i.e., that the material does not have the potential to cause undesirable effects due to bioaccumulation.

Based upon this review of the results of testing of the sediments proposed for dredging and ocean placement from SWO FNP Reach 3, the material meets the criteria for acceptability for ocean placement as described in Sections 227.6, and 227.27 of the Regulations, is Category I under EPA Region 2/CENAN guidance and is suitable for placement at the HARS under Section 228.15(d)(6) as Remediation Material.

Table 4. Bioaccumulation and Comparison Data for SWO FNP Reach 3

		PROJECT DATA										COMPARISON DATA									
Col. 1	Col. 2 [10]	Col. 3 [10]	Col. 4 [10]	Col. 5 [10]	Col. 6 [10]	Col. 7	Col. 8	Col. 9	Col. 10	Col. 11	Col. 12	Col. 13	Col. 14	Col. 15	Col. 16	Col. 17	Col. 18	Col. 19	Col. 20		
Sample I.D.	Reference	Reference	Test Sed.	Test Sed.	Test Sed.	[1]	Conv.Fac. clam/worm	Test Sed. SS	Test Sed. SS	Carcinogenic TEF [3]	Test Sed. BaP Tox. Equiv. Conc.(clam)[3] (ug/Kg)	Test Sed. BaP Tox. Equiv. Conc.(worm)[3] (ug/Kg)	Human Health Cancer (10E-4) Level[4] (ug/Kg)	Human Health Non-Cancer Level (HQ=1) (ug/Kg)[9]	Background (clam) (ug/Kg)	Background (worm) (ug/Kg)	FDA Limits [6] (ug/Kg)	Ecological Non-Specific Effects Level (ug/Kg)	Regional Matrix (ug/Kg)		
Compound	(clam) (ug/Kg)	(worm) (ug/Kg)	(clam) (ug/Kg)	(clam) (ug/Kg)	(worm) (ug/Kg)		[2]	(clam) (ug/Kg)	(worm) (ug/Kg)												
PAHs																					
Acenaphthene	0.144	0.115	0.41	*	0.244	*	1/1	0.41	0.244					8,775,000	8.1	0.5	0.5		[13]		
Acenaphthylene	0.084	0.078	0.312	*	0.160	*	1/1	0.312	0.160					[5]	4.6	1.3	1.3		[13]		
Anthracene	0.266	0.029	0.80	*	0.134	*	1/1	0.80	0.134					43,605,000	10.0	1.6	1.6		3,750		
Fluorene	0.266	0.066	0.561	*	0.113	*	1/1	0.561	0.113					5,805,000	7.4	0.3	0.3		[13]		
Naphthalene	0.549	0.61	0.94	*	0.729	*	1/1	0.94	0.729					[5]	26.4	4.5	4.5		[13]		
Phenanthrene	1.83	0.226	3.74	*	0.386	*	1/1	3.74	0.386					43,605,000	32.7	4.7	4.7		[13]		
Benzo(a)anthracene**	0.767	0.030	4.05	*	0.348	*	2/2	8.10	0.696	0.1	0.81	0.070		[5]	25.1	4.8	4.8		[13]		
Benzo(b)pyrene**	0.773	0.094	4.99	*	0.465	*	2/2	9.98	0.930	1	10.0	0.930		[5]	24.2	7.6	7.6		8,000 [7]		
Benzo(g,h,i)perylene	0.344	0.057	2.80	*	0.331	*	3/3	8.40	0.993					[5]	22.2	7.6	7.6		[13]		
Benzo(k)fluoranthene**	0.760	0.063	4.74	*	0.372	*	2/2	9.48	0.744	0.1	0.95	0.074		[5]	40.6	16.5	16.5		[13]		
Benzo(k)fluoranthene**	0.987	0.072	5.37	*	0.585	*	2/2	10.70	1.170	0.01	0.107	0.012		[5]	81.6	5.6	5.6		[13]		
Chrysene**	1.85	0.237	6.57	*	1.71	*	2/2	13.1	3.42	0.001	0.0131	0.0034		[5]	29.4	6.7	6.7		[13]		
Dibenzo(a,h)anthracene**	0.059	0.047U	0.457	*	0.05	*	2/2	0.914	0.106	1	0.91	0.11		[5]	8.5	1.1	1.1		[13]		
Fluoranthene	3.81	0.439	10.7	*	2.20	*	1/1	10.7	2.20			1		5,805,000	43.8	10.6	10.6		[13]		
Indeno(1,2,3-cd)pyrene**	0.223	0.053U	2.18	*	0.154	*	3/3	6.54	0.462	0.1	0.654	0.046		[5]	16.6	4.4	4.4		[13]		
Pyrene	2.77	0.375	14.2	*	3.03	*	1/1	14.2	3.03					4,387,000	51.3	26.6	26.6		[13]		
TOTAL PAHs																					
								99	15.5		14.4	1.2	2,000 (BaP eqv.)			104.60			40,000 [8]		
PESTICIDES																					
Aldrin	0.014U	0.014U	0.014U		0.014U		2/2	0.014	0.014					33	167	0.9	0.1	300	299 [9]		
Dieldrin	0.026	0.069	0.089	*	0.175	*	2/2	0.178	0.350					65	518	0.1	0.1	300	4.37 [9]		
a-Chlordane	0.017	0.034	0.039	*	0.083	*	2/2	0.078	0.166							0.7	0.7	300			
Trans nonachlor	0.028	0.131	0.022		0.145	*	2/2	0.044	0.290							0.5	0.5				
Heptachlor	0.011U	0.011U	0.071		0.175		2/2	0.142	0.350							0.05	0.05	300			
Heptachlor epoxide	0.077U	0.025	0.017U		0.017U		1/1	0.009	0.009							0.2	0.2	300			
Total Residual Chlordane/Heptachlor																					
								0.273	0.82				114	135		1.7		64# [9]			
Endosulfan I																					
Endosulfan I	0.015U	0.015U	0.066	*	0.011		1/1	0.066	0.011					[5]		0.2	0.2				
Endosulfan II	0.022	0.051	0.085	*	0.077	*	1/1	0.085	0.077					[5]		0.1	0.1				
Endosulfan sulfate	0.012	0.040	0.038	*	0.075	*	1/1	0.038	0.075					[5]		0.1	0.1				
Total Endosulfans																					
								0.189	0.163						87,000	0.4	0.4		2.86# [9]		
4,4-DDT																					
4,4-DDT	0.062	0.015	0.011		0.012U		11/11	0.121	0.066							0.6	1.0				
2,4-DDT	0.017U	0.017U	0.017U		0.017U		2/2	0.017	0.017								0.4				
4,4-DDD	0.065	0.067	0.201	*	0.235	*	3/3	0.60	0.71							3.9	3.9				
2,4-DDD	0.040	0.072	0.128	*	0.169	*	2/2	0.256	0.338							1.4	1.4				
4,4-DDE	0.126	0.047	0.832	*	0.306	*	2/2	1.66	0.612							3.5	4.3				
2,4-DDE	0.009U	0.009U	0.048	*	0.007		2/2	0.096	0.014							0.1	0.1				
Total DDT																					
								2.75	1.75							11.1			40		
TOTAL PCBs																					
1,4-Dichlorobenzene	2.78	9.8	15.8	*	21.2	*	1/2	15.8	42.4						106.6	88.1	2,000		100 (clam) 113 (worm)		
	0.101	0.109	0.229	*	0.096		1/1	0.229	0.096				60,000	[5]					11,820 [9]		

FOOTNOTES FOR TABLE 4:

**: Carcinogenic PAHs.

#: Levels represent the conservative level of protection for the sum of the related compounds and their metabolites.

na: Not Available

1. An “*” in this column indicates that the analyte concentration in the test sediment-exposed organisms are statistically greater than in those exposed to reference sediment. Means and statistical comparisons were determined using 1/2 the detection limit to estimate concentrations of analytes that were below the detection limit.
2. Conversion factors from 28-day bioaccumulation results to steady state (see discussion in Appendix)
3. Toxic equivalencies (TEFs) for the carcinogenic PAHs are from EPA (1993); Dioxin TEFs are from: EPA (1989).
4. This value represents the 10⁻⁴ cancer risk level for the carcinogenic PAHs. The total concentration of carcinogenic PAHs is expressed in BaP equivalents (see discussion in the text of the memo).
5. Cancer risk factor or reference dose are not assigned by EPA in IRIS (EPA 1995).
6. FDA limits are from EPA/USACE (1991).
7. This value represents the benthic level expected to result in a no-effect level for possible mutagenic and teratogenic effects in fish from exposure to BaP, which is the most toxic PAH.
8. This value represents the non-specific narcosis effects level (see discussion in Appendix). This value is compared to the sum of all PAHs measured.
9. Calculations are included in Appendix.
10. Means of five tissue replicates calculated using 1/2 detection limits to estimate concentrations of analytes that were not detected; “U” indicates that all five replicates were not detected.
11. Levels are based on the Regional Dioxin Values (EPA 1997a).
12. Level is the sum of TEQs of all dioxin congeners other than 2,3,7,8-TCDD.
13. For this PAH, the no-effect level for possible mutagenic and teratogenic effects in fish is estimated from exposure to BaP, which is the most toxic PAH.
14. Cancer and non-cancer protection levels, based on inorganic arsenic as contained in EPA’s IRIS database, are not appropriate for evaluating the potential human health impacts of arsenic bioaccumulation from dredged material, and therefore, are not included in Table 4 (see discussion in Appendix).

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Appendix

I. CORRECTION OF 28-DAY BIOACCUMULATION RESULTS TO ACCOUNT FOR STEADY STATE (Column 8-10)

Bioaccumulation tests were conducted using 28-day exposure of appropriate sensitive benthic marine organisms to sediment. Following placement of the material at the HARS, exposures to any contaminants would continue beyond 28 days, therefore, assessment of risks associated with accumulations must consider the extent to which the 28-day measured tissue concentration approximate the 'steady state' concentration of each contaminant that could occur in organisms at the HARS after extended exposure. Therefore, comparisons to guidance values or criteria for given contaminants that reflect 'steady state' conditions (as opposed to comparisons to other 28-day tissue concentrations such as comparison to reference) may require that a correction factor be applied to the 28-day concentration to account for incomplete attainment of steady state accumulation.

Steady state may be defined operationally as the lack of any significant difference (ANOVA, $\alpha = 0.05$) among tissue residues taken at three consecutive sampling intervals (Lee, *et al.* 1989). The 28-day test exposure period was selected as appropriate because most chemicals of concern will reach at least 80% of steady state in benthic marine organisms within that time frame (Boese and Lee 1992). For the few chemicals that may not meet steady state tissue concentrations in 28 days, a factor is used to adjust the data to steady state.

Steady state considerations for specific contaminants (or contaminant classes) and the derivation of correction factors (listed in Column 8) to account for incomplete uptake are discussed in the remainder of this section. Steady state-adjusted concentrations for clams and worms are presented in Columns 9 and 10, respectively.

A. Metals

In general, metals bioaccumulate more rapidly than organics and 28-day tests are sufficient to evaluate potential effects (see EPA/USACE 1991), for example, arsenic (Naqvi, *et al.* 1990; Riedel, *et al.* 1987; Oladimeji, *et al.* 1984).

Cadmium and mercury are not regulated in marine organisms as are essential metals, and, thus, no adjustment for steady state is applicable. Bioaccumulation of these metals is affected by many complex factors, and is essentially linear (Dethlefsen 1978; Giesy, *et al.* 1980; V-Balogh and Salanka 1984). Therefore, there are no adjustments that can be made to reproduce steady state. The Regional Matrix Values for cadmium and mercury, therefore, do not represent steady state conditions and so 28-day test results are used to compare to the Regional Matrix Values.

B. Organic Contaminants

a. Pesticides and 1,4-Dichlorobenzene

Uptake of non-polar organic contaminants from food is highly dependent on their hydrophobicity, a property measured by the octanol/water partition coefficient, K_{ow} . The higher the value of K_{ow} , the longer it takes to reach steady state in benthic marine organisms. Organochlorine compounds that have $\log K_{ow} > 6$ (i.e., aldrin, dieldrin, heptachlor, trans nonachlor, and alpha-chlordane), may not attain steady-state within 28 days of exposure to sediment.

Information contained in Boese and Lee (1992) indicates that aldrin, dieldrin, and α -chlordane achieve at least 50% of steady-state in 28-day bioaccumulation tests. Heptachlor and trans nonachlor, were not addressed in Boese and Lee (1992). However, the fraction of steady state achieved for these compounds after 28-days was estimated using the equations contained in McFarland (1995). Results of such calculations indicate heptachlor and trans nonachlor reach approximately 50% of steady state after 28 days. Calculating the fraction of steady state achieved for the remaining pesticides with $\log K_{ow} < 6$ (heptachlor epoxide) and for 1,4-dibenzochlorene ($\log K_{ow} = 3.44$) indicates that these compounds reach steady state within 28 days, so no adjustment is necessary.

The degree to which DDT and its metabolites reach steady state in test organisms after 28 days was evaluated using data reported by Lee *et al.* (1994). To assess the rate of bioaccumulation, they exposed the clam *Macoma nasuta*, to sediments collected from the vicinity of the United Heckathorn Superfund site in Richmond California to study tissue residues and uptake kinetics associated with exposure to pesticide-contaminated sediments. Results of the study indicate that one parent compound, 4,4-DDT, bioaccumulates much more slowly than 2,4-DDT and the DDT metabolites. The results range from approximately 9 percent of steady state after 28 days for 4,4-DDT, to 55 percent of steady state after 28 days for 2,4-DDT (Lee, et al. 1994). Based on these results, 28-day test tissue concentrations of DDT compounds and their metabolites are adjusted by multiplying the 28-day result by a factor of 11 for 4,4-DDT, a factor of three for 4,4-DDD, and a factor of two for 2,4-DDT and the remaining DDT metabolites before they are compared to Regional Matrix Values (USACE 1981).

b. PCBs

The degree to which PCBs reach steady state in test organisms after 28 days was evaluated using data reported by Lee, *et al.* (1994). To assess the rate of bioaccumulation of PCBs and other compounds, Rubinstein, *et al.* (1990) and Pruell, *et al.* (1993) exposed three species of organisms, the grass shrimp *Palaemonetes pugio*; the sandworm *Nereis virens*, and the clam *Macoma nasuta*, to sediments collected from the Passaic River, N.J. Sub-samples of the exposed organisms were removed on various days into the study including days 0, 10, 28, 42, 84, and 180.

For the clam tissue, the variance in the concentrations on day 28 and day 84 (by which point the maximum concentration had been reached) overlap, thus indicating that the two are not statistically different and the bioaccumulation on day 28 is at or very close to steady state. Thus, the clam bioaccumulation for the project sediments using 28-day exposures is acceptable for use as steady state tissue levels.

For the worm tissue, variances for days 28 and 180 do not overlap, thus indicating that steady state was probably not reached in 28 days. Variance in the data makes it difficult to quantify a real difference; means for days 28 and 180 from Rubinstein *et al.* (1990) (approximately 1,750 ng/g (nanograms per gram or parts per billion, ppb) for 28 days, and 3,000 ng/g for 180 days) suggest approximately 58% of steady state would have been reached in 28 days. On this basis, worm project data are conservatively adjusted by a factor of two to calculate a steady state tissue concentration for comparison to the HARS-Specific PCB Worm Tissue Criterion (40 CFR 228.15(d)(6)(v)(E)).

c. PAHs

The degree to which PAHs reach steady state in test organisms after 28 days was evaluated using data reported by McFarland (1995) and Meador, *et al.* (1995). The time required for a given PAH to attain a steady-state concentration following exposure to bedded sediments (t_{ss}) is determined primarily by the log K_{ow} of the compound in question (McFarland 1995; Meador, *et al.* 1995). Meador, *et al.* (1995) reviewed nine studies that investigated the attainment of steady-state tissue concentrations of PAHs by various marine invertebrates. In each case, tissue concentrations approached steady state within several days to two weeks after initiating exposure to both low molecular weight PAHs and high molecular weight PAHs. McFarland (1995) estimated t_{ss} values ranging from 3.5 to 326 days for 15 PAHs based on their hydrophobicities. The estimated steady-state concentration of the sum total of the 15 PAHs analyzed by McFarland for sediments collected from typical harbor areas revealed that the mean concentration attained after 28-day bioaccumulation tests was approximately 86% of steady-state. McFarland (1995) concluded that 28-day tests are likely to reflect steady-state. Despite this conclusion, steady state total PAH concentrations were estimated by summing individual PAH concentrations that were conservatively adjusted by a factor of 1, 2, or 3 based on their hydrophobicities and McFarland (1995).

II. CONSIDERATION OF FOOD-CHAIN TRANSFER

To properly assess risks to upper trophic level species (including humans) the efficiency with which a contaminant is transferred through the diet and the propensity of those compounds to accumulate to higher concentrations in the tissues of the consumer must be considered. The potential movement of contaminants through the food chain was considered and appropriate trophic transfer factors applied to adjust the data accordingly to reflect the propensity of specific contaminants or contaminant groups to accumulate to higher (or lower) concentrations in upper trophic level organisms (including humans). Data and/or methods used to develop these trophic transfer adjustment factors are discussed in the remainder of this section.

Metals

Trophic transfer of most metals is not sufficient to qualify as biomagnification (Brown and Neff 1993). The lack of observed biomagnification for such metals as arsenic, cadmium, lead, mercury, and zinc is the result of incomplete absorption of metals across the gut, rapid excretion, and dilution in muscle, which represents a large part of the total body weight of most marine animals (Fowler 1982; Suedel *et al.* 1994). For purposes of conducting the human health and ecological evaluations below, a conservative trophic transfer coefficient equal to one will be used for these non-biomagnifying metals (Suedel *et al.* 1994 and references cited therein).

Pesticides and Industrial Chemicals

The potential for these chemicals to biomagnify was also evaluated. Although organic contaminants with values of $\log K_{ow} > 4$ tend to biomagnify in the marine food chain, studies (USACE 1995) have shown that this is not true for higher molecular weight compounds such as the most highly chlorinated PCBs or for easily metabolized compounds such as PAHs. Those organic compounds which are not efficiently excreted, such as certain pesticides, can biomagnify in the food chain. For the organic constituents with a potential to biomagnify in the marine food chain, trophic transfer factors were calculated, using the approach described by Gobas (1993). The values are summarized in Attachment B.

PAHs

With regard to the potential for biomagnification of PAHs, feeding studies show that assimilation rates from ingested food are extremely low, e.g., more than 98% of the target contaminant remained in an undigested form in fish gut 48 hours after feeding squid containing radio-labeled benzo[a]pyrene to young cod (Corner, *et al.* 1976) and juvenile Atlantic herring (Whittle, *et al.* 1977). PAH metabolites are also transferred through the marine food chain; however, they are absorbed even less efficiently than their parent compounds (McElroy and Sisson 1989; McElroy, *et al.* 1991). Up to 99% of the PAH compounds taken up by fish are metabolized and excreted into bile, the usual elimination mode, within 24 hours of uptake (Varanasi, *et al.* 1989). Similar results are described in Brown and Neff (1993) who evaluated various studies describing trophic transfer. The studies cited in Brown and Neff (1993) indicate a trophic transfer rate for BaP from invertebrates to fish of between 0.02 and 0.23 times the concentration in the ingested invertebrates (Corner, *et al.* 1976, O'Connor, *et al.*, 1988, McElroy, *et al.* 1991). This was considered when assessing the ecological and human health effects of the project material as discussed below.

III. CONSIDERATION OF ECOLOGICAL EFFECTS

A. Potential for ecological effects based on Water Quality Criteria (Column 19)

The potential for ecological impacts due to bioaccumulation of several compounds of concern was evaluated by calculating a Water Quality Criterion Tissue Level (WQCTL). The WQCTL is calculated by multiplying the Clean Water Act Section 304(a)(1) Federal water quality criterion chronic value (CV) for the chemical by the empirically determined bioconcentration factor (BCF) for the chemical for a representative marine organism (Lee, *et al.* 1989). A BCF is the ratio of the concentration of a contaminant in an organism to the concentration of the contaminant in water. Thus, the WQCTL represents the tissue concentration that would be expected in an organism exposed to water containing the chemical at the CV concentration. The CV level is set to protect 95% of all tested organisms included in the water quality criterion database, thus representing a conservative level of protection (EPA 1985b). Table 4 lists the calculated WQCTLs. Sources of CVs and BCFs are from EPA ambient water quality criteria documents (EPA 1980b, 1980c, 1980d, 1980e, 1980f, 1984a, 1984b, 1985a, 1985c, 1986, 1987b and 1992a) and Calabrese (1984; for silver). Calculations are shown in Attachment A.

Several pesticides were evaluated based on the sum of their primary constituents and associated metabolites (e.g., total chlordane, total endosulfan, and total DDT). Alpha(trans)-chlordane, trans

nonachlor, heptachlor and heptachlor epoxide represent the primary components of technical chlordane and its metabolites found in the tissue of aquatic organisms (Jarman, *et. al.* 1996; Verschueren 1983; Sweeney, *et. al.* 1993). These constituents are summed as total chlordane as is consistent with current practice for chlordane (Jarman, *et. al.* 1996) and total DDT. The WQCTL for total chlordane was calculated using the WQC for chlordane as a conservative level of protection. While water quality criteria exist, and WQCTLs can be calculated, for heptachlor (133 ppb) and chlordane (64 ppb), the total chlordane is compared to the WQCTL for chlordane in order to be more environmentally conservative. The chlordane WQCTL provides a conservative level of protection as indicated by published residue effects levels (Sweeney, *et. al.* 1993; Bauman, *et. al.* 1987; Feroz, *et. al.* 1979; Parrish, *et. al.* 1976). Consistent with the above approach, the tissue concentration for endosulfan I, endosulfan II and endosulfan sulfate were also summed as total endosulfan and compared to the WQCTL for total endosulfan.

The WQCTLs were also calculated for all metals of concern which do not have Matrix values. For total chromium, the WQCTL was calculated based on chromium (VI), which is substantially more toxic than chromium (III) and elemental chromium in order to provide a conservative level of environmental protection.

B. Potential for ecological effects based on PAH toxicity (Column 19).

The Critical Body Residue (CBR) approach described by McCarty (1991) was used to derive values for use in evaluating the potential impacts of PAHs accumulated in the dredged material bioaccumulation test organisms. CBRs are concentrations of chemical residues in organisms which elicit a deleterious biological response associated with narcosis, which is the primary non-cancer effect of PAHs. Narcotic responses measured can be acute (e.g., immobilization or death) or chronic endpoints (e.g., reduced reproduction, fecundity or growth). CBRs are represented as the ratio of the mass of toxicant to the mass of the organism, such as millimoles or micrograms of toxicant per kilogram (mmole or ug/kg) of organism. For the narcosis endpoint, each molecule of individual PAH congeners is generally equipotent, thus the total PAH concentration is compared to the CBR. A 400-ppb dose of naphthalene would elicit a similar toxicity response as a 400-ppb dose of fluorene; if both chemicals are present together at these concentrations, then the combined dose would be approximately equal to 800 ppb of either contaminant alone.

McCarty (1991) states that an average critical body residue of 400,000 - 1,200,000 ppb can be used as an estimate for acute effects for a narcosis-producing chemical (e.g., PAHs) on fish populations. (Note: McCarty reports the CBR in units of millimoles per kilogram; this value has been converted to ppb for PAHs using the average molecular weight of the PAHs analyzed in the bioaccumulation test). Chronic effect critical body residues can be estimated by applying an acute to chronic ratio of 10 to the acute CBR (McCarty 1986; Call, *et al.* 1985). Therefore, the chronic critical body residue for PAHs can be estimated at 40,000 - 120,000 ppb of PAHs in organism tissue, and Table 4 thus uses the 40,000-ppb level.

These CBRs were based on fish data. The use of CBRs based on fish toxicity represents a conservative estimate of potential toxicity due to exposure to dredged material because: (1) it is extremely unlikely that a fish would get its whole diet from the HARS, and (2) fish are generally more sensitive than the benthic organisms in direct contact with the dredged material placed at

the HARS (e.g., Landrum, *et al.* (1988) estimated an acute CBR for crustaceans of 800,000 ppb - 42,000,000 ppb).

C. Potential ecological impacts of mutagenic, carcinogenic and teratogenic PAHs (Column 19)

EPA and the USACE reviewed eleven scientific journal articles that reported the results of laboratory experiments that sought to relate the concentration of a PAH contaminant(s) in water, as injected doses, or tissue concentrations, to mutagenic, carcinogenic, teratogenic and/or reproductive effects to fish. These studies all used fish species which are among the most sensitive organisms in the marine environment to exhibit the above effects (EPA 1996c). In addition, most of these studies focused on the PAH most believed to cause such effects for which there is data, benzo(a)pyrene (BaP). One study (Breteler 1984) discussed the possible sources and distribution of PAHs in the Hudson/Raritan estuary, and ranked the threat of PAHs to aquatic biota and humans. The main threat was believed to be carcinogenicity, with a greater threat ranking assigned to humans than biota. However, Breteler (1984) did not provide specific effects-based levels that could be used in the following analysis. Two articles evaluated the effects of crude oil, and thus were not useful for evaluating the effects of specific PAHs measured in the bioaccumulation test (Rice, *et al.* 1987; Lee, *et al.* 1981). Three studies considered the effects of specific PAHs, but did not synoptically measure tissue concentrations in the organisms (Ward, *et al.* 1981; Holcombe, *et al.* 1983; Finger, *et al.* 1985) and were not used, because the lack of tissue data for these studies makes their utility in evaluating the tissue concentration resulting from the dredged material bioaccumulation tests highly uncertain.

The remaining five papers reported measured tissue concentrations and observed reproductive effects in organisms exposed to PAH-spiked water. One article reported the tissue concentrations of adult fish and the observed effect on survival and health of the fish's offspring. Hose, *et al.* (1981) reported that adult English sole injected with benzo(a)pyrene (BaP) accumulated the chemical in the gonad and mature gametes. The amount of BaP taken up by the ovary ranged from 16,800 to 49,700 ppb. Two samples of ripe eggs contained 51,200 and 263,000 ppb of BaP and its metabolites. No adverse effects were reported for these concentrations. Hose, *et al.* (1981) also reported the results of injecting female flathead sole with BaP. Adverse effects to egg hatching success were reported for each female. The paper does not report tissue concentrations in either the parent fish or the egg of the flathead sole. Effects on reproductive success were reported but could only be correlated with the external dose injected into the parent. Therefore, since concentrations and effects were not synoptic in this report, it was not useful in the evaluation of the dredged material bioaccumulation results.

Three papers reported the results of experiments which measured fish egg or alevin concentrations of BaP and associated reproductive or carcinogenic effects (Hose, *et al.* 1982; Hannah, *et al.* 1982; Black, *et al.* 1988). Hose, *et al.* (1982) exposed three species of sole, sand sole, English sole, and flathead sole, to BaP-spiked water. Tissue concentrations of 2,100 ppb were measured in sand sole on day 6 (24 hours after hatching) and were associated with reduced hatching success. However, we did not consider the results to be appropriate for use in setting effects levels because they may have been compromised by the methods of replication used in the experimental design.

Another study considered the carcinogenicity of BaP in rainbow trout resulting from embryo microinjection (Black, *et al.* 1988). A statistically significant number of liver neoplasms was found at a concentration of approximately 200,000 ppb, with non-significant effects at up to one half that concentration. Therefore, using the above across-species UF of 10 and trophic transfer factor of 0.1 results in an aquatic no-effect level of 100,000 ppb.

Hannah, *et al.* (1982) estimated a concentration of BaP in tissue that caused abnormalities in development of rainbow trout eggs, using aqueous exposures and actual measured tissue concentrations in alevin tissues. An exposure to a 2.4 ppb mean aqueous BaP concentration accumulated an average of 12,340 ppb in alevins. This concentration was associated with an increase in percentage of abnormalities from approximately 6% at lower water concentrations (0.08, 0.21, 0.37 and 1.48 ppb) to approximately 13% at higher concentrations. From 0.08 to 1.48 ppb in the water, there were no increasing effects exhibited; therefore, the effects were apparently “real” (i.e., significantly greater than the threshold effect level of 6%) only at the aqueous 2.4 ppb concentration. The Hannah, *et al.* (1982) study is considered the most reliable study for this evaluation since it used exposure series and measured tissue concentrations associated with observed effects, and therefore allows for the calculation of a no-effects level directly from the measured results.

In applying these studies to evaluations of dredged material, consideration must be given to uncertainties in converting these kinds of results to concentrations protective of other biota. Three uncertainties needing to be considered are: (1) those associated with converting effect to no-effect concentrations, (2) across-species uncertainties, and (3) uncertainties in estimating the dose of contaminants to which the organism is exposed. These uncertainties are discussed below.

With respect to uncertainty when converting effect to no-effect concentrations, an uncertainty factor of one order of magnitude is often used when only an effect measure is reported. However, in Hannah, *et al.* (1982), the no-effect level can be estimated to be the next lowest concentration below the lowest-observed effect level, since the range of concentrations below this level did not exhibit significantly different responses. In this case, the no-effect level occurred at the water exposure concentration of 1.48 ppb. Although a tissue concentration was not measured at the 1.48 ppb water concentration, it can be calculated from the concentration measured at the effect level (i.e., the no-effect water concentration (1.48 ppb) is close to 65 percent of the observed effect concentration (2.4 ppb) so the no-effect tissue concentration should be about 65 percent of the lowest-observable effect tissue concentration ($0.65 \times 12,340 \text{ ppb} = 8,021 \text{ ppb}$)). Thus, a factor to adjust these data from lowest observed effect tissue concentration to the calculated no-observed effect tissue concentration is obtained directly from the data.

There can also be uncertainty as to the proximity to the site of toxic action in the organism that a dose or concentration is measured, and with respect to species-to-species variability. Hannah (1982) reported dose concentrations in the tissue and, therefore, there is no need to account for variability associated with the large uncertainties encountered in typical water-only exposure studies where the actual concentration at the site of toxic action is unknown. When measured in the tissues, as was done for this project, concentrations of narcotic chemicals causing effects (i.e., critical body residues, CBRs) in aquatic organisms are reported to range only from 1.4 to 21 umoles/g wet weight (a factor of about one order of magnitude) for organisms as diverse as insects, crustaceans, and fish (McCarty, *et al.* 1992). Therefore, from a tissue concentration

perspective, the species-to-species uncertainty factor appropriate for both total PAHs operating as narcotics and individual PAHs having teratogenic effects would be one order of magnitude, or a value of 10 (EPA 1996d).

In summary, a factor of 10 (representing species to species uncertainty) is an appropriate UF to use in these evaluations. Also, as described Appendix Sections I and II, Brown and Neff (1993) show that trophic transfer of PAHs up the food chain to fish decrease tissue levels by over an order of magnitude. Given this data and the fact that these studies included fish that spent 100% of their time feeding in the test sediment, whereas this would be highly unlikely to occur at an ocean site, a trophic transfer factor of 0.1 is used in this analysis. Applying this UF of 10 and a trophic transfer factor of 0.1 to the no-effects level for BaP calculated from Hannah, *et al.* (1982), as discussed above (8,021 ppb) results in a no-effect level for BaP of approximately 8,000 ppb in benthic tissue.

IV. CONSIDERATION OF POTENTIAL EFFECTS ON HUMAN HEALTH (Columns 14 and 15)

Human effects screening levels were developed with risk-based methods using conservative estimates of exposure to assess whether these contaminants would accumulate to levels in fish and shellfish that could lead to significant adverse effects to humans. The approach assessed consumption of fish and shellfish to derive conservative estimates of contaminant concentrations in benthic tissue protective of human health using the following EPA standard risk-assessment assumptions: 70-kilogram adult eats 6.5 grams of fish and shellfish per day over a 70-year lifetime. This assessment considered potential for both cancer and non-cancer effects in humans. EPA IRIS (EPA 1995) and effects information from EPA's National Toxics Rule (EPA 1992b) were used in the human health assessment to calculate acceptable levels in fish and shellfish to protect human health. Trophic transfer factors, as discussed earlier, were then used to convert these fish and shellfish levels into benthic tissue concentrations.

For regulatory purposes, EPA utilizes 10^{-4} to 10^{-6} (one in ten thousand to one in one million) as an acceptable incremental risk range for activities with potential for causing cancer in human beings (EPA 1980a; EPA 1988; EPA 1987a; Thomas 1987; EPA 1991). EPA considers a cancer risk within this range to be safe and protective of public health. This is supported by the World Health Organization Guidelines for Drinking Water Quality (WHO 1993), where it selected a 10^{-5} guideline value, and then explained that the application could vary by a factor of ten (e.g., 10^{-4} to 10^{-6}). Since this analysis uses conservative methods, the results represent conservative estimates of risk, or what are in effect plausible upper-bound estimates. Thus, the true risk is highly unlikely to be greater than estimated and could be much lower.

Table 4, Column 14 lists human cancer protection levels in benthic organisms for chemicals which are known or suspected carcinogens that would lead to a human cancer risk level of 10^{-4} . For PAHs, this analysis used BaP-equivalents derived from the toxic equivalence factor for each carcinogenic PAH (from EPA (1993); note: these factors are listed in Column 11 of Table 4 for each of the compounds).

The potential for non-cancer impacts can be expressed as a hazard quotient (HQ), which is the ratio of the average daily intake divided by the toxicological reference dose for the chemical. If the HQ is less than unity (e.g., 1), an adverse noncarcinogenic effect is highly unlikely to occur. If the HQ exceeds unity, an adverse health impact may occur. The higher the HQ, the more likely that an adverse noncarcinogenic effect will occur as a result of exposure to the contaminant in the dredged material after placement. Table 4, Column 15 lists noncancer protection levels in benthic organisms for the chemicals that are known to cause, or suspected of causing, non-carcinogenic effects, that would result in a human HQ equal to unity. Those numbers were derived using the conservative assumptions and source materials described in the introductory paragraph to this section.

For the following compounds, the following special considerations were used in evaluating the results in Table 4.

a. Metals

No reference dose has been established for lead. EPA has adopted a blood lead level of 10ug/dl as the level of concern and EPA policies are that regulatory programs should seek to minimize the number of children with blood lead levels above a target of 10 ug/dl (Final Rule for Lead and Copper NPDWR, 56FR26468, June 7, 1991), and this value was used to calculate the effects level in Table 4 (see EPA 1996e).

When interpreting the importance of arsenic tissue concentrations for human health, consideration was given to the arsenic form present (i.e., inorganic vs. organic). Arsenic is found in marine organisms as an organic complex which includes such compounds as arsenobetaine and arsenocholine (Abel and Axiak 1991). Organic arsenic in the tissues of aquatic organisms is not metabolized by predators or humans and is readily eliminated from the body through excretion (Hrudey *et al.* 1995). As a result, the toxicity of organic arsenic ingested from seafood is low and appears to pose no significant hazard (Abernathy and Ohanian 1992). For this reason, cancer and non-cancer protection levels, based on inorganic arsenic as contained in EPA's IRIS database, are not appropriate for evaluating the potential human health impacts of arsenic bioaccumulation from dredged material, and therefore, are not included in Table 4.

b. Pesticides

Alpha(trans)-chlordane, trans nonachlor, heptachlor and heptachlor epoxide represent the primary components of technical chlordane and its metabolites found in the tissue of aquatic organisms (Jarman, *et. al.* 1996; Verschueren 1983; Sweeney, *et. al.* 1993). These constituents are summed as total chlordane as is consistent with current practice for chlordane (Jarman, *et. al.* 1996) and other pesticides (e.g., total DDT). Total chlordane is evaluated using the 10^{-4} cancer risk level and non-cancer level for heptachlor epoxide, which has the greatest potency of the chlordane constituents or metabolites. Similarly, endosulfan I, endosulfan II, and endosulfan sulfate are summed and the total is compared to the conservative non-cancer protection level for endosulfan.

c. PAHs

For PAHs, this analysis used BaP-equivalents derived from the toxic equivalence factor for each carcinogenic PAH (from EPA (1993); note: these factors are listed in column 11 of Table 4 for each of the compounds).

ATTACHMENT A

Tissue concentration is calculated using BCF * Water Quality Criteria (WQC) ambient aqueous concentration, and assuming the conversion factor of 1 kg=1L.

Compound	Ambient Conc. (ug/L) ¹	BCF	Tissue Conc. (ug/Kg)	Remarks
Aldrin	0.13	2,300	299	WQC was reduced by a factor of 10 to account for chronic effects; BCF estimate is based on Dieldrin since Aldrin rapidly transformed to Dieldrin in the
Dieldrin	0.0019	2,300	4.37	BCF is based on 1.1% lipid level for marine fish, Spot (<i>Leiostomus xanthurus</i>).
Total Chlordane	0.004	16,000	64	Total Chlordane includes alpha-chlordane, trans nonachlor, heptachlor, heptachlor epoxide; WQC for Chlordane is used for Total Chlordane; BCF is based
Total Endosulfan	0.0087	328	2.85	Total Endosulfan includes endosulfan I, endosulfan II and endosulfan sulfate; WQC for Endosulfan is used for Total Endosulfan; BCF is based on 3.6% lipid
1,4-Dichlorobenzene	197	60	11820	Ambient conc. is based on lowest observed effect level (LOEL) for saltwater species from WQC, and reduced by a factor of 10 to account for chronic
Arsenic	36	350	12600	Ambient conc. is based on the saltwater criteria continuous conc. for arsenic (III); BCF is based on the Eastern Oyster (<i>Crassostrea virginica</i>).
Chromium	50	236	11800	Ambient conc. is based on Chromium (VI) since it is substantially more toxic than Chromium (III); BCF is based on polychaete worm.
Copper	2.9	3,300	9570	WQC is based on a hardness value of 100; BCF is based on soft shell clam.
Lead	8.5	1,400	11900	Ambient conc. is based on saltwater criteria continuous conc.; BCF is based on the Eastern Oyster (<i>Crassostrea virginica</i>).
Nickel	8.3	458	3802	Ambient conc. is based on saltwater criteria continuous conc.; BCF is based on the Eastern Oyster (<i>Crassostrea virginica</i>).
Silver	0.23	6,500	1495	Water Quality Criterion (WQC) was reduced by a factor of 10 to account for chronic effects; BCF is based on the Blue Mussel (<i>Mytilus edulis</i>).
Zinc	86	17,640	1517040	Ambient conc. is based on saltwater criteria continuous conc.; BCF is based on Eastern Oyster (<i>Crassostrea virginica</i>).

¹.The following documents were used to obtain the water quality criteria values.

Calabrese, A. 1984. "Effects of Long-Term Exposure to Silver and Copper on Growth, Bioaccumulation and Histopathology in the Blue Mussel (*Mytilus edulis*).” *Mar. Envir. Res.* 1, 253-274.

EPA. 1980b. Ambient Water Quality Criteria for Aldrin/Dieldrin; EPA 440/5-80-019; December 1980.

EPA. 1980c. Ambient Water Quality Criteria for Chlordane; EPA 440/5-80-027; October 1980.

EPA. 1980d. Ambient Water Quality Criteria for Heptachlor; EPA 440/5-80-052; October 1980.

EPA. 1980e. Ambient Water Quality Criteria for Endosulfan; EPA 440/5-80-046; October 1980.

EPA. 1980f. Ambient Water Quality Criteria for Dichlorobenzenes; EPA 440/5-80-039; October 1980.

EPA. 1984a. Ambient Water Quality Criteria for Lead - 1984; EPA 440/5-84-027; January 1985.

EPA. 1984b. Ambient Water Quality Criteria for Copper - 1984; EPA 440/5-84-031; January 1985.

EPA. 1985a. Ambient Water Quality Criteria for Chromium - 1984; EPA 440/5-84-029; January 1985.

EPA. 1985c. Ambient Water Quality Criteria for Arsenic - 1984; EPA 440/5-84-033; January 1985.

EPA. 1986. Ambient Water Quality Criteria for Nickel - 1986; EPA 440/5-86-004; September 1986.

EPA. 1987b. Ambient Water Quality Criteria for Zinc - 1987; EPA 440/5-87-003.

EPA. 1992a. Draft Ambient Water Quality Criteria for Silver.

ATTACHMENT B

Benthic Cancer Protection Level Calculations for the Protection of Human Health from the Consumption of Fish Exposed to Dredged Material at the Historic Area Remediation Site

Basis^{fn1}: $10^{-4} \text{ Benthic Tissue Level (ug/kg)} = \frac{[10^{-4} \text{ Conc. in Fish}] \times [\text{Whole Body/fillet Factor (1.35)}]^{fn5}}{\text{Trophic Transfer Factor}^{fn2}}$ $10^{-4} \text{ Conc. in Fish (ug/kg)} = \frac{\text{Toxicological Dose (ug/day)}}{[\text{Seafood Consumption (6.5 g/day)}]^{fn3} \times [10^{-3} \text{ kg/g}]}$ $\text{Toxicological Dose (ug/day)} = \frac{[\text{Risk Level (10}^{-4}\text{)}] \times [\text{Body Weight (70 kg)}]^{fn3} \times [10^3 \text{ ug/mg}]}{\text{Potency Factor, } q_1^* \text{ (kg-day/mg)}^{fn4}}$				
	Cancer Potency Factor q_1^* (kg-day/mg)	Acceptable Concentration in Fish (ug/kg)	Trophic Transfer Factor	Benthic Protection Level (ug/kg)
Pesticides				
Aldrin	17	63	2.6	33
Chlordane	1.3	828	2.3	486
Dieldrin	16	67	1.4	65
Heptachlor	4.5	239	2.7	120
Heptachlor epoxide	9.1	118	1.4	114
Industrial Organics				
1,4- Dichlorobenzene	0.024	44,872	1	60,577
PAHs				
Benzo(a)pyrene	7.3	147	0.1	2,000
METALS				
Arsenic	1.5	718	3	323

ATTACHMENT C

**Benthic Non-Cancer Protection Level Calculations for the Protection
of Human Health from the Consumption of Fish Exposed to Dredged Material
at the Historic Area Remediation Site**

Basis ^{fn1} :				
$\text{Benthic Tissue Level (ug/kg)} = \frac{[\text{Conc. in Fish}] \times [\text{Whole Body/fillet Factor (1.35)}]^{fn5}}{\text{Trophic Transfer Factor}^{fn2}}$ $\text{Conc. in Fish (ug/kg)} = \frac{\text{Toxicological Dose (ug/day)}}{[\text{Seafood Consumption (6.5 g/day)}^{fn3}] \times [10^{-3}\text{kg/g}]}$ $\text{Toxicological Dose (ug/day)} = [\text{Reference dose}^{fn4}] \times [\text{Body Weight (70 kg)}^{fn3}]$				
	Reference Dose RfD (ug/kg-day)	Acceptable Concentration in Seafood (ug/kg)	Trophic Transfer Factor	Benthic Protection Level (ug/kg)
Metals				
Arsenic	0.3	3,231	3	1,454
Chromium	5	54,000	1	73,000
Copper	37.1	400,000	1	540,000
Nickel	20	215,000	1	290,000
Silver	5	54,000	1	73,000
Zinc	300	3,230,769	1	4,361,538
Pesticides				
Aldrin	0.03	323	2.6	167
Chlordane	0.06	592	2.3	350
Dieldrin	0.05	538	1.4	518
Endosulfan	6	64,615	1	87,231
Heptachlor	0.5	5,385	2.7	2,692
Heptachlor epoxide	0.013	140	1.4	135
PAHs				
Acenaphthene	60	650,000	0.1	8,775,000
Fluorene	40	430,000	0.1	5,805,000
Phenanthrene	300	3,230,000	0.1	43,605,000
Anthracene	300	3,230,000	0.1	43,605,000
Fluoranthene	40	430,000	0.1	5,805,000
Pyrene	30	325,000	0.1	4,387,000

NOTES:

^{fn1} Human health cancer and non-cancer assessments adapted from *Guidance for Assessing Chemical Contaminant Data for use in Fish Advisories: Volume II: Risk Assessment and Fish Consumption Limits*. U.S. Environmental Protection Agency, EPA823-B-94-004, Office of Science and Technology, Washington, DC, June 1994.

^{fn2} Trophic transfer factors were calculated by Mr. Lawrence Burkhard, EPA Duluth, using the food chain transfer model developed by Gobas (1993).

^{fn3} Default values were taken from EPA's national toxics rule for setting water quality criteria, EPA (1992b).

^{fn4} Cancer potency factors and non-cancer reference doses are taken from EPA (1995).

^{fn5} The acceptable concentration in seafood is defined on the basis of the fillet or edible portion for humans. Trophic transfer, however, was defined based on whole-body characteristics, including lipid concentrations. Experience in New York State indicates a whole body to fillet ration ranging from 1.2 to 1.5 is applicable to lipophilic substances. The mid-range value of 1.35 is used in this analysis.

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FEDERAL AVIATION ADMINISTRATION

OE/AAA®

OBSTRUCTION EVALUATION / AIRPORT AIRSPACE ANALYSIS

DESK REFERENCE GUIDE

SUBJECT: Add a New Case (Off Airport)

**You are required to have a registered e-filing account*

All references to software products remain the protected trademarks of their manufacturers. The instructions in this document may reference Microsoft application(s). This is not meant in any way to express a preference for any particular product since there are many different browsers, programs, and operating systems available to the user. For simplicity only, one brand/product is used in the examples that follow.

If you've successfully registered, you can use your OE/AAA account to file your Notice of Proposed Construction or Alteration.

Note1: Exit this guide if you are filing an Off Airport Notice of Proposed Construction or Alteration for Wind Turbine /Met Tower (w/WT Farm) or Wind Turbine-Barge Crane structures.

Note2: Use this DRG to Add a New Case (Off Airport) to include:

- E-file a crane mounted on a barge vessel for construction - Barge Crane (not associated with a wind turbine).
- E-file a Met Tower, not associated with a wind turbine farm, select structure type 'Met Tower (non-WT Farm)'.
- E-file a building with an auxiliary wind turbine mounted on a building or structure attached to a building, not associated with a wind turbine farm; select 'Building w/Wind Turbine'.
- For a wind turbine (not associated with a wind turbine farm), select "Wind Turbine."

Note3: To e-file **Wind Turbine /Met Tower (w / WT Farm) / Wind Turbine-Barge Crane** structure types, refer to the "Add a New Case (Off Airport) for Wind Turbine /Met Tower (w/WT Farm) /WT-Barge Crane" desk reference guide or the "Add Multiple Cases (Off Airport) for Wind Turbine /Met Tower (w/WT Farm)/ WT-Barge Crane" desk reference guide.

Note4: If you're e-filing a **large Off Airport project** with the following eligible Structure Types:

- Antenna Tower
- Billboard
- Bridge
- Building
- Building w/Wind Turbine
- Catenary Wire
- Cell-On-Wheels
- Chimney
- Drilling Rig
- Feasibility Study
- Flagpole
- High Mast Illumination
- Landfill
- Light Pole
- Lighting Study
- Met Tower (non-WT Farm)
- Monopole
- Other w/Antenna
- Other w/o Antenna
- Power Line
- Sign
- Solar Panel
- Solar Tower
- Stack
- Tower
- Transmission Line
- Utility Pole
- Waste Management Facility
- Water Tank
- Workover Rig

Review the “Add Multiple Cases (Off Airport)” desk reference guide (DRG) to consider e-filing via the OE/AAA data import feature.

Add a New Case (Off Airport)

The OE/AAA electronic filing (e-file) system allows you to:

- Submit an FAA Form 7460-1 via an electronic data screen.
- Generate a map directly from your account to be submitted electronically with your filing.
- Track the status of your case as it moves through the study process.

From your OE/AAA Portal Page you have:

- Instant access to your determination, requests for additional information, etc... as they are issued by the FAA.
- The ability to attach surveys, and additional background information directly to your electronic case file(s).

Create a New Case

To create a new case, click the **Add New Case (Off Airport)** link. This will bring up the *Notice of Proposed Construction or Alteration* Page. Complete each section according to the instructions below.

OE/AAA Portal Page

My Account	Off Airport Construction (includes on Military Airport)
Name: User Name: Login Time: IP Address: Actions:	My Cases (Off Airport) Add New Case (Off Airport) Add Multiple Cases (Off Airport) Add Supplemental Notice (7460-2 Form) My Sponsors Add New Sponsor Off Airport Contacts My Circ Comments

Important: You must complete all required fields (indicated with an asterisk *) to successfully save your case. Missing data will result in a warning message at the top of your page identifying the required information.

Notice of Proposed Construction or Alteration - Off Airport

Sponsor (person, company, etc. proposing this action)						
* Sponsor:						
Construction / Alteration Information						
* Notice Of:						
* Duration:						
if Temporary : Months: Days:						
Work Schedule - Start:	(mm/dd/yyyy)					
Work Schedule - End:	(mm/dd/yyyy)					
*For temporary cranes-Does the permanent structure require separate notice to the FAA? To find out, use the Notice Criteria Tool. If separate notice is required, please ensure it is filed. If it is not filed, please state the reason in the Description of Proposal.						
State Filing:						
Structure Details						
* Latitude:	N					
* Longitude:	W					
* Horizontal Datum:	NAD83					
* Site Elevation (SE):	(nearest foot)					
* Structure Height (AGL):	(nearest foot)					
* Current Height (AGL):	(nearest foot)					
* For notice of alteration or existing provide the current AGL height of the existing structure. Include details in the Description of Proposal						
Minimum Operating Height (AGL):	(nearest foot)					
* For aeronautical study of a crane or construction equipment the maximum height should be listed above as the Structure Height (AGL). Additionally, provide the minimum operating height to avoid delays if impacts are identified that require negotiation to a reduced height. If the Structure Height and minimum operating height are the same enter the same value in both fields.						
* Requested Marking/Lighting:	None					
Other :						
Aircraft Detection Lighting System(ADLS): * Only check this box if you are proposing the installation and use of an Aircraft Detection Lighting System						
* Current Marking/Lighting:	Select One					
Other :						
* Nearest City:						
* Nearest State:						
* Description of Location:						
On the Project Summary page upload any certified survey.						
* Description of Proposal:						
Additional Location(s)						
Add New Location(s)						
Structure Summary						
* Structure Type:						
* Structure Name:						
NOTAM Number:						
FCC Number:						
Prior ASN:	- - - OE Validate Prior					
Proposed Frequency Bands						
Select any combination of the applicable frequencies/powers identified in the Colo Void Clause Coalition, Antenna System Co-Location, Voluntary Best Practices, effective 21 Nov 2007, to be evaluated by the FAA with your filing. If not within one of the frequency bands listed below, manually input your proposed frequency(ies) and power using the Add Specific Frequency link.						
Add Specific Frequency						
☐	Low Freq	High Freq	Freq Unit	ERP	ERP Unit	
☐	6	7	GHz	55	dBW	
☐	6	7	GHz	42	dBW	
☐	10	11.7	GHz	55	dBW	
☐	10	11.7	GHz	42	dBW	
☐	17.7	19.7	GHz	55	dBW	
☐	17.7	19.7	GHz	42	dBW	
☐	21.2	23.6	GHz	42	dBW	
☐	21.2	23.6	GHz	55	dBW	
☐	698	806	MHz	1000	W	
☐	806	901	MHz	500	W	
☐	806	824	MHz	500	W	
☐	824	849	MHz	500	W	
☐	851	866	MHz	500	W	
☐	869	894	MHz	500	W	
☐	896	901	MHz	500	W	
☐	901	902	MHz	7	W	
☐	929	932	MHz	3500	W	
☐	930	931	MHz	3500	W	
☐	931	932	MHz	3500	W	
☐	932	932.5	MHz	17	dBW	
☐	935	940	MHz	1000	W	
☐	940	941	MHz	3500	W	
☐	1670	1675	MHz	500	W	
☐	1710	1755	MHz	500	W	
☐	1850	1910	MHz	1640	W	
☐	1850	1990	MHz	1640	W	
☐	1930	1990	MHz	1640	W	
☐	1990	2025	MHz	500	W	
☐	2110	2200	MHz	500	W	
☐	2305	2360	MHz	2000	W	
☐	2305	2310	MHz	2000	W	
☐	2345	2360	MHz	2000	W	
☐	2496	2690	MHz	500	W	
Clone Prior ASN frequencies						
*Note: Selecting this link will only add frequency (ies)/power from the prior ASN listed in Structure Summary. Additional frequency (ies)/power must be manually added before submitting to the FAA if they are to be considered with your new filing.						
☐ I hereby certify that all of the above statements made by me are true, complete, and correct to the best of my knowledge. In addition, I agree to mark and/or light the structure in accordance with established marking and lighting standards as necessary.						
Save Cancel						

- A. ***Sponsor:** Select the Sponsor from the dropdown menu. This menu is populated from your *My Sponsors* list. The registered information will automatically display in your electronic public record as the Sponsor's Representative once the case has been completed and a valid FAA Determination is issued.
- B. ***Notice Of:** Select the type of proposal. New Construction would be a structure that has not yet been built. Alteration is a change to an existing structure such as the addition of a side mounted antenna, a change to the marking and/or lighting, a change to power and/or frequency, or a change to the height. Existing would be a correction to the latitude and/or longitude, a correction to the existing height, or if filing for an existing structure that has never been studied by the FAA.
- C. ***Duration:** If Permanent, so indicate. If Temporary, enter the estimated length of time the temporary structure will be up in Months/Days.
- D. **Work Schedule:** *(Not a Required Field)* Using the calendar icons next to the fields select the date that construction is expected to start and the date that construction should be completed.
- E. **State Filing:** *(Not a Required Field)* Indicate if the case has been filed with the state.
- F. ***Structure Type:** Select the type of structure from the Structure Type drop down list. "Note: Frequencies will not be accepted if your Structure Type is 'Lighting Study'."
- G. ***Structure Name:** Enter a name for the structure (e.g. 50 Ton Crane, Hotel, Tower, etc...)
- H. **FCC Number:** *(Not a Required Field)* If this is an existing tower that has been registered with the FCC, enter the Antenna Structure Registration number.
- I. **Prior ASN:** *(Not a Required Field)* If an FAA aeronautical study was previously conducted, enter the prior Aeronautical Study Number.

Prior ASN data can be pre-populated into the Notice of Proposed Construction or Alteration-Off Airport form data fields. When the e-filer confirms the Prior ASN data, the following data fields are available for pre-population:

- Latitude/Longitude
- Site Elevation
- Above Ground Level Height (determined AGL from valid prior ASN)
- Marking/Lighting (Recommended Marking /Lighting from valid prior ASN to requested Marking /Lighting)

J. ***Latitude /Longitude:** Latitude and Longitude must be precise geographic coordinates entered in Degrees, Minutes, and Seconds to the hundredth of a second (e.g. 25-47-4.75 N, 80-19-7.26 W).

K. ***Horizontal Datum:** Select either NAD83 or NAD27. North American Datum is a reference from which latitude/longitude measurements are made.

L. ***Site Elevation:** Enter the site elevation above mean sea level expressed in whole feet rounded to the nearest foot (e.g. 12' 3" should be entered as 12). This data should match the ground contour elevations for the site.

- **Add New Case (Off Airport) – Single e-file case entry:**
The OE/AAA system validates the Site Elevation (SE) of e-filer's entered location; if it passes the National Elevation Data (NED) terrain elevation validation check, the System takes the e-filer to the Map verify step. E-filers are alerted if the SE does not Pass the NED check. If this occurs, you must either adjust the SE or check the "SE comments provided in Additional Info" checkbox and provide SE comments in the Additional Info text box to explain the discrepancy.

- **Add New Case (Off Airport) - Additional Location(s) - Batch e-file entry:**
The OE/AAA system validates the SE entered on all Rows added when Save is selected; if they all pass the NED terrain elevation validation check, the system takes the e-filer back to the external e-Filing Form to certify their data entry and move to the Map verify step. The System validates all of the rows entered and alerts filers when the SE does not Pass the NED Data check for the location. If this occurs, you must either adjust the SE or check the "SE comments provided in Additional Info" checkbox and provide SE comments in the Additional Info text box to explain the discrepancy.

- M. * **Structure Height:** (AGL): Only for Structure Types that **ARE NOT** a **traverseway**. Your structure's height is the height above ground level in whole feet rounded to the next highest foot (e.g. 12' 3" should be entered as 13). The structure height includes anything mounted on top of the structure such as antennas, lightning rods, obstruction lights, etc.
- N. * **Unadjusted Structure Height:** (AGL): Only for Structure Types that **ARE** a **traverseway**. Your structure's height is the unadjusted structure height. Enter the unadjusted structure height above ground level in whole feet rounded to the next highest foot (e.g. 12' 3" should be entered as 13). The unadjusted structure height includes anything mounted on top of the structure such as antennas, lightning rods, obstruction lights, etc.
- O. * **Height Adjustment:**
Only for Structure Types that **ARE** a **traverseway**. The Unadjusted Structure Height AGL is adjusted upward by the system to account for the expected height of vehicles (or the highest mobile object [as applicable]) using the traverseway selected from the Structure Type drop down list in the Structure Summary section of the data entry screen.
- o For Structure Type "**Waterway**" and "**Other Traverseway**"
The Unadjusted Structure Height AGL is adjusted upward one (1) foot (default) by the system. **Enter the height** of the highest mobile object or vehicle expected to use the traverseway into the Height Adjustment field.
 - For Structure Type "**Private Road**"
The Unadjusted Structure Height AGL is adjusted upward ten (10) feet (default) by the system. **Enter the height** of the highest vehicle expected to use the traverseway into the Adjustment field.
- P. * **Total Structure Height** (AGL): Only for Structure Types that **ARE** a **traverseway**. The total of both the Unadjusted Structure Height and the Height Adjustment above ground level in whole feet rounded to the next highest foot (e.g. 12' 3" should be entered as 13).
- Q. * **Requested Marking and Lighting:** (Indicate the type Desired). The FAA Advisory Circular 70/7460-1 – Obstruction Marking and Lighting is recommended for determining the proper way to light and mark structures affecting navigable airspace. The AC can be accessed from the *Information*

Resources section of the website using the *Relevant Advisory Circulars* link.

Requested Marking/Lighting options:

- None
- Red lights
- Red lights and paint
- Red lights and flags
- Paint and 24-hour med-strobes
- Paint and a med-dual system
- Spherical markers and red lights
- Flag Marker
- Spherical Markers
- Dual-red and medium intensity
- Dual-red and high intensity white
- White-medium intensity
- White-high intensity
- White Paint/Synchronized Red Lights
- White Paint Only
- Dual medium catenary
- Dual high catenary
- White-medium catenary
- White-high catenary
- Paint day, red flashing twilight & night
- Paint day, med-strobes twilight & night
- Paint day, hi-strobes twilight & night
- Other – (if selected from the dropdown, enter the marking/lighting type in the “Other” field)

R. **Aircraft Detection Lighting System” (ADLS):** *(Not a Required Field)* Control device to operate marking/lighting systems on structures.

S. ***Current Marking /Lighting:** Indicate the current M/L on the structure; if a new structure, select N/A Proposed Structure.

T. **Current AGL:** Required for structures being e-filed as existing or alteration.

- U. **Min Operating Height** (AGL): * For aeronautical study of a crane or construction equipment the maximum height should be listed above as the Structure Height (AGL). Additionally, provide the minimum operating height to avoid delays if impacts are identified that require negotiation to a reduced height. If the Structure Height and minimum operating height are the same enter the same value in both fields.
- V. *** Nearest City/State**: Enter the name of the nearest city and the actual state where the site will be located.
- W. *** Description of Location**: Enter a brief description of the actual location of the site including the address or the relationship of the structure to roads, airports, prominent terrain, existing structures, etc.
- X. *** Description of Proposal**: Enter a complete description that details the nature of the filing.
- Y. **Add new location**: When submitting more than one case (e.g. a crane and a building or four building points) the following required fields indicated with an asterisk (*) must be completed to successfully save additional locations: J, K, L, M, G, Q. Additional rows may be added in increments of 1 thru 5. To remove an additional row, select the Delete link.
- Z. **Proposed Frequency Bands**: *(Not a Required Field)* Check any that apply. "Note: Frequencies will not be accepted if your Structure Type is 'Lighting Study'."
- AA. **Specific Frequencies**: *(Not a Required Field)* any frequency band not listed in the Proposed Frequency Bands should be added here. Select the Add Specific Frequency link and enter the Low Frequency, High Frequency, Frequency Unit, Effective Radiated Power (ERP), and ERP Unit. Select [Save] or [Cancel] to be returned to the *Case Data Entry* page. If an e-filer intends to overlap protected FAA frequencies, specific coordination with the FAA Spectrum Engineering Group will be required. A textbox allows filers to submit rationale for the frequency overlap in the e-filed Notice of Proposed Construction or Alteration-Off Airport form. "Note: Frequencies will not be accepted if your Structure Type is 'Lighting Study'."

BB. Clone Prior ASN frequencies – (Not a Required Field) The Prior ASN field must be filled before entering frequencies. This link is displayed after the Specific Frequency Bands section. This link is only available if the e-filer adds a Prior ASN that has frequencies included in the case. When selected the applicable Proposed Frequency Bands and/or Specific Frequencies from the prior ASN auto populate and are available for edit by the e-filer prior to saving the draft. Once the e-filer saves this data, it becomes part of the current filing and is transmitted to the FAA with the new ASN. The e-filer is permitted to add additional frequencies if necessary after cloned frequencies are pre-populated but duplicate entries are not allowed. “Note: Frequencies will not be accepted if your Structure Type is ‘Lighting Study’.”

Selecting the **checkbox** to accept the certify statement.

When all required fields are completed, select the **[Save] button**. This will save the case data as a draft and take you to the *Project Summary* screen.

After case data has been saved as a draft, filers are taken to the Map Verification screen that displays all cases created on the previous Add New Case(s) Off Airport screen and require Map verification before submission.

Map Verification
Mapping - Draft Reference Guide

To submit your project to the FAA, you must verify the coordinates of each case listed below:
 - Select the link labeled "View Map" to verify for each case.
 - Review the plotted location on the Map.
 - Select "Verify" once you have confirmed the location or click on the structure name link to go back to the Structure Name and review your entered coordinates.
 - NOTE: You may continue to the Project Summary page without verifying your map(s), however, before your case is eligible for submission to the FAA, you will be required to verify your plotted location. This function will also be available on the next screen for you to complete later.

Project Name:	Sponsor:	Map
Structure	City, State	Lat/Long
test1 Draft	test, CA	34° 12' 8.54" N 118° 29' 21.10" W
test2 Draft	test, CA	34° 12' 8.54" N 118° 29' 21.10" W
test3 Draft	test, CA	34° 12' 8.54" N 118° 29' 21.10" W
test4 Draft	test, CA	34° 12' 8.54" N 118° 29' 21.10" W
test5 Draft	test, CA	34° 12' 8.54" N 118° 29' 21.10" W

To submit this project you must verify the coordinates of each case listed above by verifying the map

Structure Name: test1
Latitude: 34° 12' 8.54" N
Longitude: 118° 29' 21.10" W

By verifying the coordinates represented on this map, you agree that the location of the case you have entered is correct to the best of your knowledge.

Aerial Photography Transparency Setting (1% increments): 0%

[Verify Map](#)

[Continue - Map/Verify Later](#)

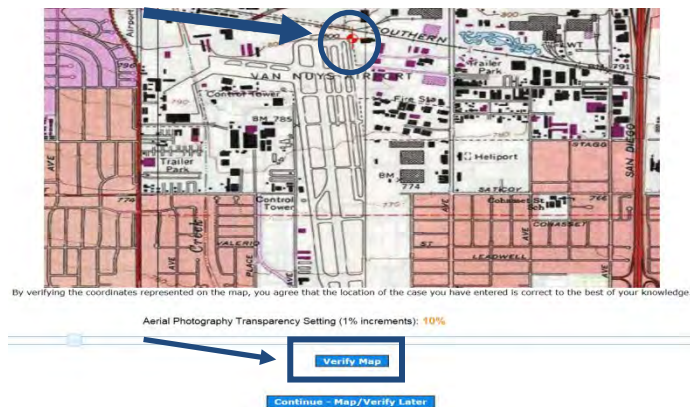
Enlarged View

Structure	City, State	Lat/Long	Map
test1 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.20" W	View Map to verify
test2 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.19" W	View Map to verify
test3 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.18" W	View Map to verify
test4 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.17" W	View Map to verify
test5 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.16" W	View Map to verify

To submit this project you must verify the coordinates of each case listed above by verifying the map

Structure Name: test5
Latitude: 34° 13' 8.54" N
Longitude: 118° 29' 21.16" W

- Select the link labeled **View Map to Verify** (displayed above the map) when more than one map needs to be verified or click the **Verify Map** button (displayed below the map) to view a single case map.
 - Review the plotted structure location on the Map (red bullseye) to verify the crosshairs on the map match with your proposed structure location.
 - Select **"Verify Map"** (at the bottom of the map) once you have confirmed the structure location. This will save the verified map but will NOT submit the case to the FAA.



- It will return you to the Project Summary screen, where if needed, the **“Structure”** link is/are available to display saved draft(s) of the data form if entered case coordinates need to be revised.

NOTE: Once a map is verified, if the e-filer returns to the saved Off Airport data entry form draft and re-certifies and saves the form data, the filer is required to re-verify the map location prior to submission to the FAA.

Project Name:		Sponsor:	
Structure	City, State	Lat/Long	Map
test1 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.20" W	View Map to verify
test2 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.19" W	View Map to verify
test3 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.18" W	View Map to verify
test4 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.17" W	View Map to verify
test5 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.16" W	View Map to verify

To submit this project you must

Structure Name: test5
Latitude: 34° 13' 8.54" N
Longitude: 118° 29' 21.16" W

If the e-filer returns to the saved Off Airport data entry form draft [view data] but cancels (does not re-certify the entered data), they won't be required to re-verify the Map.

NOTE: You may continue to the Project Summary screen without verifying your map(s), however, before your case is eligible for submission to the FAA you will be required to verify your plotted location. This function will also be available on the next screen for you to complete later.



When the only or last listed map in a project on the Map Verification screen is verified or if you continue to the Project Summary screen without verifying your map(s) the *Project Summary* screen will be displayed. Towards the right side of the page there will be a Map column and an Actions column. The Actions column contains the **Clone**, **Delete**, and **Upload a PDF** links. The Map column contains the **Verify Map** link.

Structure	City, State	Lat/Long	Map	Actions
test1 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.20" W	Show Map (Re-Verify)	Clone Delete Upload a PDF
test2 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.19" W	Verify Map	Clone Delete Upload a PDF

From the Project Summary page filers can verify or re-verify plotted location(s). To submit your project you must verify the coordinates of each case listed below by verifying the map".

To **verify** or re-verify your plotted map location from the Project Summary screen once you've confirmed the structure location; select the **"Verify Map"** or **"Re-Verify"** under the "Map" column header. On the map, click the "Verify Map" button at the bottom of the map or click on the **"Cancel"** button to return to Project Summary screen.

Notice of Proposed Construction or Alteration - Off Airport

[Add a new Case Off Airport - Desk Reference Guide](#)

[Add a New Case Off Airport for Wind Turbines - Met Towers - Desk Reference Guide](#)

Project Name:

Sponsor:

Project Summary :

[Add Another Case to this Project](#) [Change the sponsor for this Project](#)

Structure	City, State	Lat/Long	Map	Actions
test1 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.20" W	 Show Map (Re-Verify)	Clone Delete Upload a PDF
test2 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.19" W	 Verify Map	Clone Delete Upload a PDF
test3 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.18" W	 Verify Map	Clone Delete Upload a PDF
test4 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.17" W	 Verify Map	Clone Delete Upload a PDF
test5 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.16" W	 Verify Map	Clone Delete Upload a PDF

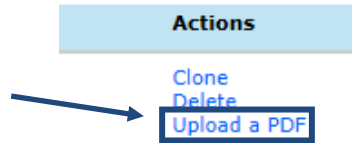
[Mapping - Desk Reference Guide](#)

[Attaching Documents - Desk Reference Guide](#)

[Upload a PDF to the Project](#)
Please upload all supporting case documentation
including the latest certified survey, if available.

To submit this project, you must verify the coordinates for each case listed above.

Attach Documents to Cases



For Off Airport cases you can upload PDF documents before and after submitting your case if needed.

Projects

One or more cases can be grouped into a Project. For example, each of the four building corner points can be a Case of a building Project. Project makes it easier to file, evaluate, manage, and approve related cases.

A screenshot of a "Project Summary" screen. At the top, there is a "Project Summary" header with two links: "Add Another Case to this Project" and "Change the sponsor for this Project". Below the header is a table with five columns: "Structure", "City, State", "Lat/Long", "Map", and "Actions". The table contains three rows of data, each representing a case. The "Actions" column for each row contains links: "Clone", "Delete", and "Upload a PDF". A blue arrow points from the left towards the "Project Summary" header.

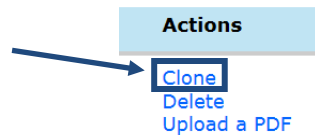
Structure	City, State	Lat/Long	Map	Actions
test1 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.20" W	Show Map (Re-Verify)	Clone Delete Upload a PDF
test2 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.19" W	Verify Map	Clone Delete Upload a PDF
test3 Draft	test, CA	34° 13' 8.54" N 118° 29' 21.18" W	Verify Map	Clone Delete Upload a PDF

Add a Case



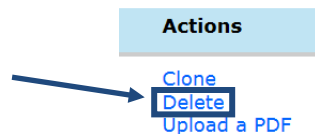
On the *Project Summary* screen you may select the **Add Another Case to this Project** link to add another case to this project. The cases entered this way will have the same project number.

Clone a Case



Another way to add a case to the project is to clone a new case from an existing case. E-filers can clone cases from the Project Summary screen of cases in their account regardless of the status (i.e. Draft/Submitted). To clone a case, click the **Clone** link. The cloning feature will copy most of the information over into a new *Case Data Entry* screen and link the cases together in a project. You may add as many cloned cases to your project as necessary. Once all of the maps for the project have been verified, the **[Submit]** button will appear on the *Project Summary* screen so that the entire project can be submitted to the FAA.

Delete a Case



You may only delete cases in Draft status. To delete a single case or a case from a project, select the **Delete** link located under the Actions header on the Project Summary screen. This will display the *Confirm Case Deletion* screen. To continue with the delete, select the **[I Confirm]** button to execute the deletion.

Submit to FAA

Note: Before submitting your case/project to the FAA, determine if you need to use the Clone or Delete features.

After the case data has been saved and map(s) verified, the **[Submit]** button will appear on the *Project Summary* screen to allow you to submit the case to the FAA. If you have provided all the information about your case or project, select the **[Submit]** button. This will take you to the *Confirm Project Submission* screen.

Project Name: _____ Sponsor: _____

Project Summary :

[Add Another Case to this Project](#) [Change the sponsor for this Project](#)

Structure	City, State	Lat/Long	Map	Actions
Test Draft	Ashburn, PA	40° 24' 40.80" N 79° 15' 32.92" W	Show Map (Re-Verify)	Clone Delete Upload a PDF

[Mapping - Desk Reference Guide](#) [Attaching Documents - Desk Reference Guide](#)

You may submit your Project to the FAA.

Submit

[Upload a PDF to the Project](#)
Please upload all supporting case documentation including the latest certified survey, if available.

Select the **[I Confirm]** button to submit the case or project to the FAA. When the submission is done, OE/AAA will display the *Project Submission Success* screen.

Confirm Project Submission

Project Name: _____

Please confirm you would like to submit Project _____ and associated cases to the FAA for processing.

I Confirm **Back**

The Aeronautical Study Number (ASN) assigned to your filed case(s) and other submission information is displayed. The Project Submission Success screen includes a link to a **state aviation contacts** map to determine if coordination of your proposed activity is necessary with your state aviation department.

Project Submission Success

Project Name: _____

Project BRICK-00065811-18 has been submitted successfully to the FAA.

Your filing is assigned Aeronautical Study Number (ASN):

- 2075-AWP-8894-OE
- 2075-AWP-8895-OE
- 2025-AWP-8896-OE
- 2025-AWP-8897-OE
- 2025-AWP-8898-OE
- 2025-AWP-8899-OE

Please refer to the assigned ASN on all future inquiries regarding this filing.

Please return to the system at a later date for status updates.

It is the responsibility of each e-filer to exercise due diligence to determine if coordination of the proposed construction or alteration is necessary with their state aviation department. Please use the link below to contact your state aviation department to determine their requirements.

[State Aviation Contacts](#)

To ensure e-mail notifications are delivered to your inbox please add noreply@faa.gov to your address book. Notifications sent from this address are system generated FAA e-mails and replies to this address will NOT be read or forwarded for review. Each system generated e-mail will contain specific FAA contact information in the text of the message.

Return to Portal

Please return to the system at a later date for status updates.

ATTACHMENT M: XYZ DATA FOR NY&NJ CHANNELS-SEGUINE/WARD/OUTERBRIDGE

Direct Link to Plans and Specifications Survey XYZ Data:

https://ehydroprod.blob.core.usgovcloudapi.net/ehydro-surveys/CENAN/CENAN_DIS_NJ_03_SWO_20251217_OT_5639_60.ZIP

For additional Survey data sets:

<https://Navigation.usace.army.mil/>

SECTION TABLE OF CONTENTS

DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

SECTION 00 90 02

WATER QUALITY CERTIFICATES

02/09

PART 1 GENERAL

PART 2 PRODUCTS

PART 3 EXECUTION

3.1 DOCUMENTS, EXHIBITS, AND OTHER ATTACHMENTS

3.1.1 WATER QUALITY CERTIFICATE(S) / FEDERAL CONSISTENCY
DETERMINATION(S) / ATTACHMENT N

-- End of Section Table of Contents --

SECTION 00 90 02

WATER QUALITY CERTIFICATES
02/09

PART 1 GENERAL

PART 2 PRODUCTS

Not Used.

PART 3 EXECUTION

3.1 DOCUMENTS, EXHIBITS, AND OTHER ATTACHMENTS

The following attachments have been provided for reference to the bidder for use in performing the contract requirements.

3.1.1 WATER QUALITY CERTIFICATE(S) / FEDERAL CONSISTENCY DETERMINATION(S) / ATTACHMENT N

The New Jersey Department of Environmental Protection (NJDEP) Federal Consistency Determination/Water Quality Certificate (FCD/WQC) for the maintenance dredging of NY&NJ Channels, Seguine/Ward/Outerbridge Reaches is enclosed.

The New York State Department of Environmental Conservation (NYSDEC) FCD/WQC shall be provided prior to the start of dredging.

The Contractor must comply with all conditions within both the NJDEP WQC and NYSDEC WQC.

A copy of both WQC's shall be displayed at a highly visible location on the dredge at all times for the duration of the project.

-- End of Section --



State of New Jersey

MIKIE SHERRILL
Governor

DEPARTMENT OF ENVIRONMENTAL PROTECTION
Division of Land Resource Protection
Mail Code 501-01A
P.O. Box 420
Trenton, New Jersey 08625-0420
dep.nj.gov/wlm/

ED POTOSNAK
Acting Commissioner

DR. DALE G. CALDWELL
Lt. Governor

February 23, 2026

Michael Oseback, Chief, Operations Support Branch
NY District Army Corps of Engineers
26 Federal Plaza
New York, NY 102780090

RE: Project Description: Federal Consistency
File and Activity No.: 0000-03-0035.1 CDT250003
Applicant: NY District Army Corps of Engineers
Project: Maintenance dredging of Seguine/Ward/Outerbridge Reaches

Dear Mr. Oseback:

This letter is in response to your request, dated December 10, 2025, for a Federal Consistency (FC), as required by Section 307 of the Federal Coastal Zone Management Act (16 USC 1451 *et seq.*) and Water Quality Certification (WQC) as required by Section 401 of the federal Clean Water Act (33 USC 1251 *et seq.*). The request was for the proposed maintenance dredging of selected critical shoals in the New York and New Jersey channels, specifically the Seguine/Ward/Outerbridge channels. Approximately 330,000 cubic yards of material will be dredged, to the project depth of -35 feet mean low water plus two feet allowable over dredge, with subsequent placement of the dredge material at the Historic Area Remediation Site (HARS). The request included a Memorandum from EPA Region 2 confirming that the sediments from this maintenance dredging are suitable for placement at the HARS as Remediation Material, specifications for the maintenance dredging, and a New Jersey Coastal Zone Management Rules compliance statement.

Federal Consistency and Water Quality Certification

The Coastal Zone Management Rules (N.J.A.C. 7:7) constitute New Jersey's enforceable policies under its federally approved Coastal Zone Management Program. The Maintenance Dredging of Ward Point Bend Federal Navigation Project has been reviewed under the following rules on Coastal Zone Management: Finfish migratory pathways (7:7-9.5), Navigation Channels (7:7-9.7), Ports (7:7-9.11), Submerged Infrastructure Routes (7:7-9.12), Historic and Archaeological Resources (7:7-9.36), Maintenance Dredging (7:7-12.8), Dredged Material Disposal (7:7-12.9), Marine Fish and Fisheries (7:7-16.2), and Water Quality (7:7-16.3). **Based on the materials submitted within the subject federal consistency request**, the Division of Land Resource Protection (Division) has determined that this dredging and placement project is consistent with the Coastal Zone Management Rules and New Jersey's federally approved Coastal Management Program.

The Division has also reviewed this contract for potential water quality impacts. Provided that the following conditions are met, it has been determined that this project is not likely to cause a violation of

New Jersey's Surface Water Quality Standards (N.J.A.C. 7:9B-1.1 et seq.). Therefore, this determination includes the State's Water Quality Certification pursuant to Section 401 of the federal Water Pollution Control Act (33 USC 1251 et seq.) subject to the following conditions. It is noted that this Water Quality Certificate is valid for a term of five years and expires on February 22, 2031.

Project Specific Conditions:

1. Dredging is prohibited from March 1 through June 30th in order to protect spawning and migrating anadromous fish.
2. Dredging is prohibited within the areas where water depths are -20 feet MLW or less from January 1 to May 31 to protect winter flounder early life stages and their Essential Fish Habitat.
3. The contractor shall immediately inform the Department of any unanticipated adverse effects on the environment not described in the application or in the conditions of this permit.
4. All necessary local, Federal, and other State approvals must be obtained by the applicant prior to the commencement of the herein-permitted activities.
5. The sediments shall be removed using a closed clamshell environment bucket.
6. The dredge shall be operated so as to control the rate of descent of the bucket so as to maximize the vertical cut of the clamshell bucket while not penetrating the sediment beyond the vertical dimension of the open bucket (i.e. overfilling the bucket). This will reduce the amount of free water in the dredged material, will avoid overfilling the bucket, and minimize the number of dredge bucket cycles needed to complete the dredging contract. The dredging contractor shall use appropriate software and sensors on the dredging equipment to ensure consistent compliance with this condition during the entire dredging operation.
7. The closed clamshell environmental bucket shall be equipped with sensors to ensure complete closure of the bucket before lifting the bucket. Said sensors shall be operational during the entire dredging operation.
8. Where a closed clamshell environmental bucket is required, it shall be lifted slowly through the water, at a rate of 2 feet per second or less.
9. Dredged material shall be placed deliberately in the barge in order to prevent spillage of material overboard.
10. "Barge Overflow" is permitted for this dredging project.
11. All barges or scows used to transport sediment shall be of solid hull construction or be sealed with concrete.
12. The gunwales of the dredge scows shall not be rinsed or hosed during dredging except to the extent necessary to ensure the safety of workers maneuvering on the dredge scow.

13. All decant water holding scows shall be water tight and of solid hull construction.
14. Decant water from this project may only be discharged within the channel from where the sediments originated, in close proximity to the dredging contract area. Discharge to another receiving waterbody requires prior approval from the Department, and may require a New Jersey Discharge Pollutant Elimination System/Discharge to Surface Water (NJDPES/DSW) permit.
15. All decant water shall be held in the decant holding scow a minimum of 24 hours after the last addition of water to the decant holding scow. Said water contained in the decant holding scow may only be discharge after this mandatory 24-hour retention time.
 - a. Should the contractor wish to reduce the required holding time, the contractor shall demonstrate that the reduced holding time is sufficient to meet a total suspended solids (TSS) background value of 30 mg/L. This TSS action level is consistent with the ambient TSS results presented in the NY District study entitled "NY and NJ Harbor Deepening Project - Total Suspended Solids (TSS) Monitoring, Interim Report" (January 2006). The total suspended solids shall be determined through gravimetric analysis. No discharge shall be permitted from the decant holding scow until the results of the gravimetric analysis have confirmed that the 30 mg/L background level has been achieved. No additional water shall be added to the decant holding scow between the time of sample acquisition and discharge. Upon successful demonstration that the reduced holding time is sufficient to meet the TSS background level of 30 mg/L, the monitoring of TSS may be suspended and the demonstrated settling time shall replace the 24-hour minimum. A successful demonstration of the reduced holding time efficiency shall be determined once three consecutive TSS analyses have confirmed that the 30 mg/L action level has been achieved by the reduced holding time.
16. Should the contractor wish to demonstrate this reduced holding time, all records including time of last addition of decant water into the scow, time of TSS sampling and the results of TSS sampling shall be submitted to the NJDEP as soon as they become available, together with a request for a reduced holding time.
17. During pumping of the decant water from the holding scow, great care shall be taken to avoid re-suspending or pumping sediment which has settled in the decant holding scow.

Approved Plans:

The drawings hereby approved consist of eight sheets prepared by the New York Army Corps of Engineers, dated March 25, 2025, and entitled:

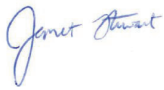
“NEW YORK AND NEW JERSEY CHANNELS
SEGUINE POINT TO OUTERBRIDGE CROSSING
CONDITION SURVERY”

- a. “VH-102,” sheet 2 of 9
- b. “VH-103,” sheet 3 of 9
- c. “VH-104,” sheet 4 of 9
- d. “VH-105,” sheet 5 of 9
- e. “VH-106,” sheet 6 of 9
- f. “VH-107,” sheet 7 of 9

- g. "VH-108," sheet 8 of 9
- h. "VH-109," sheet 9 of 9

If you have any questions regarding this letter, please contact Jeff Thein by email at Jeff.Thein@dep.nj.gov or by phone at (609) 940-5338. Please reference the Division's file number in all communication.

Sincerely,

A handwritten signature in blue ink, appearing to read "Janet Stewart".

Digitally signed
by Janet Stewart
Date: 2026.02.23
15:39:10 -05'00'

Janet Stewart, Assistant Director
Division of Land Resource Protection

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DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 11 00

SUMMARY OF WORK

06/07

PART 1 GENERAL

- 1.1 FAR 52.211-10, COMMENCEMENT, PROSECUTION AND COMPLETION OF WORK
- 1.2 WORK COVERED BY THE CONTRACT DOCUMENTS
 - 1.2.1 Project Description
- 1.3 PERFORMANCE OF WORK BY THE CONTRACTOR
- 1.4 DFARS 236.204, DISCLOSURE OF THE MAGNITUDE OF CONSTRUCTION PROJECTS
- 1.5 DATUM AND BENCH MARKS
- 1.6 MISPLACED MATERIAL
- 1.7 AUTHORIZED CONSTRUCTION AREA AND TRESPASSING
- 1.8 EXISTING WORK

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 11 00

SUMMARY OF WORK
06/07

PART 1 GENERAL

1.1 FAR 52.211-10, COMMENCEMENT, PROSECUTION AND COMPLETION OF WORK

a. The Contractor shall be required to (i) commence work under this contract within five (5) calendar days after the date the Contractor receives the Notice to Proceed, (ii) prosecute the work diligently, and (iii) complete the entire basic work ready for use, not later than **54 calendar days**, after the date the Contractor receives the notice to proceed. The time stated for completion shall include final cleanup of the premises. (FAR 52.212-3)

b. All work shall be in accordance with the drawings and specifications or instructions attached hereto and made a part thereof, or to be furnished hereafter by the Contracting Officer or Contracting Officer Representative (COR) and subject, in every detail, to his supervision, direction, and instructions.

1.2 WORK COVERED BY THE CONTRACT DOCUMENTS

1.2.1 Project Description

The work to be performed under this project consists of providing all materials, plant, labor, equipment and services (except those furnished by the Government) to perform maintenance dredging of New York and New Jersey Channel, Seguine/Ward/Outerbridge Reaches, Federal Navigation Project, with placement of the material at the Historic Area Remediation Site (HARS). The Contractor will be solely responsible for coordinating, obtaining, and be in compliance with all requirements for transportation, placement, and monitoring of the dredge material. See Section 35 20 23.10 18 entitled "DREDGED MATERIAL TRANSPORT, AND PLACEMENT MONITORING" for more information.

The maintenance for this project consists of dredging approximately 330,690 CY of material based on a plans and specifications survey dated 30 December 2025. For more details, see "ESTIMATED QUANTITIES" in Section 00 30 00 "ADMINISTRATIVE REQUIREMENTS" of these Specifications. The dredge material has been determined to be composed of sand, silt, and clay throughout the reaches. This work is subject to availability of funds.

A summary of the areas to be dredged, with associated quantities is shown in the table below:

DREDGING AREAS	TOTAL ESTIMATED QUANTITY (CY)
BASIC WORK - Reach 1: (-35' MLLW + 2' OD)	16,385
BASIC WORK- Reach 2: (-35' MLLW + 2' OD)	259,580

BASIC WORK- Reach 3: (-35' MLLW + 2' OD)	54,725
--	--------

Contract time for completion is based on a minimum average production rate of 8,000 cubic yards (CY) per day of dredging, transportation, and placement at the HARS. Should the total quantity of material exceed the estimated quantities with the Solicitation Contract Form, additional time will be allowed at the rate of one (1) calendar day for each 8,000 CY in excess of the established limit. The time stated for completion shall include mobilization, demobilization, and final clean up.

Scow monitoring is required for this project. The contractor is required to comply with the Section 35 20 23.10 18 entitled "DREDGE MATERIAL TRANSPORT PLACEMENT AND MONITORING" of these specifications.

Sampling and testing of the maintenance dredging area has been characterized by twenty-one (21) sediment core samples taken to a depth of -35 feet plus 2 feet allowable overdepth MLLW and the test results are included in the specifications in Section 00 90 01 as Attachment K.

The New York State Department of Environmental Conservation (NYSDEC) Water Quality Certificate (WQC) is currently underway. The WQC will be attached as an amendment after solicitation. The New Jersey Department of Environmental Protection (NJDEP) Federal Consistency Determination (FCD) is attached for NY&NJ Channels- Seguine/Ward/Outerbridge Reaches.

No dredging activities shall occur from 01 October 2026 to 30 November 2026 for dredging areas in sheets six (6) to ten (10) as indicated in the plan sheets.

The Contractor must comply with all conditions of the NYSDEC and NJDEP WQC/FCD. A copy of the NYSDEC WQC/ NJDEP FCD shall be displayed on the dredge at all times for the duration of the project.

The current Notice to Proceed (NTP) is scheduled for July. The work is anticipated to be performed with the approximate time period of summer/fall to winter 2026.

1.3 PERFORMANCE OF WORK BY THE CONTRACTOR

The Contractor shall perform on the site, and with its own organization and equipment, work equivalent to at least forty (40) percent of the total amount of work to be performed under the contract. This percentage may be reduced by a supplemental agreement to this contract if, during performance of the work, the Contractor requests a reduction and the Contracting Officer and/or his Representative determines that the reduction would be to the advantage of the Government. (FAR 52.236-01)

1.4 DFARS 236.204, DISCLOSURE OF THE MAGNITUDE OF CONSTRUCTION PROJECTS

The entire work (Mob/Demob and Basic Work) is estimated to cost approximately \$5,000,000.00 to \$15,000,000.00.

1.5 DATUM AND BENCH MARKS

Soundings and Elevations shown in the contract drawings are in US Survey Feet and refer to Mean Lower Low Water (MLLW) [epoch 1983-2001]. The Plane

of MEAN LOWER LOW WATER (MLLW) is 3.0-3.2 FEET BELOW NAVD88 AS DETERMINED USING VDATUM (VER. 4.6.1). The following Benchmark information should be referenced for defining local datum and required project depth.

New York and New Jersey Channels-Seguine/Ward/Outerbridge Reaches Federal Navigation Project

NSRS Benchmark:

Benchmark: B 5

PID: KV0006

Elevation: 63.49' (NAVD 1988)

Local Control Data:

Benchmark: PAA 1

Elevation: 7.39' (NAVD 1988)

All control should be verified prior to utilization. The contractor may contact the New York District Survey Branch at (201) 333-0284 for assistance in bench mark discrepancies or clarification in benchmark and / or tide gage location.

1.6 MISPLACED MATERIAL

Should the Contractor during the progress of the work, lose, dump, throw overboard, sink, or misplace any material, plant, machinery, or appliance, which in the opinion of the Contracting Officer and/or his Representative may be dangerous to or obstruct navigation, the Contractor shall recover and remove the same with the utmost dispatch. The Contractor shall give immediate notice, with description and location of such obstructions, to the Contracting Officer and/or his Representative or inspector, and when required shall mark or buoy such obstruction until the same are removed. Should he refuse, neglect, or delay compliance with the above requirements, such obstructions may be removed by the Contracting Officer and/or his Representative, and the cost of such removal may be deducted from any money due or to become due to the Contractor, or may be recovered under his bond. The liability of the Contractor for the removal of a vessel wrecked or sunk without fault or negligence shall be limited to that provided in Sections 15, 19, and 20 of the Rivers and Harbors Act of March 3, 1899 (33 USC. 410 et seq.) (DOD FARS SUPPL. 52.236-7006).

1.7 AUTHORIZED CONSTRUCTION AREA AND TRESPASSING

Do not inflict damage upon land and properties outside the authorized construction area by unwarranted entry upon, passage through, damage to, or disposal of, material on such land or property. The Contractor may make a separate agreement with any other party, regarding the use of, or right to, land or facilities outside the contract area. If such an agreement is made, it must be in writing and a copy shall be provided to the Contracting Officer. The Contractor must hold and save the Government, its officers, and agents free from liability arising from trespassing or damage occasioned by his operations.

1.8 EXISTING WORK

In addition to FAR 52.236-9 Protection of Existing Vegetation, Structures, Equipment, Utilities, and Improvements:

- a. Remove or alter existing work in such a manner as to prevent injury or damage to any portions of the existing work which remain.
- b. Repair or replace portions of existing work which have been altered during construction operations to match existing or adjoining work, as approved by the Contracting Officer. At the completion of operations, existing work must be in a condition equal to or better than that which existed before new work started.
- c. The responsibility for damage to the permanent work must be as set forth in the article of the contract clause entitled "PERMITS AND RESPONSIBILITIES". Except as herein provided, damage to all work, utilities, materials, structures, equipment, and plant, including temporary construction and utilities, pavements, vehicles, vessels, and other property along the routes used by the Contractor's plant, vessels, equipment, pipelines and/or vehicles must be repaired to the satisfaction of the Contracting Officer, at the Contractor's expense, regardless of the cause of such damage.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

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SECTION 01 20 00

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11/20, CHG 5: 02/25

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SECTION 01 20 00

PRICE AND PAYMENT PROCEDURES

11/20, CHG 5: 02/25

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals; G

1.2 SINGLE JOB PAYMENT ITEMS

Payment items for the work of this Contract for which Contract job payments will be made are listed in the SCHEDULE and described below. All costs for items of work, which are not specifically mentioned to be included in a particular job or unit price payment item, are included in the listed job item most closely associated with the work involved. The job price and payment made for each item listed constitutes full compensation for furnishing all plant, labor, materials, and equipment, and performing any associated Contractor quality control, environmental protection, meeting safety requirements, tests and reports, and for performing all work required for which separate payment is not otherwise provided.

1.2.1 Mobilization and Demobilization

1.2.1.1 Payment

Payment will be made for costs associated with mobilization and demobilization, as defined in Special Clause PAYMENT FOR MOBILIZATION AND DEMOBILIZATION.

Mobilization shall include all costs for operations accomplished prior to commencement of actual dredging operations, such as transfer of dredge and attendant plant and equipment to site, initial installations of a pipeline, preparation of a beach fill area, and other incidentals in advance of the actual dredging. Demobilization shall include general preparation for transfer of plant to home base, removal of pipelines, cleanup of beach fill areas, and transfer of a plant to its home base. The cost of work other than mobilization and demobilization of the Contractor's dredging plant and equipment shall not be included in this item.

All costs connected with the mobilization and demobilization of all of the contractor's dredging plant and equipment will be paid for at the contract lump sum price for this item. 60 percent of the lump sum price will be paid to the Contractor upon completion of his mobilization to the work site. The remaining forty percent 40 percent will be included in the final payment for work under this contract.

In the event the Contracting Officer and/or his Representative considers that the amount in this item 60 percent, which represents mobilization, does not bear a reasonable relation to the cost of the work in this contract, the Contracting Officer and/or his Representative may require

the Contractor to justify this portion of the bid. Failure to justify such price to the satisfaction of the Contracting Officer and/or his Representative will result in payment of actual mobilization costs as determined by the Contracting Officer and/or his Representative, at the completion of mobilization and payment of the remainder of this item in the final payment under this contract. The determination of the Contracting Officer and/or his Representative is not subject to appeal.

1.2.1.2 Unit of Measure

Unit of measure: job.

1.2.2 "Black Box" Monitoring Unit

1.2.2.1 Payment

Payment will be made for costs associated with operations necessary for furnishing, and equipping all container vessels used to transport dredged material to designated placement or treatment locations with container Vessel Monitoring Equipment ("Black Box") and software.

1.2.2.2 Unit of Measure

Unit of measure: job.

1.3 UNIT PRICE PAYMENT ITEMS

Payment items for the work of this Contract on which the Contract unit price payments will be made are listed in the SCHEDULE and described below. The unit price and payment made for each item listed constitutes full compensation for furnishing all plant, labor, materials, and equipment, and performing any associated Contractor quality control, environmental protection, meeting safety requirements, tests and reports, and for performing all work required for each of the unit price items.

1.3.1 DREDGING

1.3.1.1 Payment

Payment will be made for costs associated with dredging of the project, which includes performing required dredging and other operations incidental thereto Contractor-furnished disposal area(s) and disposition of dredged material.

1.3.1.2 Unit of Measure

Unit of measure: cubic yard (CY).

1.3.1.3 Measurement

The total amount of material removed and paid for under the contract will be measured by the cubic yard dredged. Measurement of the number of cubic yards dredged will be made by computing the volume between the bottom surface shown by soundings of the last survey made before dredging and the bottom surface shown by soundings of the survey made as soon as practicable after the entire work specified has been completed. The volume for measurement shall include the material within the limits of the overdepth, side slopes described in SECTION 35 20 23, Paragraph 1.5 Quantity of Material, 1.6 Overdepth Dredging, and 1.7 Side Slopes.

The sounding maps and/or drawings accurately represent the channel conditions existing in December 2025. The soundings shown on the contract drawings were taken with an electronic depth recorder and/or multi-beam surveying system with a transducer frequency of 200-400 KHz. New soundings will be taken generally two weeks prior to dredging. Soundings after dredging will be made by an electronic depth recorder with a transducer frequency of 200-400 KHz or may be made by using a multi-beam surveying system of the same frequency. Weather dependent, after-dredging-surveys will commence generally within five (5) days of completion of the project. Determination of quantities to be paid in the area specified after having been made once, will not be reopened, except on evidence of collusion, fraud or obvious error. Computations for payment purposes will generally be made by the Corps of Engineers utilizing the Triangulated Irregular Network (TIN) method.

Shoaling in the areas to be dredged on this project does not occur at a rapid rate from existing depths. However, the unit price for dredging based on the above method of computation of the quantity for which payment is to be made shall include the Contractor evaluation of shoaling, changes in the regime of the waterway, or changes caused by the Contractor's operations that may occur during the period between the survey before dredging and the survey for acceptance of work.

Scows and Hopper Dredges: Where scows and hopper dredges are used in performance of work and where the Contractor maintains load measuring devices on such scows or hopper dredges, copies of all recorded measurements made in connection with the work shall be furnished to the Contracting Officer and/or his/her representative. If as the work progresses the Contractor determines that a significant difference exists between the recorded bin or scow quantities and the survey quantities computed from the check surveys described above, he shall immediately notify the the Contracting Officer and/or his/her representative of any such discrepancies. Such notification shall include a description of the Contractor's methods of measuring scow or bin quantities, certification of measuring techniques, documentation of material sampling and methods used to convert bin or scow measurements to in-place density.

It is emphasized that bin or scow measurements shall not be used as basis for partial or final payments.

14-day partial payments will be based on approximate quantities determined by soundings or sweepings taken behind the dredge by the Contractor. The Contractor soundings are to be used for partial payments only and will not form a basis for the acceptance of the work.

The Contractor must inform the Government, in writing, within three calendar days of completing the work.

The total quantity of excavated material for which payment will be made will be the theoretical quantity between the ground surface as determined by a survey and the grade and slope of the theoretical cross sections indicated. No allowance will be made for the removal of any material outside the required slope lines unless authorized.

When the Government performs surveys using multibeam, the data will be binned using the median soundings for contours and volumes. The maximum bin size will not exceed 3'x3'.

In the event that unforeseen obstructions or hard material is discovered, shoals in these areas will be assessed using the minimum sounding within the bin. This method will not be used for computation.

1.4 CONTRACT MODIFICATIONS

In conjunction with the Contract Clause "DFARS 252.236-7000, Modification Proposals-Price Breakdown," and where actual ownership and operating costs of construction equipment cannot be determined from Contractor accounting records, equipment use rates shall be based upon the applicable provisions of the EP-1110-1-8.

1.5 PAYMENTS TO THE CONTRACTOR

Payments will be made on submission of itemized requests by the Contractor which comply with the requirements of this section, and will be subject to reduction for overpayments or increase for underpayments made on previous payments to the Contractor.

1.5.1 Obligation of Government Payments

The obligation of the Government to make payments required under the provisions of this contract will, at the discretion of the Contracting Officer, be subject to reductions and/or suspensions permitted under the FAR and agency regulations including the following in accordance with "FAR 32.503-6:

- a. Reasonable deductions due to defects in material or workmanship;
- b. Claims which the Government may have against the Contractor under or in connection with this contract;
- c. Unless otherwise adjusted, repayment to the Government upon demand for overpayments made to the Contractor; and
- d. Failure to provide up to date record drawings not current as stated in Contract Clause "FAC 5252.236-9310, Record Drawings.

1.6 SHOALING

If, before the contract is completed, shoaling occurs in the finished channel because of the natural lowering of the side slopes, re-dredging at the contract price, within the limit of available funds, may be done if agreeable to both the Contractor and the Contracting Officer and/or his Representative.

1.7 CONTINUITY OF WORK

No payment will be made for work done in any area designated by the Contracting Officer and/or his Representative until the full depth required under the contract is secured in the whole of such area, unless prevented by ledge rock; nor will payment be made for excavation in any area not adjacent to and in prolongation of areas where full depth has been secured except by decision of the Contracting Officer or Contracting Officer's Representative. Should any such nonadjacent area be excavated to full depth during the operations carried on under the contract, payment for all work therein may be deferred until the required depth has been made in the area intervening. The Contractor may be required to suspend dredging at any time when for any reason the gauges or ranges cannot be seen or properly

followed.

PART 2 PRODUCTS

2.1 QUANTITY SURVEYS

a. Quantity surveys shall be conducted, and the data derived from these surveys shall be used in computing the quantities of work performed and the actual construction completed and in place.

b. The Government shall conduct the before-dredge and after-dredge surveys and make the computations based on them. The Contractor shall conduct the survey for any periods for which progress payments are requested and shall make the computations based on these surveys. These surveys conducted by the Contractor shall be conducted under the direction of a representative of the Contracting Officer, unless the Contracting Officer waives this requirement in a specific instance.

c. Promptly upon completing a survey, the Contractor shall furnish the originals of all field notes and all other records relating to the survey or to the layout of the work to the Contracting Officer and/or his Representative, who shall use them as necessary to determine the amount of progress payments. Raw data, edited survey data with all supporting files and documentation, and all data generated during such work shall be furnished by the Contractor to the Contracting Officer and/or his Representative. The Contractor shall retain copies of all such material furnished to the Contracting Officer and/or his Representative. (FAR 52.236-16). The Government will not commence work on the after-dredge survey until the Contractor has furnished the originals of all field notes, all other records relating to the Contractor's final survey or to the layout of the work, and all data generated during such work, to the Contracting Officer and/or his Representative.

d. Contractor's participation in surveys: In accordance with agency policy guidance, the preferred methods of performance of all hydrographic quantity surveys shall be (in descending order):

(1) The Government will perform quantity surveys by using qualified in-house crews, if available.

(2) The Government will provide quantity surveys by contracting directly with qualified independent hydrographic survey contractors.

(3) The Government will permit, only in exceptional circumstances, the use of dredging contractor's surveys if the Contracting Officer and/or his Representative determine that such surveys are adequate and reasonable for payment purposes and a Government inspector, qualified in hydrographic surveying, is present during the collection of the survey data. Raw data, edited survey data with all supporting files and documentation, and all data generated during such work shall be furnished by the Contractor to the Contracting Officer and/or his Representative.

If it is determined by the Contracting Officer that it is impractical for Government personnel to perform the before-dredge and after-dredge surveys and the Government wishes the Contractor's surveys to be used, the following will be required from the Contractor:

- (1) Acceptance of all monumentation and transponder locations.
- (2) Witness by a Corps of Engineers representative of all calibration procedures such as vertical and horizontal control.
- (3) Corps of Engineers observation of raw data collection.
- (4) Submittal of both unedited as well as edited versions of survey maps.
- (5) Subsequent acceptance of edited survey map after determination by Corps survey personnel that editing was conducted in accordance with U.S. Army Corps of Engineers specifications and guidelines.
(FAR 52.236-16 - ALTERNATE I, APR 1984)

e. Any disagreements with the data collection process, and / or any data generated during such work shall be brought to the attention of the Contracting Officer or his representative within five (5) business days of the data collection process. The Contracting Officer or his representative will then determine the necessary corrective measures, if any. Failure to notify the Contracting Officer or his representative within 5 business days of the data collection process, disqualifies the Contractor from any such related claims.

f. Any survey related disagreement(s) as it pertains to this contract must be submitted in writing to the Contracting Officer or his representative within 5 business days of occurrence of such survey related event. Failure to notify the Contracting Officer or his representative within 5 business days of such survey related event disqualifies the Contractor from any such related claims.

PART 3 EXECUTION

Not used.

-- End of Section --

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-- End of Section Table of Contents --

SECTION 01 30 00

ADMINISTRATIVE REQUIREMENTS

05/21

PART 1 GENERAL

1.1 SUMMARY

The requirements of these section apply to, and are component part of, each section of the project specifications.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation, submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Pre-Construction Submittals, G

Preconstruction Submittals shall be submit to the Contracting Officer or Contracting Officer Representative at least seven (7) days prior to the preconstruction conference. Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following:

Certification of Insurance; G

Surety Bond; G

List Of Proposed Subcontractors; G

Order of Work; G The Contractor's order of work for dredging and performing other work, including disposal area plan.

Work To Be Performed By The Contractor; G,

Subcontracting Plan; G

Progress Schedule; G

Submittal Register; G

Letter Appointing Superintendent; G

Activity Hazard Analysis; G See Section 01 35 26, GOVERNMENTAL SAFETY REQUIREMENTS

Accident Prevention Plan; G See Section 01 35 26, GOVERNMENTAL SAFETY REQUIREMENTS

Work Plan; G

Quality Control (QC) plan, G See Section 01 45 00.00 20

Environmental Protection Plan; G

Notice to Mariners; G

SD-03 Product Data G

Monthly Progress Report

Project Schedule Status Report

The Contractor shall submit an updated Monthly Progress Report at the first meeting of each month, including a Project Schedule Status Report and if no meeting is in the agenda, these items shall be submitted to the Contracting Officer Representative when requested.

SD-11 Closeout Submittals

As-Built Drawings; G

1.3 PRECONSTRUCTION CONFERENCE

Work shall not commence prior to the conference.

After award of contract, a pre-construction conference will be held at such time and location as determined by the Contracting Officer for purposes of discussing and developing mutual understanding between the Contracting Officer or his authorized representative and the Contractor's Representatives regarding the terms, conditions, and requirements of the contract. Members of the conference from the Government will include the quality assurance staff, the Contracting Officer or his authorized representative, and construction staff. Members from the Contractor shall include the site superintendent, project safety officer, and the quality control staff. The discussion will include, and may not be limited to, the following:

- Contractor Supervisory and Quality Control staff.
- Correspondence Procedures
- Accident Prevention Plan
- Pollution Control Plan
- Quality Control
- Project Scheduling and Payment Procedures
- Other subjects that may be of interest to the contracting parties
- Schedule of Weekly Meeting

1.4 PROJECT MEETINGS

The Contractor and/or Contractor Representatives shall attend Weekly Project Meetings scheduled by the Government in accordance with Section 01 32 01.00 10 PROJECT SCHEDULE. The Contractor's Superintendent shall be in attendance at these scheduled project meetings.

A Monthly Progress Report shall include a Project Schedule Status Report per requirement of Section 01 32 01.00 10 PROJECT SCHEDULE.

1.5 PERFORMANCE OF WORK BY THE CONTRACTOR

The Contractor shall perform on the site, and with its own organization and equipment, work equivalent to at least forty (40) percent of the total amount of work to be performed under the contract. This percentage may be reduced by a supplemental agreement to this contract if, during performance of the work, the Contractor requests a reduction and the Contracting Officer and/or his Representative determines that the reduction would be to the advantage of the Government. (FAR 52.236-01)

1.6 SUBMITTAL OF WORK TO BE PERFORMED BY THE CONTRACTOR

The Contractor shall furnish the Contracting Officer and/or his Representative, within five (5) days after award, items of work he will perform with his own forces and the estimated cost of those items. The percentage of work that must be performed by the Contractor is stated in paragraph 1.5, entitled: "PERFORMANCE OF WORK BY THE CONTRACTOR."

1.7 LIQUIDATED DAMAGES-CONSTRUCTION

If the Contractor fails to complete the work within the time specified in the contract, the Contractor shall pay liquidated damages to the Government in the amount of \$2,987 for each calendar day of delay until the work is completed or accepted.

If the Government terminates the Contractor's right to proceed, liquidated damages will continue to accrue until the work is completed. These liquidated damages are in addition to excess costs of repurchase under the Termination clause.

1.8 PHYSICAL CONDITIONS

The information and data furnished or referred to below are not intended as representations or warranties but are furnished for information only. It is expressly understood that the Government will not be responsible for the accuracy thereof or for any deduction, interpretation or conclusion drawn by the Contractor.

a. WEATHER CONDITIONS. The site of the work is not exposed to severe wind and wave action and the Contractor's operations are not likely to be affected by ordinary storms. Real-time data are available on the World Wide Web at <http://www.ndbc.noaa.gov/index.shtml>.

(1) Fog. An examination of the records of the U.S. Weather Bureau at Newark International Airport, New York, discloses an average of 105 days per year on which fog occurs in the New York Harbor area. The maximum number of days in which dense fog (visibility <1/4 mile) occurred was 3 days.

(2) Winds and Waves. Wind and wave climate in the New York Harbor area, based upon hourly data obtained from NOAA's National Buoy Center (NDBC) observation at Station SDHN4 - Sandy Hook, NJ (8531680) is provided below:

MONTH	AVE WIND SPEED (MPH)	PEAK WIND GUST (MPH)
JAN	12.1	52.8
FEB	12.3	53.5
MAR	11.3	60.6

APR	10.1	48.3
MAY	8.9	49.0
JUN	8.3	42.2
JUL	8.1	55.7
AUG	8.2	57.5
SEP	9.1	108.5
OCT	10.7	149.7
NOV	10.5	52.6
DEC	12.7	96.4

(3) Weather conditions exceeding in severity the fog conditions described above or average wind speeds exceeding 25 mph, or wave heights exceeding 3 feet will be considered unusually severe if delays are caused thereby to operations for more than 50% of the day under this contract.

(4) ICE. Icing within the project vicinity during the contract period is not anticipated. However, in the event that icing does occur, this will be considered to be unusually severe weather if delays are caused thereby to operations for more than 50% of the day under this contract.

(5) TIDES The mean range of tide at Keyport, Raritan Bay, New Jersey (8531545) is 5.05 feet MLLW and the mean range of spring tides is 6.06 feet MLLW. Irregular fluctuations due to wind and atmospheric pressure have varied from 12.60 feet above MLLW and 2.20 feet below MLLW for the dredging area.

b. BRIDGE AND UTILITY CROSSINGS.

UTILITY CROSSINGS:

A search of the regulatory files within New York District, United States Army Corps of Engineers disclosed the following structures crossing the New York and New Jersey Channels - Seguine/Ward/Outerbridge Federal Navigation Project in the vicinity of the area to be dredged:

ITEM NO.	PERMIT NO.	DESCRIPTION
1	2017-00886-ESW	06 December 2018: Construction of Pile supported fender system around three of the existing mid river piers supporting the Outerbridge Crossing. The pile supported fender system consists of approximately two hundred and forty six d foot diameter steel pipes. The piles will be installed around a 75 foot standoff perimeter around each of the bridge piers.

ITEM NO.	PERMIT NO.	DESCRIPTION
2	2016-0098	20 December 2025: The Raritan Bay Loop work involves a 26 inch diameter natural gas pipeline beginning in Sayreville, Middlesex County, NJ. The pipeline will extend offshore, approximately 23.33 miles across Raritan Bay and Lower New York to Atlantic Ocean
3	2017-00253	20 February 2019: Installation of a single 141 inch diameter steel mooring buoy, a 30,000-lb Danforth anchor and associated hardware adjacent to the South Amboy Reach and the Raritan River Cutoff channels. On scene water depth is 12 feet MLLW. The barges would measure 40 foot wide by 150 foot long. Anchor Position 40.491683 N 74.265533 W
4	2017-00425	30 January 2018: NJ Transit construction of new replacement submarine fiber optic cables under the Raritan River. The activities include construction of 3241 linear feet of two inch diameter fiber optic cables beneath waters. Horizontal directional drilling, beginning upland at 40.492636, -74.282739 will be conducted to install the new fiber that will also terminate upland of the water at 40.5000953, -74.277976. The cables will extend from the City of South Amboy to the City of Perth Amboy, Middlesex County, NJ.
5	2002-00134	16 February 2005: A high-voltage direct-current submarine transmission cable roughly 4 to 6 feet below the seabed in Raritan Bay, south of Ward Point bend. The Department of the Army permit No. 2002-00134 dated 16 February 2005 issued to Neptune Regional Transmission System, LLC.

ITEM NO.	PERMIT NO.	DESCRIPTION
6	7509 & 7595	29 June 1967: A cathodic protected submarine cable in Raritan Bay. The Department of the Army permit No. 7509 dated 29 June 1967 & permit No. 7595 dated 20 December 1967 issued to the Transcontinental Gas Pipe Line Corporation.
7	--	5 March 1968: A submarine cable crossing from the foot of Smith Street, Perth Amboy, Middlesex County, N.J., to Tottenville, Staten Island, Richmond County, N.Y. New Jersey Telephone Company.
8	8875	28 June 1974: Install two power cables and one hose on the existing river bottom; under Arthur Kill, at the Outerbridge Crossing, Staten Island, N.Y., in accordance with the plans and drawings attached hereto and marked 1522-15 (Arthur Kill - Interstate Sanitation Commission - submarine cables and hose @ Outerbridge Crossing). Permit issued to: Interstate Sanitation Commission, 10 Columbus Circle, Room 1620, New York, New York 10019.
9	238/28	August 1928: Two 3-1/2" submarine telephone cables on the nature bottom crossing the Arthur Kill between the foot of Commerce Street in Perth Amboy, NJ and a point about 300' north of the foot of Amboy Avenue in Tottenville, Staten Island, NY. The NJ Bell Telephone, 1060 Broad Street, Newark, NJ.

ITEM NO.	PERMIT NO.	DESCRIPTION
10	Nation Wide General Permit No. 12	22 April 1999: Install fiber optic cable crossing the Arthur Kill between Fayette Street, Perth Amboy, NJ and Amboy Road, Staten Island, NY at depth of 80' below MLW. Bell Atlantic-NJ, c/o Federic R. Harris, Inc., 485B Route 1 South, Iselin, NJ 08830.
11	--	18 September 1964: To lay a 3" telephone cable crossing Arthur Kill on natural bottom between Smith Street, Perth Amboy, NJ and Amboy Avenue, S.I., NY. NJ Bell Telephone Co., 540 Broad Street, Newark, NJ.
12	260/13	14 September 1925: Install a 26,000 volt submarine cable crossing Arthur Kill between Perth Amboy, NJ and Tottenville, SI, NY. Public Service Production Co., Newark, NJ.
13	3816/5	1940: Install 2 submarine cables at a trench at a minimum depth of 42' in Arthur Kill between Bentley Ave, Tottenville, SI and Buckingham Ave, Perth Amboy, NJ.
14	849	18 September 1927: Lay two power submarine cables in Arthur kill between Buckingham Ave Perth Amboy NJ and N of Ferry Slip @ SI Rapid Transit, Tottenville SI, - Public Service Production Co. Both cables removed in 1945.
15	760	29 April 1924: Lay two telephone cables in Arthur kill on the nature bottom, between foot of Commerce Street, Perth Amboy NJ and Foot of Amboy Ave, SI, NY.

The Contractor shall take all necessary precautions to prevent damage to these structures. The Contractor is required to coordinate with the owners to mark the structures or move the cables in order to avoid interference with dredging operations. Copies of permits are available upon request. Please allow sufficient lead time.

c. LOCATION. The location of the work to be done under these specifications is as follows:

1. New York and New Jersey Channels - Seguine/Ward/Outerbridge is located along the south western shores of Staten Island, Richmond County, NY. The channel begins at Seguine Point and continues around Ward's Point, terminating at Outerbridge Crossing.

2. The Historic Area Remediation Site (HARS) is a 15.7 square nautical mile area located approximately 3.5 nautical miles east of Highlands, New Jersey and 18.9 nautical miles south of Red Hook, Kings County, New York.

d. CHANNEL AND TRANSPORTATION FACILITIES. NY & NJ Channels, Seguine Point to Outerbridge Crossing: Based on a plans and specifications survey dated 30 December 2025, a controlling depth of approximately -27.9 feet below MLLW is present at the site of the work. Channel users consist of deep draft seagoing bulk and general cargo freighters, tankers, and miscellaneous small craft vessels. Vessel traffic may cause some delay to the dredging required in these specifications.

e. CHANNEL TRAFFIC. Vessel traffic in the Seguine/Ward/Outerbridge reaches is heavy. Traffic consists of seagoing bulk and general cargo freighters, tankers, tows and scows and barges, and miscellaneous small craft. There is also potential for other construction activity ongoing in the vicinity, which may cause some delay in the dredging required in these specifications. Vessel traffic may cause some delay to the dredging required in these specifications.

e. OBSTRUCTION OF CHANNEL. The Government will not undertake to keep the area free from vessels or other obstructions except to the extent of such regulations, if any, as may be prescribed by the Secretary of the Army in accordance with the provisions of section 7 of the Rivers and Harbors Act in such manner as to obstruct navigation as little as possible, and in case the contractor's plant so obstructs the channel as to make difficult or endanger the passage of vessels, said plant shall be promptly moved on the approach of any vessel to such an extent as may be necessary to afford a practicable passage. Upon the completion of the work the Contractor shall promptly remove his plant, including ranges, buoys, piles and other marks placed by him under the contract in navigable waters or on shore.

f. BRIDGE-TO-BRIDGE RADIO-TELEPHONE EQUIPMENT. In order that radio-telephone communication may be made with passing vessels, all dredges engaged in work under the contract shall be equipped with and operate bridge-to-bridge radio-telephone equipment on VHF Channel 13 (156.65 MHz with low power output having a communication range of approximately ten (10) miles). The Federal Communication Commission has approved the frequency.

1.9 ESTIMATED QUANTITIES

The total estimated quantities of material necessary to be removed from the specified limits to complete the work described in this specifications are computed from the plans and specifications survey performed in the NY & NJ Channels - Seguine Point to Outer Bridge Crossing area completed on 30 December 2025.

Distribution of volumes is summarized below:

New York and New Jersey Channels - SeguineW/Ward/Outerbridge Federal Navigation Project

WORK AREA	REQUIRED DREDGING DEPTH (FT.) MEAN LOWER LOW WATER (MLLW)	CHANNEL TO GRADE (CY)	CHANNEL OVERDEPTH 2 FEET (CY)	SLOPE TO GRADE (CY)	SLOPE OVERDEPTH 2 FEET (CY)	ESTIMATED TOTAL (CY)
BASIC WORK	35	99,360	160,460	40,895	29,975	330,690

Total Estimated Quantity Available: 330,690 CY

The Government will survey the project prior to starting work.

Above estimated quantities were computed from the data shown on the contract drawings. The quantities that may be removed and paid for will be computed by the Triangulated Irregular Network (TIN) method or average end area method as indicated in the Technical Specifications, Section 01 20 00., entitled "PRICE & PAYMENT PROCEDURES."

1.10 VARIATION IN ESTIMATED QUANTITIES

Where the quantity of a pay item in this contract is an estimated quantity and where the actual quantity of material within the required dredging prism varies more than fifteen percent (15%) above or below the stated estimated quantity within the required dredging prism, an equitable adjustment in the contract unit price will be made upon the demand of either party. The equitable adjustment will be based upon any increase or decrease in costs due solely to the variations above one-hundred fifteen percent (115%) or below eighty-five percent (85%) of the estimated quantity. An adjustment in the contract unit price will also apply to that part of the actual quantity of allowable overdepth material above one-hundred fifteen percent (115%) or below eighty-five percent (85%) of the estimated quantity. (UAI 52.211-5001 Variation in Estimated Quantities MAR 1995)

1.11 FINAL EXAMINATION AND ACCEPTANCE

a. As soon as practicable after the completion of the entire work, as in the opinion of the Contracting Officer and/or his Representative, will not be subject to damage by further operation under the contract, such work will be thoroughly examined at the cost and expense of the Government by sounding or by sweeping, or both, as determined by the Contracting Officer and/or his Representative. Only after the contractor has submitted a complete survey of the entire contract area(s), and submitted mapping and all data acquired to create such map, will the government perform its survey. Raw data, edited survey data with all supporting files and documentation, and all data generated during such work shall be furnished by the Contractor to the Contracting Officer and/or his Representative. Should any shoals, lumps, or other lack of contract depth be disclosed by this examination, the Contractor will be required to remove same by dredging at the contract rate for dredging or by other method approved by the contracting officer, but if the bottom is soft and the shoal areas are

small and form no material obstruction to navigation, the removal of such shoals may be waived by the discretion of the Contracting Officer and/or his Representative. The Contractor or his authorized representative will be notified when soundings and/or sweepings are to be made, and will be required to accompany the survey party. When the entire contract area(s) is found to be in a satisfactory condition, it will be accepted as final. Should more than two soundings or sweeping operations by the Government over an area be necessary by reason of work for the removal of shoals disclosed at a prior sounding or sweeping operations, this work will be charged against the Contractor. The rate for each day in which the Government plant is engaged in such sounding or sweeping operations and/or is en route to or from the site or held, for the Contractor's convenience, at or near the site for these operations shall be N/A.

b. Final acceptance of the entire work and the deductions or corrections of deductions made thereon will not be reopened after having once been made, except on evidence of collusion, fraud, or obvious error, and the acceptance of the entire work shall not change the time of payment of the retained percentages of the whole or any part of the work.

c. The entire area within the contract limits will be surveyed for final examination and acceptance. This contract has no partial acceptance areas.

d. All hydrographic survey data is to be collected in accordance with the Army Corps of Engineers manual EM-1110-2-1003.

e. Any survey related disagreement(s) as it pertains to this contract must be submitted in writing to the Contracting Officer or his Representative within 5 business days of occurrence of such survey related event. Failure to notify the Contracting Officer or his representative within 5 business days of such survey related event disqualifies the Contractor from any such related claims.

1.12 PERFORMANCE EVALUATION OF CONTRACTOR

a. As a minimum, the Contractor's performance will be evaluated upon final acceptance of the work. However, interim evaluation may be prepared at any time during contract performance when determined to be in the best interest of the Government.

b. The Contractor will be rated either Exceptional, Very Good, Satisfactory, Marginal, or Unsatisfactory in the areas of Contractor Quality Control, Timely Performance, Effectiveness of Management, Compliance with Labor Standards, and Compliance with Safety Standards. The Contractor will be advised of any unsatisfactory rating, either in an individual element or in the overall rating, prior to completing the evaluation, and all Contractor comments will be maintained as part of the official record. Performance Evaluation Reports will be available to all DOD contracting offices for their future use in determining Contractor responsibility, in compliance with FAR 36.201 (c).

c. The Federal Acquisition Regulation (FAR) requires that Past Performance Information (PPI) be collected on DoD contracts. The Contractor Performance Assessment Reporting System (CPARS) is a web-enabled application the supports the completion, distribution and retrieval of all contract performance evaluations. Contractor Representative is responsible for reviewing and commenting on evaluations submitted on the Contractor Performance Assessment Reporting System (CPARS) website at <https://www.cpars.gov/index.htm>.

1.13 CERF IMPLEMENTATION

If the work specified in this contract is performed by a hopper dredge(s), the owner must have an active Basic Ordering Agreement (BOA) for the hopper dredge(s) on file with the Corps of Engineers. The Contractor shall be obligated to make the hopper dredge(s) available to serve in the Corps of Engineers Reserve Fleet (CERF) at any time that the hopper dredge(s) is performing work under this contract. When the Contracting Officer and/or his Representative is notified of the decision to activate the dredge(s) into the CERF, he shall take appropriate action to release the dredge(s). He may then extend or terminate the contract to implement whichever action is in the best interest of the Government. The CERF contract shall also be subject to the following condition:

- a. The Director of Civil Works may require the contractor to perform emergency dredging at another CONUS (48 contiguous states) site for a period of time equal to the remaining time under this contract at the date of notification plus up to ninety (90) days at the previously negotiated rate which appears on the schedule of prices in the BOA.
- b. The Chief of Engineers may require the contractor to perform emergency dredging at an OCONUS (Outside CONUS which includes Alaska, Hawaii, Puerto Rico, the Virgin Islands, or U.S. Trust Territories) site for a period of time equal to the time remaining under this contract at the date of notification plus up to one hundred eighty (180) days at the negotiated rate which appears on the schedule of prices in the BOA.
- c. The CERF shall be activated by the Chief of Engineers or the Director of Civil Works; then the Ordering Contracting Officer and/or his Representative will notify the contractor. From the time of notification, the selected hopper dredge(s) must depart for the emergency assignment within seventy-two (72) hours for CONUS or ten (10) days for OCONUS assignments.
- d. A confirming delivery order will be issued pursuant to the Basic Ordering Agreement (BOA) by the Ordering Contracting Officer and/or his Representative. Such delivery order shall utilize the schedule of rates in the BOA for the specific hopper dredges(s).
- e. If during the time period specified in a, b, or c, above, a CERF vessel(s) is still required, the contract performance may be continued for additional time by mutual agreement.

1.14 THE SEAGOING BARGE ACT (46 USC 395 ET SEQ)

The Seagoing Barge Act applies to this project. In the event the Apparent Low Bidder contemplates using plant that requires U.S. Coast Guard certification to comply with this Act, the Apparent Low Bidder shall within five (5) calendar days after bid opening submit a copy of said certificate to the Contracting Officer and/or his Representative. Failure to produce the certificate within the required time shall subject the bidder to a determination of non-responsibility.

1.15 PRICING OF ADJUSTMENTS

When costs are a factor in any determination of a contract price adjustment pursuant to the Changes clause or any other clause of this contract, such costs shall be in accordance with Part 31 of the Federal Acquisition Regulation and the DOD FAR Supplement in effect on the date of this

contract (DOD FAR SUPP 52.243.7001) costs shall be in accordance with Part 31 of the Federal Acquisition Regulation.

1.16 BID GUARANTEE

a. The offeror (bidder) shall furnish a separate bid bond, or United States bonds, Treasury notes or other public debt obligation of the United States, in the proper amount, by the time set for opening of bids. Failure to do so may be cause for rejection of the bid. The Contracting Officer and/or his Representative will return bid guarantees, other than bid bonds, (1) to unsuccessful bidders as soon as practicable after the opening of bids, and (2) to the successful bidder upon execution of contractual documents and bonds (including any necessary coinsurance or reinsurance agreements), as required by the bid as accepted.

b. If the successful bidder, upon acceptance of its bid by the Government within the period specified for acceptance, fails to execute all contractual documents or give a bond(s) as required by the solicitation within the time specified, the Contracting Officer and/or his Representative may terminate the contract for default.

c. Unless otherwise specified in the bid, the bidder will (1) allow the number of days as indicated in paragraph 13D of Standard Form 1442 for acceptance of its bid and (2) give bond within 5 days after receipt of the forms by the bidder.

d. In the event the contract is terminated for default, the bidder is liable for any cost of acquiring the work that exceeds the amount of its bid. The bid bond, or bonds or notes of the United States, is available to offset the difference.

1.17 EQUAL OPPORTUNITY PREAWARD CLEARANCE OF SUBCONTRACTORS

Notwithstanding the clause of this contract entitled "Subcontracts," the Contractor shall not enter into a first-tier subcontract for an estimated or actual amount of \$1 million or more without obtaining in writing from the Contracting Officer and/or his Representative a clearance that the proposed subcontractor is in compliance with the equal opportunity requirements and therefore is eligible for award.

1.18 SPECIAL PROHIBITION ON EMPLOYMENT

a. Definitions: "Arising out of a contract with the Department of Defense," as used in this clause, means any act in connection with (1) attempting to obtain, (2) obtaining, or (3) performing a contract or subcontract of any agency, department or component of the Department of Defense.

b. "Convictions of fraud or any other felony," as used in this clause, means any conviction for fraud or a felony in violation of state or Federal criminal statutes, whether entered on a verdict or plea, including a plea of nol contendere, for which sentence has been imposed.

c. Section 941, Title IX, Pub. L. 99-500 (10 U.S.C. 2408) prohibits a person who is convicted of fraud or any other felony arising out of a contract with the Department of Defense from working in a management or supervisory capacity on any defense contract, or serving on the board

of directors of any defense contractor, for a period, as determined by the Secretary of Defense, of not less than one (1) year from the date of conviction. Defense contractors are subject to a criminal penalty of not more than \$500,000 if they are convicted of knowingly employing a person under a prohibition or allowing such person to serve on their board of directors.

d. The Contractor agrees not to knowingly employ any person, convicted of fraud or any other felony arising out of a contract with the Department of Defense contract or subcontract or allow such person to serve on its board of directors from the date the Contractor learns of the conviction until one (1) year has expired from the date of conviction. However, if the person has also been debarred pursuant to FAR subpart 9.4, the above prohibition shall extend for the period of debarment, but in no event shall the prohibition be less than (1) year from the date of conviction.

e. If the Contractor knowingly employs a convicted person in a management or supervisory capacity on any defense contract or subcontract or knowingly allows such person to serve on its board of directors within the prohibited period, the Government may consider, in addition to the criminal penalties contained in Section 941 of Pub. L. 99-500, other available remedies, such as suspension or debarment and may direct the cancellation of this contract at no cost to the Government, or terminate this contract for default.

f. The Contractor agrees to include the substance of this clause, including this paragraph (e), appropriately modified to reflect the identity and relationship of the parties, in all subcontracts exceeding \$25,000.

1.19 SUPERINTENDENCE OF SUBCONTRACTS

a. The Contractor shall be required to furnish the following, in addition to the superintendence required by the Contract Clauses entitled: "Superintendence by the Contractor."

(1) If more than 50% and less than 70% of the value of the contract work is subcontracted, one superintendent shall be provided at the site and on the Contractor's payroll to be responsible for coordinating, directing, inspecting and expediting the subcontract work.

(2) If 70% or more of the value of the work is subcontracted, the Contractor shall be required to furnish two such superintendents to be responsible for coordinating, directing, inspecting and expediting the subcontract work.

b. If the Contracting Officer, at any time after 50% of the subcontracted work has been completed, finds that satisfactory progress is being made, he may waive all or part of the above requirement for additional superintendence subject to the right of the Contracting Officer to reinstate such requirement if at any time during the progress of the remaining work he finds that satisfactory progress is not being made. (DOD FAR Supplement 52.236-7008)

1.20 PROCEDURES FOR SUBMISSION & CONTRACT PAYMENTS

In addition to the requirements contained in the Contract Clause entitled "Payments Under Fixed-Price Construction Contracts" and to implement the requirements of the Prompt Payment Act Amendments of 1988, P.L. 100-496, the following shall apply to all payments made under this contract:

a. At the time of submission of the progress chart, the contractor shall submit for approval by the Contracting Officer or his authorized representative a breakdown of the contract work which shall be to the degree of detail required by the Contracting Officer or his representative to effect reasonable progress payments. The Contracting Officer or his representative shall review this breakdown within 30 calendar days after receipt and either advise the contractor that it is approved or disapproved, and if disapproved the reasons for disapproval. Only after the breakdown is approved shall any payment invoice be accepted from the contractor and any payment made to him. The Contracting Officer can determine it is in the best interest of the Government to make payment without an approved breakdown, however, in no case shall more than 10% of the contract amount be paid unless the breakdown is approved.

b. The contractor shall submit his request for payment by submission of a proper invoice to the office or person(s) designated in paragraph "c". For purposes of payment a "proper invoice" is defined as the following:

- (1) An estimate of the work completed in accordance with the approved breakdown indicating the percentage of work of each item and the associated costs.
- (2) A properly completed ENG Form 93 and 93a (where required).
- (3) All contractual submissions indicated elsewhere in this contract to be submitted with payment, such as updated progress schedules, updated submittal registers, etc.

The following certification executed by a responsible official of the organization authorized to bind the firm. A "responsible official" would be either a corporate officer, partner, or owner, in the case of a sole proprietorship:

I hereby certify, to the best of my knowledge and belief, that:

- (1) The amounts requested are only for performance in accordance with the specifications, terms and conditions of the contract;
- (2) Payments to subcontractors and suppliers have been made from previous payments received under the contract and timely payments will be made from the proceeds of the payment covered by this certification, in accordance with subcontract requirements and the requirements of Chapter 39 of Title 31, United States Code; and
- (3) This request for progress payments does not include any amounts, which the prime contractor intends to withhold or retain from a subcontractor or supplier in accordance with the terms and conditions of the subcontract.
- (4) All required prime and subcontractor payrolls have been

submitted.

(Name)

(Title)

(Date)

c. The Government shall designate the office or person(s) who shall first receive the invoice submissions and the Contractor shall be notified at the preconstruction conference. In addition, the contractor shall at the same time submit one copy of the detailed breakdown and the ENG Form 93 and 93a Form to the Project Engineer.

d. The Government representative shall return any request for payment, which is deemed defective within 7 days of receipt and shall specify the defects. Telephone calls with MFR or E-mail notification of the contractor about the defective invoice shall be considered sufficient notice to the contractor. If the defect concerns a disagreement as to the amount of work performed and or the amount of the payment being submitted, the Government and the contractor's representative should meet to resolve the differences and reach agreement. Upon agreement, the contractor shall submit a new breakdown and ENG Form 93 (and 93a) and any other submissions requiring correction. These will be incorporated with the previous submittal and will then constitute a proper invoice.

e. If agreement cannot be reached, the Government shall determine the proper amount per Contract Clause entitled: "Payments Under Fixed-Price Construction Contracts" and process the payment accordingly. In this event, "a proper invoice" for Prompt Payment Act purposes will not have been submitted to the Government.

f. The Government shall pay the contractor in accordance with the following time frames:

(1) Progress Payments. From the date a "Proper invoice" is received, in accordance with subparagraphs "b" and "d" of this clause, the Government will issue a check within 14 calendar days.

(2) Reduction in Retaining Payment. If during the course of the contract, a reduction in retainage payment is required, the Government shall issue a check within 30 calendar days after the approval of the release to the contractor by the Contracting Officer or his authorized representative.

(3) Final Payment. A final payment request shall not be considered valid until the contractor has fulfilled all contract requirements including all administrative items, payrolls, warranties, etc. and has submitted a release of claims. When the contractor has fulfilled all contract requirements and a "proper invoice" has been submitted, the Government shall issue a check within 30 days from the date of acceptance of the project by the

Contracting Officer.

1.21 SUBMISSION OF CLAIMS

The following shall be submitted to the Contracting Officer at the following address: U.S. Army Corps of Engineers, New York District, 26 Federal Plaza, New York, New York 10278-0090:

- a. Claims referencing or mentioning the Contract Disputes Act of 1978.
- b. Requests for a written decision by the Contracting Officer.
- c. Claims certified in accordance with the Contract Disputes Act of 1978.

No other Government representative is authorized to accept such requests. A copy shall also be provided to the Authorized Representative of the Contracting Officer.

The Contractor shall also provide the Contracting Officer with a copy of any requests for additional time, money or interpretation of contract requirements which were provided to the Authorized Representative of the Contracting Officer and which have not been resolved after 90 days.

1.22 OPERATOR QUALIFICATIONS FOR FLOATING CRANES & FLOATING DERRICKS

All floating cranes and floating derricks may be only operated by qualified operators. Operators shall be designated as qualified by a source which qualifies crane and derrick operators (e.g. an independent testing and qualifying company, a union, a governmental agency, a qualified consultant (can be an in-house resource)). Qualification shall be written (or oral) and practical operating examination unless the operator is licensed by a state or city licensing agency for the particular type of crane or derrick. Proof of qualification of crane and derrick operators shall be provided to the Contracting Officer's representative prior to the start of work.

1.23 EQUIPMENT OWNERSHIP & OPERATING EXPENSE SCHEDULE

a. Allowable costs for construction of marine plant and equipment in sound workable condition owned or controlled and furnished by a contractor or subcontractor at any tier shall be based on actual cost data when the Government can determine both ownership and operating costs for each piece of equipment or equipment groups of similar serial and series from the contractor's accounting records. When both ownership and operating costs cannot be determined from the contractor's accounting records, equipment costs shall be based upon the applicable provision of EP 1110-1-8, "Construction Equipment Ownership and Operating Expense Schedule," Region I. Working conditions shall be considered to be average for determining equipment rates using the schedule unless specified otherwise by the contracting officer. For equipment not included in the schedule, rates for comparable pieces of equipment may be used or a rate may be developed using the formula provided in the schedule. For forward pricing, the schedule effect at the time of negotiations shall apply. For retrospective pricing, the schedule in effect at the time the work was performed shall apply.

b. Equipment rental costs are allowable, subject to the provisions of

FAR 31.105(d)(ii) and FAR 31.205-36 substantiated by certified copies of paid invoices. Rates for equipment rented from an organization under common control, lease-purchase or sale-leaseback arrangements will be determined using the schedule except that rental costs leased from an organization under common control that has an established practice of leasing the same or similar equipment to unaffiliated leases are allowable. Costs for major repairs and overhaul are unallowable.

c. When actual equipment costs are proposed and the total amount of the pricing action is over \$25,000, cost or pricing data shall be submitted on Standard Form 1411, "Contract Pricing Proposal Cover Sheet." By submitting cost or pricing data, the contractor grants to the contracting officer or an authorizing representative the right to examine those books, records, documents and other supporting data that will permit evaluation of the proposed equipment costs. After price agreement the contractor shall certify that the equipment costs or pricing data submitted are accurate, complete and current. (UAI 31.105-101 Equipment Ownership & Operating Expense Schedule MAR 1995)

1.24 WAGE RATES

The attached schedule of wage rates contains rates applicable to all work under this contract (see Section 00 90 00).

1.25 LABOR-ADDITIONAL REQUIREMENTS

Fringe benefits statement: The method of payment of applicable fringe benefits will be indicated on DD Form 879, Statement of Compliance, and attached to each weekly payroll.

1.26 NOTICE TO MARINERS

The Contractor shall, prior to commencement of work, notify the 1st Coast Guard District, Aids to Navigation Officer, of his proposed operations including location and duration of operations and request that the information be published in the "Notice to Mariners." This notification must be in sufficient time so that it appears in the Notice at least 30 days prior to the commencement of the dredging operation.

1.27 ENVIRONMENTAL LITIGATION

a. If the performance of all or any part of the work is suspended, delayed, or interrupted due to an order of a court of competent jurisdiction as a result of environmental litigation, as defined below, the Contracting Officer and/or his Representative, at the request of the Contractor, shall determine whether the order is due in any part to the acts or omissions of the Contractor or a Subcontractor at any tier not required by the terms of this contract. If it is determined that the order is not due in any part to acts or omissions of the Contractor or a Subcontractor at any tier other than as required by the terms of this contract, such suspension, delay, or interruption shall be considered as if ordered by the Contracting Officer and/or his Representative in the administration of this contract under the terms of Section I, contract clause entitled "SUSPENSION OF WORK." The period of such suspension, delay or interruption shall be considered unreasonable, and an adjustment shall be made for any increase in the cost of performance of this contract (excluding profit) as provided in that clause, subject to all the provisions thereof.

b. The term "environmental litigation," as used herein, means a lawsuit alleging that the work will have an adverse effect on the environment or that the Government has not duly considered, either substantively or procedurally, the effect of the work on the environment.

1.28 PATENTS

All potential bidders need to be aware of the following patents, which may be related to performance of this contract.

Patent 6,293,731
Patent 5,007,590
Patent 5,028,010
Patent 5,353,998
Patent 5,542,614
Patent 5,794,862
Patent 5,931,605
Patent 6,017,169

The foregoing patents, and any others which may be applicable to activities performed under this contract, are neither required nor prohibited. However, if you use the methods or technology described in these or other patents, please factor license costs into your bid. The successful bidder needs to obtain the proper license(s) if he wishes to use the methods, processes, or technology described in any applicable patents.

1.29 VERIFICATION OF SMALL BUSINESS UTILIZATION

a. This clause is applicable to small business concerns whose contracts exceed \$1,000,000.

b. In accordance with the clause at FAR 52.219-8, entitled Utilization of Small Business Concerns, in effect on the date of this contract, the Contracting Officer may survey the extent of small and small disadvantaged business utilization under this contract. The Contractor may be required to report to the Contracting Officer statistical data on the number and dollar amounts of subcontracting awards with small businesses and small disadvantaged businesses.

c. As appropriate, the Contracting Officer may require one or more follow-up reports to the initial report.

d. The Contractor agrees to insert this clause in any subcontract that may exceed \$1,000,000, including this subparagraph (d).

1.30 CONTRACT INFORMATION RELEASE

This constitutes notification that, pursuant to CECW-OD Memo, 5 February 1996, "Release of Information - Dredging Projects", plant operation report including; Engineer Form 27/27A - Daily Report of Operations Hopper Dredges; and Engineer Form 4267 - Daily Report of Operations Pipeline, Dipper or Bucket Dredges, are public information and shall be provided upon proper FOIA request for active and completed contract information.

1.31 ALTERATIONS IN CONTRACT

Portions of the contract are altered as follows:

a. The following clause is incorporated as part of the contract: "By entering into this contract the Contractor certifies that neither it, nor any person or firm who has an interest in the Contractor's firm, is a person, or firm ineligible in accordance with FAR subpart "9.4" (DEAN-PRP EFAR Acquisition Letter 85-1 dated 11 Sept 85)

1.32 CONTRACTOR SUPPLY AND USE OF ELECTRONIC SOFTWARE FOR PROCESSING
DAVIS-BACON ACT CERTIFIED LABOR PAYROLLS

The contractor is encouraged to use a commercially-available electronic system to process and submit certified payrolls electronically to the Government. The requirements for preparing, processing and providing certified labor payrolls are established by the Davis-Bacon Act as stated in FAR 52.222-8, PAYROLLS AND BASIC RECORDS and FAR 52.222-13, COMPLIANCE WITH DAVIS-BACON AND RELATED ACT REGULATIONS. If the contractor elects to use an electronic Davis-Bacon payroll processing system, then the contractor shall be responsible for obtaining and providing for all access, licenses, and other services required to provide for receipt, processing, certifying, electronically transmitting to the Government, and storing weekly payrolls and other data required for the contractor to comply with Davis-Bacon and related Act regulations. When the contractor uses an electronic Davis-Bacon payroll system, the electronic payroll service shall be used by the contractor to prepare, process, and maintain the relevant payrolls and basic records during all work under this construction contract and the electronic payroll service shall be capable of preserving these payrolls and related basic records for the required 3 years after contract completion. If the contractor chooses to use an electronic Davis-Bacon payroll system, then the contractor shall obtain and provide electronic system access to the Government, as required to comply with the Davis-Bacon and related Act regulations over the duration of this construction contract. The access shall include electronic review access by the Government contract administration office to the electronic payroll processing system used by the contractor. The contractor's provision and use of an electronic payroll processing system shall meet the following basic functional criteria: commercially available; compliant with appropriate Davis Bacon Act payroll provisions in the FAR; able to accommodate the required numbers of employees and subcontractors planned to be employed under the contract; capable of producing an Excel spreadsheet compatible electronic output of weekly payroll records (format at <http://www.rmssupport.com/guides.aspx>) for export in an Excel spreadsheet to be imported into the contractor's Quality Control System (QCS) version of Resident Management System (RMS), that in turn shall export payroll data to the Government's Resident Management System (RMS); demonstrated security of data and data entry rights; ability to produce contractor-certified electronic versions of weekly payroll data; ability to identify erroneous entries and track the data/time of all versions of the certified Davis Bacon payrolls submitted to the government over the life of the contract; capable of generating a durable record copy, that is, a CD or DVD and PDF file record of data from the system database at end of the contract closeout. This durable record copy of data from the electronic Davis-Bacon payroll processing system shall be provided to the Government during contract closeout. All contractor-incurred costs related to the contractor's provision and use of an electronic payroll processing service shall be included in the contractor's price for the overall work under the contract. The costs for Davis-Bacon Act compliance using electronic payroll processing services shall not be a separately bid/proposed or reimbursed item under this contract.

1.33 FUEL USAGE

The Contractor shall furnish the Contracting Officer and/or his Representative a report, to be received on or before the last day of the calendar month, listing the totals of fuels consumed by the dredging plant, supporting vessels, and other construction equipments. The report shall list the quantities of different fuels separately. The report shall cover the period from the 25th of the preceding month to the 25th of the current month.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

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PROJECT SCHEDULE
08/23, CHG 1: 08/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AACE INTERNATIONAL (AACE)

AACE 29R-03	(2011) Forensic Schedule Analysis
AACE 52R-06	(2006) Time Impact Analysis - As Applied in Construction
AACE 84R-13	(2015) Planning and Accounting for Adverse Weather

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 67-17	(2017) Schedule Delay Analysis
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U.S. ARMY CORPS OF ENGINEERS (USACE)

ER 1-1-11	(2017) Administration -- Project Schedules
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1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals; G

Preliminary Project Schedule; G

Initial Project Schedule; G

SD-03 Product Data

Periodic Schedule Update; G

1.3 PROJECT SCHEDULER QUALIFICATIONS

Designate an authorized representative to be responsible for the preparation of the schedule and all required updating and production of reports. The Designated Project Scheduler must have prepared and maintained previous construction schedules for projects of similar size and complexity to this contract, using Primavera P6. Representative must have a suitable knowledge of CPM scheduling principles and application.

PART 2 PRODUCTS

2.1 SOFTWARE

The scheduling software utilized to produce and update the schedules required herein must be capable of meeting all requirements of this specification.

2.1.1 Government Default Software

The Government uses Primavera P6. Ensure exported schedule files are compatible with the version of P6 used by the Government.

2.1.2 Contractor Software

Scheduling software used by the contractor must be commercially available from the software vendor for purchase with vendor software support agreements available. The software routine used to create the required sdef file must be created and supported by the software manufacturer.

2.1.2.1 Primavera

If Primavera P6 is selected for use, provide the "xer" export file in a version of P6 importable by the Government system. Verify at the SEKO meeting which version of P6 is in use by the Government. Export the schedule in a version of P6 no newer than that used by the Government.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

Prepare for approval a Project Schedule, as specified herein, pursuant to FAR Clause 52.236-15 Schedules for Construction Contracts. Show in the schedule the proposed sequence to perform the work and dates contemplated for starting and completing all schedule activities. The scheduling of the entire project is required. The scheduling of construction is the responsibility of the Contractor. Contractor management personnel must actively participate in its development. Subcontractors, and suppliers working on the project must also contribute in developing and maintaining an accurate Project Schedule. Provide a schedule that is a forward planning as well as a project monitoring tool. Use the Critical Path Method (CPM) of network calculation to generate all Project Schedules. Prepare each Project Schedule using the Precedence Diagram Method (PDM).

3.2 BASIS FOR PAYMENT AND COST LOADING

The schedule is the basis for determining contract earnings during each update period and therefore the amount of each progress payment. The aggregate value of all activities coded to a contract CLIN must equal the value of the CLIN. Match the progress payment "Pay Period Thru" date in RMS to the schedule data date.

3.2.1 Activity Cost Loading

Activity cost loading must be reasonable and without front-end loading. Activities with a negative cost loading is not allowed. Provide additional documentation to demonstrate reasonableness if requested by the Contracting Officer.

3.2.2 Cost Loading of Submittals

No costs are to be assigned to activities for the preparation, review, or approval of submittals, except as described under paragraph COST LOADING OF CLOSEOUT ACTIVITIES, paragraph AS-BUILT DRAWINGS, and paragraph O&M MANUALS.

3.2.3 Withholdings / Payment Rejection

Failure to meet the requirements of this specification may result in the disapproval of the preliminary, initial, or periodic schedule updates and subsequent rejection of payment requests until compliance is met.

In the event that the Contracting Officer directs schedule revisions and those revisions have not been included in subsequent Project Schedule revisions or updates, the Contracting Officer may withhold 10 percent of pay request amount from each payment period until such revisions to the project schedule have been made.

3.3 PROJECT SCHEDULE DETAILED REQUIREMENTS

3.3.1 Level of Detail Required

Develop the Project Schedule to the appropriate level of detail to address major milestones and to allow for satisfactory project planning and execution. Failure to develop the Project Schedule to an appropriate level of detail will result in its disapproval. The Contracting Officer will consider, but is not limited to, the following characteristics and requirements to determine appropriate level of detail:

3.3.2 Activity Durations

Reasonable activity durations are those that allow the progress of ongoing activities to be accurately determined between update periods. Non-procurement activities Original Durations (OD) are not to exceed 20 workdays or 30 calendar days.

3.3.3 Procurement Activities

Include in the schedule activities associated with the critical submittals and their approvals, procurement, fabrication, and delivery of long lead materials, equipment, fabricated assemblies, and supplies. Long lead procurement activities are those with an anticipated procurement sequence of over 90 calendar days.

3.3.4 Mandatory Activities

Include the following activities in the initial project schedule and all updates.

- a. Submission, review and acceptance of SD-01 Preconstruction Submittals (individual activity for each).
- b. Mobilization Completion
- c. Individual Dredging Area Completion (e.g. CLIN X - Dredging Area Y Completion)
- d. Project Physical Completion

- e. Final inspection.
- f. Demobilization Completion

3.3.5 Government Activities

Show in schedule, Government and other agency activities that could impact progress. These activities include, but are not limited to environmental permit approvals by State regulators, inspections, utility tie-in, Government Furnished Equipment (GFE) and Notice to Proceed (NTP) for phasing requirements.

3.3.6 Standard Activity Coding Dictionary

Use the activity coding structure defined in the Standard Data Exchange Format (SDEF) in ER 1-1-11. This exact structure is mandatory. Develop and assign all Activity Codes to activities as detailed herein.

The SDEF format is as follows:

Field	Activity Code	Length	Description
1	WRKP	3	Workers per day
2	RESP	4	Responsible party
3	AREA	4	Area of work
4	MODF	6	Modification Number
5	BIDI	6	Bid Item (CLIN)
6	PHAS	2	Phase of work
7	CATW	1	Category of work
8	FOW	20	Feature of work*
*Some systems require that FEATURE OF WORK values be placed in several activity code fields. The notation shown is for Primavera P6. Refer to the specific software guidelines with respect to the FEATURE OF WORK field requirements.			

3.3.6.1 Workers Per Day (WRKP)

Assign Workers per Day for all field construction or direct work activities, if directed by the Contracting Officer. Workers per day is based on the average number of workers expected each day to perform a task for the duration of that activity.

3.3.6.2 Responsible Party Coding (RESP)

Assign responsibility code for all activities to the Prime Contractor, Subcontractor(s) or Government agency(ies) responsible for performing the

activity.

- a. Activities coded with a Government Responsibility code include, but are not limited to: Government approvals, Government design reviews, environmental permit approvals by State regulators, Government Furnished Property/Equipment (GFP) and Notice to Proceed (NTP) for phasing requirements.
- b. Activities cannot have more than one Responsibility Code. Examples of acceptable activity code values are: DOR (for the designer of record); ELEC (for the electrical subcontractor); MECH (for the mechanical subcontractor); and GOVT (for USACE).

3.3.6.3 Area of Work Coding (AREA)

Assign Work Area code to activities based upon the work area in which the activity occurs. Define work areas based on resource constraints or space constraints that would preclude a resource, such as a particular trade or craft work crew from working in more than one work area at a time due to restraints on resources or space.

3.3.6.4 Modification Number (MODF)

Assign a Modification Number Code to any activity or sequence of activities added to the schedule as a result of a Contract Modification, when approved by Contracting Officer. Key all Code values to the Government's modification numbering system. An activity can have only one Modification Number Code.

3.3.6.5 Bid Item Coding (BIDI)

Assign a Bid Item Code to all activities using the Contract Line Item Number (CLIN) to which the activity belongs, even when an activity is not cost loaded. An activity can have only one BIDI Code.

3.3.6.6 Phase of Work Coding (PHAS)

Assign Phase of Work Code to all activities. Examples of phase of work are procurement phase and construction phase. Each activity can have only one Phase of Work code.

- a. Code proposed fast track design and construction phases proposed to allow filtering and organizing the schedule by fast track design and construction packages.
- b. If the contract specifies phasing with separately defined performance periods, identify a Phase Code to allow filtering and organizing the schedule accordingly.

3.3.6.7 Category of Work Coding (CATW)

Assign a Category of Work Code to all activities. Category of Work Codes include, but are not limited to construction submittal, procurement, fabrication, weather sensitive installation, non-weather sensitive installation, start-up, and testing activities. Each activity can have no more than one Category of Work Code.

3.3.6.8 Feature of Work Coding (FOW)

Assign a Feature of Work Code to appropriate activities based on the Definable Feature of Work to which the activity belongs based on the approved QC plan.

Definable Feature of Work is defined in Section 01 45 00 QUALITY CONTROL. An activity can have only one Feature of Work Code.

3.3.7 Contract Milestones and Constraints

Milestone activities are to be used for significant project events including, but not limited to, project phasing, project start and end activities, or interim completion dates. The use of artificial float constraints such as "zero free float" or "zero total float" are prohibited.

Mandatory constraints that ignore or affect network logic are prohibited. No constrained dates are allowed in the schedule other than those specified herein. Submit additional constraints to the Contracting Officer for approval on a case by case basis.

3.3.7.1 Project Start Date Milestone and Constraint

The first activity in the project schedule must be a start milestone titled "NTP Acknowledged," which must have a "Start On" constraint date equal to the date that the NTP is acknowledged.

3.3.7.2 End Project Finish Milestone and Constraint

The last activity in the schedule must be a finish milestone titled "End Project."

Constrain the project schedule to the Contract Completion Date in such a way that if the schedule calculates an early finish, then the float calculation for "End Project" milestone reflects positive float on the longest path. If the project schedule calculates a late finish, then the "End Project" milestone float calculation reflects negative float on the longest path.

3.3.7.3 Interim Completion Dates and Constraints

Constrain contractually specified interim completion dates to show negative float when the calculated late finish date of the last activity in that phase is later than the specified interim completion date.

3.3.7.3.1 Start Phase

Use a start milestone as the first activity for a project phase. Call the start milestone "Start Phase X" where "X" refers to the phase of work.

3.3.7.3.2 End Phase

Use a finish milestone as the last activity for a project phase. Call the finish milestone "End Phase X" where "X" refers to the phase of work.

3.3.8 Adverse Weather

Ensure anticipated adverse weather is accounted for in the Project Schedule. AACE 84R-13 provides methodologies for techniques in planning

for adverse weather. The preferred methodology is the use of a weather calendar. If a method other than a weather calendar is proposed, discuss the reason during the SEKO meeting.

3.3.9 Calendars

Schedule activities on a Calendar to which the activity logically belongs. Develop calendars to accommodate any contract defined work period such as a 7-day calendar for Government Acceptance activities, concrete cure times, etc. Develop the default Calendar to match the physical work plan with non-work periods identified including weekends and holidays. Develop Seasonal Calendar(s) and assign to seasonally affected activities as applicable.

If a weather calendar is used its implementation and execution is to be in accordance with AACE 84R-13

3.3.10 Open Ended Logic

Only two open ended activities are allowed: the first activity "NTP Acknowledged" is to have no predecessor logic, and the last activity -"End Project" is to have no successor logic. Except as noted above, each activity must have at least a start-to-start or finish-to-start relationship with its predecessor and a finish-to-finish or finish-to-start relationship with its successor.

Predecessor open ended logic may be allowed in a time impact analyses upon the Contracting Officer's approval.

3.3.11 Default Progress Data Disallowed

Actual Start and Finish dates must not automatically update with default mechanisms included in the scheduling software. Updating of the percent complete and the remaining duration of any activity must be independent functions. Disable program features that calculate one of these parameters from the other. Activity Actual Start (AS) and Actual Finish (AF) dates assigned during the updating process must match those dates provided in the Contractor Quality Control Reports. Failure to document the AS and AF dates in the Daily Quality Control report will result in disapproval of the Contractor's schedule.

3.3.12 Out-of-Sequence Progress

Activities that have progressed before all preceding logic has been satisfied (Out-of-Sequence Progress) will be allowed only on a case-by-case basis subject to approval by the Contracting Officer. Propose logic corrections to eliminate out of sequence progress or justify not changing the sequencing for approval prior to submitting an updated project schedule. Address out of sequence progress or logic changes in the Narrative Report and in the periodic schedule update meetings.

3.3.13 Added and Deleted Activities

Do not delete activities from the project schedule or add new activities to the schedule without approval from the Contracting Officer. Activity ID and description changes are considered new activities and cannot be changed without Contracting Officer approval.

3.3.14 Original Durations

Activity Original Durations (OD) must be reasonable to perform the work item. OD changes are prohibited unless justification is provided and approved by the Contracting Officer.

3.3.15 Leads, Lags, and Start to Finish Relationships

Lags must be reasonable as determined by the Government and not used in place of realistic original durations, must not be in place to artificially absorb or create float, or to replace proper schedule logic.

- a. Leads (negative lags) are prohibited.
- b. Start to Finish (SF) relationships are prohibited.

3.3.16 Retained Logic

Schedule calculations must retain the logic between predecessors and successors ("retained logic" mode) even when the successor activity(s) starts and the predecessor activity(s) has not finished (out-of-sequence progress). Software features that in effect sever the tie between predecessor and successor activities when the successor has started and the predecessor logic is not satisfied ("progress override") are not be allowed.

3.3.17 Percent Complete

Update the percent complete for each activity started, based on the realistic assessment of earned value. Activities which are complete but for remaining minor punch list work and which do not restrain the initiation of successor activities may be declared 100 percent complete to allow for proper schedule management. Budgeted cost of activity is to be reduced by the amount needed to correct deficiency. The activity referenced in paragraph COST LOADING of CLOSEOUT ACTIVITIES must have its budgeted cost increased by this amount.

3.3.18 Remaining Duration

Update the remaining duration for each activity based on the number of estimated workdays it will take to complete the activity. Remaining duration may not mathematically correlate with percentage found under paragraph entitled Percent Complete. The remaining duration for unstarted activities are not to be less than its original duration.

3.3.19 Early Completion Schedule and the Right to Finish Early

An Early Completion Schedule is an Initial Project Schedule (IPS) that indicates all scope of the required contract work will be completed before the contractually required completion date.

- a. No IPS indicating an Early Completion will be accepted without being fully resource-loaded (including crew sizes and hours) and the Government agreeing that the schedule is reasonable and achievable.
- b. The Government is under no obligation to accelerate work items it is responsible for to ensure that the early completion is met nor is it responsible to modify incremental funding (if applicable) for the project to meet the contractor's accelerated work.

3.4 PROJECT SCHEDULE SUBMISSIONS

Provide the submissions as described below. The data reports and network diagrams required for each submission are contained in paragraph SUBMISSION REQUIREMENTS. If the Contractor fails or refuses to furnish the information and schedule updates as set forth herein, the Contractor will be deemed not to have provided an estimate upon which a progress payment can be made.

Review comments made by the Government on the schedule(s) do not relieve the Contractor from compliance with requirements of the Contract Documents.

3.4.1 Preliminary Project Schedule Submission

Submit the Preliminary Project Schedule in either completed or draft form no later than 7 days prior to the Preconstruction Conference; defining the planned operations detailed for the first 90 calendar days for approval. The approved Preliminary Project Schedule will be used for payment purposes not to exceed 90 calendar days after NTP. Completely cost load the Preliminary Project Schedule to balance the contract award CLINS shown on the Price Schedule. The Preliminary Project Schedule may be summary in nature for the remaining performance period. It must be early start and late finish constrained and logically tied as specified. The Preliminary Project Schedule forms the basis for the Initial Project Schedule specified herein and must include all of the required plan and program preparations, submissions and approvals identified in the contract (for example, Quality Control Plan, Safety Plan, and Environmental Protection Plan) as well as design activities, planned submissions of all early design packages, permitting activities, design review conference activities, and other non-construction activities intended to occur within the first 90 calendar days. Government acceptance of the associated design package(s) and all other specified Program and Plan approvals must occur prior to any planned construction activities. Code activities that are summary in nature, after the first 90 calendar days, with the following activity codes: Bid Item (CLIN) code (BIDI), Responsibility Code (RESP), and Feature of Work code (FOW)..

3.4.2 Initial Project Schedule Submission

Submit the Initial Project Schedule for approval within 5 calendar days after notice to proceed is issued. The schedule must demonstrate a reasonable and realistic sequence of activities which represent all work through the entire contract performance period. No payment will be made for work items not fully detailed in the Project Schedule.

3.4.3 Periodic Schedule Updates

Update the Project Schedule on a regular basis, monthly at a minimum. Provide a draft Periodic Schedule Update for review at the schedule update meetings as prescribed in the paragraph PERIODIC SCHEDULE UPDATE MEETINGS. These updates will enable the Government to assess Contractor's progress.

- a. Update information including Actual Start Dates (AS), Actual Finish Dates (AF), Remaining Durations (RD), and Percent Complete. Updated information is subject to the approval of the Government at the meeting.
- b. AS and AF dates must match the date(s) reported on the Contractor's Quality Control Report for an activity start or finish.

3.5 SUBMISSION REQUIREMENTS

Submit the following items for the Preliminary Schedule, Initial Schedule, and every Periodic Schedule Update throughout the life of the project:

3.5.1 Electronic Scheduling Data

Provide electronic scheduling data containing the current project schedule and all previously submitted schedules in the format of the scheduling software (e.g. .xer). Also include the Narrative Report and all required Schedule Reports. The electronic file is to identify the type of schedule (Preliminary, Initial, Update), full contract number, Data Date and schedule file name. Each schedule must have a unique file name and use project specific settings.

3.5.2 Narrative Report

Provide a Narrative Report with each schedule submission. The Schedule Narrative Report is a "stand alone" report in PDF or Word format. Title the report "Schedule Narrative Report". Show the Report date, schedule name, and contract number on the first page. Indicate who is responsible for the schedule update and narrative report. Provide a brief description of the Project Scope. The Narrative Report is expected to communicate to the Government the thorough analysis of the schedule output and the plans to compensate for problems, either current or potential, which are revealed through that analysis. Include the following information at a minimum in the Narrative Report:

- a. Identify and discuss the work scheduled to start in the next update period.
- b. Describe activities along the critical path in addition to activities on the next float path after the critical path.
- c. Describe current and anticipated problem areas, delaying factors and their impact, and an explanation of corrective actions taken or required to be taken. Identify the party responsible for delay, if applicable.
- d. Identify and explain why activities, based on their calculated late dates, should have either started or finished during the update period but did not.
- e. Identify and discuss all schedule changes by activity ID and activity name including what specifically was changed and why the change was needed. Include at a minimum new and deleted activities, logic changes, duration changes, calendar changes, lag changes, resource changes, and actual start and finish date changes.
- f. Identify and discuss out-of-sequence work.
- g. If the longest path changes from a monthly update to the next, provide an explanation of the factors driving the change.
- h. Respond to USACE Schedule Review Comments from the previous update(s).

3.5.3 Schedule Reports

The format, filtering, organizing, and sorting for each schedule report will be as directed by the Contracting Officer. Typically, reports contain Activity Numbers, Activity Description, Original Duration, Remaining Duration, Early Start Date, Early Finish Date, Late Start Date, Late Finish Date, Total Float, Actual Start Date, Actual Finish Date, and Percent Complete. Provide the reports electronically in .pdf format. The following lists typical reports that will be requested:

3.5.3.1 Activity Report

List of all activities sorted according to activity number.

3.5.3.2 Logic Report

List of detailed predecessor and successor activities for every activity in ascending order by activity number.

3.5.3.3 Total Float Report

A list of all incomplete activities sorted in ascending order of total float. List activities which have the same amount of total float in ascending order of Early Start Dates. Do not show completed activities on this report.

3.5.3.4 Earnings Report by CLIN

A compilation of the Total Earnings on the project from the NTP to the data date, which reflects the earnings of activities based on the agreements made in the schedule update meeting defined herein. Provided a complete schedule update has been furnished, this report serves as the basis of determining progress payments. Group activities by CLIN number and sort by activity number. Provide a total CLIN percent earned value, CLIN percent complete, and project percent complete. The printed report must contain the following for each activity: the Activity Number, Activity Description, Original Budgeted Amount, Earnings to Date, Earnings this period, Total Quantity, Quantity to Date, and Percent Complete (based on cost).

3.5.3.5 Scheduling and Leveling Report

Provide a Scheduling/Leveling Report generated from the current project schedule being submitted.

3.5.4 Network Diagram

The Network Diagram is required for the Preliminary, Initial and Periodic Updates. Depict and display the order and interdependence of activities and the sequence in which the work is to be accomplished. The Contracting Officer will use, but is not limited to, the following conditions to review compliance with this paragraph:

3.5.4.1 Continuous Flow

Show a continuous flow from left to right with no arrows from right to left. Show the activity number, description, duration, and estimated earned value on the diagram.

3.5.4.2 Project Milestone Dates

Show dates on the diagram for start of project, any contract required interim completion dates, and contract completion dates.

3.5.4.3 Critical Path

Show all activities on the critical path. The critical path is defined as the longest path.

3.5.4.4 Grouping and Sorting

Group and sort activities as directed to assist in the understanding of the activity sequence. Typically, this flow will group activities by major elements of work, category of work, work area and/or responsibility.

3.5.4.5 Cash Flow / Schedule Variance Control (SVC) Diagram

With each schedule submission, provide a SVC diagram showing 1) Cash Flow S-Curves indicating planned project cost based on projected early and late activity finish dates, and 2) Earned Value to-date.

3.6 PERIODIC SCHEDULE UPDATE

3.6.1 Periodic Schedule Update Meetings

Conduct periodic schedule update meetings for the purpose of reviewing the proposed Periodic Schedule Update, Narrative Report, Schedule Reports, and progress payment. Conduct meetings at least monthly (or as otherwise mutually agreed to). Following the schedule update meeting, make corrections to the schedule and resubmit. Include only those changes approved by the Government in the schedule review meeting. Only approved schedules may be used to generate invoices for payment.

3.6.2 Update Submission Following Progress Meeting

Submit the complete Periodic Schedule Update of the Project Schedule containing all approved progress, revisions, and adjustments, pursuant to paragraph SUBMISSION REQUIREMENTS not later than 4 workdays after the periodic schedule update meeting.

3.7 WEEKLY PROGRESS MEETINGS

Conduct a weekly meeting with the Government (or as otherwise mutually agreed to) between the meetings described in paragraph entitled PERIODIC SCHEDULE UPDATE MEETINGS for the purpose of jointly reviewing the actual progress of the project as compared to the as planned progress and to review planned activities for the upcoming two weeks. Use the current approved schedule update for the purposes of this meeting and for the production and review of reports. At the weekly progress meeting, address the status of RFIs, RFPs and Submittals.

3.8 REQUESTS FOR TIME EXTENSIONS

ASCE 67-17 provides delay analysis guidelines. If there is a conflict between the contract and guidance provided in ASCE 67-17, the contract will govern. Provide justification of delay to the Contracting Officer in accordance with the contract provisions and clauses for approval, within 10 days of a delay occurring. Also prepare a time impact analysis for each

Government request for proposal (RFP).

3.8.1 Time Extension for Unusually Severe Weather

This provision specifies the procedure for determination of time extensions for unusually severe weather in accordance with the Contract Clause entitled "Default (Fixed-Price Construction)".

Upon acknowledgment of the notice to proceed and continuing throughout the contract on a monthly basis, adverse weather days shall be recorded by the contractor on a calendar day basis (include weekends and holidays) within the Daily QC Report in RMS (SECTION 01 45 00.15 10 par. 1.6.3.1). The number of adverse weather days shall be calculated chronologically from the first to the last day in each month. At the end of each calendar month, the Contracting Officer or his/her Representative will examine the recorded adverse weather days to determine whether a contractor is entitled to a time extension. These adverse weather days must prevent work for 50 percent or more of the Contractor's work day and delay work critical to the timely completion of the project. The Contracting Officer or his/her Representative will convert any delays meeting the above requirement to calendar days and issue a modification in accordance with the general provision entitled "Termination for Default - Damages for Delay - Time Extensions".

The Contractor's schedule must reflect anticipated adverse weather delays on all weather dependent activities.

3.8.2 Justification of Delay

Provide a description of the event(s) that caused the delay and/or impact to the work. As part of the description, identify all schedule activities impacted. Show that the event that caused the delay/impact was the responsibility of the Government. Provide a time impact analysis that demonstrates the effects of the delay or impact on the project completion date, or interim completion date(s). Evaluate multiple impacts chronologically; each with its own justification of delay. With multiple impacts consider any concurrency of delay. A time extension and the schedule fragnet becomes part of the project schedule and all future schedule updates upon approval by the Contracting Officer.

3.8.3 Changes That Do Not Cause Delay

If it is determined that a change does not constitute a schedule time extension, a fragnet as described in paragraph FRAGMENTARY NETWORK, is to be created and inserted into the Project schedule.

3.8.4 Time Impact Analysis (Prospective Analysis)

Submit requests for time extensions based on a prospective analysis if the work involved or the impact identified has not already occurred. Where the impact is ongoing a prospective analysis may be considered. However, the Government reserves the right to require a retrospective analysis be prepared after the impact has ended.

Prepare a time impact analysis for approval by the Contracting Officer based on industry recommended practice AACE 52R-06. Utilize a copy of the last approved schedule prior to the first day of the impact or delay for the time impact analysis. If Contracting Officer determines the time frame between the last approved schedule and the first day of impact is too

great, prepare an interim updated schedule to perform the time impact analysis. Unless approved by the Contracting Officer, no other changes may be incorporated into the schedule being used to justify the time impact.

3.8.5 Forensic Schedule Analysis (Retrospective Analysis)

Submit requests for time extensions based on a retrospective analysis if the work involved or the impact identified has already occurred. The analysis must account for the actual performance of both the impacted work and all other contract work in the schedule.

If a methodology is chosen from AACE 29R-03, the method must adhere to the principles identified in ASCE 67-17. If there is a conflict with the methodology chosen from AACE 29R-03 and ASCE 67-17, ASCE 67-17 will govern. Choice of methodology should be discussed prior to submission to Government.

3.8.6 Fragmentary Network (Fragnet)

Prepare a proposed fragnet for time impact analysis consisting of a sequence of new activities that are proposed to be added to the project schedule to demonstrate the influence of the delay or impact to the project's contractual dates. Clearly show how the proposed fragnet is to be tied into the project schedule including all predecessors and successors to the fragnet activities. The proposed fragnet must be approved by the Contracting Officer prior to incorporation into the project schedule.

3.8.7 Time Extension

The Contracting Officer must approve the Justification of Delay including the time impact analysis before a time extension will be granted. No time extension will be granted unless the delay consumes all available Project Float and extends the projected finish date ("End Project" milestone) beyond the Contract Completion Date. The time extension will be in calendar days.

Actual delays that are found to be caused by the Contractor's own actions, which result in a calculated schedule delay, are not a cause for an extension to the performance period, completion date, or any interim milestone date.

3.8.8 Impact to Early Completion Schedule

No extended overhead will be paid for delay prior to the original Contract Completion Date for an Early Completion IPS unless the Contractor actually performed work in accordance with that Early Completion Schedule. The Contractor must show that an early completion was achievable had it not been for the impact.

3.9 FAILURE TO ACHIEVE PROGRESS

Should the progress fall behind the approved project schedule for reasons other than those that are excusable within the terms of the contract, the Contracting Officer may require provision of a written recovery plan for approval. The plan must detail how progress will be made-up to include which activities will be accelerated by adding additional crews, longer work hours, extra workdays, etc.

3.9.1 Artificially Improving Progress

Artificially improving progress by means such as, but not limited to, revising the schedule logic, modifying, or adding constraints, shortening activity durations, or changing calendars in the project schedule is prohibited. Indicate assumptions made and the basis for any logic, constraint, duration, and calendar changes used in the creation of the recovery plan. Any additional resources, manpower, or daily and weekly work hour changes proposed in the recovery plan must be evident at the work site and documented in the daily report along with the Schedule Narrative Report.

3.9.2 Failure to Perform

Failure to perform work and maintain progress in accordance with the supplemental recovery plan may result in an interim and final unsatisfactory performance rating and may result in corrective action directed by the Contracting Officer pursuant to FAR 52.236-15 Schedules for Construction Contracts, FAR 52.249-10 Default (Fixed-Price Construction), and other contract provisions.

3.9.3 Recovery Schedule

Should the Contracting Officer find it necessary, submit a recovery schedule pursuant to FAR 52.236-15 Schedules for Construction Contracts.

3.10 OWNERSHIP OF FLOAT

Except for the provision given in the paragraph IMPACT TO EARLY COMPLETION SCHEDULE, float available in the schedule, at any time, belongs to the Project and is available for Contractor and Government use. This includes activity and project float. Activity float is the number of workdays that an activity can be delayed without causing a delay to the "End Project" finish milestone. Project float (if applicable) is the number of work days between the projected early finish and the "End Project" finish.

3.11 TRANSFER OF SCHEDULE DATA INTO RESIDENT MANAGEMENT SYSTEM

Ensure schedule data is uploaded to RMS. This data is additional supporting data in a form and detail required by the Contracting Officer pursuant to FAR 52.232-5 Payments under Fixed-Price Construction Contracts. The receipt of a proper payment request pursuant to FAR 52.232-27 Prompt Payment for Construction Contracts is contingent upon the Government receiving both acceptable and approvable hard copies and matching electronic versions of the application for progress payment.

3.12 PRIMAVERA P6 MANDATORY REQUIREMENTS

Ensure Primavera P6 settings provide a schedule capable of fulfilling the requirements of the contract. The following settings are mandatory and required in all schedule submissions to the Government:

- a. Activity Codes must be Project Level, not Global or EPS level.
- b. Calendars must be Project Level, not Global or Resource level.
- c. Activity Duration Types must be set to "Fixed Duration & Units".
- d. Percent Complete Types must be set to "Physical".

- e. Time Period Admin Preferences must remain the default "8.0 hr/day, 40 hr/week, 172 hr/month, 2000 hr/year". Set Calendar Work Hours/Day to 8.0 Hour days.
- f. Set Schedule Option for defining Critical Activities to "Longest Path".
- g. Set Schedule Option for defining progressed activities to "Retained Logic".
- h. Set up cost loading using a single lump sum non-labor resource. The Price/Unit must be \$1/hr, Default Units/Time must be "8h/d", and settings "Auto Compute Actuals" and "Calculate costs from units" un-selected.
- i. Activity ID's must not exceed 10 characters.
- j. Activity Names must have a verb-noun structure and contain the most defining and detailed description within the first 30 characters.
- k. The daily ending hour for all work calendars must be the same.

-- End of Section --

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08/18, CHG 4: 02/21

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to establish conformance with the design concept and contract documents

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ENG Form 4025-R

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SECTION 01 33 00

SUBMITTAL PROCEDURES

08/18, CHG 4: 02/21

PART 1 GENERAL

1.1 SUMMARY

1.1.1 Submittal Information

The Contracting Officer may request submittals in addition to those specified when deemed necessary to adequately describe the work covered in the respective sections. Each submittal is to be complete and in sufficient detail to allow ready determination of compliance with contract requirements.

Units of weights and measures used on all submittals are to be the same as those used in the contract drawings.

1.1.2 Submission of Submittals

Schedule and provide submittals requiring Government approval before acquiring the material or equipment covered thereby. Pick up and dispose of samples not incorporated into the work in accordance with manufacturer's Safety Data Sheets (SDS) and in compliance with existing laws and regulations.

1.2 DEFINITIONS

1.2.1 Submittal Descriptions (SD)

Submittal requirements are specified in the technical sections. Examples and descriptions of submittals identified by the Submittal Description (SD) numbers and titles follow:

SD-03 Product Data

Catalog cuts, illustrations, schedules, diagrams, performance charts, instructions and brochures illustrating size, physical appearance and other characteristics of materials, systems or equipment for some portion of the work.

Samples of warranty language when the contract requires extended product warranties.

SD-11 Closeout Submittals

Documentation to record compliance with technical or administrative requirements or to establish an administrative mechanism.

Submittals required for Guiding Principle Validation (GPV) or Third Party Certification (TPC).

Special requirements necessary to properly close out a construction contract. For example, Record Drawings and as-built drawings. Also, submittal requirements necessary to properly close out a major phase of

construction on a multi-phase contract.

1.2.2 Approving Authority

Office or designated person authorized to approve the submittal.

1.2.3 Work

As used in this section, on-site and off-site construction required by contract documents, including labor necessary to produce submittals, construction, materials, products, equipment, and systems incorporated or to be incorporated in such construction. In exception, excludes work to produce SD-01 submittals.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals; G

Preconstruction Submittals shall be submitted to the Contracting Officer or Contracting Officer Representative **within eighteen (18) days after the opening of bids**. Submit the following:

Certification of Insurance; G

Surety Bond; G

List Of Proposed Subcontractors; G

Order of Work; G

The Contractor's order of work for dredging and performing other work, including disposal area plan.

Work To Be Performed By The Contractor; G,

Subcontracting Plan; G

Project Schedules; G

A Preliminary Project Schedule and an Initial Project Schedule shall be submitted in accordance with Section 01 32 01.00 10 PROJECT SCHEDULE.

Submittal Register; G

Letter Appointing Superintendent; G

Activity Hazard Analysis; G

See Section 01 35 26, GOVERNMENTAL SAFETY REQUIREMENTS

Accident Prevention Plan; G

See Section 01 35 26, GOVERNMENTAL SAFETY REQUIREMENTS

Work Plan; G

Quality Control (QC) plan, G See Section 01 45 00.00 20

Environmental Protection Plan; G

Notice to Mariners; G

Determination of No Hazard to Air Navigation; G
See Section 00 90 01, Attachment L

SD-03 Product Data

Periodic Schedule Update; G
Submitted in accordance with Section 01 32 01.00 10 PROJECT SCHEDULE.

SD-11 Closeout Submittals

As-Built Drawings; G

1.4 SUBMITTAL CLASSIFICATION

1.4.1 Government Approved (G)

Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, submittals are considered to be "shop drawings."

1.4.2 For Information Only

Submittals not requiring Government approval will be for information only. Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, they are not considered to be "shop drawings."

1.5 PREPARATION

1.5.1 Transmittal Form

Use the ENG Form 4025-R transmittal form for submitting both Government-approved and information-only submittals. Submit in accordance with the instructions on the reverse side of the form. These forms or similar forms are included in the RMS CM software that the Contractor is required to use for this contract. Properly complete this form by filling out all the heading blank spaces and identifying each item submitted. Exercise special care to ensure proper listing of the specification paragraph and sheet number of the contract drawings pertinent to the data submitted for each item.

1.5.2 Submittal Format

1.5.2.1 Format of SD-01 Preconstruction Submittals

When the submittal includes a document that is to be used in the project, or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.5.2.2 Format for SD-02 Shop Drawings

Provide shop drawings not less than 8 1/2 by 11 inches nor more than 30 by 42 inches, except for full-size patterns or templates. Prepare drawings to accurate size, with scale indicated, unless another form is required.

Ensure drawings are suitable for reproduction and of a quality to produce clear, distinct lines and letters, with dark lines on a white background.

- a. Include the nameplate data, size, and capacity on drawings. Also include applicable federal, military, industry, and technical society publication references.
- b. Dimension drawings, except diagrams and schematic drawings. Prepare drawings demonstrating interface with other trades to scale. Use the same unit of measure for shop drawings as indicated on the contract drawings. Identify materials and products for work shown.

1.5.2.2.1 Drawing Identification

Include on each drawing the drawing title, number, date, and revision numbers and dates, in addition to information required in paragraph IDENTIFYING SUBMITTALS.

Number drawings in a logical sequence. Each drawing is to bear the number of the submittal in a uniform location next to the title block. Place the Government contract number in the margin, immediately below the title block, for each drawing.

1.5.2.3 Format of SD-03 Product Data

Present product data submittals for each section. Include a table of contents, listing the page and catalog item numbers for product data.

Indicate, by prominent notation, each product that is being submitted; indicate the specification section number and paragraph number to which it pertains.

1.5.2.3.1 Product Information

Supplement product data with material prepared for the project to satisfy the submittal requirements where product data does not exist. Identify this material as developed specifically for the project, with information and format as required for submission of SD-07 Certificates.

Provide product data in units used in the Contract documents. Where product data are included in preprinted catalogs with another unit, submit the dimensions in contract document units, on a separate sheet.

1.5.2.3.2 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.2.3.3 Data Submission

Collect required data submittals for each specific material, product, unit of work, or system into a single submittal that is marked for choices, options, and portions applicable to the submittal. Mark each copy of the product data identically. Partial submittals will [not] be accepted for expedition of the construction effort.

Submit the manufacturer's instructions before installation.

1.5.2.4 Format of SD-04 Samples

1.5.2.4.1 Sample Characteristics

Furnish samples in the following sizes, unless otherwise specified or unless the manufacturer has prepackaged samples of approximately the same size as specified:

- a. Sample of Equipment or Device: Full size.
- b. Sample of Materials Less Than 2 by 3 inches: Built up to 8 1/2 by 11 inches.
- c. Sample of Materials Exceeding 8 1/2 by 11 inches: Cut down to 8 1/2 by 11 inches and adequate to indicate color, texture, and material variations.
- d. Sample of Linear Devices or Materials: 10 inch length or length to be supplied, if less than 10 inches. Examples of linear devices or materials are conduit and handrails.
- e. Sample Volume of Nonsolid Materials: Pint. Examples of nonsolid materials are sand and paint.
- f. Color Selection Samples: 2 by 4 inches. Where samples are specified for selection of color, finish, pattern, or texture, submit the full set of available choices for the material or product specified. Sizes and quantities of samples are to represent their respective standard unit.
- g. Sample Panel: 4 by 4 feet.
- h. Sample Installation: 100 square feet.

1.5.2.4.2 Sample Incorporation

Reusable Samples: Incorporate returned samples into work only if so specified or indicated. Incorporated samples are to be in undamaged condition at the time of use.

Recording of Sample Installation: Note and preserve the notation of any area constituting a sample installation, but remove the notation at the final clean-up of the project.

1.5.2.4.3 Comparison Sample

Samples Showing Range of Variation: Where variations in color, finish, pattern, or texture are unavoidable due to nature of the materials, submit sets of samples of not less than three units showing extremes and middle of

range. Mark each unit to describe its relation to the range of the variation.

When color, texture, or pattern is specified by naming a particular manufacturer and style, include one sample of that manufacturer and style, for comparison.

1.5.2.5 Format of SD-05 Design Data

Provide design data and certificates on 8 1/2 by 11 inch paper.

1.5.2.6 Format of SD-06 Test Reports

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.5.2.7 Format of SD-07 Certificates

Provide design data and certificates on 8 1/2 by 11 inch paper.

1.5.2.8 Format of SD-08 Manufacturer's Instructions

Present manufacturer's instructions submittals for each section. Include the manufacturer's name, trade name, place of manufacture, and catalog model or number on product data. Also include applicable federal, military, industry, and technical-society publication references. If supplemental information is needed to clarify the manufacturer's data, submit it as specified for SD-07 Certificates.

Submit the manufacturer's instructions before installation.

1.5.2.8.1 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.2.9 Format of SD-09 Manufacturer's Field Reports

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.5.2.10 Format of SD-10 Operation and Maintenance Data (O&M)

Comply with the requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA for O&M Data format.

1.5.2.11 Format of SD-11 Closeout Submittals

When the submittal includes a document that is to be used in the project or

is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.5.3 Source Drawings for Shop Drawings

1.5.3.1 Source Drawings

The entire set of source drawing files (DWG) will not be provided to the Contractor. Request the specific Drawing Number for the preparation of shop drawings. Only those drawings requested to prepare shop drawings will be provided. These drawings are provided only after award.

1.5.3.2 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction data for the referenced project. Any other use or reuse is at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim, and waives to the fullest extent permitted by law any claim or cause of action of any nature against the Government, its agents, or its subconsultants that may arise out of or in connection with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities, or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic source drawing files are not construction documents. Differences may exist between the source drawing files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic source drawing files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. The Contractor is responsible for determining if any conflict exists. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished source drawing files, the signed and sealed construction documents govern. Use of these source drawing files does not relieve the Contractor of the duty to fully comply with the contract documents, including and without limitation the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic source drawing files for use in producing construction data related to this contract, remove all previous indication of ownership (seals, logos, signatures, initials and dates).

1.6 QUANTITY OF SUBMITTALS

1.6.1 Number of SD-01 Preconstruction Submittal Copies

Unless otherwise specified, submit two sets of administrative submittals.

1.6.2 Number of SD-04 Samples

- a. Submit two samples, or two sets of samples showing the range of variation, of each required item. One approved sample or set of samples will be retained by the approving authority and one will be

returned to the Contractor.

- b. Submit one sample panel or provide one sample installation where directed. Include components listed in the technical section or as directed.
- c. Submit one sample installation, where directed.
- d. Submit one sample of nonsolid materials.

1.7 INFORMATION ONLY SUBMITTALS

Submittals without a "G" designation must be certified by the QC manager and submitted to the Contracting Officer for information-only. Provide information-only submittals to the Contracting Officer a minimum of 14 calendar days prior to the Preparatory Meeting for the associated Definable Feature of Work (DFOW). Approval of the Contracting Officer is not required on information only submittals. The Contracting Officer will mark "receipt acknowledged" on submittals for information and will return only the transmittal cover sheet to the Contractor. Normally, submittals for information only will not be returned. However, the Government reserves the right to return unsatisfactory submittals and require the Contractor to resubmit any item found not to comply with the contract. This does not relieve the Contractor from the obligation to furnish material conforming to the plans and specifications; will not prevent the Contracting Officer from requiring removal and replacement of nonconforming material incorporated in the work; and does not relieve the Contractor of the requirement to furnish samples for testing by the Government laboratory or for check testing by the Government in those instances where the technical specifications so prescribe.

1.8 PROJECT SUBMITTAL REGISTER

A sample Project Submittal Register showing items of equipment and materials for when submittals are required by the specifications is provided as an attachment to Section 00 09 01.

1.8.1 Submittal Management

Prepare and maintain a submittal register, as the work progresses. Do not change data that is output in columns (c), (d), (e), and (f) as delivered by Government; retain data that is output in columns (a), (g), (h), and (i) as approved. As an attachment, provide a submittal register showing items of equipment and materials for which submittals are required by the specifications. This list may not be all-inclusive and additional submittals may be required. Maintain a submittal register for the project in accordance with Section 01 45 00.15 10 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE(RMS CM).

Column (c): Lists specification section in which submittal is required.

Column (d): Lists each submittal description (SD Number. and type, e.g., SD-02 Shop Drawings) required in each specification section.

Column (e): Lists one principal paragraph in each specification section where a material or product is specified. This listing is only to facilitate locating submitted requirements. Do not

consider entries in column (e) as limiting the project requirements.

Thereafter, the Contractor is to track all submittals by maintaining a complete list, including completion of all data columns and all dates on which submittals are received by and returned by the Government.

1.8.2 Preconstruction Use of Submittal Register

Submit the submittal register. Include the QC plan and the project schedule. Verify that all submittals required for the project are listed and add missing submittals. Coordinate and complete the following fields on the register submitted with the QC plan and the project schedule:

Column (a) Activity Number: Activity number from the project schedule.

Column (g) Contractor Submit Date: Scheduled date for the approving authority to receive submittals.

Column (h) Contractor Approval Date: Date that Contractor needs approval of submittal.

Column (i) Contractor Material: Date that Contractor needs material delivered to Contractor control.

1.8.3 Contractor Use of Submittal Register

Update the following fields in the Government-furnished submittal register program or equivalent fields in the program used by the Contractor with each submittal throughout the contract.

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (j) Action Code (k): Date of action used to record Contractor's review when forwarding submittals to QC.

Column (l) Date submittal transmitted.

Column (q) Date approval was received.

1.8.4 Approving Authority Use of Submittal Register

Update the following fields:

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (l) Date submittal was received.

Column (m) through (p) Dates of review actions.

Column (q) Date of return to Contractor.

1.8.5 Action Codes

1.8.6 Delivery of Copies

Submit an updated electronic copy of the submittal register to the Contracting Officer with each invoice request. Provide an updated Submittal Register monthly regardless of whether an invoice is submitted.

1.9 VARIATIONS

Variations from contract requirements require Contracting Officer approval pursuant to contract Clause FAR 52.236-21 Specifications and Drawings for Construction, and will be considered where advantageous to the Government.

1.9.1 Considering Variations

Discussion of variations with the Contracting Officer before submission [of a variation submittal] will help ensure that functional and quality requirements are met and minimize rejections and resubmittals. For variations that include design changes or some material or product substitutions, the Government may require an evaluation and analysis by a licensed professional engineer hired by the contractor.

Specifically point out variations from contract requirements in a [transmittal letter] [variation submittal]. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

1.9.2 Proposing Variations

Check the column "variation" of ENG Form 4025 for submittals that include variations proposed by the Contractor. Set forth in writing the reason for any variations and note such variations on the submittal. The Government reserves the right to rescind inadvertent approval of submittals containing unnoted variations.

1.9.3 Warranting that Variations are Compatible

When delivering a variation for approval, the Contractor warrants that this contract has been reviewed to establish that the variation, if incorporated, will be compatible with other elements of work.

1.9.4 Review Schedule Extension

In addition to the normal submittal review period, a period of 14 calendar days will be allowed for the Government to consider submittals with variations.

1.10 SCHEDULING

Schedule and submit concurrently product data and shop drawings covering component items forming a system or items that are interrelated. Submit pertinent certifications at the same time. No delay damages or time extensions will be allowed for time lost in late submittals.

- a. Coordinate scheduling, sequencing, preparing, and processing of submittals with performance of work so that work will not be delayed by submittal processing. The Contractor is responsible for additional

time required for Government reviews resulting from required resubmittals. The review period for each resubmittal is the same as for the initial submittal.

- b. Submittals required by the contract documents are listed on the submittal register. If a submittal is listed in the submittal register but does not pertain to the contract work, the Contractor is to include the submittal in the register and annotate it "N/A" with a brief explanation. Approval by the Contracting Officer does not relieve the Contractor of supplying submittals required by the contract documents but that have been omitted from the register or marked "N/A."
- c. Resubmit the submittal register and annotate it monthly with actual submission and approval dates. When all items on the register have been fully approved, no further resubmittal is required.

Contracting Officer review will be completed within 7 calendar days after the date of submission.

1.11 GOVERNMENT APPROVING AUTHORITY

When the approving authority is the Contracting Officer, the Government will:

- a. Note the date on which the submittal was received.
- b. Review submittals for approval within the scheduling period specified and only for conformance with project design concepts and compliance with contract documents.
- c. Identify returned submittals with one of the actions defined in paragraph REVIEW NOTATIONS and with comments and markings appropriate for the action indicated.

Upon completion of review of submittals requiring Government approval, stamp and date submittals. A copy of the submittal will be retained by the Contracting Officer and a copy of the submittal will be returned to the Contractor.

1.11.1 Review Notations

Submittals will be returned to the Contractor with the following notations:

- a. Submittals marked "approved" or "accepted" authorize proceeding with the work covered.
- b. Submittals marked "approved as noted" or "approved, except as noted, resubmittal not required," authorize proceeding with the work covered provided that the Contractor takes no exception to the corrections.
- c. Submittals marked "not approved," "disapproved," or "revise and resubmit" indicate incomplete submittal or noncompliance with the contract requirements or design concept. Resubmit with appropriate changes. Do not proceed with work for this item until the resubmittal is approved.
- d. Submittals marked "not reviewed" indicate that the submittal has been previously reviewed and approved, is not required, does not have evidence of being reviewed and approved by Contractor, or is not

complete. A submittal marked "not reviewed" will be returned with an explanation of the reason it is not reviewed. Resubmit submittals returned for lack of review by Contractor or for being incomplete, with appropriate action, coordination, or change.

- e. Submittals marked "receipt acknowledged" indicate that submittals have been received by the Government. This applies only to "information-only submittals" as previously defined.

1.12 DISAPPROVED SUBMITTALS

Make corrections required by the Contracting Officer. If the Contractor considers any correction or notation on the returned submittals to constitute a change to the contract drawings or specifications, give notice to the Contracting Officer as required under the FAR clause titled CHANGES. The Contractor is responsible for the dimensions and design of connection details and the construction of work. Failure to point out variations may cause the Government to require rejection and removal of such work at the Contractor's expense.

If changes are necessary to submittals, make such revisions and resubmit in accordance with the procedures above. No item of work requiring a submittal change is to be accomplished until the changed submittals are approved.

1.13 APPROVED SUBMITTALS

The Contracting Officer's approval of submittals is not to be construed as a complete check, and indicates only that

Approval or acceptance by the Government for a submittal does not relieve the Contractor of the responsibility for meeting the contract requirements or for any error that may exist, because under the Quality Control (QC) requirements of this contract, the Contractor is responsible for ensuring information contained within each submittal accurately conforms with the requirements of the contract documents.

After submittals have been approved or accepted by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment will be considered unless accompanied by an explanation of why a substitution is necessary.

1.14 APPROVED SAMPLES

Approval of a sample is only for the characteristics or use named in such approval and is not to be construed to change or modify any contract requirements. Before submitting samples, provide assurance that the materials or equipment will be available in quantities required in the project. No change or substitution will be permitted after a sample has been approved.

Match the approved samples for materials and equipment incorporated in the work. If requested, approved samples, including those that may be damaged in testing, will be returned to the Contractor, at its expense, upon completion of the contract. Unapproved samples will also be returned to the Contractor at its expense, if so requested.

Failure of any materials to pass the specified tests will be sufficient cause for refusal to consider, under this contract, any further samples of

the same brand or make as that material. The Government reserves the right to disapprove any material or equipment that has previously proved unsatisfactory in service.

Samples of various materials or equipment delivered on the site or in place may be taken by the Contracting Officer for testing. Samples failing to meet contract requirements will automatically void previous approvals. Replace such materials or equipment to meet contract requirements.

1.15 WITHHOLDING OF PAYMENT

Payment for materials incorporated in the work will not be made if required approvals have not been obtained.

1.16 CERTIFICATION OF SUBMITTAL DATA

Certify the submittal data as follows on Form ENG 4025: "I certify that the above submitted items had been reviewed in detail and are correct and in strict conformance with the contract drawings and specifications except as otherwise stated.

_____ NAME OF CONTRACTOR _____ SIGNATURE OF CONTRACTOR

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

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GOVERNMENTAL SAFETY REQUIREMENTS

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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B30.3	(2020) Tower Cranes
ASME B30.5	(2021) Mobile and Locomotive Cranes
ASME B30.7	(2021) Winches
ASME B30.8	(2020) Floating Cranes and Floating Derricks
ASME B30.9	(2018) Slings
ASME B30.20	(2018) Below-the-Hook Lifting Devices
ASME B30.22	(2023) Articulating Boom Cranes
ASME B30.23	(2022) Personnel Lifting Systems Safety Standard for Cableways, Cranes, Derricks, Hoists, Hooks, Jacks, and Slings
ASME B30.26	(2015; R 2020) Rigging Hardware

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.22	(2007; R 2017) Safety Requirements for Rope-Guided and Non-Guided Workers' Hoists
ASSP A10.34	(2021) Protection of the Public on or Adjacent to Construction Sites
ASSP A10.44	(2020) Control of Energy Sources (Lockout/Tagout) for Construction and Demolition Operations
ASSP Z244.1	(2016) The Control of Hazardous Energy Lockout, Tagout and Alternative Methods
ASSP Z359.0	(2018) Definitions and Nomenclature Used for Fall Protection and Fall Arrest
ASSP Z359.1	(2020) The Fall Protection Code
ASSP Z359.2	(2017) Minimum Requirements for a

Comprehensive Managed Fall Protection Program

ASSP Z359.3	(2019) Safety Requirements for Lanyards and Positioning Lanyards
ASSP Z359.4	(2013) Safety Requirements for Assisted-Rescue and Self-Rescue Systems, Subsystems and Components
ASSP Z359.6	(2016) Specifications and Design Requirements for Active Fall Protection Systems
ASSP Z359.7	(2019) Qualification and Verification Testing of Fall Protection Products
ASSP Z359.11	(2014) Safety Requirements for Full Body Harnesses
ASSP Z359.12	(2019) Connecting Components for Personal Fall Arrest Systems
ASSP Z359.13	(2013) Personal Energy Absorbers and Energy Absorbing Lanyards
ASSP Z359.14	(2014) Safety Requirements for Self-Retracting Devices for Personal Fall Arrest and Rescue Systems
ASSP Z359.15	(2014) Safety Requirements for Single Anchor Lifelines and Fall Arresters for Personal Fall Arrest Systems
ASSP Z359.16	(2016) Safety Requirements for Climbing Ladder Fall Arrest Systems
ASSP Z359.18	(2017) Safety Requirements for Anchorage Connectors for Active Fall Protection Systems

ASTM INTERNATIONAL (ASTM)

ASTM F855	(2019) Standard Specifications for Temporary Protective Grounds to Be Used on De-energized Electric Power Lines and Equipment
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INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1048	(2016) Guide for Protective Grounding of Power Lines
IEEE C2	(2023) National Electrical Safety Code

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 10	(2022; ERTA 1 2021) Standard for Portable Fire Extinguishers
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NFPA 51B	(2024) Standard for Fire Prevention During Welding, Cutting, and Other Hot Work
NFPA 70	(2023; ERTA 4 2023; ERTA 5 2023; ERTA 6 2023) National Electrical Code
NFPA 70E	(2024) Standard for Electrical Safety in the Workplace
NFPA 241	(2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations
NFPA 306	(2024) Standard for the Control of Gas Hazards on Vessels

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-222	(2018H; Add 1 2019) Structural Standard for Antenna Supporting Structures and Antennas and Small Wind Turbine Support Structures
TIA-1019	(2012; R 2016) Standard for Installation, Alteration and Maintenance of Antenna Supporting Structures and Antennas

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety and Health Requirements Manual
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U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

10 CFR 20	Standards for Protection Against Radiation
29 CFR 1910	Occupational Safety and Health Standards
29 CFR 1910.146	Permit-required Confined Spaces
29 CFR 1910.147	The Control of Hazardous Energy (Lock Out/Tag Out)
29 CFR 1910.333	Selection and Use of Work Practices
29 CFR 1915	Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment
29 CFR 1915.89	Control of Hazardous Energy (Lockout/Tags-Plus)
29 CFR 1926	Safety and Health Regulations for Construction
29 CFR 1926.16	Rules of Construction
29 CFR 1926.450	Scaffolds

29 CFR 1926.500	Fall Protection
29 CFR 1926.552	Material Hoists, Personal Hoists, and Elevators
29 CFR 1926.553	Base-Mounted Drum Hoists
29 CFR 1926.1400	Cranes and Derricks in Construction
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
CPL 02-01-056	(2014) Inspection Procedures for Accessing Communication Towers by Hoist
CPL 2.100	(1995) Application of the Permit-Required Confined Spaces (PRCS) Standards, 29 CFR 1910.146

1.2 DEFINITIONS

1.2.1 Competent Person (CP)

The CP is a person designated in writing, who, through training, knowledge and experience, is capable of identifying, evaluating, and addressing existing and predictable hazards in the working environment or working conditions that are dangerous to personnel, and who has authorization to take prompt corrective measures with regards to such hazards.

1.2.2 Competent Person, Confined Space

The CP, Confined Space, is a person meeting the competent person requirements as defined EM 385-1-1 Chapter 34, with thorough knowledge of OSHA's Confined Space Standard, 29 CFR 1910.146, and designated in writing to be responsible for the immediate supervision, implementation and monitoring of the confined space program, who through training, knowledge and experience in confined space entry is capable of identifying, evaluating and addressing existing and potential confined space hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.3 Competent Person, Cranes and Rigging

The CP, Cranes and Rigging, as defined in EM 385-1-1 Chapters 15 and 16, is a person meeting the competent person requirements, who has been designated in writing to be responsible for the immediate supervision, implementation and monitoring of the Crane and Rigging Program, who through training, knowledge and experience in crane and rigging is capable of identifying, evaluating and addressing existing and potential hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.4 Competent Person, Excavation/Trenching

A CP, Excavation/Trenching, is a person meeting the competent person requirements as defined in EM 385-1-1 Chapter 25 and 29 CFR 1926, who has been designated in writing to be responsible for the immediate supervision, implementation and monitoring of the excavation/trenching program, who through training, knowledge and experience in excavation/trenching is

capable of identifying, evaluating and addressing existing and potential hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.5 Competent Person, Fall Protection

The CP, Fall Protection, is a person meeting the competent person requirements as defined in EM 385-1-1 Chapter 21 and in accordance with ASSP Z359.0, who has been designated in writing by the employer to be responsible for immediate supervising, implementing and monitoring of the fall protection program, who through training, knowledge and experience in fall protection and rescue systems and equipment, is capable of identifying, evaluating and addressing existing and potential fall hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.6 Competent Person, Scaffolding

The CP, Scaffolding is a person meeting the competent person requirements in EM 385-1-1 Chapter 22, and designated in writing by the employer to be responsible for immediate supervising, implementing and monitoring of the scaffolding program. The CP for Scaffolding has enough training, knowledge and experience in scaffolding to correctly identify, evaluate and address existing and potential hazards and also has the authority to take prompt corrective measures with regard to these hazards. CP qualifications must be documented including experience on the specific scaffolding systems/types being used, assessment of the base material that the scaffold will be erected upon, load calculations for materials and personnel, and erection and dismantling. The CP for scaffolding must have a documented minimum of 8-hours of scaffold training to include training on the specific type of scaffold being used (e.g. mast-climbing, adjustable, tubular frame), in accordance with EM 385-1-1 paragraph 22-2.a.

1.2.7 Competent Person (CP) Trainer

A competent person trainer as defined in EM 385-1-1 Chapter 2, who is qualified in the training material presented, and who possesses a working knowledge of applicable technical regulations, standards, equipment and systems related to the subject matter on which they are training Competent Persons. A competent person trainer must be familiar with the typical hazards and the equipment used in the industry they are instructing. The training provided by the competent person trainer must be appropriate to that specific industry. The competent person trainer must evaluate the knowledge and skills of the competent persons as part of the training process.

1.2.8 High Risk Activities

High Risk Activities are activities that involve work at heights, crane and rigging, excavations and trenching, scaffolding, electrical work, and confined space entry.

1.2.9 High Visibility Accident

A High Visibility Accident is any mishap which may generate publicity or high visibility.

1.2.10 Load Handling Equipment (LHE)

LHE is a term used to describe cranes, hoists and all other hoisting equipment (hoisting equipment means equipment, including crane, derricks, hoists and power operated equipment used with rigging to raise, lower or horizontally move a load).

1.2.11 Medical Treatment

Medical Treatment is treatment administered by a physician or by registered professional personnel under the standing orders of a physician. Medical treatment does not include first aid treatment even when provided by a physician or registered personnel.

1.2.12 Near Miss

A Near Miss is a mishap resulting in no personal injury and zero property damage, but given a shift in time or position, damage or injury may have occurred (e.g., a worker falls off a scaffold and is not injured; a crane swings around to move the load and narrowly misses a parked vehicle).

1.2.13 Operating Envelope

The Operating Envelope is the area surrounding any crane or load handling equipment. Inside this "envelope" is the crane, the operator, riggers and crane walkers, other personnel involved in the operation, rigging gear between the hook, the load, the crane's supporting structure (i.e. ground or rail), the load's rigging path, the lift and rigging procedure.

1.2.14 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the work, or the project.

1.2.15 Qualified Person, Fall Protection (QP for FP)

A QP for FP is a person meeting the definition requirements of EM 385-1-1 Chapter 21, and ASSP Z359.2 standard, having a recognized degree or professional certificate and with extensive knowledge, training and experience in the fall protection and rescue field who is capable of designing, analyzing, and evaluating and specifying fall protection and rescue systems.

1.2.16 Recordable Injuries or Illnesses

Recordable Injuries or Illnesses are any work-related injury or illness that results in:

- a. Death, regardless of the time between the injury and death, or the length of the illness;
- b. Days away from work (any time lost after day of injury/illness onset);
- c. Restricted work;
- d. Transfer to another job;

- e. Medical treatment beyond first aid;
- f. Loss of consciousness; or
- g. A significant injury or illness diagnosed by a physician or other licensed health care professional, even if it did not result in (a) through (f) above

1.2.17 Government Property and Equipment

Interpret "USACE" property and equipment specified in USACE EM 385-1-1 as Government property and equipment.

1.2.18 Load Handling Equipment (LHE) Accident or Load Handling Equipment Mishap

A LHE accident occurs when any one or more of the eight elements in the operating envelope fails to perform correctly during operation, including operation during maintenance or testing resulting in personnel injury or death; material or equipment damage; dropped load; derailment; two-blocking; overload; or collision, including unplanned contact between the load, crane, or other objects. A dropped load, derailment, two-blocking, overload and collision are considered accidents, even though no material damage or injury occurs. A component failure (e.g., motor burnout, gear tooth failure, bearing failure) is not considered an accident solely due to material or equipment damage unless the component failure results in damage to other components (e.g., dropped boom, dropped load, or roll over). Document an LHE mishap using the Crane High Hazard working group mishap reporting form.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Accident Prevention Plan (APP); G

APP - Construction; G

Dive Operations Plan; G

SD-06 Test Reports

Monthly Exposure Reports

Notifications and Reports

Accident Reports; G

LHE Inspection Reports

SD-07 Certificates

Crane Operators/Riggers
Standard Lift Plan; G
Critical Lift Plan ; G
Naval Architecture Analysis; G
Activity Hazard Analysis (AHA)
Confined Space Entry Permit
Hot Work Permit
Certificate of Compliance
License Certificates
Radiography Operation Planning Work Sheet; G
Portable Gauge Operations Planning Worksheet; G

1.4 MONTHLY EXPOSURE REPORTS

Provide a Monthly Exposure Report and attach to the monthly billing request. This report is a compilation of employee-hours worked each month for all site workers, both Prime and subcontractor. Failure to submit the report may result in retention of up to 10 percent of the voucher.

1.5 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this Contract, comply with the most recent edition of USACE EM 385-1-1, and the following federal, state, and local laws, ordinances, criteria, rules and regulations. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

1.6 SITE QUALIFICATIONS, DUTIES, AND MEETINGS

1.6.1 Personnel Qualifications

1.6.1.1 Site Safety and Health Officer (SSHO)

Provide an SSHO that meets the requirements of EM 385-1-1 Chapter 2 paragraph 3. The SSHO must ensure that the requirements of 29 CFR 1926.16 are met for the project. Provide a Safety oversight team that includes a minimum of one person at each project site to function as the Site Safety and Health Officer (SSHO). The SSHO or an equally-qualified Alternate SSHO must be at the work site at all times to implement and administer the Contractor's safety program and Government-accepted Accident Prevention Plan. The SSHO and Alternate SSHO must have the required training, experience, and qualifications in accordance with EM 385-1-1 Chapter 2 paragraph 3, and all associated sub-paragraphs.

1.6.1.1.1 Additional Site Safety and Health Officer (SSHO) Requirements and Duties

The SSHO may not serve as the Quality Control Manager. The SSHO may not serve as the Superintendent.

1.6.1.2 Competent Person Qualifications

Provide Competent Persons in accordance with EM 385-1-1, Chapter 2 paragraph 3 and herein. Competent Persons for high risk activities include confined space, cranes and rigging, excavation/trenching, fall protection, and electrical work. The CP for these activities must be designated in writing, and meet the requirements for the specific activity (i.e. competent person, fall protection).

The Competent Person identified in the Contractor's Safety and Health Program and accepted Accident Prevention Plan, must be on-site at all times when the work that presents the hazards associated with their professional expertise is being performed. Provide the credentials of the Competent Persons(s) to the Contracting Officer for information in consultation with the Safety Office.

1.6.1.2.1 Competent Person for Confined Space Entry

Provide a Confined Space (CP) Competent Person who meets the requirements of EM 385-1-1, Chapter 34, and herein. The CP for Confined Space Entry must supervise the entry into each confined space in accordance with EM 385-1-1, Chapter 34.

Since this work involves operations that handle combustible or hazardous materials, this person must have the ability to understand and follow through on the air sampling, Personal Protective Equipment (PPE), and instructions of a Marine Chemist, Coast Guard authorized persons, or Certified Industrial Hygienist. Confined space and enclosed space work must comply with NFPA 306, OSHA 29 CFR 1915, Subpart B, "Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment," or as applicable, 29 CFR 1910.146 for general industry.

1.6.1.2.2 Competent Person for Scaffolding

Provide a Competent Person for Scaffolding who meets the requirements of EM 385-1-1, Chapter 22 and herein.

1.6.1.2.3 Competent Person for Fall Protection

Provide a Competent Person for Fall Protection who meets the requirements of EM 385-1-1, Chapters 21 and 22, and herein.

1.6.1.3 Qualified Trainer Requirements

Individuals qualified to instruct the 40 hour contract safety awareness course, or portions thereof, must meet the definition of a Competent Person Trainer, and, at a minimum, possess a working knowledge of the following subject areas: EM 385-1-1, Electrical Standards, Lockout/Tagout, Fall Protection, Confined Space Entry for Construction; Excavation, Trenching and Soil Mechanics, and Scaffolds in accordance with 29 CFR 1926.450, Subpart L.

Instructors are required to:

- a. Prepare class presentations that cover construction-related safety requirements.
- b. Ensure that all attendees attend all sessions by using a class roster signed daily by each attendee. Maintain copies of the roster for at least five years. This is a certification class and must be attended 100 percent. In cases of emergency where an attendee cannot make it to a session, the attendee can make it up in another class session for the same subject.
- c. Update training course materials whenever an update of the EM 385-1-1 becomes available.
- d. Provide a written exam of at least 50 questions. Students are required to answer 80 percent correctly to pass.
- e. Request, review and incorporate student feedback into a continuous course improvement program.

1.6.1.4 Dredging Contract Requirements

1.6.1.4.1 Dredging Safety Personnel Requirements

- a. Provide a minimum of one full-time SSHO assigned per project site for the primary working shift.
- b. For a project involving multiple work shifts, provide one full-time SSHO for each additional shift.
- c. An example of one dredging project site is reflected in each of the following:
 - (1) a mechanical dredge, tug(s) and scow(s), scow route, and material placement site; or
 - (2) a hydraulic pipeline dredge, attendant plant, and material placement site; or,
 - (3) a hopper dredge (include land-based material placement site - if applicable.)

1.6.1.4.2 SSHO Requirements for Dredging

- a. In addition to requirements stated elsewhere in this specification, an individual serving as a SSHO must be present at the construction work site, located so that they have full mobility and access to all major work operations when work is being performed. The SSHO must be available during their shift for immediate verbal consultation and notification, either by phone or radio.
- b. The SSHO must report to a senior project (or corporate) official.
- c. The SSHO must inspect all work areas and operations during initial set-up. For additional shifts, the SSHO must inspect all work areas and operations and provide personal oversight during dredge operations.

1.6.1.4.3 Collateral Duty Safety Officer (CDSO) Requirements for Dredging

- a. A CDSO is an individual who is assigned collateral duty safety responsibilities in addition to their full-time occupation, and who supports and supplements the SSHO efforts in managing, implementing and enforcing the Contractor's Safety and Health Program. An assigned CDSO must be an individual(s) with work oversight responsibilities, such as master, mate, fill foreman, or superintendent. A CDSO must not be an employee responsible for continuous mechanical or equipment operations, such as an equipment operator.
- b. A CDSO must perform safety program tasks as assigned by the SSHO and must report safety findings to the SSHO. The SSHO must document results of safety findings and provide information for inclusion in the CQC reports to the Contracting Officer.

1.6.1.4.4 Safety Personnel Training Requirements for Dredging

A SSHO and a CDSO for dredging Contracts must take either a formal classroom or online OSHA 30-hour Construction Safety Course, or an equivalent 30 hours of formal classroom or online safety and health training covering the subjects of the OSHA 30-hour Course in accordance with EM 385-1-1 section 2-3, applicable to dredging work, and given by qualified instructors. In exception to EM 385-1-1 paragraph 2-4.b, comply with the following:

- a. The SSHO must maintain competency through having taken 24 hours of formal classroom or online safety and health related coursework every year for, at minimum, the last 3 years. Hours spent as an instructor in such courses will be considered the same as attending them, but each course only gets credit once (for example, instructing a 1-hour asbestos awareness course five times in a year provides one hour credit for training).
- b. The SSHO and a CDSO must have a minimum of five years of experience within the past ten years in one of the following:
 - (1) Supervising/managing dredging activities (managing safety programs or processes and conducting hazard analyses and developing controls in activities or environments with similar hazards)
 - (2) Supervising/managing marine construction activities (managing safety programs or processes and conducting hazard analyses and developing controls in activities or environments with similar hazards)
 - (3) Supervising/managing land-based construction activities (managing safety programs or processes and conducting hazard analyses and developing controls in activities or environments with similar hazards)

1.6.1.5 Crane Operators/Riggers

Provide Operators, Signal Persons, and Riggers meeting the requirements in EM 385-1-1, Chapter 15 for Riggers and Chapter 16 for Crane Operators and Signal Persons. In addition, for mobile cranes with Original Equipment Manufacturer (OEM) rated capacities of 50,000 pounds or greater, designate crane operators qualified by a source that qualifies crane operators (i.e., union, a Government agency, or an organization that tests and qualifies

crane operators). Provide proof of current qualification.

1.6.2 Personnel Duties

1.6.2.1 Duties of the Site Safety and Health Officer (SSHO)

The SSHO must:

- a. Conduct daily safety and health inspections and maintain a written log which includes area/operation inspected, date of inspection, identified hazards, recommended corrective actions, estimated and actual dates of corrections. Attach safety inspection logs to the Contractors' daily production report.
- b. Conduct mishap investigations and complete required accident reports. Report mishaps and near misses.
- c. Use and maintain OSHA's Form 300 to log work-related injuries and illnesses occurring on the project site for Prime Contractors and subcontractors, and make available to the Contracting Officer upon request. Post and maintain the Form 300A on the site Safety Bulletin Board.
- d. Maintain applicable safety reference material on the job site.
- e. Attend the pre-construction conference, pre-work meetings including preparatory meetings, and periodic in-progress meetings.
- f. Review the APP and AHAs for compliance with EM 385-1-1, and approve, sign, implement and enforce them.
- g. Establish a Safety and Occupational Health (SOH) Deficiency Tracking System that lists and monitors outstanding deficiencies until resolution.
- h. Ensure subcontractor compliance with safety and health requirements.
- i. Maintain a list of hazardous chemicals on site and their material Safety Data Sheets (SDS).
- j. Maintain a weekly list of high hazard activities involving energy, equipment, excavation, entry into confined space, and elevation, and be prepared to discuss details during QC Meetings.
- k. Provide and keep a record of site safety orientation and indoctrination for Contractor employees, subcontractor employees, and site visitors.

Superintendent, QC Manager, and SSHO are subject to dismissal if the above or any other required duties are not being effectively carried out. If either the Superintendent, QC Manager, or SSHO are dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out.

1.6.3 Meetings

1.6.3.1 Preconstruction Conference

- a. Contractor representatives who have a responsibility or significant

role in accident prevention on the project must attend the preconstruction conference. This includes the project superintendent, Site Safety and Occupational Health Officer, quality control manager, or any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program and procedures associated with it).

- b. Discuss the details of the submitted APP to include incorporated plans, programs, procedures and a listing of anticipated AHAs that will be developed and implemented during the performance of the Contract. This list of proposed AHAs will be reviewed and an agreement will be reached between the Contractor and the Contracting Officer as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, and Government review of AHAs to preclude project delays.
- c. Deficiencies in the submitted APP, identified during the Contracting Officer's review, must be corrected, and the APP re-submitted for review prior to the start of construction. Work is not permitted to begin until an APP is established that is acceptable to the Contracting Officer.

1.6.3.2 Safety Meetings

Conduct safety meetings to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHA (by trade), establish safe working procedures for anticipated hazards, and provide pertinent Safety and Occupational Health (SOH) training and motivation. Conduct meetings at least once a month for all supervisors at the project location. The SSHO, supervisors, foremen, or CDSOs must conduct meetings at least once a week for the trade workers. Document meeting minutes to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and furnish copies to the Contracting Officer on request. Notify the Contracting Officer of all scheduled meetings 7 calendar days in advance.

1.6.3.3 Daily Toolbox Safety Talks

Conduct daily "Toolbox Safety Talks" or "Toolbox Talks" to reinforce key safety procedures and involve workers in discussion of safety and health procedures and program requirements. Document meeting with meeting agenda or discussion summary, to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and attach documentation to the Contractors' daily safety inspection logs in accordance with paragraph 1.6.2.1.

1.7 ACCIDENT PREVENTION PLAN (APP)

1.7.1 APP - Construction

A qualified person must prepare the written site-specific APP. Prepare the APP in accordance with the format and requirements of EM 385-1-1, and as supplemented herein. Cover all paragraph and subparagraph elements in EM 385-1-1. The APP must be job-specific and address any unusual or unique aspects of the project or activity for which it is written. The APP must interface with the Contractor's overall safety and health program referenced in the APP in the applicable APP element, and made site-specific. Describe the methods to evaluate past safety performance of potential subcontractors in the selection process. Also, describe

innovative methods used to ensure and monitor safe work practices of subcontractors. The Government considers the Prime Contractor to be the "controlling authority" for all work site safety and health of the subcontractors. Contractors are responsible for informing their subcontractors of the safety provisions under the terms of the Contract and the penalties for noncompliance, coordinating the work to prevent one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting subcontractor operations to ensure that accident prevention responsibilities are being carried out. The APP must be signed by an officer of the firm (Prime Contractor senior person), the individual preparing the APP, the on-site superintendent, the designated SSHO, the Contractor Quality Control Manager, and any designated Certified Safety Professional (CSP) or Certified Health Physicist (CIH). The SSHO must provide and maintain the APP and a log of signatures by each subcontractor foreman, attesting that they have read and understand the APP, and make the APP and log available on-site to the Contracting Officer. If English is not the foreman's primary language, the Prime Contractor must provide an interpreter.

Submit the APP to the Contracting Officer within 15 calendar days of Contract award and not less than 10 calendar days prior to the date of the preconstruction conference for acceptance. Work cannot proceed without an accepted APP. Once reviewed and accepted by the Contracting Officer, the APP and attachments will be enforced as part of the Contract. Disregarding the provisions of this Contract or the accepted APP is cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified. Continuously review and amend the APP, as necessary, throughout the life of the Contract. Changes to the accepted APP must be made with the knowledge and concurrence of the Contracting Officer, project superintendent, SSHO and Quality Control Manager. Incorporate unusual or high-hazard activities not identified in the original APP as they are discovered. Should any severe hazard exposure (i.e. imminent danger) become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the Contracting Officer within 24 hours of discovery. Eliminate and remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ASSP A10.34), and the environment.

1.7.2 Names and Qualifications

Provide plans in accordance with the requirements outlined in section 2-7 of EM 385-1-1, including the following:

- a. Names and qualifications (resumes including education, training, experience and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer and other competent and qualified personnel to be used. Specify the duties of each position.
- b. Qualifications of competent and of qualified persons. As a minimum, designate and submit qualifications of competent persons for each of the following major areas: excavation; scaffolding; fall protection; hazardous energy; confined space; health hazard recognition, evaluation and control of chemical, physical and biological agents; and personal protective equipment and clothing to include selection, use and maintenance.

1.7.3 Plans

Provide plans in the APP in accordance with the requirements outlined in Appendix A of EM 385-1-1, including the following:

1.7.3.1 Confined Space Entry Plan

Develop a confined or enclosed space entry plan in accordance with EM 385-1-1, applicable OSHA standards 29 CFR 1910, 29 CFR 1915, and 29 CFR 1926, OSHA Directive CPL 2.100, and any other federal, state and local regulatory requirements identified in this Contract. Identify the qualified person's name and qualifications, training, and experience. Delineate the qualified person's authority to direct work stoppage in the event of hazardous conditions. Include procedure for rescue by Contractor personnel and the coordination with emergency responders. (If there is no confined space work, include a statement that no confined space work exists and none will be created.)

1.7.3.2 Standard Lift Plan (SLP)

Plan lifts to avoid situations where the operator cannot maintain safe control of the lift. Prepare a written SLP in accordance with EM 385-1-1, Chapter 16 paragraph 7, using ENG Form 6203 (16-10.a) for every lift or series of lifts (if duty cycle or routine lifts are being performed). The SLP must be developed, reviewed and accepted by all personnel involved in the lift in conjunction with the associated AHA. Signature on the AHA constitutes acceptance of the plan. Maintain the SLP on the LHE for the current lift(s) being made. Maintain historical SLPs for a minimum of three months.

1.7.3.3 Critical Lift Plan - Crane or Load Handling Equipment

Provide a Critical Lift Plan as required by EM 385-1-1, Chapter 16 paragraph 7, using ENG Form 6213 (16-10.c). In addition, Critical Lift Plans are required for the following:

- a. Lifts over 50 percent of the capacity of barge mounted mobile crane's hoist.
- b. When working around energized power lines where the work will get closer than the minimum clearance distance in EM 385-1-1 Table 16-1.
- c. For lifts with anticipated binding conditions.
- d. When erecting cranes.

1.7.3.3.1 Critical Lift Plan Planning and Schedule

Critical lifts require detailed planning and additional or unusual safety precautions. Develop and submit a critical lift plan to the Contracting Officer 30 calendar days prior to critical lift. Comply with load testing requirements in accordance with EM 385-1-1, paragraph 16-7.d.

1.7.3.3.2 Lifts of Personnel

In addition to the requirements of EM 385-1-1, paragraph 16-8.hh, for lifts of personnel, demonstrate compliance with the requirements of 29 CFR 1926.1400 and EM 385-1-1.

1.7.3.4 Barge Mounted Mobile Crane Lift Plan

Provide a Naval Architecture Analysis and include an LHE Manufacturer's Floating Service Load Chart in accordance with EM 385-1-1, paragraph 16-8.z(3).

1.7.3.5 Multi-Purpose Machines, Material Handling Equipment, and Construction Equipment Lift Plan

Multi-purpose machines, material handling equipment, and construction equipment used to lift loads that are suspended by rigging gear, require proof of authorization from the machine OEM that the machine is capable of making lifts of loads suspended by rigging equipment. Written approval from a qualified registered professional engineer, after a safety analysis is performed, is allowed in lieu of the OEM's approval. Demonstrate that the operator is properly trained and that the equipment is properly configured to make such lifts and is equipped with a load chart.

1.7.3.6 Fall Protection and Prevention (FP&P) Plan

The plan must be in accordance with the requirements of EM 385-1-1, Chapter 21 and ASSP Z359.2, be site specific, and address all fall hazards in the work place and during different phases of construction. Address how to protect and prevent workers from falling to lower levels when they are exposed to fall hazards above 6 feet. A competent person or qualified person for fall protection must prepare and sign the plan documentation. Include fall protection and prevention systems, equipment and methods employed for every phase of work, roles and responsibilities, assisted rescue, self-rescue and evacuation procedures, training requirements, and monitoring methods. Review and revise, as necessary, the Fall Protection and Prevention Plan documentation as conditions change, but at a minimum every six months, for lengthy projects, reflecting any changes during the course of construction due to changes in personnel, equipment, systems or work habits. Keep and maintain the accepted Fall Protection and Prevention Plan documentation at the job site for the duration of the project. Include the Fall Protection and Prevention Plan documentation in the Accident Prevention Plan (APP).

1.7.3.7 Rescue and Evacuation Plan

Provide a Rescue and Evacuation Plan in accordance with EM 385-1-1 Chapter 21 and ASSP Z359.2, and include in the FP&P Plan and as part of the APP. Include a detailed discussion of the following: methods of rescue; methods of self-rescue; equipment used; training requirement; specialized training for the rescuers; procedures for requesting rescue and medical assistance; and transportation routes to a medical facility.

1.7.3.8 Hazardous Energy Control Program (HECP)

Develop a HECP in accordance with EM 385-1-1 Chapter 12, 29 CFR 1910.147, 29 CFR 1910.333, 29 CFR 1915.89, ASSP Z244.1, and ASSP A10.44. Submit this HECP as part of the Accident Prevention Plan (APP). Conduct a preparatory meeting and inspection with all effected personnel to coordinate all HECP activities. Document this meeting and inspection in accordance with EM 385-1-1, Chapter 12. Ensure that each employee is familiar with and complies with these procedures.

1.7.3.9 Excavation Plan

Identify the safety and health aspects of excavation, and provide and prepare the plan in accordance with EM 385-1-1, paragraph 25-8.

1.7.3.10 Lead, Cadmium, and Chromium Compliance Plan

Identify the safety and health aspects of work involving lead, cadmium and chromium, and prepare in accordance with Section 02 83 00 LEAD REMEDIATION.

1.7.3.11 Asbestos Hazard Abatement Plan

Identify the safety and health aspects of asbestos work, and prepare in accordance with Section 02 82 00 ASBESTOS REMEDIATION.

1.7.3.12 Site Safety and Health Plan

Identify the safety and health aspects, and prepare in accordance with Section 01 35 29.13 HEALTH, SAFETY, AND EMERGENCY RESPONSE PROCEDURES FOR CONTAMINATED SITES.

1.7.3.13 Polychlorinated Biphenyls (PCB) Plan

Identify the safety and health aspects of Polychlorinated Biphenyls work, and prepare in accordance with Sections 02 84 33 REMOVAL AND DISPOSAL OF POLYCHLORINATED BIPHENYLS (PCBs) and 02 61 23 REMOVAL AND DISPOSAL OF PCB CONTAMINATED SOILS.

1.7.3.14 Site Demolition Plan

Identify the safety and health aspects, and prepare in accordance with Section 02 41 00 DECONSTRUCTION and referenced sources.

1.8 ACTIVITY HAZARD ANALYSIS (AHA)

Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor, subcontractor, or supplier performing the work, and provided for Prime Contractor review and approval before submitting to the Contracting Officer. AHAs must be signed by the SSHO, Superintendent, QC Manager and the subcontractor Foreman performing the work. Format the AHA in accordance with EM 385-1-1, Chapter 2 or as directed by the Contracting Officer. Submit the AHA for review at least 15 working days prior to the start of each activity task, or DFOW. The Government reserves the right to require the Contractor to revise and resubmit the AHA if it fails to effectively identify the work sequences, specific anticipated hazards, site conditions, equipment, materials, personnel and the control measures to be implemented.

AHAs must identify competent persons required for phases involving high risk activities, including confined entry, crane and rigging, excavations, trenching, electrical work, fall protection, and scaffolding.

1.8.1 AHA Management

Review the AHA list periodically (at least monthly) at the Contractor supervisory safety meeting, and update as necessary when procedures,

scheduling, or hazards change. Use the AHA during daily inspections by the SSHO to ensure the implementation and effectiveness of the required safety and health controls for that work activity.

1.8.2 AHA Signature Log

Each employee performing work as part of an activity, task or DFOV must review the AHA for that work and sign a signature log specifically maintained for that AHA prior to starting work on that activity. The SSHO must maintain a signature log on site for every AHA. Provide employees whose primary language is other than English, with an interpreter to ensure a clear understanding of the AHA and its contents.

1.9 DISPLAY OF SAFETY INFORMATION

1.9.1 Safety Bulletin Board

Prior to commencement of work, erect a safety bulletin board at the job site. Where size, duration, or logistics of project do not facilitate a bulletin board, an alternative method, acceptable to the Contracting Officer, that is accessible and includes all mandatory information for employee and visitor review, may be deemed as meeting the requirement for a bulletin board. Include and maintain information on safety bulletin board as required by EM 385-1-1, Chapter 1. Additional items required to be posted include:

- a. Confined space entry permit.
- b. Hot work permit.

1.9.2 Safety and Occupational Health (SOH) Deficiency Tracking System

Establish a SOH deficiency tracking system that lists and monitors the status of SOH deficiencies in chronological order. Use the tracking system to evaluate the effectiveness of the APP. A monthly evaluation of the data must be discussed in the QC or SOH meeting with everyone on the project. The list must be posted on the project bulletin board and updated daily, and provide the following information:

- a. Date deficiency identified;
- b. Description of deficiency;
- c. Name of person responsible for correcting deficiency;
- d. Projected resolution date;
- e. Date actually resolved.

1.10 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

1.11 EMERGENCY MEDICAL TREATMENT

Contractors must arrange for their own emergency medical treatment in accordance with EM 385-1-1. Government has no responsibility to provide

emergency medical treatment.

1.12 NOTIFICATIONS and REPORTS

1.12.1 Mishap Notification

Notify the Contracting Officer as soon as practical, but no more than twenty-four hours, after any mishaps, including recordable accidents, incidents, and near misses, as defined in EM 385-1-1 Chapter 2, any report of injury, illness, or any property damage. For LHE or rigging mishaps, notify the Contracting Officer as soon as practical but not more than four hours after mishap. The Contractor is responsible for obtaining appropriate medical and emergency assistance and for notifying fire, law enforcement, and regulatory agencies. Immediate reporting is required for electrical mishaps, to include Arc Flash; shock; uncontrolled release of hazardous energy (includes electrical and non-electrical); load handling equipment or rigging; fall from height (any level other than same surface); and underwater diving. These mishaps must be investigated in depth to identify all causes and to recommend hazard control measures.

Within notification include Contractor name; Contract title; type of Contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any mishap.

1.12.2 Accident Reports

- a. Conduct an accident investigation for recordable injuries and illnesses, property damage, and near misses as defined in EM 385-1-1, to establish the root cause(s) of the accident. Complete the applicable USACE Accident Report Form 3394, and provide the report to the Contracting Officer within 5 calendar days of the accident. The Contracting Officer will provide copies of any required or special forms.
- b. Near Misses: Report all "Near Misses" within 24 hrs. The Contracting Officer will provide the Contractor the required forms. Near miss reports are considered positive and proactive Contractor safety management actions.
- c. Conduct an accident investigation for any load handling equipment accident (including rigging accidents) to establish the root cause(s) of the accident. Complete the LHE Accident Report (Crane and Rigging Accident Report) form and provide the report to the Contracting Officer within 30 calendar days of the accident. Do not proceed with crane operations until cause is determined and corrective actions have been implemented to the satisfaction of the Contracting Officer. The Contracting Officer will provide a blank copy of the accident report form.

1.12.3 LHE Inspection Reports

Submit LHE inspection reports required in accordance with EM 385-1-1 and as specified herein with Daily Reports of Inspections.

1.12.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging

Provide a ENG Form 6209 Certificate of Compliance for LHE entering an activity under this Contract and in accordance with EM 385-1-1. Post certifications on the crane.

Develop a Standard Lift Plan (SLP) in accordance with EM 385-1-1, Chapter 16 using ENG Form 6203 Crane Standard Lift Plan for each lift planned. Submit SLP to the Contracting Officer for approval within 15 calendar days in advance of planned lift.

1.13 HOT WORK

1.13.1 Permit and Personnel Requirements

Submit and obtain a written permit prior to performing "Hot Work" (i.e. welding or cutting) or operating other flame-producing/spark producing devices, from the Fire Division. A permit is required from the Explosives Safety Office for work in and around where explosives are processed, stored, or handled. CONTRACTORS ARE REQUIRED TO MEET ALL CRITERIA BEFORE A PERMIT IS ISSUED. Provide at least two 20 pound 4A:20 BC rated extinguishers for normal "Hot Work". The extinguishers must be current inspection tagged, and contain an approved safety pin and tamper resistant seal. It is also mandatory to have a designated FIRE WATCH for any "Hot Work" done at this activity. The Fire Watch must be trained in accordance with NFPA 51B and remain on-site for a minimum of one hour after completion of the task or as specified on the hot work permit.

When starting work in the facility, require personnel to familiarize themselves with the location of the nearest fire alarm boxes and place in memory the emergency Fire Division phone number. REPORT ANY FIRE, NO MATTER HOW SMALL, TO THE RESPONSIBLE FIRE DIVISION IMMEDIATELY.

1.13.2 Work Around Flammable Materials

Obtain permit approval from a NFPA Certified Marine Chemist, or Certified Industrial Hygienist for "HOT WORK" within or around flammable materials (such as fuel systems or welding/cutting on fuel pipes) or confined spaces (such as sewer wet wells, manholes, or vaults) that have the potential for flammable or explosive atmospheres.

Whenever these materials, except beryllium and chromium (VI), are encountered in indoor operations, local mechanical exhaust ventilation systems that are sufficient to reduce and maintain personal exposures to within acceptable limits must be used and maintained in accordance with manufacturer's instruction and supplemented by exceptions noted in EM 385-1-1, Chapter 6.

1.14 RADIATION SAFETY REQUIREMENTS

Submit License Certificates, employee training records, and Leak Test Reports for radiation materials and equipment to the Contracting Officer and Radiation Safety Office (RSO) for all specialized and licensed material and equipment proposed for use on the construction project (excludes portable machine sources of ionizing radiation including moisture density and X-Ray Fluorescence (XRF)). Maintain on-site records whenever licensed radiological materials or ionizing equipment are on Government property.

Protect workers from radiation exposure in accordance with 10 CFR 20, ensuring any personnel exposures are maintained As Low As Reasonably Achievable.

1.14.1 Radiography Operation Planning Work Sheet

Submit a Gamma and X-Ray Radiography Operation Planning Work Sheet to Contracting Officer 14 days prior to commencement of operations involving radioactive materials or radiation generating devices. For portable machine sources of ionizing radiation, including moisture density and XRF, use and submit the Portable Gauge Operations Planning Worksheet instead. The Contracting Officer will review the submitted worksheet and provide questions and comments.

Contractors must use primary dosimeters process by a National Voluntary Laboratory Accreditation Program (NVLAP) accredited laboratory.

1.14.2 Site Access and Security

Coordinate site access and security requirements with the Contracting Officer for all radiological materials and equipment containing ionizing radiation that are proposed for use on a government facility. For gamma radiography materials and equipment, a Government escort is required for any travels on the Installation. The Government authorized representative will meet the Contractor at a designated location outside the Installation, ensure safety of the materials being transported, and will escort the Contractor for gamma sources onto the Installation, to the job site, and off the Installation. For portable machine sources of ionizing radiation, including moisture density and XRF, the Government authorized representative will meet the Contractor at the job site.

Provide a copy of all calibration records, and utilization records for radiological operations performed on the site.

1.14.3 Loss or Release and Unplanned Personnel Exposure

Loss or release of radioactive materials, and unplanned personnel exposures must be reported immediately to the Contracting Officer, RSO, and Base Security Department Emergency Number.

1.14.4 Site Demarcation and Barricade

Properly demark and barricade an area surrounding radiological operations to preclude personnel entrance, in accordance with EM 385-1-1, Nuclear Regulatory Commission, and Applicable State regulations and license requirements, and in accordance with requirements established in the accepted Radiography Operation Planning Work Sheet.

Do not close or obstruct streets, walks, and other facilities occupied and used by the Government without written permission from the Contracting Officer.

1.14.5 Security of Material and Equipment

Properly secure the radiological material and ionizing radiation equipment at all times, including keeping the devices in a properly marked and locked container, and secondarily locking the container to a secure point in the Contractor's vehicle or other approved storage location during

transportation and while not in use. While in use, maintain a continuous visual observation on the radiological material and ionizing radiation equipment. In instances where radiography is scheduled near or adjacent to buildings or areas having limited access or one-way doors, make no assumptions as to building occupancy. Where necessary, the Contracting Officer will direct the Contractor to conduct an actual building entry, search, and alert. Where removal of personnel from such a building cannot be accomplished and it is otherwise safe to proceed with the radiography, position a fully instructed employee inside the building or area to prevent exiting while external radiographic operations are in process.

1.14.6 Transportation of Material

Comply with 49 CFR 173 for Transportation of Regulated Amounts of Radioactive Material. Notify Local Fire authorities and the site Radiation Safety Officer (RSO) of any Radioactive Material use.

1.14.7 Schedule for Exposure or Unshielding

Actual exposure of the radiographic film or unshielding the source must not be initiated until after 5 p.m. on weekdays.

1.14.8 Transmitter Requirements

Adhere to the base policy concerning the use of transmitters, such as radios and cell phones. Obey Emissions control (EMCON) restrictions.

1.15 CONFINED SPACE ENTRY REQUIREMENTS

Confined space entry must comply with Chapter 34 of EM 385-1-1, OSHA 29 CFR 1926, OSHA 29 CFR 1910, OSHA 29 CFR 1910.146, and OSHA Directive CPL 2.100. Any potential for a hazard in the confined space requires a permit system to be used.

1.15.1 Entry Procedures

Prohibit entry into a confined space by personnel for any purpose, including hot work, until the qualified person has conducted appropriate tests to ensure the confined or enclosed space is safe for the work intended and that all potential hazards are controlled or eliminated and documented. Comply with EM 385-1-1, Chapter 34 for entry procedures. Hazards pertaining to the space must be reviewed with each employee during review of the AHA.

1.15.2 Forced Air Ventilation

Forced air ventilation is required for all confined space entry operations and the minimum air exchange requirements must be maintained to ensure exposure to any hazardous atmosphere is kept below its action level.

1.15.3 Sewer Wet Wells

Sewer wet wells require continuous atmosphere monitoring with audible alarm for toxic gas detection.

1.15.4 Rescue Procedures and Coordination with Local Emergency Responders

Develop and implement an on-site rescue and recovery plan and procedures. The rescue plan must not rely on local emergency responders for rescue from

a confined space.

1.16 DIVE SAFETY REQUIREMENTS

Develop a Dive Operations Plan, AHA, emergency management plan, and personnel list that includes qualifications, for each separate diving operation. Submit these documents to the District Dive Coordinator (DDC) via the Contracting Officer or Government Designated Authority (GDA), for review and approval at least 15 working days prior to commencement of diving operations. These documents must be at the diving location at all times. Provide each of these documents as a part of the project file.

1.17 SEVERE STORM PLAN

In the event of a severe storm warning, the Contractor must comply with the applicable Storm Plan and:

- a. Secure outside equipment and materials and place materials that could be damaged in protected areas.
- b. Check surrounding area, including roof, for loose material, equipment, debris, and other objects that could be blown away or against existing facilities.
- c. Ensure that temporary erosion controls are adequate.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 CONSTRUCTION AND OTHER WORK

Comply with EM 385-1-1, NFPA 70, NFPA 70E, NFPA 241, the APP, the AHA, Federal and State OSHA regulations, and other related submittals and activity fire and safety regulations. The most stringent standard prevails.

PPE is governed in all areas by the nature of the work the employee is performing. Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn or carried/available on each person. Mandatory PPE includes:

- a. Hard Hat
- b. Long Pants
- c. Appropriate Safety Shoes
- d. Appropriate Class Reflective Vests

3.1.1 Worksite Communication

Employees working alone in a remote location or away from other workers must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, land-line telephones or other acceptable means). The selected communication must be readily available (easily

within the immediate reach) of the employee and must be tested prior to the start of work to verify that it effectively operates in the area/environment. Develop an employee check-in/check-out communication procedure to ensure employee safety.

3.1.2 Hazardous Material Exclusions

Notwithstanding any other hazardous material used in this Contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury or polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. Notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on base.

3.1.3 Unforeseen Hazardous Material

Contract documents identify materials such as PCB, lead paint, and friable and non-friable asbestos and other OSHA regulated chemicals (i.e. 29 CFR Part 1910.1000). If material(s) that may be hazardous to human health upon disturbance are encountered during construction operations, stop that portion of work and notify the Contracting Officer immediately. Within 14 calendar days the Government will determine if the material is hazardous. If material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification pursuant to FAR 52.243-4 Changes and FAR 52.236-2 Differing Site Conditions.

3.2 UTILITY OUTAGE REQUIREMENTS

Apply for utility outages at least 10 days in advance. At a minimum, the written request must include the location of the outage, utilities being affected, duration of outage, any necessary sketches, and a description of the means to fulfill energy isolation requirements in accordance with EM 385-1-1, Chapter 11. Some examples of energy isolation devices and procedures are highlighted in EM 385-1-1, Chapter 12. In accordance with EM 385-1-1, Chapter 12, where outages involve Government or Utility personnel, coordinate with the Government on all activities involving the control of hazardous energy.

These activities include, but are not limited to, a review of HECF and HEC procedures, as well as applicable Activity Hazard Analyses (AHAs). In accordance with EM 385-1-1, Chapter 11 and NFPA 70E, work on energized electrical circuits must not be performed without prior Government authorization. Government permission is considered through the permit process and submission of a detailed AHA. Energized work permits are considered only when de-energizing introduces additional or increased hazard or when de-energizing is infeasible.

3.3 OUTAGE COORDINATION MEETING

After the utility outage request is approved and prior to beginning work on

the utility system requiring shut-down, conduct a pre-outage coordination meeting in accordance with EM 385-1-1, paragraph 12-8. This meeting must include the Prime Contractor, the Prime and subcontractors performing the work, the Contracting Officer, and the Public Utilities representative. All parties must fully coordinate HEC activities with one another. During the coordination meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HECP training in accordance with EM 385-1-1, paragraph 12-3. Clarify when personal protective equipment is required during switching operations, inspection, and verification.

3.4 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

Provide and operate a Hazardous Energy Control Program (HECP) in accordance with EM 385-1-1 Chapter 12, 29 CFR 1910.333, 29 CFR 1915.89, ASSP A10.44, NFPA 70E, and paragraph HAZARDOUS ENERGY CONTROL PROGRAM (HECP).

3.4.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer's Representative, and required by EM 385-1-1, paragraph 12-5.a. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

3.4.2 Lockout/Tagout Isolation

Where the Government or Utility performs equipment isolation and lockout/tagout, the Contractor must place their own locks and tags on each energy-isolating device and proceed in accordance with the HECP. Before any work begins, both the Contractor and the Government or Utility must perform energy isolation verification testing while wearing required PPE detailed in the Contractor's AHA and required by EM 385-1-1, Chapters 2, 5, and 11. Install personal protective grounds, with tags, to eliminate the potential for induced voltage in accordance with EM 385-1-1, Chapter 12.

3.4.3 Lockout/Tagout Removal

Upon completion of work, conduct lockout/tagout removal procedure in accordance with the HECP. In accordance with EM 385-1-1, Chapter 12, each lock and tag must be removed from each energy isolating device by the authorized individual or systems operator who applied the device. Provide formal notification to the Government (by completing the Government form if provided by Contracting Officer's Representative), confirming that steps of de-energization and lockout/tagout removal procedure have been conducted and certified through inspection and verification. Government or Utility locks and tags used to support the Contractor's work will not be removed until the authorized Government employee receives the formal notification.

3.5 FALL PROTECTION PROGRAM

Establish a fall protection program, for the protection of all employees exposed to fall hazards. Within the program include company policy, identify roles and responsibilities, education and training requirements, fall hazard identification, prevention and control measures, inspection, storage, care and maintenance of fall protection equipment and rescue and evacuation procedures in accordance with ASSP Z359.2 and EM 385-1-1, Chapter 21.

3.5.1 Training

Institute a fall protection training program. As part of the Fall Protection Program, provide training for each employee who might be exposed to fall hazards and using personal fall protection equipment. Provide training by a competent person for fall protection in accordance with EM 385-1-1, Chapter 21. Document training and practical application of the competent person in accordance with EM 385-1-1, Chapter 21 and ASSP Z359.2 in the AHA.

3.5.2 Fall Protection Equipment and Systems

Enforce use of personal fall protection equipment and systems designated (to include fall arrest, restraint, and positioning) for each specific work activity in the Site Specific Fall Protection and Prevention Plan and AHA at all times when an employee is exposed to a fall hazard. Protect employees from fall hazards as specified in EM 385-1-1, Chapter 21.

Provide personal fall protection equipment, systems, subsystems, and components that comply with EM 385-1-1 Chapter 21, 29 CFR 1926.500 Subpart M, ASSP Z359.0, ASSP Z359.1, ASSP Z359.2, ASSP Z359.3, ASSP Z359.4, ASSP Z359.6, ASSP Z359.7, ASSP Z359.11, ASSP Z359.12, ASSP Z359.13, ASSP Z359.14, ASSP Z359.15, ASSP Z359.16 and ASSP Z359.18.

3.5.2.1 Additional Personal Fall Protection Measures

In addition to the required fall protection systems, other protective measures such as safety skiffs, personal floatation devices, and life rings, are required when working above or next to water in accordance with EM 385-1-1, Chapter 21. Personal fall protection systems and equipment are required when working from an articulating or extendible boom, swing stages, or suspended platform. In addition, personal fall protection systems are required when operating other equipment such as scissor lifts. The need for tying-off in such equipment is to prevent ejection of the employee from the equipment during raising, lowering, travel, or while performing work.

3.5.2.2 Personal Fall Protection Equipment

Only a full-body harness with a shock-absorbing lanyard or self-retracting lanyard is an acceptable personal fall arrest body support device. The use of body belts is not acceptable. Harnesses must have a fall arrest attachment affixed to the body support (usually a Dorsal D-ring) and specifically designated for attachment to the rest of the system. Snap hooks and carabineers must be self-closing and self-locking, capable of being opened only by at least two consecutive deliberate actions and have a minimum gate strength of 3,600 lbs in all directions. Use webbing, straps, and ropes made of synthetic fiber. The maximum free fall distance when using fall arrest equipment must not exceed 6 feet, unless the proper

energy absorbing lanyard is used. Always take into consideration the total fall distance and any swinging of the worker (pendulum-like motion), that can occur during a fall, when attaching a person to a fall arrest system. Equip all full body harnesses with Suspension Trauma Preventers such as stirrups, relief steps, or similar in order to provide short-term relief from the effects of orthostatic intolerance in accordance with EM 385-1-1, Chapter 21.

3.5.3 Fall Protection for Roofing Work

Implement fall protection controls based on the type of roof being constructed and work being performed. Evaluate the roof area to be accessed for its structural integrity including weight-bearing capabilities for the projected loading.

a. Low Sloped Roofs:

(1) For work within 6 feet from unprotected edge of a roof having a slope less than or equal to 4:12 (vertical to horizontal), protect personnel from falling by the use of conventional fall protection systems (personal fall arrest/restraint systems, guardrails, or safety nets) in accordance with EM 385-1-1, Chapter 21 and 29 CFR 1926.500. A safety monitoring system is not adequate fall protection and is not authorized.

(2) For work greater than 6 feet from the unprotected roof edge, addition to the use of conventional fall protection systems the use of a warning line system is also permitted, in accordance with 29 CFR 1926.500 and EM 385-1-1, Chapter 21.

b. Steep-Sloped Roofs: Work on a roof having a slope greater than 4:12 (vertical to horizontal) requires a personal fall arrest system, guardrails with toe-boards, or safety nets. This requirement also applies to residential or housing type construction.

3.5.4 Horizontal Lifelines (HLL)

Provide HLL in accordance with EM 385-1-1, Chapter 21. Commercially manufactured horizontal lifelines (HLL) must be designed, installed, certified and used, under the supervision of a qualified person, for fall protection as part of a complete fall arrest system which maintains a safety factor of 2 (29 CFR 1926.500). The competent person for fall protection may (if deemed appropriate by the qualified person) supervise the assembly, disassembly, use and inspection of the HLL system under the direction of the qualified person. Locally manufactured HLLs are not acceptable unless they are custom designed for limited or site specific applications by a Registered Professional Engineer who is qualified in designing HLL systems.

3.5.5 Guardrails and Safety Nets

Design, install and use guardrails and safety nets in accordance with EM 385-1-1, Chapter 21 and 29 CFR 1926 Subpart M.

3.5.6 Rescue and Evacuation Plan and Procedures

When personal fall arrest systems are used, ensure that the mishap victim can self-rescue or can be rescued promptly should a fall occur. Prepare a Rescue and Evacuation Plan and include a detailed discussion of the

following: methods of rescue; methods of self-rescue or assisted-rescue; equipment used; training requirement; specialized training for the rescuers; procedures for requesting rescue and medical assistance; and transportation routes to a medical facility. Include the Rescue and Evacuation Plan within the Activity Hazard Analysis (AHA) for the phase of work, in the Fall Protection and Prevention (FP&P) Plan, and the Accident Prevention Plan (APP). The plan must be in accordance with the requirements of EM 385-1-1, ASSP Z359.2, and ASSP Z359.4.

3.6 WORK PLATFORMS

3.6.1 Scaffolding

Provide employees with a safe means of access to the work area on the scaffold. Climbing of any scaffold braces or supports not specifically designed for access is prohibited. Comply with the following requirements:

- a. Scaffold platforms greater than 20 feet in height must be accessed by use of a scaffold stair system.
- b. Ladders commonly provided by scaffold system manufacturers are prohibited for accessing scaffold platforms greater than 20 feet maximum in height.
- c. An adequate gate is required.
- d. Employees performing scaffold erection and dismantling must be qualified.
- e. Scaffold must be capable of supporting at least four times the maximum intended load, and provide appropriate fall protection as delineated in the accepted fall protection and prevention plan.
- f. Stationary scaffolds must be attached to structural building components to safeguard against tipping forward or backward.
- g. Special care must be given to ensure scaffold systems are not overloaded.
- h. Side brackets used to extend scaffold platforms on self-supported scaffold systems for the storage of material are prohibited. The first tie-in must be at the height equal to 4 times the width of the smallest dimension of the scaffold base.
- i. Scaffolding other than suspended types must bear on base plates upon wood mudsills (2 in x 10 in x 8 in minimum) or other adequate firm foundation.
- j. Scaffold or work platform erectors must have fall protection during the erection and dismantling of scaffolding or work platforms that are more than 6 feet.
- k. Delineate fall protection requirements when working above 6 feet or above dangerous operations in the Fall Protection and Prevention (FP&P) Plan and Activity Hazard Analysis (AHA) for the phase of work.

3.6.2 Elevated Aerial Work Platforms (AWPs)

Workers must be anchored to the basket or bucket in accordance with

manufacturer's specifications and instructions (anchoring to the boom may only be used when allowed by the manufacturer and permitted by the CP). Lanyards used must be sufficiently short to prohibit worker from climbing out of basket. The climbing of rails is prohibited. Lanyards with built-in shock absorbers are acceptable. Self-retracting devices are not acceptable. Tying off to an adjacent pole or structure is not permitted unless a safe device for 100 percent tie-off is used for the transfer.

Use of AWP's must be operated, inspected, and maintained as specified in the operating manual for the equipment and delineated in the AHA. Operators of AWP's must be designated as qualified operators by the Prime Contractor. Maintain proof of qualifications on site for review and include in the AHA.

3.7 EQUIPMENT

3.7.1 Material Handling Equipment (MHE)

- a. Material handling equipment such as forklifts must not be modified with work platform attachments for supporting employees unless specifically delineated in the manufacturer's printed operating instructions. Material handling equipment fitted with personnel work platform attachments are prohibited from traveling or positioning while personnel are working on the platform.
- b. The use of hooks on equipment for lifting of material must be in accordance with manufacturer's printed instructions. Material Handling Equipment Operators must be trained in accordance with OSHA 29 CFR 1910, Subpart N.
- c. Operators of forklifts or power industrial trucks must be licensed in accordance with OSHA.

3.7.2 Load Handling Equipment (LHE)

The following requirements apply. In exception, these requirements do not apply to commercial truck mounted and articulating boom cranes used solely to deliver material and supplies (not prefabricated components, structural steel, or components of a systems-engineered metal building) where the lift consists of moving materials and supplies from a truck or trailer to the ground; to cranes installed on mechanics trucks that are used solely in the repair of shore-based equipment; to crane that enter the activity but are not used for lifting; nor to other machines not used to lift loads suspended by rigging equipment. However, LHE accidents occurring during such operations must be reported.

- a. Equip cranes and derricks as specified in EM 385-1-1, Chapter 16.
- b. Notify the Contracting Officer 15 working days in advance of any LHE entering the activity, in accordance with EM 385-1-1, Chapter 21, so that necessary quality assurance spot checks can be coordinated. Contractor's operator must remain with the crane during the spot check. Rigging gear must be in accordance with OSHA, ASME B30.9 Standards safety standards.
- c. Comply with the LHE manufacturer's specifications and limitations for erection and operation of cranes and hoists used in support of the work. Perform erection under the supervision of a designated person (as defined in ASME B30.5). Perform all testing in accordance with the manufacturer's recommended procedures.

- d. Comply with ASME B30.5 for mobile and locomotive cranes, ASME B30.22 for articulating boom cranes, ASME B30.3 for construction tower cranes, ASME B30.8 for floating cranes and floating derricks, ASME B30.9 for slings, ASME B30.20 for below the hook lifting devices and ASME B30.26 for rigging hardware.
- e. When operating in the vicinity of overhead transmission lines, operators and riggers must be alert to this special hazard and follow the requirements of EM 385-1-1 Chapter 11, and ASME B30.5 or ASME B30.22 as applicable.
- f. Do not use crane suspended personnel work platforms (baskets) unless the Contractor proves that using any other access to the work location would provide a greater hazard to the workers or is impossible. Do not lift personnel with a line hoist or friction crane. Additionally, submit a specific AHA for this work to the Contracting Officer. Ensure the activity and AHA are thoroughly reviewed by all involved personnel.
- g. Inspect, maintain, and recharge portable fire extinguishers as specified in NFPA 10, Standard for Portable Fire Extinguishers.
- h. All employees must keep clear of loads about to be lifted and of suspended loads, except for employees required to handle the load.
- i. Use cribbing when performing lifts on outriggers.
- j. The crane hook/block must be positioned directly over the load. Side loading of the crane is prohibited.
- k. A physical barricade must be positioned to prevent personnel access where accessible areas of the LHE's rotating superstructure poses a risk of striking, pinching or crushing personnel.
- l. Maintain inspection records in accordance by EM 385-1-1, Chapter 16, including shift, monthly, and annual inspections, the signature of the person performing the inspection, and the serial number or other identifier of the LHE that was inspected. Records must be available for review by the Contracting Officer.
- m. Maintain written reports of operational and load testing in accordance with EM 385-1-1, Chapter 16, listing the load test procedures used along with any repairs or alterations performed on the LHE. Reports must be available for review by the Contracting Officer.
- n. Certify that all LHE operators have been trained in proper use of all safety devices (e.g. anti-two block devices).
- o. Take steps to ensure that wind speed does not contribute to loss of control of the load during lifting operations. At wind speeds greater than 20 mph, the operator, rigger and lift supervisor must cease all crane operations, evaluate conditions and determine if the lift may proceed. Base the determination to proceed or not on wind calculations per the manufacturer and a reduction in LHE rated capacity if applicable. Include this maximum wind speed determination as part of the activity hazard analysis plan for that operation.
- q. Follow FAA guidelines when required based on project location.

3.7.3 Machinery and Mechanized Equipment

- a. Proof of qualifications for operator must be kept on the project site for review.
- b. Manufacture specifications or owner's manual for the equipment must be on-site and reviewed for additional safety precautions or requirements that are sometimes not identified by OSHA or USACE EM 385-1-1. Incorporate such additional safety precautions or requirements into the AHAs.

3.7.4 Base Mounted Drum Hoists

- a. Operation of base mounted drum hoists must be in accordance with EM 385-1-1 and ASSP A10.22.
- b. Rigging gear must be in accordance with applicable ASME/OSHA standards.
- c. When used on telecommunication towers, base mounted drum hoists must be in accordance with TIA-1019, TIA-222, ASME B30.7, 29 CFR 1926.552, and 29 CFR 1926.553.
- d. When used to hoist personnel, the AHA must include a written standard operating procedure. Operators must have a physical examination in accordance with EM 385-1-1 paragraph 16-3.f and trained, at a minimum, in accordance with EM 385-1-1 paragraph 16-8.hh and 16-1.b. The base mounted drum hoist must also comply with OSHA Instruction CPL 02-01-056 and ASME B30.23.
- e. Material and personnel must not be hoisted simultaneously.
- f. Personnel cage must be marked with the capacity (in number of persons) and load limit in pounds.
- g. Construction equipment must not be used for hoisting material or personnel or with trolley/tag lines. Construction equipment may be used for towing and assisting with anchoring guy lines.

3.7.5 Use of Explosives

Explosives must not be used or brought to the project site without prior written approval from the Contracting Officer. Such approval does not relieve the Contractor of responsibility for injury to persons or for damage to property due to blasting operations.

Storage of explosives, when permitted on Government property, must be only where directed and in approved storage facilities. These facilities must be kept locked at all times except for inspection, delivery, and withdrawal of explosives.

3.8 EXCAVATIONS

Soil classification must be performed by a competent person in accordance with 29 CFR 1926 and EM 385-1-1.

3.8.1 Utility Locations

Provide a third party, independent, private utility locating company to positively identify underground utilities in the work area in addition to

any station locating service and coordinated with the station utility department.

3.8.2 Utility Location Verification

Physically verify underground utility locations, including utility depth, by hand digging using wood or fiberglass handled tools when any adjacent construction work is expected to come within 3 feet of the underground system.

3.8.3 Utilities Within and Under Concrete, Bituminous Asphalt, and Other Impervious Surfaces

Utilities located within and under concrete slabs or pier structures, bridges, parking areas, and the like, are extremely difficult to identify. Whenever Contract work involves chipping, saw cutting, or core drilling through concrete, bituminous asphalt or other impervious surfaces, the existing utility location must be coordinated with station utility departments in addition to location and depth verification by a third party, independent, private locating company. The third party, independent, private locating company must locate utility depth by use of Ground Penetrating Radar (GPR), X-ray, bore scope, or ultrasound prior to the start of demolition and construction. Outages to isolate utility systems must be used in circumstances where utilities are unable to be positively identified. The use of historical drawings does not alleviate the Contractor from meeting this requirement.

3.9 ELECTRICAL

Perform electrical work in accordance with EM 385-1-1, Chapters 11 and 12.

3.9.1 Conduct of Electrical Work

As delineated in EM 385-1-1, electrical work is to be conducted in a de-energized state unless there is no alternative method for accomplishing the work. In those cases obtain an energized work permit from the Contracting Officer. The energized work permit application must be accompanied by the AHA and a summary of why the equipment/circuit needs to be worked energized. Underground electrical spaces must be certified safe for entry before entering to conduct work. Cables that will be cut must be positively identified and de-energized prior to performing each cut. Attach temporary grounds in accordance with ASTM F855 and IEEE 1048. Perform all high voltage cable cutting remotely using hydraulic cutting tool. When racking in or live switching of circuit breakers, no additional person other than the switch operator is allowed in the space during the actual operation. Plan so that work near energized parts is minimized to the fullest extent possible. Use of electrical outages clear of any energized electrical sources is the preferred method.

When working in energized substations, only qualified electrical workers are permitted to enter. When work requires work near energized circuits as defined by NFPA 70, high voltage personnel must use personal protective equipment that includes, as a minimum, electrical hard hat, safety shoes, insulating gloves and electrical arc flash protection for personnel as required by NFPA 70E. Insulating blankets, hearing protection, and switching suits may also be required, depending on the specific job and as delineated in the Contractor's AHA. Ensure that each employee is familiar with and complies with these procedures and 29 CFR 1910.147.

3.9.2 Qualifications

Electrical work must be performed by QP with verifiable credentials who are familiar with applicable code requirements. Verifiable credentials consist of State, National and Local Certifications or Licenses that a Master or Journeyman Electrician may hold, depending on work being performed, and must be identified in the appropriate AHA. Journeyman/Apprentice ratio must be in accordance with State, Local requirements applicable to where work is being performed.

3.9.3 Arc Flash

Conduct a hazard analysis/arc flash hazard analysis whenever work on or near energized parts greater than 50 volts is necessary, in accordance with NFPA 70E.

All personnel entering the identified arc flash protection boundary must be QPs and properly trained in NFPA 70E requirements and procedures. Unless permitted by NFPA 70E, no Unqualified Person is permitted to approach nearer than the Limited Approach Boundary of energized conductors and circuit parts. Training must be administered by an electrically qualified source and documented.

3.9.4 Grounding

Ground electrical circuits, equipment and enclosures in accordance with NFPA 70 and IEEE C2 to provide a permanent, continuous and effective path to ground unless otherwise noted by EM 385-1-1.

Check grounding circuits to ensure that the circuit between the ground and a grounded power conductor has a resistance low enough to permit sufficient current flow to allow the fuse or circuit breaker to interrupt the current.

3.9.5 Testing

Temporary electrical distribution systems and devices must be inspected, tested and found acceptable for Ground-Fault Circuit Interrupter (GFCI) protection, polarity, ground continuity, and ground resistance before initial use, before use after modification and at least monthly. Monthly inspections and tests must be maintained for each temporary electrical distribution system, and signed by the electrical CP or QP.

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SOURCES FOR REFERENCE PUBLICATIONS

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PART 1 GENERAL

1.1 REFERENCES

Various publications are referenced in other sections of the specifications to establish requirements for the work. These references are identified in each section by document number, date and title. The document number used in the citation is the number assigned by the standards producing organization (e.g., ASTM B564 Standard Specification for Nickel Alloy Forgings). However, when the standards producing organization has not assigned a number to a document, an identifying number has been assigned for reference purposes.

1.2 ORDERING INFORMATION

The addresses of the standards publishing organizations whose documents are referenced in other sections of these specifications are listed below, and if the source of the publications is different from the address of the sponsoring organization, that information is also provided.

AACE INTERNATIONAL (AACE)
726 East Park Avenue #180
Fairmont, WV 26554 USA
Ph: 304-296-8444
Fax: 304-291-5728
Internet: <https://web.aacei.org/>

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)
1801 Alexander Bell Drive
Reston, VA 20191
Ph: 800-548-2723; 703-295-6300
Email: customercare@asce.org
Internet: <https://www.asce.org/>

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Two Park Avenue
New York, NY 10016-5990
Ph: 800-843-2763
Fax: 973-882-1717
E-mail: customercare@asme.org
Internet: <https://www.asme.org/>

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)
520 N. Northwest Highway
Park Ridge, IL 60068
Ph: 847-699-2929
E-mail: customerservice@assp.org
Internet: <https://www.assp.org/>

ASTM INTERNATIONAL (ASTM)
100 Barr Harbor Drive, P.O. Box C700
West Conshohocken, PA 19428-2959

Ph: 610-832-9500
 Fax: 610-832-9555
 E-mail: service@astm.org
 Internet: <https://www.astm.org/>

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)
 445 and 501 Hoes Lane
 Piscataway, NJ 08854-4141
 Ph: 732-981-0060 or 800-701-4333
 Fax: 732-981-9667
 E-mail: onlinesupport@ieee.org
 Internet: <https://www.ieee.org/>

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
 1 Batterymarch Park
 Quincy, MA 02169-7471
 Ph: 800-344-3555
 Fax: 800-593-6372
 Internet: <https://www.nfpa.org>

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)
 1320 North Courthouse Road, Suite 200
 Arlington, VA 22201
 Ph: 703-907-7700
 Fax: 703-907-7727
 E-mail: marketing@tiaonline.org
 Internet: <https://www.tiaonline.org/>

U.S. ARMY CORPS OF ENGINEERS (USACE)
 CRD-C DOCUMENTS available on Internet:
<http://www.wbdg.org/ffc/army-coe/standards>
 Order Other Documents from:
 Official Publications of the Headquarters, USACE
 E-mail: hqpublications@usace.army.mil
 Internet: <http://www.publications.usace.army.mil/>
 or
<https://www.hnc.usace.army.mil/Missions/Engineering-Directorate/TECHINFO/>

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)
 8601 Adelphi Road
 College Park, MD 20740-6001
 Ph: 866-272-6272
 Internet: <https://www.archives.gov/>
 Order documents from:
 Superintendent of Documents
 U.S. Government Publishing Office (GPO)
 732 N. Capitol Street, NW
 Washington, DC 20401
 Ph: 202-512-1800 or 866-512-1800
 Bookstore: 202-512-0132
 Internet: <https://www.gpo.gov/>

PART 2 PRODUCTS

Not used

PART 3 EXECUTION

Not used

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SECTION 01 45 00

QUALITY CONTROL
08/23, CHG 1: 05/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D3740 (2019) Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction

ASTM E329 (2023) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety -- Safety and Occupational Health (SOH) Requirements

1.2 PAYMENT

Separate payment will not be made for providing and maintaining an effective Quality Control program. Include all associated costs in the applicable Schedule item.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Contractor Quality Control (CQC) Plan; G

1.4 GENERAL REQUIREMENTS

Establish and maintain an effective quality control (QC) system that complies with FAR 52.246-12 Inspection of Construction. QC is comprised of plans, procedures, and organization necessary to produce an end product that complies with the Contract requirements. The QC system covers all construction operations, both onsite and offsite, and must be keyed to the proposed construction sequence. The Quality Control Manager, Superintendent, Site Safety and Health Officer (SSHO), and all on-site supervisors are responsible for the quality of work and are subject to removal by the Contracting Officer for non-compliance with the quality

requirements specified in the Contract. The Quality Control Manager must maintain a physical presence at the work site at all times and is the primary individual responsible for all quality control.

1.5 QUALITY CONTROL (QC) PROGRAM REQUIREMENTS

Establish and maintain a QC program as described in this section. The QC program consists of a QC Organization, QC Plan, QC Plan Meeting(s), a Coordination and Mutual Understanding Meeting, QC meetings, three phases of control, submittal review and approval, testing, completion inspections, QC certifications, and documentation necessary to provide materials, equipment, workmanship, fabrication, construction and operations that comply with the requirements of this Contract. The QC program must cover on-site and off-site work and be keyed to the work sequence. No construction work or testing may be performed unless the QC Manager is on the work site. The QC Manager must report to an officer of the firm and not be subordinate to the Project Superintendent or the Project Manager. The QC Manager, Project Superintendent and Project Manager must work together effectively. Although the QC Manager is the primary individual responsible for quality control, all individuals will be held responsible for the quality of work on the job.

1.5.1 Meetings

1.5.1.1 Quality Control Plan Meeting

Prior to submission of the QC Plan, the Contractor may request a meeting with the Contracting Officer to discuss the QC Plan requirements of this Contract.

The purpose of this meeting is to develop a mutual understanding of the QC Plan requirements prior to plan development and submission and to agree on the Contractor's list of Definable Feature of Work (DFOW).

1.5.1.2 Coordination and Mutual Understanding Meeting

After the before start of construction, and prior to acceptance by the Government of the CQC Plan, meet with the Contracting Officer and discuss the Contractor's quality control system. During the meeting, a mutual understanding of the system details must be developed, including the forms for recording the CQC operations, control activities, testing, administration of the system for both onsite and offsite work, and the interrelationship of Contractor's Management and control with the Government's Quality Assurance. Minutes of the meeting will be prepared by the QC Manager and signed by the Contractor and the Government. Provide a copy of the signed minutes to all attendees[and include in the QC Plan]. At a minimum the Coordination and Mutual Understanding Meeting must be repeated when a new QC Manager is appointed. There can be other occasions when subsequent conferences will be called by either party to reconfirm mutual understandings or address deficiencies in the CQC system or procedures which can require corrective action by the Contractor.

1.5.1.2.1 Purpose

The purpose of this meeting is to develop a mutual understanding of the QC details, including documentation, administration for on-site and off-site work, design intent, environmental requirements and procedures, coordination of activities to be performed, and the coordination of the Contractor's management, production, and QC personnel. At the meeting, the

Contractor must explain in detail how three phases of control will be implemented for each DFO, as well as how each DFO will be affected by each management plan or requirement as listed below:

- a. Waste Management Plan.
- b. Procedures for noise and acoustics management.
- c. Environmental Protection Plan.
- d. Environmental regulatory requirements.

1.5.1.2.2 Coordination of Activities

Coordinate activities included in various sections to assure efficient and orderly installation of each component. Coordinate operations included under different sections that are dependent on each other for proper installation and operation.

1.5.1.2.3 Attendees

As a minimum, the Contractor's personnel required to attend include an officer of the firm, the Project Manager, Project Superintendent, QC Manager, Alternate QC Manager, Environmental Manager, and subcontractor representatives. Each subcontractor who will be assigned QC responsibilities must have a principal of the firm at the meeting.

1.5.1.3 Quality Control (QC) Meetings

After the start of construction, conduct weekly QC meetings led by the QC Manager at the work site with the Project Superintendent, and the other personnel as necessary. The QC Manager is to prepare the minutes of the meeting and provide a copy to the Contracting Officer within 2 working days after the meeting. The Contracting Officer may attend these meetings. As a minimum, accomplish the following at each meeting:

- a. Review the minutes of the previous meeting.
- b. Review the schedule and the status of work and deficiencies/rework. Review the most current approved schedule (in accordance with schedule specification) and the status of work and deficiencies/rework.
- c. Review the status of submittals and Request For Information (RFIs).
- d. Review the work to be accomplished in the next 3 weeks as defined by the schedule section paragraph WEEKLY PROGRESS MEETINGS in Section 01 32 01.00 10 PROJECT SCHEDULE and all documentation required for that work.
- e. Review Testing Plan and Log including status of tests performed since last QC Meeting.
- f. Resolve QC and production problems. Discuss status of pending change orders.
- g. Address items that may require revising the QC Plan.
- h. Review Accident Prevention Plan (APP) and effectiveness of the safety program.

- i. Review environmental requirements and procedures.
- j. Review Environmental Management Plan.
- k. Review Waste Management Plan.
- l. Review the status of training completion.

1.5.2 Contractor Quality Control (CQC) Plan

Submit no later than 7 days prior to the Preconstruction Conference, the CQC Plan proposed to implement the requirements FAR 52.246-12 Inspection of Construction. Construction will be permitted to begin only after acceptance of the CQC Plan and other Contract requirements or acceptance of an interim plan applicable to the particular feature of work to be started. Work outside of the accepted interim plan will not be permitted to begin until acceptance of a CQC Plan or another interim plan containing the additional work.

1.5.2.1 Content of Contractor Quality Control (CQC) Plan

Provide a CQC Plan, prior to start of construction that includes a table of contents, with major sections identified, pages numbered sequentially, and that documents the proposed methods and responsibilities for accomplishing quality control during the construction of the project. The CQC Plan must at a minimum include the following sections:

- a. A description of the quality control organization and acknowledgment that the CQC staff will implement the three phase control system for all aspects of the work specified.
- b. An organizational chart showing the quality control organization with individual names and job titles and lines of authority up to an executive of the company at the home office.
- c. NAMES AND QUALIFICATIONS: Names and qualifications, in resume format, (including position titles and durations for qualifying experiences) for each person in the QC organization. Include the Construction Quality Management (CQM) for Contractors course certifications for the QC personnel as required by the paragraph CONSTRUCTION QUALITY MANAGEMENT TRAINING.
- d. DUTIES, RESPONSIBILITY AND AUTHORITY OF QC PERSONNEL: Duties, responsibilities, and authorities of each person in the QC organization.
- e. OUTSIDE ORGANIZATIONS: A listing of outside organizations, such as architectural and consulting engineering firms, that will be employed by the Contractor and a description of the services these firms will provide.
- f. APPOINTMENT LETTERS: Letters signed by an officer of the firm appointing the QC Manager and Alternate QC Manager, and stating that they are responsible for implementing and managing the QC program as described in this Contract. Include in this letter the responsibility of the QC Manager and Alternate QC Manager to implement and manage the three phases of control, and their authority to stop work that is not in compliance with the Contract. Letters of direction are to be issued by the QC Manager to all other QC Specialists or quality control

representatives outlining their duties, authorities, and responsibilities. Include copies of the letters in the QC Plan.

- g. SUBMITTAL PROCEDURES AND INITIAL SUBMITTAL REGISTER: Procedures for reviewing, approving, scheduling, and managing submittals, including those of subcontractors, offsite fabricators, suppliers, and purchasing agents. Provide the name(s) of the person(s) in the QC organization authorized to review and certify submittals prior to approval. Provide the initial submittal of the Submittal Register as specified in Section 01 33 00 SUBMITTAL PROCEDURES.
- h. TESTING LABORATORY INFORMATION: Testing laboratory information required by the paragraph ACCREDITATION REQUIREMENTS, as applicable.
- i. TESTING PLAN AND LOG: A Testing Plan and Log that includes the tests required, associated feature of work required, referenced by the specification paragraph number requiring the test, the frequency, and the person responsible for each test.
- j. Procedures to complete construction deficiencies from identification through acceptable corrective action. Establish verification procedures that identified deficiencies have been corrected. This phase is performed prior to beginning work on each definable feature of work, after all required plans, documents, materials are approved, and after copies are at the work site.
- k. Reporting procedures, including proposed reporting formats.
- l. Procedures for submitting and reviewing design changes/variations prior to submission to the Contracting Officer.
- m. LIST OF DEFINABLE FEATURES: A Definable Feature of Work (DFOW) is a task that is separate and distinct from other tasks and has control requirements and work crews unique to that task. A DFOW is identified by different trades or disciplines, or it is work by the same trade in a different environment. A DFOW is by definition any item or activity on the construction schedule, and the schedule specification provides direction regarding how the DFOWs are to be structured. Include in the list of DFOWs for all activities on the Construction Schedule. Although each section of the specifications can generally be considered as a definable feature of work, there are frequently more than one definable features under a particular section. Identify the specification section number and schedule activity ID for each DFOW listed. The DFOW list will be reviewed in coordination with the construction schedule and agreed upon during the Coordination of Mutual Understanding Meeting.
- n. PROCEDURES FOR PERFORMING AND TRACKING THE THREE PHASES OF CONTROL: Identify procedures used to ensure the three phases of control to manage the quality on this project. For each Definable Feature of Work (DFOW), a Preparatory and Initial phase checklist will be filled out during the Preparatory and Initial phase meetings. Conduct the Preparatory and Initial Phases and meetings with a view towards obtaining quality construction by planning ahead and identifying potential problems for each DFOW.
- o. PROCEDURES FOR COMPLETION INSPECTION: Procedures for identifying and documenting the completion inspection process. Include in these procedures the responsible party for punch out inspection, pre-final

inspection, and final acceptance inspection.

- p. TRAINING PROCEDURES AND TRAINING LOG: Procedures for coordinating and documenting the training of personnel required by the Contract.
- q. ORGANIZATION AND PERSONNEL CERTIFICATIONS LOG: Procedures for coordinating, tracking and documenting all certifications required for entities such as subcontractors, testing laboratories, suppliers, and personnel. The QC Manager will ensure that certifications are current, appropriate for the work being performed, and will not lapse during any period of the Contract that the work is being performed.

1.5.3 Acceptance of the Quality Control (QC) Plan

The Contracting Officer's acceptance of the Contractor QC Plan is required prior to the start of construction. The Contracting Officer reserves the right to require changes in the QC Plan and operations as necessary, including removal or addition of personnel, to ensure the specified quality of work. The Contracting Officer reserves the right to interview any member of the QC organization at any time to verify the submitted qualifications. All QC organization personnel are subject to acceptance by the Contracting Officer. The Contracting Officer may require the removal of any individual for non-compliance with quality requirements specified in the Contract.

1.5.4 Notification of Changes

Notify the Contracting Officer, in writing, of any proposed changes in the QC Plan or changes to the QC organization personnel. Proposed changes are subject to acceptance by the Contracting Officer.

1.6 QUALITY CONTROL (QC) ORGANIZATION

1.6.1 Personnel Requirements

The requirements for the CQC organization are a Site Safety and Health Officer (SSHO), QC Manager, a Design Quality Manager, and enough qualified personnel to ensure safety and Contract compliance. The SSHO reports directly to a senior project (or corporate) official independent from the QC Manager. The SSHO will also serve as a member of the CQC Staff. Personnel identified in the technical provisions as requiring specialized skills to assure the required work is being performed properly. The CQC staff always maintains a presence at the site during progress of the work and have complete authority and responsibility to take any action necessary to ensure Contract compliance. The CQC staff will be subject to acceptance by the Contracting Officer. Provide adequate office space, filing systems and other resources as necessary to maintain an effective and fully functional CQC organization. Promptly complete and furnish all letters, material submittals, shop drawing submittals, schedules, and all other project documentation to the CQC organization. The CQC organization is responsible for always maintaining these documents and records at the site, except as otherwise acceptable to the Contracting Officer.

1.6.2 Quality Control (QC) Manager

1.6.2.1 Duties

Provide a QC Manager at the work site to implement and manage the QC program. The QC Manager must be employed by the Prime Contractor. The only

duties and responsibilities of the QC Manager are to manage and implement the QC program on this Contract. The QC Manager must attend the partnering meetings, QC Plan Meetings, Coordination and Mutual Understanding Meeting, conduct the QC meetings, perform the three phases of control, perform submittal review and approval, ensure testing is performed and provide QC certifications and documentation required in this Contract. The QC Manager is responsible for managing and coordinating the three phases of control and documentation performed by testing laboratory personnel and any other inspection and testing personnel required by this Contract. The QC Manager is the manager of all QC activities.

1.6.2.2 Qualifications

The QC Manager must be an individual with a minimum of 5 years combined experience in the following positions: Project Superintendent, QC Manager, Project Manager, Project Engineer or Construction Manager on similar size and type construction contracts which included the major trades that are part of this Contract. The individual must have at least 2 years experience as a QC Manager. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification, safety compliance, and sustainability.

The QC Manager and all members of the QC organization must be capable of reading, writing, and conversing fluently in the English language.

1.6.2.3 Construction Quality Management Training

In addition to the above experience and education requirements, the QC Manager and all members of the QC team must have completed the CQM for Contractors course. If the QC Manager does not have a current certification, obtain the CQM for Contractors course certification within 90 days of award. This course is periodically offered by the Naval Facilities Engineering Systems Command and the Army Corps of Engineers. Contact the Contracting Officer for information on the next scheduled class.

The Construction Quality Management Training certificate expires after 5 years. If the QC Manager's certificate has expired, retake the course to remain current.

1.6.3 Organizational Changes

Maintain the QC staff with personnel as required by the specification section at all times. When it is necessary to make changes to the QC staff, revise the CQC Plan to reflect the changes and submit the changes to the Contracting Officer for acceptance.

1.6.4 Alternate Quality Control (QC) Manager Duties and Qualifications

Designate an alternate for the QC Manager at the work site to serve in the event of the designated QC Manager's absence. The qualification requirements for the Alternate QC Manager must be the same as for the QC Manager.

1.7 SUBMITTAL AND DELIVERABLES REVIEW AND APPROVAL

Procedures for submission, review and approval of submittals are described in Section 01 33 00 SUBMITTAL PROCEDURES. Procedures must include field verification of relevant dimensions and component characteristics by the QC organization prior to submittal being sent to the Contracting Officer. The

CQC organization is responsible for certifying that all submittals and deliverables are in compliance with the Contract.

1.8 THREE PHASES OF CONTROL

CQC enables the Contractor to ensure that the construction, including that of subcontractors and suppliers, complies with the requirements of the Contract. At least three phases of control must be conducted by the QC Manager to adequately cover both on-site and off-site work for each definable feature of the construction work as follows:

1.8.1 Preparatory Phase

Document the results of the preparatory phase actions by separate minutes prepared by the QC Manager and attach to the daily CQC report. Instruct applicable workers as to the acceptable level of workmanship required to meet Contract specifications.

Notify the Contracting Officer at least 2 business days in advance of each preparatory phase meeting. The meeting will be conducted by the QC Manager and attended by the Project Superintendent, and the foreman responsible for the DFOW. When the DFOW will be accomplished by a subcontractor, that subcontractor's foreman must attend the preparatory phase meeting. This phase is performed prior to beginning work on each definable feature of work, after all required plans/documents/materials are approved/accepted, and after copies are at the work site. Perform the following prior to beginning work on each DFOW:

- a. Review each paragraph of the applicable specification sections, reference codes, and standards. Make available during the preparatory inspection a copy of those sections of referenced codes and standards applicable to that portion of the work to be accomplished in the field. Maintain and make available in the field for use by Government personnel until final acceptance of the work.
- b. Review the Contract drawings.
- c. Verify that field measurements are as indicated on construction or shop drawings or both before confirming product orders, to minimize waste due to excessive materials.
- d. Verify that appropriate shop drawings and submittals for materials and equipment have been submitted and approved. Verify receipt of approved factory test results, when required.
- e. Review the testing plan and ensure that provisions have been made to provide the required QC testing.
- f. Examine the work area to ensure that the required preliminary work has been completed and complies with the Contract and ensure any deficiencies/rework items in the preliminary work have been corrected and confirmed by the Contracting Officer.
- g. Review coordination of product/material delivery to designated prepared areas to execute the work.
- h. Examine the required materials, equipment and sample work to ensure that they are on hand and conform to the approved shop drawings and submitted data and are properly stored.

- i. Check to assure that all materials and equipment have been tested, submitted, and approved.
- j. Discuss specific controls to be used, construction methods, construction tolerances, workmanship standards, and the approach that will be used to provide quality construction by planning ahead and identifying potential problems for each DFOW. Ensure any portion of the plan requiring separate Contracting Officer acceptance has been approved.
- k. Review the APP and appropriate Activity Hazard Analysis (AHA) to ensure that applicable safety requirements are met, and that required Safety Data Sheets (SDS) are submitted.

1.8.2 Initial Phase

Notify the Contracting Officer at least 2 business days in advance of each initial phase. When construction crews are ready to start work on a DFOW, conduct the initial phase with the Project Superintendent, and the foreman responsible for that DFOW. Observe the initial segment of the DFOW to ensure that the work complies with Contract requirements. Document the results of the initial phase in the Initial Phase Checklist. Repeat the initial phase for each new crew to work on-site when acceptable levels of specified quality are not being met. Indicate the exact location of initial phase for definable feature of work for future reference and comparison with follow-up phases. Perform the following for each DFOW:

- a. Check work to ensure that it is in full compliance with Contract requirements. Review minutes of the preparatory meeting.
- b. Verify adequacy of controls to ensure full Contract compliance. Verify required control inspection and testing comply with the Contract.
- c. Establish level of workmanship and verify that it meets the minimum acceptable workmanship standards. Compare with required sample panels as appropriate.
- d. Resolve any workmanship issues.
- e. Ensure that testing is performed by the approved laboratory.
- f. Check work procedures for compliance with the APP and the appropriate AHA to ensure that applicable safety requirements are met.
- g. Review project specific work plans (i.e., HAZMAT Abatement, Stormwater Management) to ensure all preparatory work items have been completed and documented.

1.8.3 Follow-Up Phase

Perform the following for on-going DFOW daily, or more frequently as necessary, until the completion of each DFOW. The Final Follow-Up for any DFOW will clearly note in the daily report the DFOW is completed, and all deficiencies/rework items have been completed in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST. Each DFOW that has completed the Initial Phase and has not completed the Final Follow-up must be included on each daily report. If no work was performed on that DFOW for the period of that daily report, it must be so noted. Document all Follow-Up activities

for DFOWs in the daily CQC Report:

- a. Ensure the work including control testing complies with Contract requirements until completion of that particular work feature. Record checks in the CQC documentation.
- b. Maintain the quality of workmanship required.
- c. Ensure that testing is performed by the approved laboratory.
- d. Ensure that deficiencies/rework items are being corrected. Conduct final follow-up checks and correct all deficiencies prior to the start of additional features of work which may be affected by the deficient work.
- e. Do not build upon nor conceal non-conforming work.
- f. Assure manufacturers' representatives have performed necessary inspections if required and perform safety inspections.

1.8.4 Additional Preparatory and Initial Phases

Conduct additional preparatory and initial phases on the same DFOW if the quality of on-going work is unacceptable, if there are changes in the applicable QC organization, if there are changes in the on-site production supervision or work crew, if work on a DFOW has not started within 45 days of the initial preparatory meeting or has resumed after 45 days of inactivity, or if other problems develop.

1.8.5 Notification of Three Phases of Control for Off-Site Work

Notify the Contracting Officer at least 2 weeks prior to the start of the preparatory and initial phases.

1.8.6 Deficiency/Rework Items List

The QC Manager must maintain a list of work that does not comply with the Contract, identifying what items need to be corrected, the activity ID number associated with the item, the date the item was originally discovered, the date the item will be corrected by, and the date the item was corrected.

The QC Manager reviews the list at each weekly QC Meeting:

- a. There is no requirement to report a deficiency/rework item that is corrected the same day it is discovered.
- b. No successor task may be advanced beyond the preparatory phase meeting until all deficiencies/rework items have been cleared by the QC Manager and concurred with by the Contracting Officer. This must be confirmed as part of the Preparatory Phase activities.
- c. Attach a copy of the "Deficiency/Rework Items List" to the last daily CQC Report of each month.
- d. The Contractor is responsible for including those items identified by the Contracting Officer.
- e. All deficiencies/rework items must be confirmed as corrected by the QC

Manager, and concurred by the Contracting Officer, prior to commencement of any completion inspections per paragraph COMPLETION INSPECTIONS unless specifically exempted by the Contracting Officer.

- f. Non-Compliance with these requirements is grounds for removal in accordance with paragraph ACCEPTANCE OF THE QUALITY CONTROL (QC) PLAN.
- g. All delays, concurrent or related to failure to manage, monitor, control, and correct deficiencies/rework items are entirely the responsibility of the Contractor and can not be made the subject, or any component of any request for additional time or compensation.

1.9 TESTING

Perform specified or required tests to verify that control measures are adequate to provide a product which conforms to Contract requirements. Upon request, furnish to the Government duplicate samples of test specimens for possible testing by the Government. Testing includes operation and acceptance tests when specified. Procure the services of an U.S. Army Corps of Engineers approved testing laboratory or establish an approved testing laboratory at or near the project site. Perform the following activities and record and provide the following data:

- a. Verify that testing procedures comply with Contract requirements.
- b. Verify that facilities and testing equipment are available and comply with testing standards.
- c. Check test instrument calibration data against certified standards.
- d. Verify that recording forms and test identification control number system, including all test documentation requirements, have been prepared.
- e. Record results of all tests taken, both passing and failing on the CQC report for the date taken. Specification paragraph reference, location where tests were taken, and the sequential control number identifying the test. If approved by the Contracting Officer, actual test reports are submitted later with a reference to the test number and date taken. Provide an information copy of tests performed by an offsite or commercial test facility directly to the Contracting Officer. Failure to submit timely test reports as stated results in nonpayment for related work performed and disapproval of the test facility for this Contract.

1.9.1 Laboratory Accreditation Authorities

All testing laboratories must be validated by the USACE Material Testing Center (MTC) for the tests to be performed. Information on the USACE MTC with web-links to both a list of validated testing laboratories and for the laboratory inspection request for can be found at:
<https://mtc.erd.c.dren.mil>

1.9.2 Capability Check

The Contracting Officer retains the right to check laboratory equipment in the proposed laboratory and the laboratory technician's testing procedures, techniques, and other items pertinent to testing, for compliance with the standards set forth in this Contract. Laboratories utilized for testing

soils, concrete, asphalt, and steel must meet criteria detailed in ASTM D3740 and ASTM E329.

1.9.2.1 Onsite Laboratory

The Government reserves the right to utilize the Contractor's control testing laboratory and equipment to make assurance tests, and to check the Contractor's testing procedures, techniques, and test results at no additional cost to the Government.

1.9.3 Test Results

Cite applicable Contract requirements, tests or analytical procedures used. Provide actual results and include a statement that the item tested or analyzed conforms or fails to conform to specified requirements. If the item fails to conform, notify the Contracting Officer immediately. Conspicuously stamp the cover sheet for each report in large red letters "CONFORMS" or "DOES NOT CONFORM" to the specification requirements, whichever is applicable. Test results must be signed by a testing laboratory representative authorized to sign certified test reports. Furnish the signed reports, certifications, and other documentation to the Contracting Officer via the QC Manager.

1.10 COMPLETION INSPECTIONS

1.10.1 Punch-Out Inspection

Near the completion of all work or any increment thereof, established by a completion time stated in the Contract Clause entitled "Commencement, Prosecution, and Completion of Work," or stated elsewhere in the specifications, the QC Manager must conduct an inspection of the work and develop a "punch list" of items which do not conform to the approved drawings, specifications, and Contract. Include in the punch list any remaining items on the "Deficiency/Rework Items List", that were not corrected prior to the Punch-Out Inspection as approved by the Contracting Officer in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST. Include within the punch list the estimated date by which the deficiencies will be corrected. Provide a copy of the punch list to the Contracting Officer.

The QC Manager, or staff, must make follow-on inspections to ascertain that all deficiencies have been corrected. All punch list items must be confirmed as corrected by the QC Manager and concurred by the Contracting Officer. Once this is accomplished, notify the Government that the facility is ready for the Government "Pre-Final Inspection".

1.10.2 Pre-Final Inspection

The Government and QC Manager will perform this inspection to verify that the facility is complete and ready to be occupied. A Government "Pre-Final Punch List" will be documented by the QC Manager as a result of this inspection. The QC Manager will ensure that all items on this list are corrected and concurred by the Contracting Officer prior to notifying the Government that a "Final" inspection with the Client can be scheduled. All items noted on the "Pre-Final" inspection must be corrected and concurred by the Contracting Officer in a timely manner and be accomplished before the Contract completion date for the work, or any increment thereof, if the project is divided into increments by separate completion dates unless exceptions are directed by the Contracting Officer.

1.10.3 Final Acceptance Inspection

Notify the Contracting Officer at least 14 calendar days prior to the date a final acceptance inspection can be held. State within the notice that all items previously identified on the pre-final punch list will be corrected and acceptable, along with any other unfinished Contract work, by the date of the final acceptance inspection. The Contractor must be represented by the QC Manager, the Project Superintendent, and others deemed necessary. Attendees for the Government will include the Contracting Officer, other Government QA personnel, and personnel representing the Client. Failure of the Contractor to have all Contract work acceptably complete for this inspection will be cause for the Contracting Officer to bill the Contractor for the Government's additional inspection cost in accordance with the Contract Clause entitled "Inspection of Construction."

1.11 QUALITY CONTROL (QC) CERTIFICATIONS

1.11.1 Contractor Quality Control (CQC) Report Certification

Contain the following statement within the CQC Report: "On behalf of the Contractor, I certify that this report is complete and correct and equipment and material used, and work performed during this reporting period is in compliance with the Contract drawings and specifications to the best of my knowledge, except as noted in this report."

1.11.2 Completion Certification

Upon completion of work under this Contract, the QC Manager must furnish a certificate to the Contracting Officer attesting that "the work has been completed, inspected, tested and is in compliance with the Contract." Provide a copy of this final QC Certification for completion to the preparer of the Operation & Maintenance (O&M) documentation.

1.12 DOCUMENTATION AND INFORMATION FOR THE CONTRACTING OFFICER

Maintain current and complete records of on-site and off-site QC program operations and activities.

Contact the Contracting Officer for sample forms or print from RMS-QCS as needed. Prior to commencing work on construction, the Contractor must obtain a copy set of the current report forms. The report forms will consist of the Contractor Quality Control (CQC) Report, CQC Report (Continuation Sheet), Contractor Production Report, Contractor Production Report (Continuation Sheet), Preparatory Phase Checklist, Initial Phase Checklist, Testing Plan and Log, and Rework Items List. Unless otherwise provided by the Contracting Officer, Contractor may use the forms provided as related material located at <https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-01-45-00>.

1.12.1 Construction Documentation

Reports are required for each day that work is performed and must be attached to the Contractor Quality Control Report prepared for the same day. Maintain current and complete records of on-site and off-site QC program operations and activities. Reports are required for each day work is performed. Account for each calendar day throughout the life of the Contract.

The Project Superintendent and the QC Manager must prepare and sign the Contractor Production and CQC Reports, respectively. Every space on the forms must be filled in. Use N/A if nothing can be reported in one of the spaces. The reporting of work must be identified by terminology consistent with the construction schedule. In the "Remarks" sections of the reports, enter pertinent information including directions received, problems encountered during construction, work progress and delays, conflicts or errors in the drawings or specifications, field changes, safety hazards encountered, instructions given and corrective actions taken, delays encountered, a record of visitors to the work site, quality control problem areas, deviations from the QC Plan, construction deficiencies encountered, and meetings held. For each entry in the report(s), identify the Schedule Activity No. that is associated with the entered remark.

1.12.2 Quality Control Activities

CQC and Contractor Production reports will be prepared daily to maintain current records providing factual evidence that required quality control activities and tests have been performed. Include in these records the work of subcontractors and suppliers on an acceptable form that includes, as a minimum, the following information:

- a. The name and area of responsibility of the Contractors and any subcontractors.
- b. Operating plant/equipment with hours worked, idle, or down for repair.
- c. Work performed each day, giving location, description, and by whom. When a Network Analysis Schedule (NAS) is used, identify each item of work performed each day by NAS activity number.
- d. Control phase activities performed. Preparatory, and Initial phase Checklists associated with the DFOW referenced to the construction schedule. Follow-up phase activities identified to the DFOW. If testing or specific QC Specialist activities are associated with the Follow-up phase activities for a specific DFOW note this and include those reports.
- e. Test and control activities performed with results and references to specifications and drawings requirements. Identify the control phase (Preparatory, Initial, Follow-up). List of deficiencies noted, along with corrective action in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST.
- f. Quantity of materials received at the site with statement as to acceptability, storage, and reference to specifications and drawings requirements.
- g. Submittals and deliverables reviewed, with Contract reference, by whom, and action taken.
- h. Offsite surveillance activities, including actions taken.
- i. Job safety evaluations stating what was checked, results, and instructions or corrective actions.
- j. Instructions given/received and conflicts in plans and specifications.

1.12.3 Verification Statement

Indicate a description of trades working on the project; the number of personnel working; weather conditions encountered; and any delays encountered. Cover both conforming and deficient features and include a statement that equipment and materials incorporated in the work and workmanship comply with the Contract.

Furnish the original and one copy of these records in report form to the Government by 10:00 AM the next working day after the date covered by the report. As a minimum, prepare and submit one report for every 7 days of no work and on the last day of a no work period. All calendar days need to be accounted for throughout the life of the Contract. The first report following a day of no work will be for that day only. Reports need to be signed and dated by the QC Manager. Include copies of test reports and copies of reports prepared by all subordinate quality control personnel within the QC Manager Report.

1.12.4 Quality Control Validation

Establish and maintain the following in an electronic folder. Divide folder into a series of tabbed sections as shown below. Ensure folder is updated at each required progress meeting.

- a. CQC Meeting minutes in accordance with paragraph QUALITY CONTROL (QC) MEETINGS.
- b. All completed Preparatory and Initial Phase Checklists, arranged by specification section, further sorted by DFOV referenced to the construction schedule. Submit each individual Phase Checklist the day the phase event occurs as part of the CQC daily report.
- c. All milestone inspections, arranged by Activity Number referenced to the construction schedule.
- d. An up-to-date copy of the Testing Plan and Log with supporting field test reports, arranged by specification section referenced to the DFOV to which individual reports results are associated. Individual field test reports will be submitted within 2 working days after the test is performed in accordance with the paragraph QUALITY CONTROL ACTIVITIES.
- e. Copies of all Contract modifications, arranged in numerical order. Also include documentation that modified work was accomplished.
- f. An up-to-date copy of the paragraph DEFICIENCY/REWORK ITEMS LIST.
- g. Upon commencement of Completion Inspections of the entire project or any defined portion, maintain up-to-date copies of all punch lists issued by the QC staff to the Contractor and subcontractors and all punch lists issued by the Government in accordance with the paragraph COMPLETION INSPECTIONS.

1.12.5 Testing Plan and Log

As tests are performed, the QC Manager will record on the "Testing Plan and Log" the date the test was performed and the date the test results were forwarded to the Contracting Officer. Attach a copy of the updated "Testing Plan and Log" to the last daily CQC Report of each month. Provide a copy of the final "Testing Plan and Log" to the preparer of the Operation

& Maintenance (O&M) documentation.

1.12.6 As-Built Drawings

The QC Manager must ensure the as-built drawings, required by Section 01 78 00 CLOSEOUT SUBMITTALS are kept current on a daily basis and marked to show deviations which have been made from the Contract drawings. The as-built drawings document shall commence with the QC Manager ensuring all amendments, or changes to the Contract prior to Contract award are accurately noted in the initial document set creating the accurate baseline of the Contract prior to any work starting. Ensure each deviation has been identified with the appropriate modifying documentation (e.g., PC No., Modification No., Request for Information No.). The QC Manager must initial each revision. Upon completion of work, the QC Manager will furnish a certificate attesting to the accuracy of the as-built drawings prior to submission to the Contracting Officer.

1.13 NOTIFICATION ON NON-COMPLIANCE

The Contracting Officer will notify the Contractor of any detected non-compliance with the Contract. Take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, is deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders will be made the subject of a claim for extension of time for excess costs or damages by the Contractor.

1.14 DELIVERY, STORAGE, AND HANDLING

Designate receiving/storage areas for incoming material to be delivered according to installation schedule and to be placed convenient to work area in order to minimize waste due to excessive materials handling and misapplication. Store and handle materials in a manner as to prevent loss from weather and other damage. Keep materials, products, and accessories covered and off the ground, and store in a dry, secure area. Prevent contact with material that may cause corrosion, discoloration, or staining. Protect all materials and installations from damage by the activities of other trades.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

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11/23

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SECTION 01 45 00.15 10

RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM)
11/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety -- Safety and Occupational Health (SOH) Requirements

1.2 MEASUREMENT AND PAYMENT

The work of this section is not measured for payment. The Contractor is responsible for the work of this section, without any direct compensation other than the payment received for contract items.

1.3 CONTRACT ADMINISTRATION

The Government will use the Resident Management System (RMS) to assist in its monitoring and administration of this contract. The Government accesses the system using the Government Mode of RMS (RMS GM) and the Contractor accesses the system using the Contractor Mode (RMS CM). The term RMS will be used in the remainder of this section for both RMS GM and RMS CM. The joint Government-Contractor use of RMS facilitates electronic exchange of information and overall management of the contract. The Contractor accesses RMS to record, maintain, input, track, and electronically share information with the Government throughout the contract period in the following areas:

- Administration
- Finances
- Quality Control
- Submittal Monitoring
- Scheduling
- Closeout
- Import/Export of Data

The Contractor must be prepared to transition from the Resident Management System (RMS) to the new Construction Management Platform (CMP) at any time during the period of performance of this contract, once the USACE enterprise implements this change. The Contractor's obligation to transition from RMS to the CMP system is a requirement of this contract. Therefore, additional time or cost increase to complete this transition will not be considered. In the event of the transition, RMS project information will be migrated to CMP by the government and available in the CMP system. After the transition, the Contractor must utilize the CMP system to perform all contract administration functions previously required in RMS, including all areas listed above. After the transition, "RMS" and "CMP" will be synonymous for all "Resident Management System" and "RMS" references in this contract's specification. To

facilitate the transition from RMS to CMP, online training for the CMP system will be made available to the Contractor with enough time to train staff, prior to the transition occurring.

1.3.1 Correspondence and Electronic Communications

For ease and speed of communications, exchange correspondence and other documents in electronic format to the maximum extent feasible. Paper documents will govern in the event of discrepancy with the electronic version. Contracting Officer will determine which documents require submission of paper copies.

1.3.2 Other Factors

Other portions of this document have a direct relationship to the reporting accomplished through RMS. Particular attention is directed to FAR 52.236-15 Schedules for Construction Contracts; FAR 52.232-27 Prompt Payment for Construction Contracts; FAR 52.232-5 Payments Under Fixed-Priced Construction Contracts; Section 01 32 01.00 10 PROJECT SCHEDULE; Section 01 33 00 SUBMITTAL PROCEDURES; Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS; and Section 01 45 00 QUALITY CONTROL.

1.4 RMS SOFTWARE

RMS is a web based application. Download, install and be able to utilize the latest version of RMS within 7 calendar days of receipt of the Notice to Proceed. RMS software, user manuals, access and installation instructions, program updates and training information are available from the RMS website (<https://rms.usace.army.mil>). The Government and the Contractor will have different access authorities to the same contract database through RMS. The common database will be updated automatically each time a user finalizes an entry or change.

1.5 CONTRACT DATABASE - GOVERNMENT

The Government will enter the basic contract award data in RMS prior to granting the Contractor access. The Government entries into RMS will generally be related to submittal reviews, correspondence status, and Quality Assurance (QA) comments, as well as other miscellaneous administrative information.

1.6 CONTRACT DATABASE - CONTRACTOR

Contractor entries into RMS establish, maintain, and update data throughout the duration of the contract. Contractor entries generally include prime and subcontractor information, daily reports, submittals, RFI's, schedule updates and payment requests. RMS includes the ability to import attachments and export reports in many of the modules, including submittals. The Contractor responsibilities for entries in RMS typically include the following items:

1.6.1 Administration

1.6.1.1 Contractor Information

Enter all current Contractor administrative data and information into RMS within 7 calendar days of receiving access to the contract in RMS. This includes, but is not limited to, Contractor's name, address, telephone numbers, management staff, and other required items.

1.6.1.2 Subcontractor Information

Enter all missing subcontractor administrative data and information into RMS CM within seven calendar days of receiving access to the Contract in RMS or within seven calendar days of the signing of the subcontractor agreement for agreements signed at a later date. This includes name, trade, address, phone numbers, and other required information for all subcontractors. A subcontractor is listed separately for each trade to be performed.

1.6.1.3 Correspondence

Identify all Contractor correspondence to the Government with a serial number. Prefix correspondence initiated by the Contractor's site office with "S". Prefix letters initiated by the Contractor's home (main) office with "H". Letters are numbered starting from 0001. (e.g., H-0001 or S-0001). The Government's letters to the Contractor will be prefixed with "C" or "RFP".

1.6.1.4 Equipment

Enter and maintain a current list of equipment planned for use or being used on the jobsite, including the most recent and planned equipment inspection dates in accordance with Section 01 45 00 QUALITY CONTROL.

1.6.1.5 Reports

Track the status of the project using the reports available in RMS. The value of these reports reflects the quality of the data input. These reports include but are not limited to the Progress Payment Request worksheet, Quality Control (QC) comments, Deficiency Tracking Log, Submittal Register Status, and Three-Phase Control worksheets in accordance with Sections 01 20 00 PRICE AND PAYMENT PROCEDURES, Section 01 32 01.00 10 PROJECT SCHEDULE, 01 33 00 SUBMITTAL PROCEDURES, and 01 45 00 QUALITY CONTROL.

1.6.1.6 Request For Information (RFI)

Create and track all Requests For Information (RFI) in the RMS Administration Module for Government review and response. Provide status at weekly progress meeting in accordance with Section 01 32 01.00 10 PROJECT SCHEDULE.

1.6.2 Finances

1.6.2.1 Pay Activity Data

Develop and enter a list of pay activities in conjunction with the project schedule in accordance with Section 01 32 01.00 10 PROJECT SCHEDULE. The sum of pay activities equals the total Contract amount, including modifications. Each pay activity must be assigned to a Contract Line Item Number (CLIN). The sum of the activities assigned to a CLIN equals the amount of each CLIN.

1.6.2.2 Payment Requests

Prepare all progress payment requests using RMS. Update the work completed under the Contract at least monthly, measured as percent or as specific

quantities. After the update, generate a payment request and prompt payment certification using RMS. Submit the signed prompt payment certification and payment request as well as supporting data, either electronically or by hard copy.

1.6.3 Quality Control (QC)

Enter and track implementation of the 3-phase QC Control System, QC testing, transferred and installed property and warranties in RMS. Link each deficiency and QC test to a pay activity. Prepare daily reports, identify and track deficiencies, document progress of work, and support other Contractor QC requirements in RMS. Maintain all data on a daily basis. Ensure that RMS reflects all quality control methods, tests and actions contained within the Contractor Quality Control (CQC) Plan and Government review comments. Provide within seven calendar days of Government acceptance of the CQC Plan.

1.6.3.1 Quality Control (QC) Reports

The Contractor's Quality Control (QC) Daily Report in RMS is the official report. The Contractor can use other supplemental formats to record QC data, but information from any supplemental formats are to be consolidated and entered into the RMS QC Daily Report. Any supplemental information may be entered into RMS as an attachment to the report. QC Daily Reports must be finalized and signed in RMS within 24 hours after the date covered by the report.

1.6.3.2 Deficiency Tracking

Use the QC Daily Report Module to enter and track deficiencies. Deficiencies identified and entered into RMS by the Contractor or the Government will be sequentially numbered with a QC or QA prefix for tracking purposes. Enter each deficiency into RMS the same day that the deficiency is identified. Monitor, track and resolve all QC and QA entered deficiencies. A deficiency is not considered to be corrected until the Government indicates concurrence in RMS.

1.6.3.3 Three-Phase Control Meetings

Maintain scheduled and actual dates and times of preparatory and initial control meetings in RMS, and upload meeting minutes of the meetings. Worksheets for the three-phase control meetings are generated within RMS.

1.6.3.4 Labor and Equipment Hours

Enter labor and equipment exposure hours on a daily basis. Roll up the labor and equipment exposure data into a monthly exposure report.

1.6.3.5 Accident/Safety Reporting

Both the Contractor and the Government enter safety related comments in RMS as a deficiency. The Contractor must monitor, track and show resolution for safety issues in the QC Daily Report area of the RMS QC Module. In addition, follow all reporting requirements for accidents and incidents as required in EM 385-1-1, Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS and as required by any other applicable Federal, State or local agencies.

1.6.3.6 Definable Features of Work

Enter each feature of work, as defined in the approved CQC Plan, into the RMS QC Module. A feature of work may be associated with a single or multiple pay activities, however a pay activity is only to be linked to a single feature of work.

1.6.3.7 Activity Hazard Analysis

Import activity hazard analysis electronic document files into the RMS QC Module utilizing the document package manager. Prepare activity hazard analysis in accordance with Section 01 35 26 GOVERNMENT SAFETY REQUIREMENTS.

1.6.4 Submittal Management

Enter all current submittal register data and information into RMS within seven calendar days of receiving access to the Contract in RMS. The Contractor is responsible for entering Activity Numbers for each submittal item corresponding to their Schedule. The Contractor will import the information shown in the submittal register following the specification Section 01 33 00 SUBMITTAL PROCEDURES unless otherwise directed by the Contracting Officer. Group electronic submittal documents into transmittal packages to send to the Government, except very large electronic files, samples, spare parts, mock ups, color boards, or where hard copies are specifically required. Track transmittals and update the submittal register in RMS on a daily basis throughout the duration of the Contract.

Contractor may request submittals to be added to RMS throughout the lifecycle of the project. Approval/disapproval of Contractor requests is at the discretion of the Contracting Officer. Submittals entry into RMS must be updated daily or as practical for performance of the Contract, as approved by the Contracting Officer.

1.6.5 Schedule

Enter and update the contract project schedule in RMS by either manually entering all schedule data or by importing the Standard Data Exchange Format (SDEF) file, based on the requirements in Section 01 32 01.00 10 PROJECT SCHEDULE.

1.6.6 Closeout

Closeout documents, processes and forms are managed and tracked in RMS by both the Contractor and the Government. Ensure that all closeout documents are entered, completed and documented within RMS.

1.7 IMPLEMENTATION

Use of RMS as described in the preceding paragraphs is mandatory. Ensure that sufficient resources are available to maintain contract data within the RMS system. RMS is an integral part of the Contractor's required management of quality control.

1.8 NOTIFICATION OF NONCOMPLIANCE

Take corrective action within seven calendar days after receipt of notice of RMS non-compliance by the Contracting Officer.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

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SECTION 01 50 00

TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS

11/20, CHG 2: 08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1

(2024) Safety -- Safety and Health
Requirements Manual

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Construction Site Plan; G

1.3 CONSTRUCTION SITE PLAN

Prior to the start of work, submit for Government approval a site plan showing the locations and dimensions of temporary facilities (including layouts and details, equipment and material storage area (onsite and offsite), and access and haul routes, avenues of ingress/egress to the fenced area and details of the fence installation. Identify any areas which may have to be graveled to prevent the tracking of mud. Indicate if the use of a supplemental or other staging area is desired. Show locations of safety and construction fences, site trailers, construction entrances, trash dumpsters, temporary sanitary facilities, and worker parking areas.

1.4 NAVIGATION AIDS

Notify the Federal Aids to Navigation Officer 30 days prior to the commencement of work.

The Contractor shall not relocate or move any aids to navigation that have been established by the U.S. Coast Guard. If it becomes necessary to have any aids to navigation moved in order to complete dredging operations under this contract, the Contractor shall notify the appropriate Coast Guard Office, with a copy to the Contracting Officer and/or his Representative, not less than 30 days prior to the need for movement. The Contractor shall notify the appropriate Coast Guard Office of the approximate time of completion of dredging.

The Coast Guard has indicated that the requirements of navigation may make

it necessary to establish times other than those requested. IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY TO COORDINATE HIS PLANS WITH THE COAST GUARD SUFFICIENTLY IN ADVANCE OF THE PLANNED CLOSING TO PREVENT DELAY TO THE DREDGING OPERATIONS AND COMPLY WITH THE COAST GUARD REQUIREMENT.

1.5 OBSTRUCTION EVALUATION / AIRPORT AIRSPACE ANALYSIS

The Contractor shall file forms 7460-1 and 7460-2 if applicable and as specified in Section 00 90 01, Attachment L, Obstruction Evaluation / Airport Airspace Analysis Desk Reference Guide to ensure that 14 CFR Part 77.9 is met.

1.6 LAYOUT OF WORK

- a. The Government will establish the following base lines and bench marks at the site of the work: (Monuments and bench marks as shown on the drawings.)
- b. From the base lines and bench marks established by the Government, complete the layout of the work and provide measurements for the execution of the work to the location and limit marks indicated, subject to such modifications as the Contracting Officer may require to meet changed conditions or as a result of necessary modifications to the contract work.
- c. Provide stakes, templates, platforms, equipment, tools and material, and labor as required for laying out the work from the base lines and bench marks established by the Government. Maintain and preserve all stakes and other marks established by the Contracting Officer until authorized to remove them, and if such marks are destroyed by the Contractor or through his negligence prior to their authorized removal, they may be replaced by the Contracting Officer, at his discretion, and the expense of replacement will be deducted from any amounts due or to become due the Contractor. The Contracting Officer may require that work be suspended at any time when location and limit marks established by the Contractor are not reasonably adequate to permit checking of the work.

1.7 SUBMERGED PIPELINES

Should it become necessary in the performance of this contract to use a submerged pipelines across a navigable channel, the Contractor must notify the Contracting Officer, in writing to be received in the District Office at least ten working days prior to the desired closure date. This notification shall include the following:

- (1) Location (Channel Centerline Stationing) and depth (over the top of the pipeline) at which the submerged line will be placed;
- (2) The desired length of time the channel is to be closed for installation of the pipeline;
- (3) The date and hour placement or removal will commence;
- (4) The date and hour of anticipated completion; and
- (5) The name and telephone number of the person to be contacted for information and response to any emergency condition.

The requirements of navigation may make it necessary to establish times other than those requested. It is the contractor's responsibility to coordinate his/her plans with the Coast Guard sufficiently in advance of the planned closing to prevent delay to the dredging operations and comply with the Coast Guard requirements.

1.8 FLOATING PIPELINE

Design rubber float lines to minimize the number of connections in the discharge line and reduce the need to disconnect the float line from the submerged line in moderate storm conditions. Provide an information-only design submittal prior to starting this Definable Feature of Work. Floating pipeline, used as access ways, shall be equipped with walkways and guard rails. The walkway must be securely anchored to the pipeline and a guard rail equivalent in strength to a wood rail 2 inch by 4 inch must be provided on at least one side of the walkway and must be secured to upright at intervals of not more than 8 feet and of not less than 36 nor more than 42 inches in height above the walkway. Float pipelines not equipped with walkways must be barricaded to prevent personnel access.

1.9 BRIDGE-TO-BRIDGE RADIO-TELEPHONE EQUIPMENT

In order that radio-telephone communication may be made with passing vessels, all dredges engaged in work under the contract shall be equipped with and operate bridge-to-bridge radio-telephone equipment on VHF Channel 13 (156.65 MHz with low power output having a communication range of approximately ten (10) miles). The Federal Communication Commission has approved the frequency.

1.10 HEAD PROTECTION AREA

THE ENTIRE WORK SITE UNDER THIS CONTRACT IS DESIGNATED AS A HARD HAT AREA. The Contractor shall post the area in accordance with the requirements of EM 385-1-1, and shall ensure that all prime and subcontractor personnel, vendors, and visitors utilize hard hats while within the project area. (EM 385-1-1 2024)

1.11 SIGNAL LIGHTS

The Contractor shall display signal lights and conduct his operations in accordance with the General Regulations of the Coast Guard governing lights and day signals to be displayed by towing vessels with tows on which no light signals can be displayed. Vessels working on wrecks, dredges, and vessels engaged in laying cables or pipe or in submarine or bank protection operations, lights to be displayed on dredge pipelines, and day signals to be displayed by vessels of more than 65 feet in length moored or anchored in a fairway or channel, and the passing by other vessels or floating plant working in navigable channels, as approved by the Commandant, U.S. Coast Guard with respect to vessels on the high seas (33 CAR 81 App. A-72 COLREGS, Part C), vessels in inland waters (33-CFR 93.18 - 93.31a), and vessels in western rivers (33 CFR 95.5.51 - 95.70)), as applicable. (DAEN-PRP Ind dtd. 12 Sep. 83)

1.12 REVERSE SIGNAL ALARM

- a. All construction equipment, except light service trucks, panels, pickup trucks, station wagons, crawler-type cranes, power shovels and draglines, whether moving alone or in combination shall be equipped with reverse signal alarms. The alarm shall be mounted on the rear of

the equipment and shall be protected or constructed as to withstand severe wear and tear, adverse weather and unfavorable environmental working conditions and shall be certified by the manufacturer to fully meet the following performance standards.

- b. The alarm shall produce a relatively pure tone which shall peak within the American National Std's. Inst. standard octave pass band of 600 to 2400 cycles per second and shall produce a 0.2 to 0.5 second audible warning within the initial three (3) feet of backward movement of the vehicle on which it is mounted and at regular intervals, not to exceed three (3) seconds throughout the backward movement. The alarm shall automatically stop when backward movement ceases. The sound intensity of the alarm shall range from and not exceed 90-100 db (decibels) at a horizontal distance of five feet from the alarm.
- c. Actuation of the alarm shall be automatic by direct connections to any part of the equipment that moves or acts in a manner distinctive only to the rearward movement of the vehicle with no manual controls of any kind between the source of actuation and the alarm. Where application of this requirement to specific types of the equipment has impractical application, other means of actuation may be used upon written approval of the Contracting Officer and/or his Representative.
- d. The use of the alarm shall be in the addition to prescribed requirements for signalmen.

1.13 INSPECTION

The inspectors will assess whether the gauges, ranges, location marks and limit marks are maintained in proper order; but the presence of the inspector shall not relieve the Contractor of the responsibility for the proper execution of the work in accordance with the specifications. The Contractor will be required:

- a. To furnish, on the request of the Contracting Officer and/or his Representative or any inspector, the use of such boats, boatmen, laborers, a part of the ordinary and usual equipment and crew of the dredging plant as may be reasonably necessary in inspecting and supervising the work. However, the Contractor will not be required to furnish such facilities for the surveys prescribed in the clause entitled: "Final Examination and Acceptance."
- b. To furnish, on the request of the Contracting Officer or the Contracting Officer's Representative or any inspector, suitable transportation from all points on shore designated by the Contracting Officer and/or his Representative to and from the various pieces of plant and to and from the placement area. Should the Contractor refuse, neglect, or delay compliance with these requirements, the specific facilities may be furnished and maintained by the Contracting Officer or Contracting Officer Representative, and the cost thereof will be deducted from any amounts due or to become due to the Contractor.
- c. To provide the Contracting Officer and/or his Representative or any inspector, a contact available to assist in coordinating/conducting virtual inspections in cases when in person inspections cannot be performed.

1.14 DOD CONDITION OF READINESS

DOD will set the Condition of Readiness (COR) based on the weather forecast for sustained winds 50 knots (58 mph) or greater. Contact the Contracting Officer for the current COR setting.

Monitor weather conditions a minimum of twice a day and take appropriate actions according to the approved Emergency Plan in the accepted Accident Prevention Plan, EM 385-1-1 Emergency Plan requirements and the instructions below.

Unless otherwise directed by the Contracting Officer, comply with:

- a. Condition FOUR (Sustained winds of 58 mph or greater expected within 72 hours): Normal daily jobsite cleanup and good housekeeping practices. Collect and store in piles or containers scrap lumber, waste material, and rubbish for removal and disposal at the close of each work day. Maintain the construction site including storage areas, free of accumulation of debris. Stack form lumber in neat piles less than 3.3 feet high. Remove all debris, trash, or objects that could become missile hazards. Review requirements pertaining to "Condition THREE" and continue action as necessary to attain "Condition FOUR" readiness. Contact Contracting Officer for weather and COR updates and completion of required actions.
- b. Condition THREE (Sustained winds of 58 mph or greater expected within 48 hours): Maintain "Condition FOUR" requirements and commence securing operations necessary for "Condition ONE" which cannot be completed within 18 hours. Cease all routine activities which might interfere with securing operations. Commence securing and stow all gear and portable equipment. Make preparations for securing buildings. Reinforce or remove formwork and scaffolding. Secure machinery, tools, equipment, materials, or remove from the jobsite. Expend every effort to clear all missile hazards and loose equipment from general base areas. Contact Contracting Officer for weather and COR updates and completion of required actions. Review requirements pertaining to "Condition TWO" and continue action as necessary to attain "Condition THREE" readiness.
- c. Condition TWO (Sustained winds of 58 mph or greater expected within 24 hours): Secure the jobsite, and leave Government premises.
- d. Condition ONE. (Sustained winds of 58 mph or greater expected within 12 hours): Contractor access to the jobsite and Government premises is prohibited.

PART 2 PRODUCTS

2.1 TEMPORARY SIGNAGE

2.1.1 Bulletin Board

Prior to the commencement of work activities, provide a clear weatherproof covered bulletin board not less than 36 by 48 inches in size for displaying the Equal Employment Opportunity poster, a copy of the wage decision contained in the Contract, Wage Rate Information poster, Safety and Health Information as required by EM 385-1-1 and other information approved by the Contracting Officer. Coordinate requirements herein with 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. Locate the bulletin board at the project

site in a conspicuous place easily accessible to all employees, and in location as approved by the Contracting Officer.

2.1.2 Project Identification Sign

The requirements for the signs, their content, and location are as specified in Section 01 58 00 PROJECT IDENTIFICATION. Erect signs within 15 days after receipt of the notice to proceed. Correct the data required by the safety sign daily, with light colored metallic or non-metallic numerals.

The Contractor shall furnish and erect a project sign at a location directed by the Contracting Officer or Contracting Officer's Representative. The project sign shall conform with all requirements as specified in Section 00 90 01, Attachment C, Project Identification and Safety Signs.

The Contractor shall maintain the sign in good condition throughout the construction period. No separate payment will be made for erecting and maintaining the project sign, and all costs in connection therewith will be considered the obligation of the Contractor. Upon completion of the project, the Contractor shall remove the sign from the work site.

2.1.3 Safety Sign

The Contractor shall construct a safety sign at a location directed by the Contracting Officer and/or his Representative. The sign shall be 6 feet by 4 feet and shall conform to the requirements as specified in Section 00 90 01, Attachment C, Project Identification and Safety Signs.

The sign shall be erected as soon as possible and within 5 calendar days after the date of the receipt of the Notice to Proceed. The data required by the sign shall be corrected daily. No separate payment will be made for erecting and maintaining the safety sign and all costs in connection therewith will be considered the obligation of the contractor. Upon completion of the project, the Contractor shall remove the sign from the work site.

2.1.4 Warning Signs

Post temporary signs, tags, and labels to give workers and the public adequate warning and caution of construction hazards according to the EM 385-1-1. Attach signs to the perimeter fencing every 150 feet warning the public of the presence of construction hazards. Signs must require unauthorized persons to keep out of the construction site. Correct the data required by safety signs daily. Post signs at all points of entry designating the construction site as a hard hat area.

PART 3 EXECUTION

3.1 CONTRACTOR'S TEMPORARY FACILITIES

Contractor is responsible for security of their property. Provide adequate outside security lighting at the temporary facilities. Trailers must be anchored to resist high winds and meet applicable state or local standards for anchoring mobile trailers. Coordinate anchoring with EM 385-1-1. The Contract Clause entitled "FAR 52.236-10, Operations and Storage Areas" and the following apply:

3.2 ACCOMMODATIONS FOR GOVERNMENT INSPECTIONS

The Contractor shall furnish regularly to Government inspectors onboard the dredge or other craft upon which they are employed a suitable separate room for an office. The room shall be fully equipped and maintained to the satisfaction of the Contracting Officer and/or his Representative; it shall be properly heated, ventilated, and lighted, and shall have a desk which can be locked, and a chair for each inspector, beverage (e.g. water, coffee and tea) and bathroom conveniences. In addition, the Contractor shall provide a suitable sleeping area for use by the Government inspector when weather or sea conditions preclude safe transfer to shore. The entire cost to the Contractor for furnishing, equipping and maintaining the foregoing accommodations shall be included in the contract price. If the Contractor fails to meet these requirements, the Contracting Officer and/or his Representative will secure the facilities referred to above, and the cost thereof will be deducted from payments to the Contractor.

3.3 CLEANUP

Remove construction debris, waste materials, packaging material and the like from the work site daily. Any dirt or mud which is tracked onto paved or surfaced roadways must be cleaned away. Store all salvageable materials resulting from demolition activities within the fenced area described above or at the supplemental storage area. Neatly stack stored materials not in trailers, whether new or salvaged.

3.4 RESTORATION OF STORAGE AREA

Upon completion of the project remove the bulletin board, signs, barricades, haul roads, and all other temporary products from the site. After removal of trailers, materials, and equipment from within the fenced area, remove the fence. Restore areas used during the performance of the Contract to the original or better condition. Remove gravel used to traverse grassed areas and restore the area to its original condition, including top soil and seeding as necessary.

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ENVIRONMENTAL PROTECTION
04/06

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024 Safety and Health Requirements Manual)

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

33 CFR 328	Definitions of Waters of the United States
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 262	Standards Applicable to Generators of Hazardous Waste
40 CFR 279	Standards for the Management of Used Oil
40 CFR 302	Designation, Reportable Quantities, and Notification
40 CFR 355	Emergency Planning and Notification
40 CFR 68	Chemical Accident Prevention Provisions
49 CFR 171 - 178	Hazardous Materials Regulations

1.2 DEFINITIONS

1.2.1 Environmental Pollution and Damage

Environmental pollution and damage is the presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances of importance to human life; affect other species of importance to humankind; or degrade the environment aesthetically, culturally and/or historically.

1.2.2 Environmental Protection

Environmental protection is the prevention/control of pollution and habitat disruption that may occur to the environment during construction. The control of environmental pollution and damage requires consideration of land, water, and air; biological and cultural resources; and includes management of visual aesthetics; noise; solid, chemical, gaseous, and

liquid waste; radiant energy and radioactive material as well as other pollutants.

1.2.3 Contractor Generated Hazardous Waste

Contractor generated hazardous waste means materials that, if abandoned or disposed of, may meet the definition of a hazardous waste. These waste streams would typically consist of material brought on site by the Contractor to execute work, but are not fully consumed during the course of construction. Examples include, but are not limited to, excess paint thinners (i.e. methyl ethyl ketone, toluene etc.), waste thinners, excess paints, excess solvents, waste solvents, and excess pesticides, and contaminated pesticide equipment rinse water.

1.2.4 Surface Discharge

The term "Surface Discharge" implies that the water is discharged with possible sheeting action and subsequent soil erosion may occur. Waters that are surface discharged may terminate in drainage ditches, storm sewers, creeks, and/or "waters of the United States" and would require a permit to discharge water from the governing agency.

1.2.5 Waters of the United States

All waters which are under the jurisdiction of the Clean Water Act, as defined in 33 CFR 328.

1.2.6 Wetlands

Those areas that are inundated or saturated by surface or ground water at a frequency and duration sufficient to support, and that under normal circumstances do support, a prevalence of vegetation typically adapted for life in saturated soil conditions. Wetlands generally include swamps, marshes, and bogs. Official determination of whether or not an area is classified as a wetland must be done in accordance with WETLANDS MANUAL.

1.3 GENERAL REQUIREMENTS

Minimize environmental pollution and damage that may occur as the result of construction operations. The environmental resources within the project boundaries and those affected outside the limits of permanent work must be protected during the entire duration of this contract. The Contractor shall comply with all applicable environmental Federal, State, and local laws and regulations. The Contractor shall be responsible for any delays resulting from failure to comply with environmental laws and regulations.

1.4 OIL TRANSFER OPERATIONS

The Contractor shall assure that oil transfer operations to or from his plant comply with all Federal, state, municipal laws, codes and regulations (see Attachment D of Section 00 90 01 for regulation on "Oil and Hazardous Material Transfer Operations" - 33 CFR 156). Particular attention is invited to 33 CFR Subchapter O, Pollution. The Contractor shall incorporate in the accident prevention program, submitted in compliance with Section 00700, contract clause "ACCIDENT PREVENTION," sufficient information to demonstrate that all fuel transfers will be made in compliance with 33 CFR 156 and any other applicable laws, codes and regulations. (NAP-1/81)

1.5 NOISE LEVEL CRITERIA

Maximum sound pressure levels measured at the booster pumps, engine generators and vehicles including muffler and exhaust components, shall not exceed decibel values prescribed by Federal, State and local codes and regulations. The foregoing applies with the pumps under normal operating conditions at not less than 50% full rated load. Tests will be performed at no additional cost to the Government by the Contractor, or the pump engine and generator manufacturer or by an approved testing agency having proven capability in noise measurements and using approved measurement equipment and acceptable measurement procedures. Test data will be submitted to the Contracting Officer for approval 5 days after the commencement of dredging operations. Should the sound pressure (noise) level test data indicate a need for sound reduction, the Contractor will provide necessary enclosures, baffles or any other appropriate means around booster pumps and engine generators including muffler and exhaust components, so as to attain noise levels which do not exceed those prescribed by aforementioned regulations. Details of the proposed enclosure or baffling arrangement shall be submitted to the Contracting Officer for approval. Such details will be accomplished by calculations, which indicate anticipated compliance of sound pressure (noise) levels with codes and regulations noted herein before. All this is especially enforceable where these booster and engine generators locations are in bathing areas.

1.6 SUBCONTRACTORS

The Contractor shall ensure compliance with this section by subcontractors.

1.7 PAYMENT

No separate payment will be made for work covered under this section. Payment of fees associated with environmental permits, application, and/or notices obtained by the Contractor, and payment of all fines/fees for violation or non-compliance with Federal, State, Regional and local laws and regulations are the Contractor's responsibility. All costs associated with this section must be included in the contract price. The Contractor shall be responsible for all fines/fees for violation or non-compliance with Federal, State, Regional and local laws and regulations.

1.8 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. The following shall be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals; G

Environmental Protection Plan; G

1.9 ENVIRONMENTAL PROTECTION PLAN

Prior to commencing construction activities or delivery of materials to the site, the Contractor shall submit an Environmental Protection Plan for review and approval by the Contracting Officer. The purpose of the Environmental Protection Plan is to present a comprehensive overview of known or potential environmental issues which the Contractor must address

during construction. Issues of concern must be defined within the Environmental Protection Plan as outlined in this section. The Contractor shall address each topic at a level of detail commensurate with the environmental issue and required construction task(s). Topics or issues which are not identified in this section, but which the Contractor considers necessary, shall be identified and discussed after those items formally identified in this section. Prior to submittal of the Environmental Protection Plan, the Contractor shall meet with the Contracting Officer for the purpose of discussing the implementation of the initial Environmental Protection Plan; possible subsequent additions and revisions to the plan including any reporting requirements; and methods for administration of the Contractor's Environmental Protection Plan. The Environmental Protection Plan shall be current and maintained onsite by the Contractor.

1.9.1 Compliance

No requirement in this Section shall be construed as relieving the Contractor of any applicable Federal, State, and local environmental protection laws and regulations. During Construction, the Contractor will be responsible for identifying, implementing, and submitting for approval any additional requirements to be included in the Environmental Protection Plan.

1.9.2 Contents

The environmental protection plan shall include but shall not be limited to, the following:

- a. Name(s) of person(s) within the Contractor's organization who is(are) responsible for ensuring adherence to the Environmental Protection Plan.
- b. Name(s) and qualifications of person(s) responsible for manifesting hazardous waste to be removed from the site, if applicable.
- c. Name(s) and qualifications of person(s) responsible for training the Contractor's environmental protection personnel.
- d. Description of the Contractor's environmental protection personnel training program.
- e. An erosion and sediment control plan which identifies the type and location of the erosion and sediment controls to be provided. The plan must include monitoring and reporting requirements to assure that the control measures are in compliance with the erosion and sediment control plan, Federal, State, and local laws and regulations. A Storm Water Pollution Prevention Plan (SWPPP) may be substituted for this plan.
- f. Drawings showing locations of proposed temporary excavations or embankments for haul roads, stream crossings, material storage areas, structures, sanitary facilities, and stockpiles of excess or spoil materials including methods to control runoff and to contain materials on the site.
- g. Traffic control plans including measures to reduce erosion of temporary roadbeds by construction traffic, especially during wet weather. Plan shall include measures to minimize the amount of mud

transported onto paved public roads by vehicles or runoff.

h. Work area plan showing the proposed activity in each portion of the area and identifying the areas of limited use or nonuse. Plan should include measures for marking the limits of use areas including methods for protection of features to be preserved within authorized work areas.

i. Drawing showing the location of borrow areas.

j. The Spill Control plan shall include the procedures, instructions, and reports to be used in the event of an unforeseen spill of a substance regulated by 40 CFR 68, 40 CFR 302, 40 CFR 355, and/or regulated under State or Local laws and regulations. The Spill Control Plan supplements the requirements of EM 385-1-1. This plan shall include, as a minimum:

- (1) The name of the individual who will report any spills or hazardous substance releases and who will follow up with complete documentation. This individual will immediately notify the Contracting Officer and Facility Response Personnel in addition to the legally required Federal, State, and local reporting channels (including the National Response Center 1-800-424-8802) if a reportable quantity is released to the environment. Include in the plan a list of the required reporting channels and telephone numbers.

- (2) The name and qualifications of the individual who will be responsible for implementing and supervising the containment and cleanup.

- (3) Training requirements for Contractor's personnel and methods of accomplishing the training.

- (4) A list of materials and equipment to be immediately available at the job site, tailored to cleanup work of the potential hazard(s) identified.

- (5) The names and locations of suppliers of containment materials and locations of additional fuel oil recovery, cleanup, restoration, and material-placement equipment available in case of an unforeseen spill emergency.

- (6) The methods and procedures to be used for expeditious contaminant cleanup.

k. A Non-hazardous solid waste disposal Plan identifying methods and locations for solid waste disposal including clearing debris and schedules for disposal and the following.

- (1) The Contractor shall identify any subcontractors responsible for the transportation and disposal of solid waste. Submit licenses or permits for solid waste disposal sites that are not a commercial operating facility.

- (2) Evidence of the disposal facility's acceptance of the solid waste must be attached to this plan during the construction. The Contractor shall attach a copy of each of the Non-hazardous Solid Waste Diversion Reports to the disposal plan. The report shall be submitted on the first working day after the first quarter that

non-hazardous solid waste has been disposed and/or diverted and shall be for the previous quarter (e.g. the first working day of January, April, July, and October).

(3) The report shall indicate the total amount of waste generated and total amount of waste diverted in cubic yards (CY) or tons along with the percent that was diverted.

l. A recycling and solid waste minimization plan with a list of measures to reduce consumption of energy and natural resources. The plan shall detail the Contractor's actions to comply with and to participate in Federal, State, Regional and local Government sponsored recycling programs to reduce the volume of solid waste at the source.

m. An air pollution control plan detailing provisions to assure that dust, debris, materials, trash, etc., do not become air borne and travel off the project site.

n. A contaminant prevention plan that: identifies potentially hazardous substances to be used on the job site; identifies the intended actions to prevent introduction of such materials into the air, water, or ground; and details provisions for compliance with Federal, State, and local laws and regulations for storage and handling of these materials. In accordance with EM 385-1-1, a copy of the Material Safety Data Sheets (MSDS) and the maximum quantity of each hazardous material to be onsite at any given time must be included in the contaminant prevention plan. As new hazardous materials are brought onsite or removed from the site, the plan shall be updated.

o. A waste water management plan that identifies the methods and procedures for management and/or discharge of waste waters which are directly derived from construction activities, such as concrete curing water, clean-up water, dewatering of ground water, disinfecting of water, hydrostatic test water, and water used in flushing of lines. If a settling/retention pond is required, the plan must include the design of the pond including drawings, removal plan, and testing requirements for possible pollutants. If land application will be the method of disposal for the waste water, the plan must include a sketch showing the location for land application along with a description of the pretreatment methods to be implemented. If surface discharge will be the method of disposal, include a copy of the permit and associated documents as an attachment prior to discharging the waste water. If disposal is to a sanitary sewer, the plan must include documentation that the Waste Water Treatment Plant Operator has approved the flow rate, volume, and type of discharge.

p. A historical, archaeological, cultural resources biological resources and wetlands plan that defines procedures for identifying and protecting historical, archaeological, cultural resources, biological resources and wetlands known to be on the project site: and/or identifies procedures to be followed if historical archaeological, cultural resources, biological resources and wetlands not previously known to be onsite or in the area are discovered during construction. Include in the plan methods to assure the protection of known or discovered resources, identifying lines of communication between Contractor personnel and the Contracting Officer.

1.9.3 Appendix

Attach to the Environmental Protection Plan, as an appendix, copies of all environmental permits, permit application packages, approvals to construct, notifications, certifications, reports, and termination documents.

1.10 PROTECTION FEATURES

This paragraph supplements the Contract Clause PROTECTION OF EXISTING VEGETATION, STRUCTURES, EQUIPMENT, UTILITIES, AND IMPROVEMENTS. Prior to start of any onsite construction activities, the Contractor and the Contracting Officer will make a joint condition survey. Immediately following the survey, the Contractor will prepare a brief report including a plan describing the features requiring protection under the provisions of the Contract Clauses, which are not specifically identified on the drawings as environmental features requiring protection along with the condition of trees, shrubs and grassed areas immediately adjacent to the site of work and adjacent to the Contractor's assigned storage area and access route(s), as applicable. This survey report will be signed by both the Contractor and the Contracting Officer upon mutual agreement as to its accuracy and completeness. The Contractor must protect those environmental features included in the survey report and any indicated on the drawings, regardless of interference which their preservation may cause to the work under the contract.

1.11 SPECIAL ENVIRONMENTAL REQUIREMENTS

The Contractor shall comply with the special environmental requirements listed in the attached Federal Consistency Determination(s) / Water Quality Certificate(s) of section 00 90 02 FCD / WATER QUALITY CERTIFICATE.

1.12 ENVIRONMENTAL ASSESSMENT OF CONTRACT DEVIATIONS

Any deviations from the drawings, plans and specifications, requested by the Contractor and which may have an environmental impact, will be subject to approval by the Contracting Officer and may require an extended review, processing, and approval time. The Contracting Officer reserves the right to disapprove alternate methods, even if they are more cost effective, if the Contracting Officer determines that the proposed alternate method will have an adverse environmental impact.

1.13 NOTIFICATION

The Contracting Officer will notify the Contractor in writing of any observed noncompliance with Federal, State or local environmental laws or regulations, permits, and other elements of the Contractor's Environmental Protection plan. After receipt of such notice, the Contractor will inform the Contracting Officer of the proposed corrective action and take such action when approved by the Contracting Officer. The Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No time extensions will be granted or equitable adjustments allowed for any such suspensions. This is in addition to any other actions the Contracting Officer may take under the contract, or in accordance with the Federal Acquisition Regulation or Federal Law.

1.14 PROTECTION OF ENVIRONMENTAL RESOURCES

The Contractor shall at all times perform all work and take such steps

required to minimize interference with or disturbance to fish and wildlife. The Contractor will not be permitted to alter water flows or otherwise disturb native habitat adjacent to the work area, which in the opinion of the Contracting Officer, are critical to fish and wildlife. The Contractor shall anticipate orders to stop work at the discretion of the Contracting Officer if any fish or wildlife are endangered by construction activities. The environmental resources within the project boundaries and those affected outside the limits of permanent work under this contract shall be protected during the entire period of this contract. The Contractor shall confine his activities to areas defined by the drawings and specifications.

1.14.1 Protection of Endangered and Threatened Species

1.14.1.1 Scope of Work

Scope of work for Endangered Species Observers; applicable only for work to be performed by Hopper Dredge year-round. Endangered species observers will report to the Contracting officer (actual date(s) will depend on the contract arrangements between the USACE, NY district and the contractor).

1.14.1.2 Purpose

This scope of work (SOW) outlines the Contractor's requirements for conducting sea turtle and whale monitoring on board the Contractor's hopper dredge. The Contractor will supply endangered species observers to be placed aboard the dredging vessel to monitor for the presence of sea turtles and record the observations of whales. Endangered species observers must be certified in writing as acceptable by the National Marine Fisheries Service (NMFS) for observing and handling endangered sea turtles. The data collected during this project will be used, along with additional data, to prepare a biological assessment on endangered species occurring within the New York District. The intent of the following requirements is to monitor the effect of Hopper Dredging on sea turtles within waters of the New York District.

1.14.1.3 Requirements: The Contractor shall complete the following;

a. Site Description/Background: Based on a survey performed and finalized in 30 December 2025, , the estimated volume of available dredge material is shown on Section 00 01 00 "Solicitation, Offer and Award Form" and on Section 01 30 00 "Administrative Requirements", Paragraph 1.9 "Estimated Quantities". For more detail and clarification, see paragraph 1.2 "WORK COVERED BY THE CONTRACT DOCUMENTS" in Section 01 11 00 "SUMMARY OF WORK" of these specifications. The dredge material shall be placed at the HARS.

b. Endangered Species Protection: The Contractor shall provide education material to all dredge personnel on sea turtles and whales, and instruct the dredge operator in the proper procedures used for documenting any whale sightings. Said material will be provided by endangered species observers. The dredge operator and endangered species observers are responsible for recording the presence of any whales within and around the entire project site. The Contractor shall advise all dredge personnel that there are civil and criminal penalties for harming, harassing or killing sea turtles and whales that are protected under the Endangered Species Act.

c. Accommodation for Endangered Species Observers: The Contractor shall

provide the following to Endangered Species Observers:

- (1) A separate and private sleeping quarters, bathroom and shower facility for exclusive use of endangered species observers.
- (2) Accommodations and meals, which are equal to that which the dredge officers receive.
- (3) Transportation to and from the dredge vessel and the mainland.

d. Provisions of Endangered Species Observers Monitoring:

- (1) The Contractor will provide endangered species observers on-board the dredge vessel observation coverage one hundred (100) percent of the total dredging time year-round.
- (2) Endangered Species Observers will check for the presence of sea turtles and fragments of sea turtles entrained within the dredge material brought on board the dredge or seen in the vicinity of the dredging vessel.
- (3) Inspect brackets or screens to ensure that they are functional and meet specifications as described in paragraph g below.
- (4) The Contractor will provide additional man power resources from the dredge crew to assist endangered species observers. Additional man-power may be needed for, but is not limited to, the following
 - i. Installing and maintaining sufficient lighting.
 - ii. Cleaning fixing, deploying and emptying specimen collection.

e. Detail Criteria for Endangered Species Observers: Endangered Species observers shall:

- (1) Initiate endangered species monitoring year-round.
- (2) Conduct subsequent monitoring so that the total aggregate time onboard the dredge equals fifty (50) percent of total time of the dredging operation.
- (3) Provide inspection coverage on a rotating, six (6) hours on and six (6) off basis.
- (4) Rotate six (6) hour monitoring periods from week to week.
- (5) Fill out a turtle observation reporting log sheet. See Section 00 90 01.
- (6) Fill out an incidental report of sea turtle mortality sheet and follow procedures in paragraph number 6 below, if a sea turtle has been taken.
- (7) Observe for the presence of whales during each six (6) hours monitoring period, and fill out a daily whale reporting log.
- (8) Assemble a detailed season summary report.

f. Disposition of Turtle Parts: Should an incidental take of a sea

turtle occur, endangered species observers shall:

(1) Fill out a turtle observation reporting log sheet and an incidental report of sea turtle mortality log sheet. (See Section 00 90 01)

(2) Photograph, using color film, sea turtle and their parts.

(3) Place in plastic bags, dead sea turtle and their parts, label to note location, time and date taken, and store in a freezer, which will be supplied by the Contractor.

(4) Immediately contact the district biologist at 917-790-8617 and National Marine Fisheries Service at 978-282-8465 or 978-281-9208.

(5) Examine any live sea turtle for injury and release if not seriously injured. If an injury is such that would prohibit release, endangered species observers will contact the district biologist 917-790-8617.

g. Hopper Dredge Modifications: The Contractor Shall:

(1) Install baskets or screens over the hopper inflow and overflow, and any openings(s) that is/are present along the inflow pipe infrastructure to ensure one hundred (100) percent screening of dredge material. Baskets or screens or inflow openings shall have openings that do not exceed four (4) inches by four (4) inches. The design and method of fabrication will depend on the construction of the dredge being used and shall be subject to approval by the Contracting Officer, upon consultation with the district biologist. Baskets and/or screens shall remain in place during all dredging operations during monitoring weeks from 15 June to 30 November of the calendar year. Metal that is resisting to erosion from sandblasting will be used to fabricate baskets or screens.

(2) Replace or repair baskets or screens if openings exceed four inches by four inches.

(3) Install and maintain sufficient lighting to properly and safely examine collected specimens. Endangered species observers and the district biologist will determine the amount of light needed.

h. Special Dredge Operations Requirements: The Contractor Shall:

(1) Allocate sufficient time for endangered species observers to examine dredged material that was collected and document findings before initiating pump-out procedures of dredge material. Endangered species observers and the district biologist will determine the amount of light needed.

(2) Undertake evasive maneuvers to avoid collision with sea turtles and whales.

(3) Inspect, repair baskets or screens before the next dredge cycle begins. Endangered species observers and the district biologist will determine after inspection if repairs meet specifications as defined in section 1.12.1.3.h. above. Endangered

species observers shall notify the Contracting officer's representative and the district biologist as soon as possible, if dredging proceeds before repairs are made.

(4) The Contractor shall keep an adequate supply of repair material for baskets and or screens in stock in the vessel.

i. Dredge Requirements: Contractor is Responsible for the following:

(1) Providing endangered species observers with a statement of dangers associated with work-on-board the dredge vessel.

(2) Ensuring that endangered species observers follow safety requirements and recommendations while on board the dredge vessel and while in transit between the vessel and the mainland.

(3) Obtaining a copy of the Corps of Engineers Manual, EM 385-1-1, dated March 2024, entitled "Safety and Health Requirements Manual".

j. The Contractor Shall Follow the Reporting Procedures Listed Bellow:

(1) A sample turtle observation reporting log sheet is provided and shall be completed for every cycle (load), whether sea turtles are present or not.

(2) All data in the original form shall be submitted directly to the district biologist, Ms. Jenine Gallo, U.S. Army Corps of Engineers, CENAN-PL-EA, 26 Federal Plaza, New York City, New York 10278-0090, at the end of the each monitoring week. Weekly summaries will be emailed to jenine.gallo@usace.army.mil as soon as they become available. Copies of the data will be supplied to the district Contracting Officer. Following completion of the project, a copy of the Contractor's log regarding sea turtles and whales shall be forwarded to the district biologist.

(3) Continuous liaison with the district biologist shall be maintained to avoid problems with execution of this endangered species observation program, and to ensure compliance with prescribed Corps of Engineers policies and procedures. It will be the responsibility of the Contractor to report significant developments.

(4) The detailed season summary report shall be submitted to the district biologist by 01 December of each calendar year.

(5) Any sighting of an injured or incapacitated sea turtle or whale will be reported immediately to the district biologist.

k. Inspections by the Project Biologist: Periodically the district biologist may inspect the dredge vessel to ensure that all criteria and modifications have been met. At the district biologist discretion, he/she may make inspections to ensure that the endangered species observation program is properly executed. The Contractor will provide the district biologist with accommodations as defined in paragraph c under sub-section 1.14.1.3 above.

1.14.2 Payment

No separate payment or direct payment will be made for the cost of the work

covered under this section, and such work is to be considered as a subsidiary obligation of the Contractor.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 ENVIRONMENTAL PERMITS AND COMMITMENTS

This paragraph supplements the Contractor's responsibility under the contract clause "PERMITS AND RESPONSIBILITIES" to the extent that the Government has obtained the NJDEP/NYSDEC Water Quality Certificate and Coastal Zone Management Consistency Statement. The Contractor shall Comply with the terms and conditions of the attached NJDEP/NYSDEC Water Quality Certificate and Coastal Zone Management Consistency Statement attached as an appendix on this document.

3.2 LAND RESOURCES

The Contractor shall confine all activities to areas defined by the drawings and specifications. Prior to the beginning of any construction, the Contractor shall identify any land resources to be preserved within the work area. Except in areas indicated in the drawings or specified to be cleared, the Contractor shall not remove, cut, deface, injure, or destroy land resources including trees, shrubs, vines, grasses, topsoil, and land forms without approval. No Ropes, cables, or guys will not be fastened to or attached to any trees for anchorage unless specifically authorized. The Contractor shall provide effective protection for land and vegetation resources at all times, as defined in the following subparagraph. Stone, soil, or other materials displaced into uncleared areas shall be removed by the Contractor.

3.2.1 Work Area Limits

Prior to commencing construction activities the Contractor shall mark the area(s) that need not to be disturbed under this contract. Isolated areas within the general work area, which are not to be disturbed shall be marked or fenced. Monuments and markers shall be protected before construction operations commence. Where construction operations are to be conducted during darkness, any markers must be visible in the dark. The Contractor's personnel must be knowledgeable of the purpose for marking and/or protecting particular objects.

3.2.2 Landscape

Trees, shrubs, vines, grasses, land forms and other landscape features indicated and defined on the drawings to be preserved must be clearly identified by marking, fencing, or wrapping with boards, or any other approved techniques. The Contractor shall restore landscape features damaged or destroyed during construction operations outside the limits of the approved work area.

3.2.3 Erosion and Sediment Controls

The Contractor shall be responsible for providing erosion and sediment control measures in accordance with Federal, State, and local laws and regulations. The erosion and sediment controls selected and maintained by the Contractor shall be such that water quality standards are not violated as a result of the Contractor's construction activities. The area of bare soil exposed at any one time by construction operations should be kept to a minimum. The Contractor shall construct or install temporary and permanent erosion and sediment control Best Management Practices (BMPs). BMPs may include, but not be limited to, vegetation cover, stream bank stabilization, slope stabilization, silt fences, construction of terraces, interceptor channels, sediment traps, inlet and outfall protection, diversion channels, and sedimentation basins. Any temporary measures shall be removed after the area has been stabilized.

3.2.4 Contractor Facilities and Work Areas

The Contractor's field offices, staging areas stockpile storage, and temporary buildings shall be placed in areas designated on the drawings or as directed by the Contracting Officer. Temporary movement or relocation of Contractor facilities will be made only when approved. Erosion and sediment controls must be provided for onsite borrow and spoil areas to prevent sediment from entering nearby waters. Temporary excavation and embankments for plant and/or work areas must be controlled to protect adjacent areas.

3.3 WATER RESOURCES

The Contractor shall monitor construction activities to prevent pollution of surface and ground waters. Toxic or hazardous chemicals shall not be applied to soil or vegetation unless otherwise indicated. All water areas affected by construction activities shall be monitored by the Contractor. For construction activities immediately adjacent to impaired surface waters, the Contractor must be capable of quantifying sediment or pollutant loading to that surface water when required by State or Federally issued Clean Water Act permits.

3.3.1 Cofferdams, Diversions, and Dewatering Operations

Construction operations for dewatering, removal of cofferdams, tailrace excavation, and tunnel closure will be controlled at all times to maintain compliance with existing State water quality standards and designated uses of the surface water body. The Contractor shall comply with the state of New Jersey water quality standards and anti-degradation provisions.

3.3.2 Wetlands

The Contractor shall not enter, disturb, destroy, or allow discharge of contaminants into any wetlands, except as authorized herein. The Contractor shall be responsible for the protection of wetlands shown on the drawings in accordance with paragraph 3.1 "Environmental Permits and Commitments" above. Authorization to enter specific wetlands shall not relieve the Contractor from any obligation to protect other wetlands within, adjacent to, or in the vicinity of the construction site and associated boundaries.

3.4 AIR RESOURCES

Equipment operation, activities, or processes will be performed by the Contractor in accordance with all Federal and State air emission and performance laws and standards.

3.4.1 Particulates

Dust particles; aerosols and gaseous by-products from construction activities; and processing and preparation of materials, such as from asphaltic batch plants; must be controlled at all times, including weekends, holidays and hours when work is not in progress. The Contractor shall maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and other work areas within or outside the project boundaries free from particulates which would cause the Federal, State, and local air pollution standards to be exceeded or which would cause a hazard or a nuisance. Sprinkling, chemical treatment of an approved type, baghouse, scrubbers, electrostatic precipitators or other methods will be permitted to control particulates in the work area. Sprinkling, to be efficient, must be repeated to keep the disturbed area damp at all times. The Contractor must have sufficient, competent equipment available to accomplish these tasks. Particulate control shall be performed as the work proceeds and whenever a particulate nuisance or hazard occurs. The Contractor shall comply with all State and local visibility regulations.

3.4.2 Odors

Odors from construction activities must be controlled at all times. The odors must be in compliance with State regulations and/or local ordinances and may not constitute a health hazard.

3.4.3 Sound Intrusions

The Contractor shall keep all construction activities under surveillance and control to minimize environment damage by noise and shall comply with the provisions of the State of New Jersey rules.

3.4.4 Burning

Burning will not be allowed on the project site unless specified in other sections of the specifications or authorized in writing by the Contracting Officer. The specific time, location, and manner of burning will be subject to approval.

3.5 CHEMICAL MATERIALS MANAGEMENT AND WASTE DISPOSAL

Disposal of wastes will be as directed below, unless otherwise specified in other sections and/or shown on the drawings.

3.5.1 Solid Wastes

Solid wastes (excluding clearing debris) shall be place in containers which are emptied on a regular schedule. Handling, storage, and disposal shall be conducted to prevent contamination. Segregation measures shall be employed so that no hazardous or toxic waste will become co-mingled with solid waste.

3.5.2 Chemicals and Chemical Wastes

Chemicals shall be dispensed ensuring no spillage to the ground or water. Periodic inspections of dispensing areas to identify leakage and/or potential leakage shall be performed and documented; corrective action shall also be documented. This documentation will be periodically reviewed by the Government. Chemical waste shall be collected in corrosion resistant, compatible containers. Collection drums must be monitored and removed to a staging or storage area when contents are within 6 inches of the top. Wastes will be classified, managed, stored, and disposed of in accordance with Federal, State, and local laws and regulations.

3.5.3 Contractor Generated Hazardous Wastes/Excess Hazardous Materials

Hazardous wastes are defined in 40 CFR 261, or are as defined by applicable State and local regulations. Hazardous materials are defined in 49 CFR 171 - 178. The Contractor shall at a minimum, manage and store hazardous waste in compliance with 40 CFR 262 and shall manage and store hazardous waste in accordance with the installation hazardous waste management plan. The Contractor shall take sufficient measures to prevent spillage of hazardous and toxic materials during dispensing. The Contractor shall segregate hazardous waste from other materials and wastes, and shall take precautionary measures such as berming or other appropriate measures against accidental spillage. The Contractor shall be responsible for storage, describing, packaging, labeling, marking, and placarding of hazardous waste and hazardous material in accordance with 49 CFR 171 - 178, State, and local laws and regulations. The Contractor shall transport Contractor generated hazardous waste off Government property within 60 days in accordance with the Environmental Protection Agency(EPA)and the Department of Transportation laws and regulations. The Contractor shall dispose of hazardous waste in compliance with Federal, State and local laws and regulations. Spills of hazardous or toxic materials shall be immediately reported to the Contracting Officer and the Facility Environmental Office. Cleanup and cleanup costs due to spills are the Contractor's responsibility. The disposition of Contractor generated hazardous waste and excess hazardous materials are the Contractor's responsibility.

3.5.4 Fuel and Lubricants

Storage, fueling and lubrication of equipment and motor vehicles shall be conducted in a manner that affords the maximum protection against spill and evaporation. Fuel, lubricants and oil shall be managed and stored in accordance with all Federal, State, Regional, and local laws and regulations. Used lubricants and used oil to be discarded shall be stored in marked corrosion-resistant containers and recycled or disposed in accordance with 40 CFR 279, State, and local laws and regulations. Storage of fuel on the project site shall be in accordance with all Federal, State, and local laws and regulations.

3.5.5 Waste Water

Disposal of waste water will be as specified below.

- a. Waste water from construction activities, such as onsite material processing, concrete curing, foundation and concrete clean-up, water used in concrete trucks, forms, etc. will not be allowed to enter water ways or to be discharged prior to being treated to remove pollutants. The Contractor shall dispose of the

construction related waste water off-Government property in accordance with all Federal, State, Regional and Local laws and regulations.

- b. For discharge of ground water and run-off surface water, the Contractor shall discharge in accordance with all Federal, State, Regional, and/or Local laws and regulations.
- c. Water generated from the flushing of lines after hydrostatic testing shall be land applied in accordance with all Federal, State, and local laws and regulations for land application.

3.6 RECYCLING AND WASTE MINIMIZATION

The Contractor shall participate in State and local government sponsored recycling programs. The Contractor is further encouraged to minimize solid waste generation throughout the duration of the project.

3.7 NON-HAZARDOUS SOLID WASTE DIVERSION REPORT

The contractor shall maintain an inventory of non-hazardous solid waste diversion and disposal of construction and demolition debris. The Contractor shall submit a report to the Facility through the Contracting Officer on the first working day after each fiscal year quarter, starting the first quarter that non-hazardous solid waste has been generated. Include the following in the report:

- a. Construction and Demolition (C&D) Debris Disposed in Cubic Yards (CY) or Tons (TN), as appropriate.
- b. Construction and Demolition (C&D) Debris Recycled in Cubic Yards (CY) or Tons (TN), as appropriate.
- c. Total (C&D) Debris Generated in Cubic Yards (CY) or Tons (TN), as appropriate.
- d. Waste sent to Waste-To-Energy Incineration Plant (This amount should not be included in the recycled amount) in Cubic Yards (CY) or Tons (TN), as appropriate.

3.8 HISTORICAL, ARCHAEOLOGICAL, AND CULTURAL RESOURCES

Known existing historical, archaeological, and cultural resources within the Contractor's work area are shown on the drawings. The Contractor shall protect these resources and be responsible for their preservation during the life of the Contract. If during excavation or other construction activities any previously unidentified or unanticipated historical, archaeological, and cultural resources are discovered or found, all activities that may damage or alter such resources will be temporarily suspended. Resources covered by this paragraph include but are not limited to: any human skeletal remains or burials; artifacts; shell, midden, bone, charcoal, or other deposits; rock or coral alignments, paving's, wall, or other constructed features; and any indication of agricultural or other human activities. Upon such discovery or find, immediately notify the Contracting Officer so that the appropriate authorities may be notified and a determination made as to their significance and what, if any, special disposition of the finds should be made. The Contractor shall cease all activities that may result in impact to or the destruction of these

resources and shall secure the area and prevent employees or other persons from trespassing on, removing, or otherwise disturbing such resources.

3.9 BIOLOGICAL RESOURCES

The Contractor shall minimize interference with, disturbance to, and damage to fish, wildlife, and plants including their habitat. The protection of threatened and endangered animal and plant species, including their habitat, in accordance with Federal, State, Regional, and local laws and regulations.

3.10 TRAINING OF CONTRACTOR PERSONNEL

Contractor's personnel shall be trained in all phases of environmental protection and pollution control by means of conducting environmental protection/pollution control meetings prior to commencing construction activities. Additional meetings shall be conducted for new personnel and when site conditions change. The training and meeting agenda shall include: methods of detecting and avoiding pollution; familiarization with statutory and contractual pollution standards, familiarization and care of monitoring devices, vegetative covers in order to ensure adequate and continuous environmental/pollution control. Additionally, the meeting topics shall include information on anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants, recognition and protection of archeological sites, artifacts, wetlands and endangered species and their habitat that are known to be in the area.

3.11 POST CONSTRUCTION CLEANUP

The Contractor shall clean up all areas used for construction in accordance with Contract Clause: "Cleaning Up". The Contractor shall unless otherwise instructed in writing by the Contracting Officer, obliterate all signs of temporary construction facilities such as haul roads, work area, structures, foundations of temporary structures, stockpiles of excess or waste materials, and other vestiges of construction prior to final acceptance of the work. The disturbed area must be graded, filled and the entire area seeded unless otherwise indicated.

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08/19, CHG 5: 08/22

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SECTION 01 58 00

PROJECT IDENTIFICATION
08/19, CHG 5: 08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EP 310-1-6a (2006; 2019 Change 2) Project Operation --
Sign Standards Manual, VOL 1

EP 310-1-6b (2006) Sign Standards Manual, VOL 2,
Appendices

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Sign Legend Orders; G

1.3 PROJECT IDENTIFICATION SIGN

1.3.1 Construction Project Signs

Furnish the construction project sign package, maintain the signs during construction, and remove the signs from the job site upon completion of the project. The construction project sign package consists of two signs: one for project identification and the other to show the on-the-job safety performance of the contractor. Ensure that the package conforms to the requirements of EP 310-1-6a and EP 310-1-6b, specifically Section 16. Submit the sign legend orders as described in Section 16 of EP 310-1-6a before erecting the signs.

Utilize the Example Graphic of Signage with Dimensions for Civil Works Project and follow USACE policy provided in Engineering Pamphlet (EP) 310-1-6a, Sign Standards Manual, VOL 1 for guidelines on fabricating, locating, and mounting construction project signs
([https://www.publications.usace.army.mil/Portals/76/Users/182/86/2486/EP 310-1-6a%20change%202.pdf](https://www.publications.usace.army.mil/Portals/76/Users/182/86/2486/EP%20310-1-6a%20change%202.pdf)).

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

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05/19, CHG 1: 08/21

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CLOSEOUT SUBMITTALS
05/19, CHG 1: 08/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

ERDC/ITL TR-19-6 (2019) A/E/C Graphics Standard, Release 2.1

ERDC/ITL TR-19-7 (2019) A/E/C CAD Standard - Release 6.1

1.2 DEFINITIONS

1.2.1 As-Built Drawings

As-built drawings are the marked-up drawings, maintained by the Contractor on-site, that depict actual conditions and deviations from the Contract Documents. These deviations and additions may result from coordination required by, but not limited to: contract modifications; official responses to submitted Requests for Information (RFI's); direction from the Contracting Officer; design that is the responsibility of the Contractor, and differing site conditions. Maintain the as-builts throughout construction as red-lined PDF files. These files serve as the basis for the creation of the record drawings.

1.2.2 Record Drawings

The record drawings are the final compilation of actual conditions reflected in the as-built drawings.

1.3 SOURCE DRAWING FILES

Request the full set of electronic drawings, in the source format, for Record Drawing preparation, after award and at least 30 days prior to required use.

1.3.1 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction [drawings and]data for the referenced project. Any other use or reuse is at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim and waives to the fullest extent permitted by law, any claim or cause of action of any nature against the Government, its agents or sub consultants that may arise out of or in connection with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use

of these electronic files.

These electronic CAD drawing files are not construction documents. Differences may exist between the CAD files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic CAD files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished Source drawing files, the signed and sealed construction documents govern. The Contractor is responsible for determining if any conflict exists. Use of these Source Drawing files does not relieve the Contractor of duty to fully comply with the contract documents, including and without limitation, the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic source drawing files for use in producing construction [drawings and]data related to this contract, remove all previous indicia of ownership (seals, logos, signatures, initials and dates).

1.4 RECORD DRAWINGS

The Government will provide pdf and or program files at the preconstruction conference that contains one set of "as-designed" electronic CAD files in the specified software and format revised to reflect all amendments and the final contract PDF drawings. The CAD files are provided to enable preparation of as-built or as-constructed drawings. If discrepancies exist between the CAD files and the contract PDF drawings, correct the CAD files to show the contract PDF drawings.

1.4.1 Variation with Contract Drawings

The electronic files provided are not part of the contract documents. If there is any discrepancy between the electronic files and the contract drawings, the contract drawings govern. The Government has no responsibility to modify any GFM due to changes in the design that occur after award.

Evaluate the content and quality of the GFM upon receipt. If major discrepancies or omissions occur in the GFM, notify the Contracting Officer and indicate the nature of such variations.

1.4.2 Data Loss, Corruption, and Error

Transfer of GFM files may result in corrupted files resulting in data loss and errors. Use of GFM files at own risk. Verify data integrity upon receipt and request a replacement if necessary. Make any adjustment in file structure, format, or software version as needed to make GFM compatible with computer systems and/or software to meet the requirements of the contract.

1.4.3 Modeling Completeness and Quality

The Government makes no guarantee that the GFM provide the level of completeness or quality as required by the contract. Further, the Government makes no guarantee that identified variations will be corrected upon notification.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-11 Closeout Submittals

As-Built Drawings; G

Record Drawings; G

1.6 QUALITY CONTROL

Additions and corrections to the contract drawings must be equal in quality and detail to that of the originals. Line colors, line weights, lettering, layering conventions, and symbols must conform to ERDC/ITL TR-19-7.

PART 2 PRODUCTS

2.1 RECORD DRAWINGS

Prepare the CAD drawing files in MicroStation DGN format compatible with a Windows operating system.

2.1.1 Additional Drawings

If additional drawings are required, prepare them using the specified electronic file format applying the same graphic standards specified for original drawings. The title block and drawing border to be used for any new final record drawings must be identical to that used on the contract drawings.

2.1.1.1 Sheet Numbers and File Names

If a sheet needs to be added between two sequential sheets, append a Supplemental Drawing Designator in accordance with ERDC/ITL TR-19-7 Adding a drawing sheet, and ERDC/ITL TR-19-6 Adding or deleting drawing sheets and index sheet procedures.

PART 3 EXECUTION

3.1 AS-BUILT DRAWINGS

Provide and maintain two black line print copies of the PDF contract drawings for As-Built Drawings. Maintain the as-builts throughout construction as red-lined hard copies on site or red-lined PDF files.

3.1.1 Markup Guidelines

Make comments and markup the drawings complete without reference to letters, memos, or materials that are not part of the As-Built drawing. Show what was changed, how it was changed, where item(s) were relocated and change related details. These working as-built markup prints must be neat, legible and accurate as follows:

- a. Use base colors of red, green, and blue. Color code for changes as follows:

- (1) Special (Blue) - Items requiring special information, coordination, or special detailing or detailing notes.
 - (2) Deletions (Red) - Over-strike deleted graphic items (lines), lettering in notes and leaders.
 - (3) Additions (Green) - Added items, lettering in notes and leaders.
- b. Provide a legend if colors other than the "base" colors of red, green, and blue are used.
 - c. Add and denote any additional equipment or material facilities, service lines, incorporated under As-Built Revisions if not already shown in legend.
 - d. Use frequent written explanations on markup drawings to describe changes. Do not totally rely on graphic means to convey the revision.
 - e. Use legible lettering and precise and clear digital values when marking prints. Clarify ambiguities concerning the nature and application of change involved.
 - f. Wherever a revision is made, also make changes to related section views, details, legend, profiles, plans and elevation views, schedules, notes and call out designations, and mark accordingly to avoid conflicting data on all other sheets.
 - g. For deletions, cross out all features, data and captions that relate to that revision.
 - h. For changes on small-scale drawings and in restricted areas, provide large-scale inserts, with leaders to the applicable location.
 - i. Indicate one of the following when attaching a print or sketch to a markup print:
 - (1) Add an entire drawing to contract drawings
 - (2) Change the contract drawing to show changes on the drawing.
 - (3) Provided for reference only to further detail the initial design.
 - j. Incorporate all shop and fabrication drawings into the markup drawings.
- 3.1.2 As-Built Drawings Content

Revise As-Built Drawings or red-lined PDF files in accordance with ERDC/ITL TR-19-6 and ERDC/ITL TR-19-7. Keep these working as-built markup drawings current on a weekly basis and at least one set available on the jobsite at all times. Changes from the contract drawings which are made during construction or additional information which might be uncovered in the course of construction must be accurately and neatly recorded as they occur by means of details and notes. Submit the working as-built markup drawings for approval prior to submission of each monthly pay estimate. For failure to maintain the working and final record drawings as specified herein, the Contracting Officer may withhold a monthly progress payment until approval of updated drawings. Show on the as-built drawings, but not limited to, the following information:

- a. The location and description of any bridges, utility lines, or any other structures or or obstructions within the construction area. The location includes dimensions to permanent features.
- b. The location and dimensions of any changes with the channel.
- c. The grade or alignment of channel, structures or utilities if any changes were made from the contract plans.
- d. Correct elevations if changes were made in channel depths.
- e. All changes, which result from contract modifications.
- f. Where contract drawings or specifications allow options, only the option selected for construction shall be shown on the as-built prints.
- g. All amendments to the contract drawings issued during the solicitation period shall be posted on the as-built drawings

3.2 RECORD DRAWING FILES

If additional drawings are required, prepare them using the specified electronic file format applying the same graphic standards specified for original drawings. The title block and drawing border to be used for any new final record drawings must be identical to that used on the contract drawings. Accomplish additions and corrections to the contract drawings using CAD files. Provide all program files and hardware necessary to prepare final PDF record drawings. The Contracting Officer will review final PDF record drawings for accuracy and return them to the Contractor for required corrections, changes, additions, and deletions.

If additional drawings are required, prepare them using the specified electronic file format applying the same graphic standards specified for original drawings. the title blocks and drawing border to be used for any new records drawing must be identical to that used on the contract drawings. Accomplish additions and corrections to the contract drawings using CAD files. Provide all program files and hardware necessary to prepare final PDF record drawings. The Contracting Officer will review final PDF record drawings for accuracy and retrn them to the Contractor for required corrections, changes, additions, and deletions.

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SECTION 35 20 23

DREDGING
06/07

PART 1 GENERAL

1.1 GENERAL INFORMATION

This Section covers furnishing suitable dredging plant and performing all work required to remove the specified materials from within the prescribed work area limits as indicated, and placement of the dredge material within the prescribed material placement areas. The Contractor is responsible for making their own investigation of submerged, surface, and overhead structures in the work areas and other locations they find necessary to traverse. The exact location, depths, and heights of various structures including, but not limited to submarine cables, pipes, highlines, docks, piers, bulkheads, and bridges (as applicable), are not known and it will be necessary for the Contractor to ascertain interference problems and notify the respective owners in advance of dredging operations. The Contractor is responsible for making necessary arrangements with the respective owners of the structure(s) to assure satisfactory completion of dredging in the vicinity with a minimum interruption of service, and must perform their operations in such a manner as will avoid damage to these facilities.

1.2 DEFINITIONS

1.2.1 Maintenance Material

Maintenance material is defined as that comprising shoaling which has occurred since the channel areas were last dredged.

1.2.2 New Work Material

New work material is defined as previously undredged material.

1.2.3 Hard Material

Hard material is defined as material requiring blasting or the use of special equipment for economical removal, and includes boulders or fragments too large to be removed in one piece by the dredge.

1.2.4 Container and Transport Vessels

The term "container vessel" refers to any vessel used to contain, or hold, dredged material (scow, barge, hopper dredge, etc.). The term "transport vessel" refers to any vessel (tugboat, hopper dredge, work boat, etc.) used to move a container vessel. In the case of a hopper dredge, the hopper dredge may act as both a "container vessel" and a "transport vessel". If hopper dredge engines are not used for propulsion, and another vessel (tugboat, for example) is used to move it, then the hopper dredge would be the "container vessel" and the tugboat would be the "transport vessel".

Vessel Identification and Certification: All vessels (container and transport) that will be used, and/or may be used, during this contract must be identified prior to the first day of dredging. Vessels identified for use on the contract must be certified to be fully operational, mechanically

sound, completely seaworthy, and free of leaks or other defects.

1.2.5 Environmental Bucket

An environmental bucket is required this project. An environmental bucket is required for dredging silt or other fine-grained unconsolidated materials and/or sediments classified as Class B or Class C according to NYSDEC In-Water and Riparian Management of Sediment and dredged Material Technical & Operational Guidance Series (TOGS) 5.1.9.

- a. The bucket must be constructed with sealing gaskets or an overlapping sealed design at the jaws, and seals or flaps positioned at locations of vent to minimize the loss of material during transport through the water column and into the barge.
- b. Any seals or flaps designed and/or installed at the jaws and locations of vent openings must tightly cover these openings while the bucket is lifted through the water column and into the barge. If excessive loss of water and/or sediments from the bucket is observed from the time of its breaking the water surface to crossing the barge gunwale, the Permittee must halt dredging operations and inspect the bucket for defects. Operations must be suspended until all necessary repairs or replacements are made.
- c. Bucket hoist speed must be limited to approximately 2 feet per second. The bucket must be lifted in a continuous motion through the water column and into the barge.
- d. The bucket must be lowered to the level of the barge gunwales prior to the release of load.
- e. Where the environmental bucket repeatedly encounters resistance from consolidated materials (e.g., stone, rock), a conventional digging bucket may be used. The Department must be provided with written notice of intent to switch buckets at least 48 hours (2 business days) prior to use of the conventional bucket.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Soundings or Sweepings; G

1.4 METHOD OF COMMUNICATION

Provide a system of communication between the dredge crew and the crew at the placement / processing area. A portable two-way radio is acceptable.

1.5 OTHER DREDGING OPERATIONS

The Contractor should anticipate the possibility of concurrent dredging

operations, adjacent to or nearby the dredge and placement areas in this contract. Delays should be anticipated in transiting to, and through, the placement sites during concurrent dredging and placement operations. As a standard safety precaution, observe all placement site specific restrictions pertaining to the limit of dredges operating within at any given time. Coordination with other dredge plant in the vicinity of the dredge and placement areas is required.

1.6 ENVIRONMENTAL COMPLIANCE AND PROTECTION

Comply with conditions and requirements of State or Federal permits. The Government will secure the permit for dredging and placement of material as indicated.

During the life of the contract, provide and maintain environmental protective measures. Also, environmental protective measures required to correct conditions, such as oil spills or debris, that occur during the dredging operations, must be provided. Comply with Federal, State, and local regulations pertaining to water, air, and noise pollution.

1.7 BASIS FOR BIDS

Payment will be at the contract unit price per cubic yard, multiplied by total cubic yards of acceptable dredging. Include a bid unit price per cubic yard of dredging based on the quantity as specified. If the Contracting Officer requires an increase or a decrease in total volume of dredging, the contract price will be adjusted in accordance with the FAR 52.211-18 Variation in Estimated Quantity. Dredging conditions specified and indicated describe conditions which are known. However, the Contractor is responsible for other conditions encountered which are not unusual when compared to conditions recognized in the dredging business as usual in dredging activities such as those required under this contract.

PART 2 PRODUCTS

2.1 CHARACTER OF MATERIAL

Based on sampling and testing of the proposed dredged material in 2025, the material to be removed consists of primarily, silt, sand, and clay:

- a. REACH 1: 59.7% SAND, 26.7% SILT, & 13.6% CLAY
REACH 2: 24.7% SAND, 54.3% SILT, & 21.0% CLAY
REACH 3: 16.9% SAND, 63.2% SILT, & 19.9% CLAY
- b. The Contractor is ultimately responsible for determination of the characteristics of materials to be dredged.

PART 3 EXECUTION

3.1 INSPECTION

Inspect the work, keep records of work performed, and ensure that gages, targets, ranges, and other markers are in place and usable for the intended purpose. Provide, at the request of the Contracting Officer, boats, boatmen, laborers, and materials necessary for inspecting, supervising, and surveying the work. When required, provide transportation for the Contracting Officer and inspectors to and from the placement area and between the dredging plant and adjacent points on shore.

3.2 DREDGING

3.2.1 Order of Work

The Contractor will determine the order for this project and comply with all environmental restrictions. The Government reserves the right to change the order of work at any time.

3.2.2 Interference with Navigation

Minimize interference with the use of channels and passages. The Contractor is responsible for shifting or moving of dredges or the interruption of dredging operations to accommodate the movement of vessels and floating equipment, if necessary. Adhere to Coast Guard Regulations for passing vessels.

3.2.3 Lights

Each night, between sunset and sunrise and during periods of restricted visibility, provide lights for floating plants, pipelines, ranges, and markers. Also, provide lights for buoys that could endanger or obstruct navigation. When night work is in progress, maintain lights from sunset to sunrise for the observation of dredging operations. Lighting must conform to United States Coast Guard requirements for visibility and color.

3.2.4 Navigation Warnings

Furnish and maintain navigation warning signs along any pipelines. Provide notice to increase public awareness of potential hazards presented by dredge plant equipment by stating the location, date of construction, equipment mooring, marshaling areas, using local newspapers, radios, television, waterway users associations, or other appropriate area specific communication networks. Ensure that an announcement is made through the same networks at the beginning of the dredging operation. Make periodic updates/status announcements at intervals of not more than a month throughout the contract life.

3.2.5 Ranges, Gages, and Lines

Provide, set, and maintain ranges, buoys, and markers needed to define the work and to facilitate inspection. Establish and maintain gages in locations observable from each part of the work so that the depth may be determined. Suspend dredging when the gages or ranges cannot be seen or followed. The Contracting Officer will furnish, upon request by the Contractor, survey lines, points, and elevations necessary for the setting of ranges, gages, and buoys.

3.2.6 Dredge Plant and Equipment

Maintain all dredge plant and associated equipment such as, but not limited to, scows, coamings, barges, and pipelines, to meet the requirements of the work. Promptly repair leaks or breaks along pipelines. Remove any misplaced dredged material.

3.2.7 Layout of Work

The Contractor will be provided layout charts for all dredging areas based on the schedule of work, the Contractor's hydrosurvey requests, and/or the results of Government hydrographic survey monitoring. The most current

Government hydrosurvey information will be displayed on the dredging layout charts and provided to the Contractor. Other pertinent information to be included with each layout package will be: payment method, required and maximum dredging depths, coordinates for the material placement site, distance one-way from dredging location to placement site, specific placement site instructions, supplemental tide gauge information, horizontal control locations, and computed hydrosurvey quantities.

The total estimated quantity of material to be removed is computed as that material that is within the specified limits, including side slopes and overdepth. The quantities listed are estimates only. All estimated quantities are subject to FAR 52.211-18 Variation in Estimated Quantity.

3.2.7.1 Overdepth Dredging

To cover unavoidable inaccuracies of dredging processes, material actually removed to a depth of two (2) feet, below the required depth specified and within the dredging limits will be measured and paid for at full contract price. The quantity of overdepth dredging is based on the contour dredging method where payment is allowed for all material up to Overdepth Grade except below project grade on side slopes.

3.2.7.2 Side Slopes

Side slopes in the dredge areas within this contract will be measured and paid based on a 3 feet horizontal to 1 foot vertical ratio. Dredging on side slopes must follow, as closely as practicable, the dredge area limits indicated on associated layout drawings.

A 2 foot, allowance will be made for dredging beyond the indicated or specified side slopes, except as provided herein. The allowance will be determined by projecting a line upwards, paralleling the project design side slopes, from the intersection of the overdepth dredging limit (at a point located vertically below the limit of dredging at the top of slope). The amount of material excavated from side slopes will be determined by either cross-sections or computer, or both.

3.2.8 Obstructions and Debris

Except as indicated, the Government has no knowledge of cables, pipes, or other artificial obstructions or of any wrecks, wreckage, or other material that would necessitate the use of explosives or the employment of additional equipment for economical removal. Contractors should however exercise due diligence in determining the existence of any obstructions within proposed work areas during bid preparation.

Trash, debris and other miscellaneous man-made and natural organic objects may be encountered during the course of the dredging. This material may be present at any location above the required depth as indicated on drawings. Man-made materials may include (but are not limited to) wood, sheet metal, glass, lumber, plastics, tires, hoses, chains, cables, and hawsers and abandoned steel pipes. Natural organic objects include wooden objects such as tree branches but does not include inorganic objects such as large boulders.

3.2.9 Dredging Requirements

Dredge Areas

- 1) BASIC WORK - REACH 1: requires dredging to the plane of -35 feet MLLW plus 2 foot allowable overdepth.
- 3) BASIC WORK - REACH 2: requires dredging to the plane of -35 feet MLLW plus 2 foot allowable overdepth.
- 4) BASIC WORK - REACH 3: requires dredging to the plane of -35 feet MLLW plus 2 foot allowable overdepth.

Placement Area

Material is to be placed at the Historic Area Remediation Site (HARS). See Section 01 20 00 PRICE AND PAYMENT PROCEDURES for more details.

3.2.10 Quality Control

Establish and maintain quality control for operations to assure compliance with contractual requirements and maintain records of this quality control for dredging operations.

While performing all dredging work control the horizontal positioning of the dredge with electronic positioning.

3.2.11 Surveys during Progress of Work

Quality of dredging depth will be determined by soundings or sweepings taken behind the dredge as work progresses. The Contractor will take progress soundings or sweepings. The Contractor will submit all data files and survey information, including mapping of these surveys to the COR for their review.

3.2.12 Salvaged Material

Anchors, chains, firearms, and other articles of value, which are brought to the surface during dredging operations, must remain or become the property of the Government and will be placed on shore at a convenient location near the site of the work, as directed.

3.2.13 Safety of Structures

The prosecution of work must ensure the stability of piers, bulkheads, and other structures lying on or adjacent to the site of the work, insofar as structures may be jeopardized by dredging operations. Repair damage resulting from dredging operations is the responsibility of the Contractor, insofar as such damage may be caused by variation in locations or depth of dredging, or both, from that indicated or permitted under the contract. The Contractor is responsible for coordinating with the owner of the structure for any necessary repairs.

3.3 PLACEMENT OPERATIONS

3.3.1 Placement of Excavated Materials

Provide for safe transportation and placement of dredged materials. The placement of dredged materials in unauthorized places is forbidden. Comply with rules and regulations of all, federal, state and local authorities.

3.3.2 Charges for Material Placement

The Contractor is responsible for payment of any charges imposed by the state-permitted, contractor-furnished placement site.

3.3.3 Mislplaced Dredged Material

Any dredged materials deposited at locations other than in areas designated or approved by the Contracting Officer will be considered misplaced material and will not be paid for until the Contractor, at his own expense, removes and deposits such misplaced material where directed. This required removal and redeposit of the misplaced material and any necessary placement site restoration work is not the basis for a time extension or additional compensation under this contract.

The Contractor will immediately notify the Contracting Officer or his Representative if any material is misplaced or when infill or shoaling beyond the Contractor's control occurs.

3.4 MEASUREMENT

The Government will take soundings before and after dredging. The government will perform a before-dredge (BD) survey of the entire work area prior to the commencement of work. Once complete and the contractor has submitted a complete survey of the entire dredge area along with all associated files, the government will perform an after-dredge (AD) survey of the same area. The material removed between these two surveys within the allowable footprint and up to the maximum dredge template will be computed. This computation will be made using the TIN method.

Final quantities will be subject to deductions or correction of deductions previously made because of excessive overdepth, dredging outside or authorized areas, or placement of material in an unauthorized manner.

3.4.1 Method of Measurement

Quantity of material removed that will be paid for will be measured by cubic yard in place by means of volume difference from soundings taken before and after dredging. The drawings represent existing conditions based on current available information, but will be verified and corrected, if necessary, by soundings taken before dredging in each locality.

3.4.2 Monthly Estimates

Monthly estimates of work completed will be based on the result of soundings taken during the progress of the work. Deductions will be made for dredging and placement not in accordance with the specifications.

3.5 FINAL EXAMINATION AND ACCEPTANCE

As soon as practicable after the completion of areas, which in the opinion

of the Contracting Officer, will not be affected by further dredging operations, each area will be examined by the Government by sounding or sweeping, or both. Remove shoals and lumps as required by methods approved by the Government. However, if the bottom is soft and the shoal areas form no material obstruction to navigation, removal may be waived at the discretion of the Government. The Contractor will be notified when soundings or sweepings are to be made. When areas are found to be in a satisfactory condition, the work therein will be accepted as complete.

Re-dredging at the Contract price, within the limit of available funds, may be completed with the consent of both the Government and the Contractor when infill or shoaling beyond the Contractor's control occurs in any area previously accepted.

This contract has no partial acceptance areas. Any survey related disagreement(s) as it pertains to this contract must be submitted in writing to the Contracting Officer or his representative within 5 business days of occurrence of such survey related event. Failure to notify the Contracting Officer or his representative within 5 business days of such survey related event disqualifies the Contractor from any such related claims.

3.6 PLANT REMOVAL

Upon completion of the work, the contractor shall remove all dredging plant, including ranges, buoys, piles, and other markers or obstructions.

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SECTION 35 20 23.10 18

DREDGED MATERIAL TRANSPORT, PLACEMENT AND MONITORING
06/07

PART 1 GENERAL

1.1 DREDGED MATERIAL TRANSPORT AND PLACEMENT MONITORING

This contract section describes the requirements associated with the U.S. Army Corps of Engineers (USACE) New York District (NYD) Dredged Material Transport and Placement Monitoring program. The dredging contractor (contractor) must implement these placement requirements to ensure compliance with all applicable laws, regulations, including those requirements within the Marine Protection, Research and Sanctuaries Act (MPRSA Section 103) for the monitoring and surveillance of the transportation and ocean-placement of dredged material. The District monitoring program helps both the contractors (dredging contractors) and the District to ensure compliance with Section 404 of the Clean Water Act, Section 103 of the Marine Protection, Research, and Sanctuaries Act (MPRSA), 40 CFR Part 228 - Criteria for the Management of Disposal Sites for Ocean Dumping, in addition to all applicable state environmental laws. The basic components of the monitoring program are described below. The implementation of the program, using these components, is discussed thereafter.

All costs associated with the equipment, software, personnel, and other aspects of the monitoring program must be provided by the contractor (dredging contractor).

Where within this section, it states the contractor's responsibilities, this is inclusive of any and all agents (e.g. sub-contractors, inspectors) that may act on the contractor's behalf to ensure the placement requirements associated with this section are met. Failure of an agent to act on the contractor's behalf does not relieve the contractor of its responsibility to comply with the placement requirements.

The term "container vessel" refers to any vessel used to contain, or hold, dredged material (scow, barge, hopper dredge, etc.). The term "transport vessel" refers to any vessel (tugboat, hopper dredge, workboat, etc.) used to move a container vessel. In the case of a hopper dredge, the hopper dredge may act as both a "container vessel" and a "transport vessel". If hopper dredge engines are not used for propulsion, and another vessel (tugboat, for example) is used to move it, then the hopper dredge would be the "container vessel" and the tugboat would be the "transport vessel".

1.2 DREDGED MATERIAL TRANSPORT AND PLACEMENT MEETING

At least five (5) days prior to the initiation of all dredging, representatives of the contractor (including all of the contractor's applicable representatives, sub-contractors, and Inspectors on the project), shall meet with representatives of the USACE NY District (NYD) to discuss these placement requirements. The day and time of the meeting will be at the discretion of the NYD Operations Division and must be coordinated

by the contractor at least one week before the day of the meeting. Placement of dredged material must NOT occur until this pre-construction meeting (otherwise known as the Dredged Material Transport and Placement Meeting (DMTPM)) has occurred. The placement requirements may be subject to change at any time during the dredging project. Any changes or additional requirements, specified by NYD, must be implemented by the contractor as soon as practicable.

At the pre-construction meeting, the contractor must provide a list of all container vessels planned for use on this dredging project, which must include at least one (1) vessel designated as a backup barge. Container vessels identified for use on the project must be certified by the contractor to be fully operational, mechanically sound, completely seaworthy, and free of leaks or other defects.

At the pre-construction meeting, the contractor must also submit a list to NYD with the names of all DMIs who will be working on the project. This list must also specify and verify DMI company affiliations, DMI certification, and the expected duration of employment. DMIs who will be on duty at the beginning of the dredging project must be present at the pre-construction meeting to review the placement requirements associated with this project. Any DMIs who begin duty after the first day of dredging must also be briefed by NYD personnel on the placement requirements prior to working as a DMI on the project. Notice of replacement DMIs must be submitted to NYD at least five (5) business days prior to beginning work, unless unforeseen events prevents such notification.

The contractor must also provide NYD with documentation of National Marine Fisheries Service (NMFS) endangered species/marine mammal observer certification for each such observer who will be working on the project at the pre-construction meeting (see Part 1.4 of this section).

1.3 NEW YORK DISTRICT CERTIFIED DREDGED MATERIAL INSPECTORS (DMIs)

The contractor, at its own expense, shall have NYD certified dredged material inspectors (DMIs) monitor the transport of all dredged materials to the HARS, and/or any other open-water placement location, as described further within this section. A list of DMIs can be obtained from the NYD Technical Manager for Dredged Material Management, Tiffani M Lee at (917) 790-8539. Only DMIs on this list may be used during this project.

The contractor is responsible for ensuring that DMIs, employed either directly or through sub-contracting, are fulfilling the DMI duties described in this contract section and in the associated placement requirements.

At any time during the project, the Contract Officer or Contract Officer Representative (COR) can require the contractor to remove and replace any DMIs who are not satisfactorily fulfilling their duties/requirements. Any DMI determined to be performing unsatisfactorily may also be subject to de-certification of their DMI status by NYD.

1.3.1 RESPONSIBILITIES OF DREDGED MATERIAL INSPECTORS (DMIS)

Dredged material inspector (DMI) responsibilities are described as follows:

- DMIs will ensure that the placement requirements are being followed;
- DMIs are required to be on duty and in the transport vessel wheelhouse, to observe container vessel monitoring equipment function, watch for endangered species, if the DMIs are also NMFS certified endangered species / marine mammal observers (see Part 1.4 of this section for further explanation of these NMFS observer requirements), from the time the transport vessel departs from the dredging site, until the container vessel has completely emptied and all DMI reporting requirements have been completed for each placement trip (please note: The terms 'dredged material monitoring' and 'container vessel monitoring' are used interchangeably throughout this document);
- DMIs must estimate the volume of dredged material within each container vessel prior to the start of each trip to the designated dredged material placement location. The dredged material density and container vessel draft must be used by the DMI to estimate this volume, via use of dredging contractor-provided container vessel loading tables, unless the container vessel monitoring software provides the calculation automatically based on container vessel sensor data. This estimated volume must be recorded on the USACE (NYD) Transportation and Placement Log (TPL) form (see Part 1.8 of this section for an example of this form);
- DMIs must clearly photograph each container vessel (both in close-up and overview) with a working digital camera after loading, but prior to transport away from the dredging site. The close-up photograph must show the loaded container vessel such that the level and characteristics of material (e.g. color, texture, material composition, and approximate grain size) within the container vessel can be determined from the photograph. The overview photograph must show the relationship between the material in the container vessel and the container vessel itself (e.g. height of material in container vessel, distribution of material in container vessel, standing water next to dredged material in container vessel, etc.). The DMI must also take a photograph of the container vessel that documents the position of the plimsoll line relative to the water level;
- The minimum resolution required for each photograph is 1200 x 1600 pixels. Digital cameras used for the photography must be capable of both wide and narrow angle photographs;
- The contractor, either directly or indirectly via one or more of its agents (e.g. dredging contractor) , will be responsible for ensuring that sufficient lighting is provided to meet the container vessel photograph requirements, regardless of weather conditions or time of day. Flash photography must be used if darkness or other conditions require supplemental illumination. If the flash included with the digital camera does not provide adequate lighting, supplemental flash units and/or flood lights, may be necessary;
- Each digital photograph must be submitted, by the DMI, with the other container vessel monitoring data (TPL, checklist, etc.) for posting on the dredged material / container vessel monitoring website. Photographs must be digitally imprinted with the dredging project name, contract number, date and time of each photograph, the trip number, and the container vessel identification;

- DMIs must measure the distance from the transport vessel to the container vessel, at the time of placement, using the hand-held laser range finder (if necessary) and record the value on the TPL;
- DMIs must complete a TPL and checklist for each ocean placement trip made by a container vessel loaded with dredged material, while on duty within the transport vessel, utilizing the dredged material monitoring software on a laptop dedicated for this purpose. DMI submittals (i.e. TPL and checklist), on the basis of one (1) set of submittals for each container vessel of dredged material, must be electronically submitted to NYD (points of contact (POCs), as specified at the pre-construction meeting), within two (2) hours after each placement event at the HARS, and/or any other open-water placement location, has occurred. Emails shall include signed and dated PDF copies of TPLs with the following naming convention: Trip_###_SWO_26_TPL.pdf.
- The DMI must either select "YES" or "NO" for each checklist item as appropriate, at each major stage of a trip (i.e. at the dredging site, while underway to the open water placement site, and during placement of dredged material). The checklist is included in Part 1.9 of this contract section;
- Any items receiving a "NO" answer are considered discrepancies and must be reported immediately to NYD at (917) 790-8082 or x8539, the contractor, and a dredging contractor representative, and the Dredged Material Monitoring Equipment and Software Contractor (DMMESC). These contacts are referred to as the "Notification List". Any discrepancies must also be recorded in the 'Comments' section of the TPL;
- When DMIs place the 'Notification List' telephone calls, DMIs must state their name, the dredging project they are serving as a DMI on, the name of the transport vessel, container vessel, the date, the time, and a brief description of the checklist item and the reason for the "NO" answer(s) for any checklist items;
- In the event of container vessel monitoring equipment/software (i.e. both PDMMES and BDMMES systems) malfunction, the DMI must manually complete a map of the placement area showing the position of the container vessel at the time container vessel doors were first opened, including the distance from the transport vessel to the container vessel (as determined using the hand-held laser range finder), the transport vessel differential global positioning system (DGPS) position at the time of initial container vessel door opening, the vessel direction of travel, and the bearing to the container vessel from the transport vessel. Maps must also be submitted to NYD with the TPL and other required documents. Please refer to Section 1.7.2 for a detailed description of this procedure;
- DMIs are NOT responsible for the operation of transport vessels nor are DMIs authorized to operate any equipment used to open and/or close container vessels;
- The contractor may be required to have DMIs take on additional duties at any time during the dredging project, as specified by NYD.

1.3.2 EQUIPMENT REQUIREMENTS FOR DREDGED MATERIAL INSPECTORS (DMIS)

The contractor is responsible for ensuring that each DMI working on the dredging project is provided with the following equipment (e.g. items, provisions, accommodations, and supplies):

- A legible copy of the permit or contract specifications;
- A legible copy of the placement requirements (including the placement grid map) received at the pre-construction meeting, and any additional requirements provided by NYD during the dredging project;
- An 8" - 12" wide protractor with degrees printed or embossed on the curved surface, and dividers for scaling distances off of maps and charts;
- Container vessel loading tables for each container vessel used to transport dredged material;
- A fully operational, handheld laser range finder with a range of at least 1,000 feet, not more than ten (10) years old. Spare batteries for the laser range finder must be available at all times;
- A working digital camera for the purposes of taking container vessel load and plimsoll photographs, which meets the specifications and requirements described above in Part 1.3.1 of this contract section;
- A fully operable cellphone in each DMI's possession at all times, with active phone numbers unique to each phone, available for placing and receiving calls at all times. DMI cell phone numbers must be provided to NYD at the pre-construction meeting;
- Access to the transport vessel DGPS, fathometer, and radar;
- DMIs must be provided with a designated bunk space or other suitable sleeping location, and a suitable location for completing paperwork associated with DMIs duties, while working aboard a transport vessel as a DMI.
- DMIs are not allowed to be on duty for more than twelve (12) hours per twenty-four (24) hour day (a day is defined as the 24-hr period from 12:00 AM to 12:00 AM the following day);
- DMIs must be provided a minimum of eight (8) hours of continuous off-duty time each day to allow appropriate rest to ensure safety and competence.

1.4 NATIONAL MARINE FISHERIES SERVICE (NMFS) ENDANGERED SPECIES/MARINE MAMMAL OBSERVER

The contractor shall ensure that a National Marine Fisheries Service (NMFS) approved Observer is aboard the vessel transporting any container loaded with dredged material to the placement site. The observer will have the responsibility for determining the presence of endangered species (sea turtles and whales) during transit to, and upon arrival at the location for all placement activities. The observer will ensure that the transport and container vessels do not intentionally approach any whale or sea turtle closer than 100 feet while in transit to and from the HARS, and/or any other open-water placement location. Upon arrival at the placement site,

placement of dredged material may occur only if no endangered species are observed to be present within a 0.25 nautical mile of the placement site. If endangered species are observed to be present within 0.25 nautical mile of the designated placement location, then the placement of dredged material must not occur. Placement of the dredged material may occur only when the observed animals have moved outside the 0.25 nautical mile zone around the designated placement location, or have completely departed the site.

In all such cases where whales, sea turtles, and Sturgeons have been encountered, the observer must submit a written report incorporating the following information: animal type (Whale, sea turtle, or Sturgeon); the specific species (if known); the date, time and location of the sighting (latitude, longitude); approximate distance away from the transport vessel and/or container vessel; number of individuals observed; behavior (feeding, nursing, migrating, etc.).

DMIs who are also certified as NMFS Observers may fulfill the duties of both positions currently on the same project. If a DMI is to also serve as a NMFS Observer, he/she must possess or acquire, prior to the initiation of the project, valid certification from NMFS of training on techniques for identifying species, and preparing applicable reports for instances where endangered species are encountered. *Please note, if a DMI is not a NMFS certified endangered species/marine mammal observer, the contractor must still ensure that such an observer is aboard the vessel transporting any container loaded with dredged material to the placement site within the HARS, and/or any other open-water placement location.*

1.5 DREDGED MATERIAL MONITORING EQUIPMENT

The contractor is required, at its own expense, to have all container vessels used to transport dredged material to any and all placement or treatment locations (HARS, non-HARS, or any other locations in the New York Bight or Lower Bay or any other area) equipped with container vessel monitoring equipment and software, through a separate contractor (i.e. Dredged Material Monitoring Equipment and Software Contractor (DMMESC)) not owned or affiliated with the dredging company. Only the DMMESC may service (e.g. install, modify, program, repair/replace and/or remove) the container vessel monitoring software/equipment.

Container vessel monitoring equipment must be installed and operating correctly on any container vessel used to transport dredged material prior to ANY dredged material being placed in the container vessel. All container vessel monitoring data collection must begin prior to ANY dredged material being placed in the container vessel and continue for the duration that any dredged material is contained in a container vessel. Container vessels loaded with any quantity of dredged material transported from the dredging site, must have all monitoring equipment functioning and recording data.

The contractor shall ensure that the container vessel monitoring equipment is continuously operational twenty-four (24) hours each day, at all times any dredged material is being loaded at dredging sites, when any dredged material is contained within any container vessel, when any dredged material is transported from dredging sites, while any container vessels are returning to dredging sites after ocean placement, and when any container vessel is returning from upland facilities.

The contractor is also responsible for maintenance of all container vessel monitoring equipment and software, through the services of a DMMESC. Any problems with operation/function of the container vessel monitoring software and/or equipment must be directed to the DMMESC immediately to address, and to the NYD at (917) 790-8539 or (917) 790-8082. Any monitoring equipment that is damaged or destroyed must be replaced/repared prior to using the container vessel again.

At any time during the dredging project, the contractor must provide the DMMESC full access to all dredges, transport vessels, and/or container vessels used on this project, as the DMMESC determines necessary, to complete any services that may be required to fulfill the container vessel monitoring requirements associated with the project.

As it pertains to such services stated above, the contractor must provide the DMMESC with transportation from the shore to any vessel equipped with container vessel monitoring equipment, and any other assistance determined reasonable and/or necessary. Transportation must be provided in a timely manner such that the requirements associated with container vessel monitoring continue without interruption.

The contractor must provide the DMMESC with all plans and specs, drawings, etc., associated with the project at least ten (10) business days prior to the start of dredging;

The contractor is required to provide the DMMESC with a container vessel loading table, or ullage table, for each container vessel used to transport dredged material to upland facilities and ocean placement locations, that includes the range of possible container vessel drafts associated with the range of possible dredged material volumes contained in each container vessel;

Container vessel monitoring equipment and software must be approved by NYD prior to use. If NYD has not previously observed the operation of a specific sub-contractor's container vessel monitoring equipment and software proposed for use, the contractor is responsible for satisfactorily demonstrating the operability of this proposed equipment and software in order to receive approval by NYD. Such approval must be obtained by the contractor at least ten (10) business days prior to the start of dredging.

The following specifications are associated with, and required for, the container vessel monitoring equipment, software, and the container vessel monitoring website:

- A self-contained "black box" unit must be installed on all container vessels, and must be capable of recording on a twenty-four (24) hour basis, latitude and longitude positions of the container vessel, through DGPS (or wide area augmentation system (WAAS) enabled) technology, and container vessel draft, using a water pressure sensor, and the level of dredged material within container vessels, using a bin level sensor, both at variable sampling rates ranging from one (1) reading (i.e. of position, draft, and bin level every six (6) to nine (9) seconds to one (1) reading every one (1) minute:

- Container vessels used to transport sand, glacial till, Pleistocene clay, or rock, to an offshore placement location (HARS and/or any other open-water placement location, and/or reef) must include fore and aft draft sensors and one (1) bin level sensor.

- Container vessels used for transport of fine-grained, muddy dredged material to the HARS, and/or any other open-water placement location, must include fore and aft draft sensors and fore and aft bin level sensors.
- Container vessels used to transport fine-grained, muddy dredged material to upland placement/treatment sites must include one (1) bin level sensor and one (1) draft sensor.
- Any transport vessels used to transport dredged material to any location, and all dredges used on the project, must be equipped with satellite (or cellular) real-time tracking and messaging systems maintained by the DMMESC;
- All transport vessels used to transport dredged material to open-water placement sites (non-upland placement/treatment locations) must be equipped to automatically record bathymetric data using a fathometer on each transport vessel. Transport vessel fathometers must have digital data output that is compatible with container vessel monitoring laptop computers and software;

Laptop computer container vessel monitoring software must include the following required capabilities:

- Trip numbering assignment, to be entered by the DMI at the beginning of each placement trip;
- Electronic versions of the TPL and checklist, to be completed by the DMI during each placement trip;
- Map displays with selectable scales that allow DMIs to view the dredging site, the New York Harbor area and New York Bight, the HARS, and/or any other open-water placement location, and designated placement grids and coordinates. The container vessel monitoring software must graphically display placement grids developed for the project and allow viewing of the position of the transport vessel and/or container vessel throughout the placement trip;
- Links to websites that allow National Oceanic and Atmospheric Administration (NOAA) offshore buoy data to be viewed in real-time, along with the latest marine weather forecasts;
- Real-time plotting of container vessel draft, bin level and speed, to be used to monitor potential container vessel leakage;
- Downloading of digital camera photos for inclusion with placement trip data;
- Collection of transport vessel fathometer data for inclusion with other placement trip data. Fathometer data must be collected within the specified boundaries that define each placement site;
- Transmission, at the end of each placement trip, of electronic TPLs, checklists, container vessel position, speed, draft & bin level data, fathometer data, and digital photos, to a website described below.

A website must be maintained by the DMMESC and include the following capabilities and reporting products:

- All data collected by the container vessel monitoring equipment, including, but not limited to, trip plots, draft & bin level plots, TPLs, checklists, digital photos, must be posted to a website accessible to NYD personnel as follows:

- within two (2) hours of DMI submission of the data from the laptop computer.

- within twenty-four (24) hours of container vessel unloading at the upland/treatment facility, for each container vessel used for upland placement/treatment.

- Plotting of summary maps, illustrating recorded trip courses (i.e. trip plots) to placement site(s), as a function of trip quantity (i.e. individual, multiple or all trips), trip date range, dredge, container vessel, transport vessel, DMI, placement location, and sub-project i.d., This feature must also allow the user to sub-query and map the selected trip data sets, as a function of the parameters described above, based on trip positioning data classification, as follows: loading points, standby points, all placement points, first placement point, full trip plot (i.e. all data points).

This feature must allow the user to view trip data at various scales, by map type (i.e. areal overview (with and without labels), and road), and by pre-configured default zoom extents (i.e. initial overview (full trip plot), dredging site, and placement site). Interactive coordinate information (i.e. latitude & longitude specified in decimal degrees to six (6) decimal places) must be illustrated, as function of cursor position on the map.

This feature must also contain a measuring tool that allows the user to measure distances (in feet, yards, miles, nautical miles, meters, and kilometers), areas of any shape (in square (sq.) feet, sq. yards, sq. miles, sq. meters, sq. kilometers, acres, and hectares, and plot points (in decimals degrees, and in degrees, minutes, seconds).

- Plotting of all recorded container vessel draft, bin level, and transport vessel speed, data points, as function of time, per selected trip data set. Each data plot type must be clearly distinguished from the others, must allow for cursor selected illustration of date, time and value (i.e. draft & bin level in feet, and speed in knots, both to one (1) decimal place) for each recorded data point.

As the cursor moves across any of the data plots, the respective position as which all data was recorded, must be illustrated in the map display of the trip plot.

This measure must also allow the user to zoom in and plot sub-sets of trip data.

- Automatic generation of emails to the Notification List for the following trip incidents:

- TPL checklist items marked "No".

- Misplaced dredged material (e.g. emergency short dumps, or mis-dumps due to human and/or mechanical error, at the dredging site, in transit, or in the vicinity of the placement site).

- Container vessel losses or gains in draft and/or bin level of 1.5

feet or greater, occurring during the transit and/or standby phases of a trip (based on the DMMESC's quality assurance (QA) review of the trip data set); the standby phase includes that stage in the trip, prior to transit to the HARS, and/or any other open-water placement location, either after the loading phase at the dredging site is complete, or in between loading phases, regardless of the nature of the loss or gain.

- Malfunctioning / non-working sensors (e.g. DGPS, bin, draft/water pressure) on any container vessels.

- Observed depth within the placement grid shallower than fifty (50) feet, based on recorded fathometer values.

All of the automatic emails types described above will also be recorded on the website in a consolidated database. This database must include the following data fields: trip number, container vessel, transport vessel, DMI, alarm type, date sent, and hyperlink to the email sent. For the 'TPL checklist' emails, this database must keep a record of the cumulative "No" results for the duration of the project, including a description of and tally of those checklist items receiving a "No";

- A placement grid cell management tool that reports the following per grid cell: total container vessel placements, total estimated volume of dredged material in the cell resulting from container vessel placements, the trip numbers associated with the container vessel placement events that occurred within the cell. The placement grid cell management tool must also have the capability for NYD, or the DMMESC as directed by NYD, to enable, disable, and set the number of maximum container vessel placement events within placement grid cells;

- Positioning information, viewable in real-time, or as soon as data transmission permits, of all dredge plants, container vessels and transport vessels used in the project, based on location data acquired by the satellite (or cellular) tracking system.

The satellite (cellular) tracking system must allow NYD to communicate with DMIs stationed aboard transport vessels via a messaging tool within the website, as a backup measure, when other modes of communication are not possible (e.g. in the event transport vessels are out of cell phone range);

- A dredging production database that reports the following per trip: trip number, date of trip, dredge used, transport vessel used, container vessel used, trip destination, estimated container vessel volume, and the time taken for the following - container vessel loading, container vessel de-watering, container vessel transit, container vessel unloading, dredged material disposal from the container vessel, return trip from the placement site and container vessel idleness, as applicable. The dredging production database will also include a map of the dredging site that is updated after the completion of each trip with the location of where dredging associated with each container vessel-loading event occurred;

- Transport vessel fathometer data must be acquired during all trips, and generated into the following reporting formats, for viewing on the container vessel monitoring website:

- Trip-specific bathymetric plots (depth vs. time).

- Bathymetric map(s) of the placement site, based on the cumulative fathometer data collected throughout the duration of the dredging project;

where overlapping of data occurs, the shallowest depth must be reported. For rock projects, the above-described map must be generated and updated on a weekly basis. For non-rock projects, the above-described map must be generated at project completion.

- Provide a maintenance log on the container vessel monitoring website of the DMMESC's remote-access and on-site service visits associated with any container and/or transport vessel hardware/software installation and/or repairs.

- The DMMESC will submit all acquired monitoring data on a weekly basis, for those weeks that data acquisition occurs, no later than (NLT) the close of business (COB) of the Friday of the following week. This data will be submitted remotely, via a USACE-approved web-based file sharing system, which the DMMSEC will be provided access to by NYD at the start of the project. Should the DMMSEC not be able to transmit the data remotely via this system, it must do so by regular mail, in CD/DVD format to NYD (address specified below), and within the same timeframe described above.

The positioning data associated with each trip, must be submitted in format compatible with ArcGIS10 software, such that the user is able to view the data output as a function of loading points, all placement points, first placement point, full trip plot (i.e. all data points).

This data submission must also include all required digital photos taken by DMIs of loaded container vessels.

All information entered into TPLs must be submitted in a relational database format.

All data collected by DMMSEC as a result of this contract, shall become property of the U.S Government.

1.5.1 PRIMARY MONITORING EQUIPMENT AND SOFTWARE SYSTEMS

Primary Dredged Material Monitoring Equipment and Software (PDMMES) systems, consisting of laptop computers with container vessel monitoring software, onboard all transport vessels used to transport dredged material to all sites, must allow the data (e.g. vessel location, draft, and bin level), being collected by the "black box" unit DGPS (or WAAS enabled) receiver and sensors on the container vessels being transported, to be viewed in real-time by the DMIs and by transport vessel crew members when used for open-water placement trips. PDMMES system data must be automatically transmitted from all container vessels to the transport vessel for storage and transmission to the DMMESC's computer system for posting on the container vessel monitoring website.

1.5.2 BACKUP DREDGED MATERIAL MONITORING EQUIPMENT AND SOFTWARE SYSTEMS

Backup Dredged Material Monitoring Equipment and Software (BDMMES) systems, used for placement trips, consisting of laptop computers with container vessel monitoring software, onboard all transport vessels used to transport dredged material, must allow the DGPS (or WAAS enabled) position of the transport vessel to be viewed on the computer screen, along with an estimated position of the container vessel based on the towing distance and angular offset of the container vessel from the course line of the transport vessel.

1.6 HISTORIC AREA REMEDIATION SITE (HARS)

The HARS is a 15.7 square nautical mile area located approximately 3.5 nautical miles east of Highlands, New Jersey and 7.7 nautical miles south of Rockaway, Long Island, bounded by the following coordinates:

Point	LatitudeDMS	LongitudeDMS	LatitudeDDM	LongitudeDDM
A	40° 25' 39" N	73° 53' 55" W	40° 25.65' N	73° 53.92' W
B	40° 25' 39" N	73° 48' 58" W	40° 25.65' N	73° 48.97' W
C	40° 21' 19" N	73° 48' 57" W	40° 21.32' N	73° 48.95' W
D	40° 21' 19" N	73° 52' 30" W	40° 21.32' N	73° 52.50' W
E	40° 21' 52" N	73° 52' 30" W	40° 21.87' N	73° 52.50' W
F	40° 21' 52" N	73° 53' 55" W	40° 21.87' N	73° 53.92' W

DMS = Degrees, Minutes, Seconds
DDM = Degrees, Decimal Minutes

1.7 PLACEMENT OF DREDGED MATERIAL AT OCEAN WATER SITES

1.7.1 PLACEMENT OF DREDGED MATERIAL AT OCEAN WATER SITES, GENERAL

All placement of dredged material within the HARS must occur in accordance with the placement requirements, and with any additional requirements provided by NYD at any time during the project. Additional placement locations may be provided after dredging begins.

The contractor must receive grid coordinates for the HARS placement location(s) before dredging starts. Individual grid cells may be as small as 100 feet wide and 100 feet long. Details of the placement location within the HARS, for HARS-suitable dredged material, will be provided at the DMTPM.

Twenty-four (24) hours prior to departure of the first project vessel from port for the open water placement of any dredged material, the contractor must notify NYD by email. Emails regarding departures must be made to the Dredged Material Management Section at NYDredging@usace.army.mil, ryan.p.corbett@usace.army.mil, tiffani.m.lee@usace.army.mil, fiona.b.dunn@usace.army.mil, daniel.mcsweeney@usace.army.mil, arsheen.ehtesham@usace.army.mil, and kelly.l.vega@usace.army.mil. The contractor must furnish the contractor name, project name, DMI name and estimated time of departure.

All non-rock material approved for ocean disposal shall be transported to the HARS, unless otherwise directed. Dredged material placement must not occur anywhere outside of the HARS, in HARS buffer zones, in shipwreck buffer zones, in the HARS "No Discharge Zone", or in any other area of the HARS determined by NYD to be off-limits for dredged material placement.

All placement events must be recorded and signed by the master of the tow. Every vessel engaged in the transportation of dredged material must have its name or number, and owner's name, painted in letters and numbers at least fourteen (14) inches high, on both port and starboard sides of the vessel. These names and numbers must be kept distinctly legible at all times, and vessels not so marked, cannot be used to transport or place dredged material.

The NY District reserves the right to have NYD and/or EPA employees and other invited representatives accompany all trips to the placement site(s)

to observe placement activities and/or verify the contractor's compliance with these placement requirements.

1.7.2 PLACEMENT OF DREDGED MATERIAL AT OCEAN WATER SITES, PROTOCOLS

To help ensure proper placement of dredged material at the HARS, or other open-water placement locations, the following placement protocols must be followed:

a. DGPS navigation and fathometer equipment must be present and fully operational on board the transport vessel and must be calibrated periodically in accordance with the manufacturer's guidelines. The instrumentation must also conform to current industry standards. Re-calibration of the instrumentation will be required in instances where major modifications to the transport vessel have been made. Specific documentation certifying the accuracy of instruments may be requested by NYD when needed at any time. Fixed aids to navigation, with known latitude-longitude coordinates, must be used periodically to double-check the accuracy of navigation equipment. In addition, locations with a known depth and stable bottom must be used to periodically double-check accuracy of fathometers.

Container vessels may not be transported from the dredging site for offshore placement of dredged material unless the transport vessel DGPS navigation system, container vessel monitoring equipment and software, transport vessel fathometer, hand-held laser rangefinder, container vessel radio-control system (if used), digital camera, and backup radio on container vessel (if a scowman is used) are all in full working order and provide correct information.

b. The container vessel radio-control system (if used on the project) and the PDMMES and BDMMES systems must be inspected for correct operation. If any problems with the container vessel, radio-control system, PDMMES and BDMMES systems, video camera, or laser range finder, are encountered, corrections must be made before offshore transport of the container vessel may proceed, except when DMMESC personnel are onboard or attempting to fix a container vessel monitoring problem, in which case placement would proceed using the BDMMES. However, the BDMMES must not be used on more than two (2) consecutive placement trips.

c. Container vessels must be inspected for the presence of any conditions that may cause potential leakage. Prior to loading an empty container vessel at the dredging site, the empty container vessel must be inspected for presence of large dents or visible holes. Any holes must be repaired prior to placing any dredged material in the container vessel. Dents must be closely examined to ensure that a hole is not present, or that the hull strength is not compromised. The juncture of the two (2) split hulls, when the container vessel is closed, must form a straight line and the rubber gasket must form a tight seal. Damage to the rubber seal or juncture must be repaired prior to loading the container vessel.

Prior to leaving the dredging site, container vessels must be inspected to ensure correct operation of mechanical features. Container vessels must also be inspected for the presence of any conditions that may cause navigation problems. At least one (1) backup container vessel must be available for immediate use, if one (1) of the regularly used container vessels develops a condition that prevents it from being fully

operational, mechanically sound, completely seaworthy, and free of leaks or other defects.

d. Container vessels loaded with a total of more than 6,000 CY of dredged material and water must not be transported to the HARS. Only HARS-suitable dredged material can be transported to the HARS. Container vessels used for transporting HARS-suitable material must not be used for the transport of material that is unsuitable for the HARS; container vessels used for transporting material, unsuitable for the HARS, to upland placement/treatment sites, must not be used for the transport of HARS-suitable material to the HARS and/or other open-water placement sites.

Floatable material, and/or trash, and/or debris excavated during the dredging project, including, but not limited to, wood, tires, scrap metal, car bodies, shopping carts, etc., must never be transported to the HARS or other open-water placement locations, regardless of the source of the items/material. Only dredged material approved for open-water placement shall be transported and/or placed at designated open-water placement locations. The contractor will be responsible for retrieving non-approved materials/items from open-water placement sites if such placement occurs.

e. The contractor is responsible for determining, prior to the departure of the transport vessel from the dredging site, that the forecasted weather and sea conditions at the expected time of placement of dredged material within the HARS will:

- 1) Allow for the safeguard of personnel and property during the transporting operation and full release of material from the container vessel at the designated location and

- 2) Not result in a loss of dredged material from the container vessel due to waves breaking into the container vessel.

If weather/sea conditions will not permit placement of dredged material at the designated grid cell, the container vessel must not be towed from the dredging site until conditions improve and allow safe and accurate dredged material placement.

f. Container vessels must be observed for potential leaking of dredged material, as indicated by visible turbidity plumes (muddier water) behind the container vessel, or significant change (of greater to or equal than 1.5 feet) in the container vessel draft and/or bin level. Container vessels exhibiting gradual changes (of at least one (1) foot) in draft and/or bin level changes may be leaking, and this must be noted on the TPL, and also must be reported to the Notification List immediately.

If gradual draft and/or bin level changes appear to occur regularly, or if a significant change in container vessel draft and/or bin level occurs during any trip, the contractor must examine the container vessel as soon as possible to determine if a leak is present. If a situation arises that requires emergency dumping of dredged material, all reasonable efforts to dump outside of navigation channels must be made by the contractor.

The container vessel draft and bin level values, as recorded by the container vessel sensors, must be documented in the TPL by the DMIs reported five (5) minutes of departing from the dredge site, while in transit, and at the designated placement location, just prior to container vessel door opening.

g. Container vessels must be brought to the designated grid cell, or coordinates, of the HARS, or other designated placement location, using the DGPS navigation systems of the transport vessel and the container vessel monitoring software onboard the transport vessel. Placement in the appropriate location must be documented by the DMI using the container vessel monitoring software while the container vessel position data is automatically being recorded by the container vessel monitoring system.

Container vessels must be towed at an appropriate speed and tow length that ensures successful placement of dredged material at the placement site, and that also maintains reliable navigation under safe weather/sea conditions. For each trip, the DMI must measure the distance from the transport vessel to the container vessel at the time of placement using the hand-held laser range finder, obtain a reading, and record the value on the TPL.

During each trip to the HARS, the dredged material contained in a container vessel must be able to be placed (initiation of placement) within individual grid cells as small as 100 feet wide and 100 feet long. Dredged material placed outside of designated grid areas of the HARS must be minimized to the maximum extent practicable.

h. If the PDMMES does not show reliable DGPS (or WAAS enabled) coordinates in the vicinity of the designated placement grid or other designated placement location, or is not functional, the BDMMES must be used to locate the placement site and estimate the container vessel position during placement. Length of toelines must be measured using the hand-held laser range finder. The bearing to the container vessel from the transport vessel must also be noted at the time of placement. Vessel navigation must be maintained in the direction of the maximum grid dimension for all placements, to the greatest extent possible.

The angular displacement of the container vessel from the transport vessel course (i.e. trackline) must be estimated by sighting the container vessel behind the transport vessel while holding a protractor with the 90 degree mark pointing directly behind the transport vessel in line with the vessel track line (wake). A pencil must then be used to point at the container vessel, to the left or right of the 90° mark, to determine the angular displacement off of the transport vessel track line, recorded as degrees to the left or right when sighting the container vessel, rounded to the nearest 5°. Container vessels directly behind the transport vessel would be reported at 0° angular displacement (a container vessel displaced 10° to the left of the 90° mark on the protractor would be reported as 10° left, etc.). This angle must be recorded on the TPL, along with the following information if this option is used:

- 1) Coordinates of the transport vessel at the start and end of placement.
- 2) Length of tow line (distance from transport vessel stern to container vessel bow).
- 3) Angular displacement of container vessel from trackline of transport vessel.
- 4) Estimate of lateral displacement of container vessel from the transport vessel trackline.
- 5) Estimated longitude and latitude of container vessel at time of door opening and closing.

The lateral displacement may be estimated by the following formula (for angular displacements up to 20 degrees):

Lateral Displacement = Tow Length x Sine of Angular Displacement

The following values of sine (i.e. Sin ?) may be used:

Sin (5 degrees of angular displacement) = 0.087
Sin (10 degrees of angular displacement) = 0.174
Sin (15 degrees of angular displacement) = 0.259
Sin (20 degrees of angular displacement) = 0.342

For example, when using a 200 foot tow length, a container vessel is observed to track 15 degrees to the right of the transport vessel trackline. The estimated lateral displacement of the container vessel is:

200 feet x 0.259 = 52 feet

This means that when plotting the container vessel position on a map of the placement area, the container vessel would be plotted ~50 feet to the right and ~200 feet behind the position of the transport vessel (i.e. tow length distance). The errors in estimating increase with longer tow lengths. Because of this, it is critical to maintain as short a tow length as possible if the BDMMES or transport vessel's DGPS navigation system is used for placement. Perimeter grid cells are not permitted for use if PDMMES is not functioning. The closest adjacent grid cell toward the center of the grid must be used.

If the PDMMES system fails after leaving the dredging site, the backup system may be used for up to two (2) consecutive placement trips while awaiting DMMES personnel to service the equipment. No more than two (2) consecutive trips to ocean placement sites may be made by such container vessels without the PDMMES fully functioning.

i. If neither the PDMMES nor BDMMES show reliable DGPS (or WAAS enabled) coordinates in the vicinity of the designated placement grid or other designated placement location, or are not functional, or weather/sea conditions prevent reliable maneuvering of the container vessel, the transport vessel DGPS must be used to position the container vessel at the center of the grid, or other backup location in the grid as specified by NYD.

Length of toelines must be measured using the hand-held laser range finder. The bearing to the container vessel from the transport vessel must also be noted at the time of placement. The angular displacement of the container vessel from the transport vessel course (track line) must be estimated by sighting the container vessel behind the transport vessel while holding a protractor. This angle must be recorded on the TPL, along with the following information if this option is used:

- 1) Coordinates of the transport vessel at the start and end of placement.
- 2) Length of tow line (distance from transport vessel stern to container vessel bow).
- 3) Angular displacement of container vessel from trackline of transport vessel.

4) Estimate of lateral displacement of container vessel from the transport vessel trackline.

5) Estimated longitude and latitude of container vessel at time of door opening and closing.

j. The grid center, or other backup placement location, will only be used if steps (h) and (i) are attempted without success, or when inclement weather/sea conditions prevent reliable maneuvering of the container vessel. The grid center must not be used if inclement weather conditions persist. Placement at the grid center is an emergency procedure.

k. If weather and/or sea conditions prevent reliable measurement of transport distance using the hand-held laser range finder, the transport vessel's radar must be used to determine the distance and bearing to the container vessel.

l. If neither the PDMMES nor BDMMES systems, nor the transport vessel DGPS systems, provide navigation coordinates, the container vessel must be brought to a suitable location for correction of navigation problems. Placement of dredged material is not allowed if a reliable DGPS system is not providing coordinates at the time of container vessel door opening.

m. If radio communication with the container vessel is lost, preventing operation of radio-controlled container vessels, placement must not occur until the system is repaired. Voice contact, through radio or direct communication, must be maintained with a scowman, or other personnel, who is riding aboard a container vessel, for the duration of the placement trip. Container vessel opening must only occur when a direct, voice command has been given to personnel aboard the container vessel, or when radio communication with radio-controlled container vessels is maintained.

If the container vessel's engine cannot be operated by the radio-control system, and the container vessel is boarded to attempt to fix the engine, the container vessel must be located at the designated placement position if the container vessel's engine is started. Any problems with a radio control system must be fixed prior to subsequent use of the container vessel. The DMI must note on the TPL any time the radio-controlled container vessel system malfunctions and manual discharge is required, and immediately notify the Notification List.

n. A primary and backup radio must be onboard all manned container vessels, along with backup power supplies. Hand signals must never be used to direct the scowman regarding container vessel opening/closing. All personnel aboard container vessels, or who may board container vessels while transporting dredged material, must be informed that discharge of dredged material will only be allowed while voice communication is maintained.

o. To help ensure that dredged material is transported and placed at the HARS in accordance with the requirements described above, for each trip, the DMI must complete the required checklist (see Section 1.9). Items in the checklist must be reviewed by the DMI at the dredging site, while underway, and at the HARS. Any item on the checklist that receives a "NO" answer must be reported immediately to the Notification List. If the "NO" answer is related to the container vessel monitoring equipment and/or software, the DMMESC must also be notified immediately. These discrepancies must be noted on the TPL associated with the trip using the letter-number code associated with each item. Each placement trip to the

HARS must use a checklist, to be completed by the DMI working aboard the transport vessel, using the container vessel monitoring software or by hand.

All of the checklist items included with the "At the Dredging Site" portion of the trip MUST receive a "YES" answer before a container vessel may be towed from the dredging site for placement at the HARS. If ANY of these checklist items receives a "NO" answer on the checklist, the issue must be corrected such that the checklist item can be answered with a "YES" prior to the container vessel being used to transport dredged material away from the dredging site.

p. Two (2) exceptions to this exist: 1) If a backup container vessel is used, it must be noted on the TPL, but normal placement can continue. 2) When the PDMES systems are malfunctioning, dredged material may be transported from the dredging site if DMMESC personnel are onboard to fix/service the equipment, or if the BDMES system is functioning. If any of the items in Part A are answered "NO" by the DMI, the Notification List must be contacted immediately. Telephone numbers of personnel on the Notification List must be supplied to all DMIs working on the dredging project. Reports of discrepancies or unusual events must also be submitted by the DMI as soon as possible to the NYD. Discrepancies must be noted on the TPL using the code letter/number associated with each item in the lists.

q. Parts B and C of the checklist pertain to activities/requirements of DMI while underway to the designated placement location and at the placement location, respectively. All of these items must be verified by the DMI aboard the transportation vessel. If any of these items are answered "NO" by the DMI, the Notification List must be contacted immediately.

1.8 USACE TRANSPORTATION AND PLACEMENT LOG

1.8 USACE TRANSPORTATION AND PLACEMENT LOG



USACE Transportation and Placement Log



Log Number:

Date:

Project Information

Trip Number:

Tow Owner:

Contract:

Inspector's Name:

Project Name:

Tug's Navigational Unit:

Tug and Scow Information

Tug Name:

Scow Name/Number:

Tug Captain:

Scowman's Name:

Loading/Pre-Transit Information:

Dredge Departed From:

Material (Cubic yds.):

Time Loading Complete (hh:mm:ss):

Est. Density:

ETA to Placement Site (hh:mm:ss):

Material Description:

Aft Draft: Forward Draft:

Winds at reporting station closest to placement location are presently blowing:

Image of Material:

Material Closeup:

Image of Plimsoll Mark:

Placement Site Weather Conditions

Wind Direction (from):

Wind Speed (mph):

Observed Water Depth (ft):

Wave Swell Height (ft):

Visibility (N miles):

Weather Conditions:

Marine Mammals/Sea Turtles Sighting (Yes/No):

Transit/Placement Information

Time Scow Departed Dredge Site:

Tug position determined by (GPS/DGPS):

Time scow arrives at project placement area:

Approximate distance from scow to target at time of discharge (ft):

Length of towline at time of discharge:

Distance from tug navigational antenna to tug's towing bit (ft):

Direction of scow relative to tug towing bit (degrees):

Estimated Scow Speed (kts):

Scow Heading (Degrees):

Start Time (doors open):

Speed (kts):

GPS/DGPS Latitude:

Heading (degrees):

GPS/DGPS Longitude:

End Time (placement complete):

Speed (kts):

GPS/DGPS Latitude:

Heading (degrees):

GPS/DGPS Longitude:

Post Placement Time (doors closed):

Speed (kts):

GPS/DGPS Latitude:

Heading (degrees): 9

GPS/DGPS Longitude:

Comments

1.9 DREDGED MATERIAL INSPECTOR CHECKLIST

See attached.

1.9 DREDGED MATERIAL INSPECTOR CHECKLIST

Dredged Material Inspector (DMI) Checklist:

DREDGING PROJECT: _____

TRIP NUMBER: _____

DREDGED MATERIAL INSPECTOR NAME: _____

INSPECTOR SIGNATURE: _____

DATE: _____

Answer YES or NO to the following questions. Circle other choices and/or fill in blanks as appropriate. Any item on the checklist that receives a "NO" answer must be reported immediately to NY District at: (917) 790-8082 and NYDredging@usace.army.mil, or x8539, the contractor, and a dredging contractor representative not onboard the transport vessel. If the "NO" answer is related to the container vessel monitoring systems, the container vessel monitoring contractor must also be notified immediately. Items receiving "NO" answers must be indicated on the TPL using the letter-number code next to each item description and described on the TPL comments section.

All of the checklist items included with the "At the Dredging Site" portion of the trip must receive a "YES" answer before a vessel containing dredged material may be transported away from the dredging site for placement at the HARS. If ANY of these checklist items receives a "NO" answer on the checklist, the issue must be corrected, such that the checklist item can be answered with a "YES" prior to the container vessel that is being used for transport of dredged material departs from the dredging site.

PART A - At the Dredging Site:

- A1 For all dredging projects, a copy of the placement requirements, received at the pre-construction and/or Dredged Material Transport and Placement Meeting (DMTPM) is in the DMI's possession. For federal projects, a copy of the contract specifications is in the DMI's possession. For permitted projects, a copy of the permit special conditions is in the DMI's possession.
- A2 The container vessel being used to transport the dredged material is mechanically sound, does not leak, and has no visible damage that may cause leaking.
- A3 A regularly used container vessel was used (i.e. as opposed to a backup container vessel).
- A4 A container vessel loading table for the container vessel being towed is aboard the transport vessel and available for the DMI to use.
- A5 An estimated dredged material density has been provided by the dredging contractor and recorded on the TPL form.
- A6 The material being dredged has been observed by the DMI for general characteristics (grain size, color, consistency) and recorded on the TPL form.

- A7 For container vessels loaded with any rock (rock is defined as any stones greater than 2.5 inches in diameter), the estimated rock percent, as provided by the dredge operator, has been recorded on the TPL form.
- A8 An estimate of the volume of material in the container vessel has been calculated by the DMI using the container vessel loading table and recorded on the TPL form; if not automatically calculated by the container vessel monitoring software.
- A9 Container vessel contains less volume of dredged material than the maximum volume allowed for placement during a single trip. *Container vessels loaded with a total of more than 6,000 CY of dredged material and water must not be transported to the HARS.*
- A10 The monitoring systems (Primary and Backup) are fully operational and are functioning. Any container vessel monitoring system malfunctions must be reported immediately to the DMMEESC. Transport vessels are not allowed to leave the dredging site while transporting any container vessels with dredged material if the container vessel monitoring systems are not fully operational. If the PDMMES system is not functional, the Backup monitoring system may only be used on two (2) consecutive offshore placement trips using an affected container vessel. No more than two (2) consecutive trips without the PDMMES system can ever be made.
- A11 A fathometer is fully operational, functioning, and installed on the transport vessel.
- A12 A radio onboard the transporting vessel is operable and can receive NOAA marine weather forecasts and ocean conditions.
- A13 Current and forecasted marine weather and ocean conditions at the designated placement location have been monitored on the radio and will allow safe and accurate placement of dredged material. Conditions have been recorded on the TPL form.
- A14 DGPS navigation system is fully operational, functioning, and installed aboard the transporting vessel.
- A15 A radar system is fully operational, functioning, and installed aboard the transporting vessel.
- A16 The satellite (or cellular), vessel tracking system on the transport vessel is present and operable.
- A17 Radio-control system for container vessel operation (if scowman is not used) is fully operational and functioning.
- A18 Radio and backup radio system, for communication between container vessels and transport vessels, are aboard container vessel (if scowman is used), are fully operational and functioning.
- A19 Hand-held laser range finder, not more than ten (10) years old, with at least a 1000 foot range, is aboard transport vessel, fully operational and functioning, and available for DMI use, along with a set of backup batteries.
- A20 A fully operable cell phone that can send and receive calls is in the possession of the DMI onboard the transport vessel.
- A21 A protractor, pair of dividers (for map/chart distance scaling), and an up-to-date nautical chart that includes the placement area, are available for use by the DMI aboard the transport vessel.

- A22 DMI is provided full access to fathometer, radar, vessel DGPS, and any other equipment/information necessary to conduct DMI duties.
- A23 A digital photograph has been taken of the container vessel that documents the position of the plimsoll line relative to the water level.
- A24 A digital photograph (close-up) has been taken of the loaded container vessel such that the level and characteristics of material (color, texture, material composition, and approximate grain size) within the container vessel can be determined from the photograph. A digital photograph (overview) has been taken of the loaded container vessel such that the relationships between material and container vessel walls can be determined from the photograph.
- A25 The container vessel contains only dredged material (no trash, debris, timbers, tires, etc.) and only dredged material approved for open-water placement.
- A26 Digital photograph(s) of the container vessel have been downloaded into the container vessel monitoring software.
- A27 DMI activated the container vessel monitoring software at the dredging site.
- A28 Time of departure from dredging site has been recorded on the TPL form.
- A29 The bin-level, as indicated by the container vessel fore and aft bin sensors, has been entered into the TPL form comments section.
- A30 DMI was at the dredging site while Part A, Items 1-30, of the TPL checklist was reviewed and completed.
- A31 The container vessel draft (pressure) and bin level sensor values, as displayed by the container vessel monitoring software has been recorded on the TPL form. (This value must be noted approximately five (5) minutes after leaving the dredging site, while being towed, to account for any shifting / settling of the material in the container vessel).

PART B - While Underway to Open-Water Placement Site:

- B1 Container vessel draft is being monitored with container vessel monitoring software to detect sudden or gradual changes in draft.
- B2 In the vicinity of the Verrazzano-Narrows Bridge, radio and backup radios aboard the container vessel (if a scowman is used) have been checked and both are fully functional.
- B3 If the DMI is also a NMFS certified marine mammal/endangered species observer, observation and appropriate reporting is conducted.
- B4 Container vessel draft and/or bin level varies by less than one (1) foot, or by less than 12 points of pressure (draft only), from the value at the dredging site.
- B5 Container vessel does not appear to be listing.
- B6 Water behind container vessel has been observed, if possible, to ensure that no turbid water plumes are present.
- B7 A fixed reference position, such as a channel marker, has been used to ensure that the transport vessel DGPS and Container vessel DGPS positions agree.

- B8 Marine weather and sea conditions present and forecast to be present at the placement location are periodically monitored. The transport vessel captain, after consultation with the DMI, may decide to return to the dredging site based on an updated marine forecast.

PART C - In the Vicinity of the Designated Placement Location:

- C1 Water depths were continuously monitored (a reading observed at least every 5 seconds) with the transport vessel fathometer while navigating anywhere within the placement area boundary (transport vessel crew must also monitor water depths).
- C2 All water depths observed anywhere within the placement grid were at a depth of at least fifty (50) feet.
- C3 If depths less than fifty (50) feet were observed in the placement grid, the latitude, longitude and depth have been recorded in the TPL form comments section.
- C4 Container vessel radio control equipment operates without any problems.
- C5 Placement was initiated in the targeted grid cell and was coordinated with transport vessel crew.
- C6 Container vessel draft and/or bin level varies by less than 1.5 feet, or by less than twenty (20) points of pressure (draft only), from the value at the dredging site.
- C7 Container vessel draft information immediately prior to container vessel door opening has been recorded on the TPL form.
- C8 TPL form was completed using the container vessel monitoring software, or by hand if container vessel monitoring system malfunctions, within 30 minutes of container vessel door opening.
- C9 Container vessel monitoring equipment, transportation vessel navigation equipment, and all other equipment related to placement of dredged material worked without any problems.
- C10 All activities associated with placement of dredged materials appeared to be conducted in a safe manner.
- C11 Nothing occurred that might have resulted in incorrect placement of dredged material.
- C12 If a marine mammal or sea turtle was observed, the appropriate NMFS form was completed.
- C13 Any item in the checklist that received a "NO" answer was noted in the comments section of the TPL (letter and number).
- C14 TPL form submitted within two (2) hours of container vessel door, or hopper bin, opening, for HARS placement.
- C15 A copy of the TPL form, with checklist "NO" answer documentation in the comments section, has been signed by the DMI and placed in a file/folder to become part of a permanent record of the trip.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --